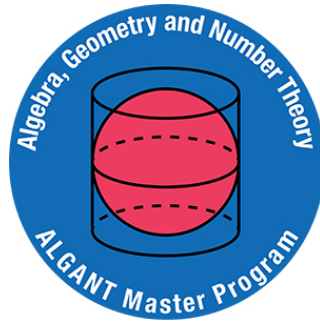




UNIVERSITÀ DEGLI STUDI DI MILANO
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THE COTANGENT COMPLEX IN DERIVED DIFFERENTIAL GEOMETRY

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Introduction

This thesis aims to study the notion of the cotangent complex in derived differential geometry, a framework that can be regarded as the natural setting for the study of quotients and intersections arising in differential geometry. It is worthwhile to briefly recall the history of the subject and to outline the motivations for this relatively recent field of research.

When one is introduced to smooth manifolds, the first thing one notices is the expressive power of the structure they encode. The condition of having smooth compatibilities allows us to use the full strength of calculus without worrying about losing continuity or differentiability. However, the category of smooth manifolds is not closed under intersections (formally, pullbacks) and quotients. To understand how we have arrived at solutions to these problems in differential geometry, it is useful to take a detour and see how algebraic geometry has faced similar issues.

Algebraic geometry began as the study of solutions to polynomial equations, but by the 1960s, through the advent of scheme theory, it had become something richer and better behaved (see [24]). Unlike differential geometry, scheme theory can handle intersections as part of the theory; however, if the intersection is not transverse, the complexity of singularities is still not fully captured intrinsically within the classical scheme-theoretic framework, one often has to appeal to homological methods. Moreover, even in this category, quotients do not always exist.

In the second half of the twentieth century, a solution to the quotient problem began to take shape, starting with the works of Giraud, Deligne, Mumford, and Artin. By introducing spaces which admitted internal symmetries, Stack theory solved the problem of taking quotients; however, the problem of singularities in intersections remained unresolved for quite some time.

Over the years, within the world of algebraic geometry, homotopical methods began to emerge as computational tools to address such difficulties, though initially without a clear geometric interpretation. This changed with the pioneering work of Toën and Vezzosi (see [59], [60], [61]), followed later by Lurie (see [34]). Nowadays, derived algebraic geometry has developed an extensive literature, and its methods have found applications in a wide range of areas, including mathematical physics and symplectic geometry.

At this point, one might expect that a similar development had taken place in differential geometry, but this is not the case. The first attempts to import these ideas from derived algebraic geometry can be found in the works of Spivak [56] and Joyce [28]. However, most of the work on derived differential geometry is fairly recent, with foundational papers published by Carchedi (e.g. [12], [9], [10]), Steffens (e.g. [57], [58]), and Nuiten [50]. At the current stage, much of the foundational framework is in place, but the gap with derived algebraic geometry has yet to be closed.

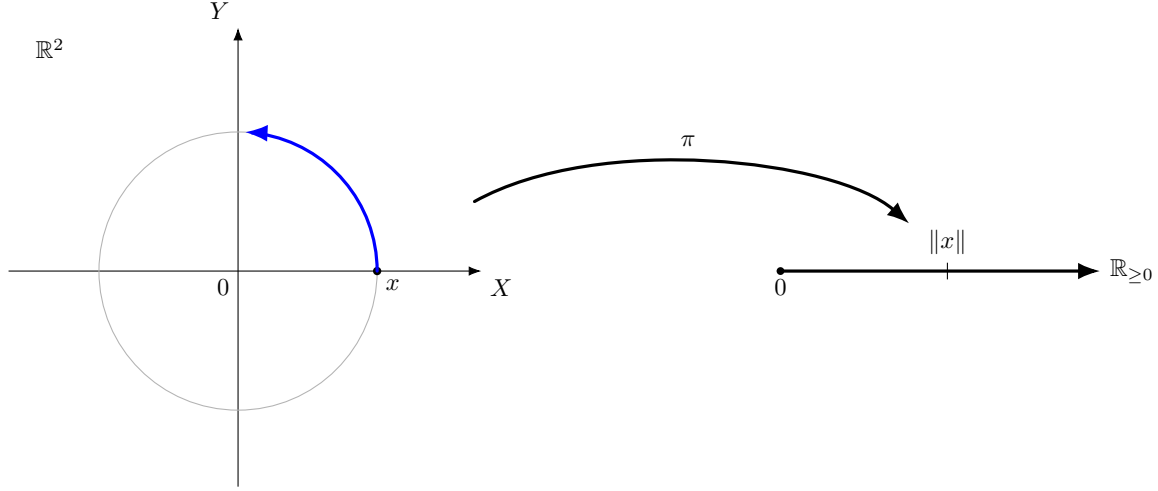
It is now time to try to intuitively approach the world of derived differential geometry, which is precisely the subject of this work. As we said, we are trying to handle singularities arising from quotients and from intersections. The first kind of singularity is treated within the framework of stacks, and the second within the framework of derived manifolds. Derived differential geometry united this to perspective within the ∞ -category of derived stacks.

Warning. To avoid technicalities in this introduction, we will slightly abuse terminology by using the terms “stacks”, “derived stacks”, and “derived manifolds” without further distinction. However, we want to make clear that the following serves to lay an intuition for the thesis. For example, we will not work with the entire ∞ -category of stacks, but rather with the full subcategory of differentiable stacks. Also the ∞ -category of derived manifold is too small for the kind of construction we want to do, so we will need to consider a *bigger* category.

Stacks

To understand stacks, we first need to understand the problem with usual quotients.

Consider $\mathbb{C} \simeq \mathbb{R}^2$ as a smooth manifold with the action of the circle group S^1 given by complex multiplication.



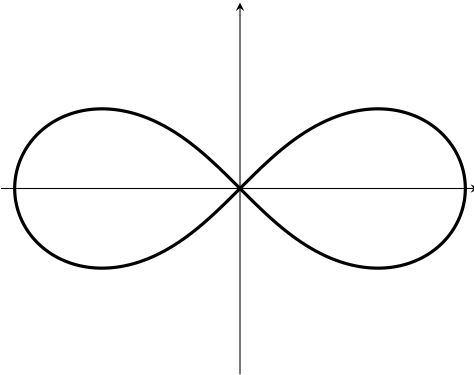
The orbits are circles centered at the origin and the singleton $\{0\}$. The quotient map $\pi : \mathbb{R}^2 \rightarrow \mathbb{R}/S^1 \simeq \mathbb{R}_{\geq 0}$ sends $x \in \mathbb{R}^2$ to its absolute value $\|x\|$. In particular, we see that $\mathbb{R}^2/S^1$ is not a smooth manifold. The issue lies at the cone point $\{0\}$: indeed, if we had considered the action of S^1 on $\mathbb{R}^2 \setminus \{0\}$, we would have obtained a smooth manifold diffeomorphic to the real line. What is happening is that, while the action of S^1 on $\mathbb{R}^2 \setminus \{0\}$ is free, at $\{0\}$ we have $z \cdot 0 = 0$ for every $z \in S^1$. When passing to the quotient $\mathbb{R}/S^1$, this self-identification of 0 is lost: we forget that under the action of S^1 , 0 is symmetric to itself. This is unsatisfactory, because we want to retain the internal symmetries of a point under a group action. The cone point shouldn't be like the other points on the half-line: it should be a *thick* point, where the *thickness* is given by all the ways in which S^1 identifies the point 0 with itself.

Stacks solve this problem precisely by allowing points to have internal structure. In other words, a point in a stack is not merely a point: it comes equipped with the data of its internal symmetries. For example, when taking the *stacky quotient* of the point pt by the action of a group G , the result is not just a point, but rather a space $\mathbb{B}G$, which can be described intuitively as a *thick* point, where the *thickness* is given by all the ways in which G identifies the point with itself.

To make this idea concrete, however, we have to move from sets to spaces, so that objects are no longer discrete and are allowed to have internal symmetries captured by the π_1 of the space. Our examples so far come from set theory, so we stop at π_1 ; however, in general, symmetries may themselves be identified, and this ambiguity is captured by the π_2 of the space, and so on.

Derived manifolds

Consider the smooth surfaces $z = (x^2 + y^2)^2 - 4(x^2 - y^2)$ and $z = 0$ in $\mathbb{R}^3$, where the first is a double saddle and the second is the x, y -plane. The intersection $0 = (x^2 + y^2)^2 - 4(x^2 - y^2)$ of these two surfaces is the Bernoulli lemniscate, which is not a smooth manifold.



This occurs because the intersection is not transverse. Derived manifolds provide a solution to this problem. More precisely, the ∞ -category of derived manifolds can be described as the “universal ∞ -category” into which the category of smooth manifolds embeds, in such a way that the embedding preserves transverse pullbacks while also adding the non-transverse ones.

Derived stacks

Finally, one is naturally interested in having both quotients and intersections. This is achieved through the ∞ -category of *derived stacks*. Conceptually, passing from derived manifolds to derived stacks is analogous to passing from smooth manifolds to stacks, namely by allowing internal symmetries.

We can summarize the situation with the following diagram:

$$\begin{array}{ccc} \text{Stack} & \longleftrightarrow & \text{DerivedStack} \\ \uparrow & & \uparrow \\ \text{Manifold} & \longleftrightarrow & \text{DerivedManifold} \end{array}$$

Here, moving from left to right corresponds to adding intersections, while moving from bottom to top corresponds to adding quotients. Part of this introduction was inspired by [2]; for a more extensive and expository discussion of the topic, we refer the reader to *loc. cit.*

Tangent and cotangent structures

Up to this point, we have avoided mentioning two of the main tools of derived geometry, if not *the* main tools, namely, the tangent and cotangent complexes. This thesis is not focused on the tangent complex, and we will not refer to it directly, although an instance of the relative tangent complex will appear in Subsection 3.5.3. Nevertheless, we find it worthwhile to say a few words about the tangent complex.

What behaves poorly in the presence of singularities in algebraic geometry, or simply fails to exist in differential geometry, is the classical notion of tangent space. Just as sets fail to capture internal symmetries, the tangent space fails to capture the complexity of singularities. In the same way that we pass from sets to spaces to better account for symmetries, in order to study the complexity of a singularity we should pass from the category of modules over a ring to the derived ∞ -category of modules over that ring. In other words, we need to replace the tangent space with its derived analogue, the *tangent complex*. In the same way that homotopy groups capture the complexity of identifications in a quotient, the homologies of the tangent complex capture the complexity of a singularity. Following this analogy, we expect that for regular points the tangent complex has homology only in degree zero, and indeed this is the case. Moreover, as we read in [2]:

The tangent structure could be encoded in a complex whose homology is concentrated in degree 0 if and only if the point is regular. The other homology groups are thus a reflection the singular structure of the point. If there exists non trivial positive homology groups, this means that some bad quotient was involved in the singular structure. Moreover, if the homology is non trivial for some $n > 0$, this means at least n bad quotients had to be involved to create the singularity. If there exist non trivial negative homology groups, this means that some bad intersection was involved in the singular structure. Moreover, if the homology is non trivial for some $n < 0$, this means at least n non transverse intersections had to be involved to create the singularity.

In sufficiently nice cases, the tangent complex is better understood as the dual of another object: the cotangent complex, which is the main object of this thesis. This module is interesting in its own right and has a long history closely related to deformation theory. The present thesis, however, will not focus on deformation theory; for a thorough and comprehensive introduction we refer to [41, Section 0.1.3]. Following the exposition of *loc. cit.*, we will briefly outline the types of problems studied in this subject and explain why the cotangent complex plays such a fundamental role in this context.

In the study of a scheme X_0 , one is sometimes interested in understanding the schemes that are “close” to X_0 in some sense. For example, suppose we have a morphism of schemes $X \rightarrow S$ and that X_0 is defined as the fiber over a point $s \in S$, i.e. $X_0 := X \times_S \{s\}$. One may then ask whether it is possible to deduce properties of other fibers $X_{s'} := X \times_S \{s'\}$ from those of X_0 . To establish such a relationship, additional assumptions are usually required: for instance, it is not reasonable to expect a connection between X_0 and $X_{s'}$ if s and s' lie in different connected components of S . Thus, some global condition on the morphism $X \rightarrow S$ or on S itself is needed.

To avoid these issues, one may restrict to the case $S = \mathbf{Spec}(R)$ for some ring R . In this setting, one is led to *formal deformation theory*, whose aim is to study schemes that are “infinitesimally close” to X_0 . In this thesis we will not explicitly use the language of formal deformation theory; however, we will discuss square-zero extensions, which provide a formal way of describing “infinitesimal thickenings”. Under suitable assumptions (see, for a formal treatment, [41, Section 17.3.1]), one can say that the cotangent complex “controls the deformation theory” of a

morphism. For a complete treatment of deformation theory and its relation to the cotangent complex we refer to other sources (see, for example, [41, Chapter 17]).

Contributions

This thesis is primarily devoted to presenting derived differential geometry, with the aim of introducing the cotangent complex and providing an application in the context of differentiable stacks. Most of the results discussed in the thesis have already appeared in the work of Nuiten [50], Carchedi [10, 12], and Steffens [57]. For these parts (see the Organization below), we have combined the three references, aiming to balance exposition and detail. Where necessary, we have added further arguments or modified the presentation. It should be understood that not all of the proofs reported here differ significantly from the original ones.

We then provide an application of the developed material. This application is motivated by the following observation: despite significant progress in the theory of stacks over manifolds, the definition of a tangent bundle in the sense of a tangent complex, as explained above, is not yet well established.¹ In the literature, one usually finds the definition of the relative tangent stack, constructed by choosing an atlas. However, this object falls short of capturing the complexity and structure of a tangent complex. In this thesis, we close this gap for representable submersions of differentiable stacks. First, we provide a definition that does not depend on the choice of an atlas. Then, we relate this definition to the tangent complex by proving that it coincides with the dual of the cotangent complex.

Motivated by the fact that for a smooth manifold M :

$$\Gamma(M, TM) \simeq \mathrm{Hom}_{\mathcal{C}^\infty(M)}(\Omega^1(M), \mathcal{C}^\infty(M)),$$

our first goal was to prove that, for a smooth manifold M , the module of differential 1-forms $\Omega^1(M)$ is its cotangent complex (the dual of the tangent complex). In this case, the tangent complex would be $\Gamma(M, TM)$. As discussed above, this is reasonable since smooth manifolds are highly regular geometric objects. Building on this, we then aimed to develop its relative version and to globalize it at the level of differentiable stacks.

Organization

The thesis is organized as follows. In Chapter 1, following the exposition of Bastiaan Cnossen [16], we introduce stacks, differentiable stacks, and the corresponding tangent bundle as presented in [16]. We conclude the chapter by providing a definition of the relative tangent bundle for a representable submersion that is independent of the choice of atlas, thus addressing the problem raised in [16, Remark 3.5.7].

In Chapter 2, we present the framework of derived differential geometry. We recall the theory of $\mathcal{C}^\infty$ -rings and introduce homotopically locally ringed spaces. We make precise the fact that the ∞ -category of derived manifolds is the universal one obtained by adjoining non-transverse pullbacks to the category of smooth manifolds. This result is due to the joint work of Steffens and Carchedi [12]. The chapter concludes with an introduction to derived stacks as presented in [50].

Chapter 3 is devoted to the study of modules, with the aim of defining the cotangent complex. More precisely, we first present classical and well-established results: the category of discrete modules, discrete square-zero extensions, and discrete derivations, providing proofs for those cases where we could not find a reference². We then present new material from Nuiten [50], Carchedi [10], and Steffens [57]: quasi-coherent modules and the cotangent complex for $\mathcal{C}^\infty$ -rings and derived stacks. This part combines the three references cited above and provides more details in selected proofs. The chapter concludes with an application, namely an explicit computation of the cotangent complex. More precisely, we relate the relative tangent bundle, as defined at the end of Chapter 1, with the cotangent complex, and we compute the underlying sheaf of the absolute cotangent complex of a differentiable stack.

Chapters 2 and 3 follow the same philosophy: we first provide an intuition for how things work in the classical (discrete) setting, and then use these results and ideas to see how they develop in the derived setting.

Notation and convention

We denote by **CRing** the category of ordinary commutative rings with unit. We denote by **Top** the category of topological spaces. Unless otherwise specified, every smooth manifold is assumed to be Hausdorff, second-countable, and finite-dimensional. Any action of a Lie group G on a smooth manifold M will be understood to be smooth.

¹See Nuiten [50, Corollary 6.4.33] for one approach to this problem.

²We do not claim that such proofs do not exist.

The symmetric group on n elements is denoted by Σ_n .

Regarding ∞ -categories, we adopt a model-independent approach, following the ongoing book project [15] on synthetic ∞ -category theory. However, since this book is not yet completed, we cannot rely on it entirely. In particular, foundational results will be taken from [37] and [40], choosing quasi-categories as our model. Moreover, every 1-category will be regarded as an ∞ -category from the outset, and we will never explicitly use the nerve construction. We will denote by $\mathbf{Cat}$ the category of 1-categories and by $\mathbf{Cat}_\infty$ the ∞ -category of ∞ -categories.

We denote by $\mathbf{Spc}$ the ∞ -category of spaces, by $\mathbf{Spt}$ the ∞ -category of spectra, and by $\mathbf{Spt}_{\geq 0}$ the ∞ -category of connective spectra.

We denote the internal Hom of a symmetric monoidal (∞ -)category $\mathcal{C}$ by $\underline{\mathrm{Hom}}_{\mathcal{C}}$.

Given a functor $F \rightarrow \mathbf{Cat}_\infty$, we denote by $\mathrm{Un}^{\mathrm{cc}}(F)$ and $\mathrm{Un}^{\mathrm{ct}}(F)$ its cocartesian and cartesian unstraightenings, respectively.

For an ∞ -category $\mathcal{C}$, we denote by $\mathrm{PSh}(\mathcal{C})$ the presheaf category (always assumed to be in spaces). We distinguish between two Yoneda functors:

$$y: \mathcal{C} \rightarrow \mathrm{PSh}(\mathcal{C}) \quad \text{and} \quad j: \mathcal{C} \rightarrow \mathrm{PSh}(\mathcal{C})^{\mathrm{op}},$$

where $y(X) = \mathrm{Hom}_{\mathcal{C}}(-, X)$ and $j(X) = \mathrm{Hom}_{\mathcal{C}}(X, -)$ for $X \in \mathcal{C}$.

Let $u: \mathcal{C} \rightarrow \mathcal{D}$ and $F: \mathcal{C} \rightarrow \mathcal{E}$ be functors between ∞ -categories. We denote the left Kan extension (when it exists) of F along u by $\mathrm{Lan}_u F$, and the right Kan extension (when it exists) of F along u by $\mathrm{Ran}_u F$.

For most of this thesis, in an attempt to remain self-contained, we avoid working explicitly with ∞ -operads, as the relevant theorems can often be formulated without directly referring to them. The drawback of this choice is that many of the relevant references in Chapter 3 are found in [40], where the language of ∞ -operads is used extensively.

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Chapter 1

Differentiable Stacks

This chapter aims to define the relative tangent bundle for differentiable stacks, as described in [16, Section 3.5]. In particular, we will introduce only the essential theory needed to formulate this definition.

For a broader treatment of stacks over manifolds, we refer the reader to both classical references (e.g. [13], [25], [7], [45]) to more recent developments involving ∞ -categorical techniques (e.g. [16], [49]).

In Section 1.1, we present some foundational definitions and properties of the theory of stacks over manifolds, focusing in particular on the notion of differentiable stacks. Section 1.2 establishes the connection between Lie groupoids (see Appendix E.1.1) and differentiable stacks. In Section 1.3, we introduce vector bundles over stacks, and in Subsection 1.3.1, we examine the relation between vector bundles over differentiable stacks and VB-groupoids. Finally, in Section 1.4, we define the relative tangent bundle and highlight the difficulties associated with the definition.

The main reference for this chapter is [16, Chapter II, Section 2].

1.1 Stacks

Let us denote with $\mathbf{Mfd}$ the category of smooth manifold and smooth maps. We give $\mathbf{Mfd}$ the open cover topology those turning it into a site. We denote by $\mathrm{Shv}(\mathbf{Mfd})$ the associated ∞ -topos (see Appendix F) of sheaves of spaces over $\mathbf{Mfd}$.

Definition 1.1.1. A *stack on $\mathbf{Mfd}$* is an object $\mathcal{X} \in \mathrm{Shv}(\mathbf{Mfd})$.

We will simply refer to stack on $\mathbf{Mfd}$ as *stack*. As already noticed in the introduction the category of stack should be an enhancement of the category of smooth manifold. In particular, we seek a fully faithful functor $\mathbf{Mfd} \hookrightarrow \mathrm{Shv}(\mathbf{Mfd})$. The easiest way to achieve this is through Yoneda embedding. Note that the site $(\mathbf{Mfd}, \mathrm{open})$ is subcanonical, i.e. the functor $\mathrm{Hom}_{\mathbf{Mfd}}(-, M) : \mathbf{Mfd} \rightarrow \mathbf{Set} \hookrightarrow \mathbf{Spc}$ is a sheaf. In the rest of the chapter we will identify a smooth manifold M with the sheaf $y(M)$ where y is the Yoneda embedding.

Definition 1.1.2. We say that a stack $\mathcal{X}$ is a *smooth manifold* if it lies in the essential image of the Yoneda embedding.

In Appendix F.2.1 we give a brief introduction to effective epimorphism in an ∞ -topos. In the case of $\mathrm{Shv}(\mathbf{Mfd})$ we can characterize the effective epimorphism as those maps admitting a *local section*:

Definition 1.1.3. Let M be a smooth manifold and let $f : \mathcal{X} \rightarrow M$ be a map of sheaves. We say that f *admits local sections* if there exists an open cover $\{U_i\}_{i \in I}$ of M such that the map $U_i \hookrightarrow M$ factors through f for each $i \in I$.

Proposition 1.1.4. *If M is a smooth manifold, then a map $f : \mathcal{X} \rightarrow M$ in $\mathrm{Shv}(\mathbf{Mfd})$ is an effective epimorphism if and only if it admits local sections.*

Proof. This is a special case of F.3.9. □

Remark 1.1.5. It often happens that, when we have a fully faithful inclusion $\mathcal{C} \hookrightarrow \mathcal{D}$, where $\mathcal{D}$ is constructed from $\mathcal{C}$ (for instance, $\mathbf{Mfd} \hookrightarrow \mathrm{Shv}(\mathbf{Mfd})$), we wish to *transfer properties of morphisms* from the category $\mathcal{C}$ to $\mathcal{D}$. This is typically achieved in the following way:

Let $f : X \rightarrow Y$ be a morphism in $\mathcal{D}$. For every morphism $g : Z \rightarrow Y$, with $Z \in \mathcal{C}$, such that the pullback $Z \times_Y X$ exists and belongs to $\mathcal{C}$, we say that f has a property P if the induced morphism $Z \times_Y X \rightarrow Z$ has property P (or some related property P').

One may require g to satisfy additional conditions, typically to ensure the existence of the pullback $Z \times_Y X$. An instance of this principle can be found in [16, Definition 2.4.1]; more broadly, we will adopt similar constructions throughout this thesis.

We will also see that when a left adjoint π_0 to the inclusion $\mathcal{C} \hookrightarrow \mathcal{D}$ is available (for instance in the derived setting), it is often convenient to *transfer properties* by requiring that the image under π_0 of a morphism satisfies the desired property.

Hopefully, the above discussion will assist us in presenting the definition of representable morphisms. Morally, we would like to say that a morphism between smooth manifolds is representable, and more generally, that a morphism of stacks is representable if its base change along a map from a smooth manifold yields a morphism of smooth manifolds. However, to make this precise, we must first address the fact that pullbacks do not generally exist in the site **Mfd**. Fortunately, differential geometry provides a well-behaved notion of pullback when we restrict to the class of smooth submersions.

Definition 1.1.6. A morphism of stack $f : \mathcal{X} \rightarrow \mathcal{Y}$ is called:

1. *representable submersion* if for any smooth manifold M and any morphism $M \rightarrow \mathcal{Y}$, the fiber product $M \times_{\mathcal{Y}} \mathcal{X}$ is a smooth manifold and the induced map $M \times_{\mathcal{Y}} \mathcal{X} \rightarrow M$ is a submersion. If the map $M \times_{\mathcal{Y}} \mathcal{X} \rightarrow M$ is also surjective we say that f is *representable surjective submersion*;
2. *representable* if for any representable submersion $M \rightarrow \mathcal{Y}$ from a smooth manifold M , the pullback $M \times_{\mathcal{Y}} \mathcal{X}$ is a smooth manifold.

The achievement of getting that every morphism $M \rightarrow N$ between smooth manifold is representable comes with the drawback that if $f : \mathcal{X} \rightarrow \mathcal{Y}$ is a morphism of stack and there are no representable submersion from smooth manifolds to $\mathcal{Y}$ then f is automatically representable.

Remark 1.1.7. In analogy with the world of smooth manifolds, where a smooth submersion $M \rightarrow N$ is surjective if and only if it admits local sections, we have from Proposition 1.1.4 that a representable submersion $\mathcal{X} \rightarrow \mathcal{Y}$ of stacks is a representable surjective submersion if and only if it is an effective epimorphism.

Definition 1.1.8 (Open substack). Let $j : \mathcal{U} \rightarrow \mathcal{X}$ be a representable submersion of stacks. We say that j exhibits $\mathcal{U}$ as an *open substack* of $\mathcal{X}$ if for every smooth manifold M and any morphism $M \rightarrow \mathcal{X}$, the induced map of smooth manifolds $M \times_{\mathcal{X}} \mathcal{U} \rightarrow M$ is an open embedding.

As we now see, a nice feature of stacks is that representability is a local condition:

Lemma 1.1.9 (cf. [16, Lemma 2.1.8]). *Let $\mathcal{X}$ be a stack and assume that there exists a collection of open substack $\mathcal{U} = \{j_\alpha : U_\alpha \hookrightarrow \mathcal{X}\}_{\alpha \in I}$ of $\mathcal{X}$ such that the map $\coprod_{\alpha \in I} U_\alpha \rightarrow \mathcal{X}$ is an effective epimorphism. If each U_α is a smooth manifold, then so is $\mathcal{X}$.*

Proof. By definition of an effective epimorphism (see Definition E.1.9) we have that $\mathcal{X}$ is equivalent to the colimit of the Čech nerve (see Definition E.1.6) of the map $\coprod_{\alpha \in I} U_\alpha \rightarrow \mathcal{X}$. Since each map $j_\alpha : U_\alpha \rightarrow \mathcal{X}$ is assumed to be a representable submersion, we have that the iterated pullbacks $\mathcal{U}_{\alpha_1} \times_{\mathcal{X}} \cdots \times_{\mathcal{X}} \mathcal{U}_{\alpha_n}$ are smooth manifolds. In particular, the Čech nerve construction $\check{C}(\mathcal{U})$ factors through the category of smooth manifold. Now call M the colimit of such diagram in **Top**, i.e.

$$M = \operatorname{coeq} \left(\coprod_{\alpha \neq \beta} U_\alpha \times_{\mathcal{X}} U_\beta \rightrightarrows \coprod_{\alpha} U_\alpha \right).$$

Notice that each map $U_\alpha \hookrightarrow M$ is an open embedding, in particular M admits an open cover of smooth manifolds U_α . Since the smooth structure are compatible on intersections the open cover $\{U_\alpha\}_{\alpha \in I}$ exhibit a smooth atlas for a differential structure on M . It follows that M is a colimit in **Mfd**. Consider $\mathcal{F}$ a stack and recall that by Proposition F.3.7 we have:

$$\operatorname{Hom}_{\operatorname{PSh}(\mathbf{Mfd})}(y(M), \mathcal{F}) \xrightarrow{\sim} \lim_{\Delta_{op}} \operatorname{Hom}_{\operatorname{PSh}(\mathbf{Mfd})}(\check{C}(\mathcal{U})_n, \mathcal{F}).$$

However the right term is also equivalent to $\operatorname{Hom}_{\operatorname{PSh}(\mathbf{Mfd})}(\mathcal{X}, \mathcal{F})$. By Yoneda's Lemma we conclude that M is equivalent to $\mathcal{X}$, proving the claim. $\square$

Corollary 1.1.10 (cf. [16, Proposition 2.1.9]). *Let $\mathcal{X} \rightarrow \mathcal{Y}$ be a morphism of stacks and assume that $\mathcal{Y}$ admits an open cover $\{V_\alpha\}_\alpha$ such that each of the pullback stacks $V_\alpha \times_{\mathcal{Y}} \mathcal{X}$ is a smooth manifold. Then $\mathcal{X}$ is a smooth manifold.*

Proof. The maps $V_\alpha \times_{\mathcal{Y}} \mathcal{X} \rightarrow \mathcal{X}$ are pullbacks of the open embedding $V_\alpha \hookrightarrow \mathcal{Y}$ and thus are open substacks of $\mathcal{X}$. Since in an ∞ -topos effective epimorphism are stable under base change F.2.7 we have that $\coprod V_\alpha \times_{\mathcal{Y}} \mathcal{X} \rightarrow \mathcal{X}$ is an effective epimorphism. By lemma 1.1.9 we conclude. $\square$

Proposition 1.1.11 (cf. [16, Proposition 2.1.10]). *Consider the pullback diagram*

$$\begin{array}{ccc} \mathcal{X}' & \longrightarrow & \mathcal{X} \\ f' \downarrow & \lrcorner & \downarrow f \\ \mathcal{Y}' & \xrightarrow{g} & \mathcal{Y} \end{array}$$

of stacks.

1. *If f is a representable submersion, then f' is representable submersion.*
2. *If g is a representable submersion and f representable then f' is representable.*
3. *If g is an effective epimorphism and f' is representable submersion, then f is a representable submersion.*
4. *If g is a representable surjective submersion and f' is a representable submersion, then f is representable.*

Proof. i) and ii) are immediate from definition and pasting law for pullback. For iii) consider M a smooth manifold, $\{U_i\}_{i \in I}$ an open covering and a morphism $M \rightarrow \mathcal{Y}$. By Proposition F.2.7 we have that the map $M \times_{\mathcal{Y}'} \mathcal{Y}' \rightarrow M$ is an effective epimorphism. By Proposition 1.1.4 it follows that every map $U_i \rightarrow M$ factors as follow:

$$\begin{array}{ccccc} & & M \times_{\mathcal{Y}'} \mathcal{Y}' & \longrightarrow & \mathcal{Y}' \\ & \nearrow & \downarrow g' & & \downarrow g \\ U_i & \hookrightarrow & M & \longrightarrow & \mathcal{Y} \end{array}$$

In light of Corollary 1.1.10 to prove that $\mathcal{X} \times_{\mathcal{Y}} M$ is a smooth manifold it is enough to prove that the fiber product $U_i \times_M (\mathcal{X} \times_{\mathcal{Y}} M) \simeq U_i \times_{\mathcal{Y}} \mathcal{X}$ is a smooth manifold. Fix i and let us denote by s the morphism $U_i \rightarrow \mathcal{Y}'$, and set $N := U_i \times_{\mathcal{Y}'} \mathcal{X}'$. Consider the following pullback squares:

$$\begin{array}{ccccc} N & \longrightarrow & \mathcal{X}' & \longrightarrow & \mathcal{X} \\ f'' \downarrow & \lrcorner & f' \downarrow & \lrcorner & \downarrow f \\ U_i & \xrightarrow{s} & \mathcal{Y}' & \xrightarrow{g} & \mathcal{Y} \end{array}$$

Since f' is a representable submersion it follow that $N \simeq \mathcal{X} \times_{\mathcal{Y}} U_i$ is a smooth manifold and the map $f'' : N \rightarrow U_i$ is a submersion. This proves that f is representable submersion.

For part iv), let $M \rightarrow \mathcal{Y}$ be a representable submersion from some smooth manifold M . We have to show that the pullback $M \times_{\mathcal{Y}} \mathcal{X}$ is again a smooth manifold. By pulling back the whole situation along $M \rightarrow \mathcal{Y}$, we obtain the following commutative diagram:

$$\begin{array}{ccccc} & & \mathcal{X}' & \longrightarrow & \mathcal{X} \\ & \nearrow & \downarrow & & \downarrow \\ (M \times_{\mathcal{Y}} \mathcal{X}) \times_{\mathcal{X}'} \mathcal{X}' & \longrightarrow & M \times_{\mathcal{Y}} \mathcal{X}' & \longrightarrow & M \times_{\mathcal{Y}} \mathcal{X} \\ \downarrow h' & & \downarrow & & \downarrow \\ & \nearrow & \mathcal{Y}' & \xrightarrow{h} & \mathcal{Y} \\ M \times_{\mathcal{Y}} \mathcal{Y}' & \xrightarrow{k} & M & \longrightarrow & \mathcal{Y} \end{array}$$

Denote $(M \times_{\mathcal{Y}} \mathcal{X}) \times_{\mathcal{X}} \mathcal{X}'$ by N , $M \times_{\mathcal{Y}} \mathcal{Y}'$ by M' and $\mathcal{X} \times_{\mathcal{Y}} M$ by $\mathcal{X}''$. By applying the pasting lemma for pullbacks, we find ourselves in the situation of the following cartesian square

$$\begin{array}{ccc} N & \xrightarrow{k'} & \mathcal{X}'' \\ h' \downarrow & & \downarrow k \\ M' & \xrightarrow{k} & M \end{array}$$

in $\mathrm{Shv}(\mathbf{Mfd})$, where the stacks M, M' and N are all smooth manifolds and k is a surjective submersion. We have to show that $\mathcal{X}''$ is a smooth manifold as well.

As k is a surjective smooth submersion, it follows from [16, Lemma C.10] that g admits local sections: there exists an open cover $\{U_i\}_{i \in I}$ of M such that for every $i \in I$ there is a smooth map $s_i : U_i \rightarrow M'$ such that the composite $k \circ s_i : U_i \rightarrow M$ is equal to the inclusion of U_i into M . By Corollary 1.1.10, it suffices to show that each of the pullback stacks $\mathcal{X}_i := \mathcal{X}'' \times_M U_i$ is a smooth manifold. By base changing the whole situation along the map $U_i \hookrightarrow M$, we may thus assume that the map $k : M' \twoheadrightarrow M$ admits a section $s : M \rightarrow M'$.

We may now define a map $t = (s \circ k, \mathrm{id}) : \mathcal{X}_i \rightarrow N = M' \times_M \mathcal{X}_i$. Since it is a section of the map $k' : N \rightarrow \mathcal{X}_i$, it follows that the stack $\mathcal{X}$ is a retract in $\mathrm{Shv}(\mathbf{Mfd})$ of the smooth manifold N . It follows from idempotent completeness of $\mathbf{Mfd}$, [16, Corollary C.5], that $\mathcal{X}_i$ is itself a smooth manifold, finishing the proof. $\square$

Stacks are, in general, too broad for the kinds of constructions we aim to perform. In particular, we would like to transfer certain standard constructions from differential geometry, such as the tangent bundle, directly to stacks. To achieve this, we must restrict our attention to stacks that are *presented* by smooth manifolds, in a way analogous to how an atlas presents a differentiable structure on a manifold.

Definition 1.1.12 (Differentiable stack). Let $\mathcal{X}$ be a stack on $\mathbf{Mfd}$. A *representable atlas* for $\mathcal{X}$ is a smooth manifold M together with a representable surjective submersion $p : M \rightarrow \mathcal{X}$. A stack which admits a representable atlas is called a *differentiable stack*. We let

$$\mathbf{DiffStk} \subseteq \mathrm{Shv}(\mathbf{Mfd})$$

denote the full subcategory spanned by the differentiable stacks.

For a list of nice properties of differentiable stack we refer to [16, Section II,2.2]. We decided to select two of these properties to show the good behavior of differentiable stack.

Lemma 1.1.13 (cf. [16, Lemma 2.2.3]). *Let $f : \mathcal{X} \rightarrow \mathcal{Y}$ be a representable morphism, and assume that $M \twoheadrightarrow \mathcal{Y}$ is a representable atlas for $\mathcal{Y}$. Then the pullback $N := M \times_{\mathcal{Y}} \mathcal{X} \rightarrow \mathcal{X}$ is a representable atlas for $\mathcal{X}$.*

Proof. The map $N \rightarrow \mathcal{X}$ is a representable submersion and effective epimorphism since it is a base change of the map $M \twoheadrightarrow \mathcal{Y}$ (Proposition 1.1.11). Furthermore, the assumption that f is representable submersion guarantees that N is a smooth manifold. $\square$

Proposition 1.1.14 (cf. [16, Corollary 2.2.8]). *Let $\mathcal{X}$ be a differentiable stack. Then any morphism $N \rightarrow \mathcal{X}$ from a smooth manifold N is representable.*

Proof. Let $M \twoheadrightarrow \mathcal{X}$ be representable atlas for $\mathcal{X}$. Since any morphism of smooth manifold is representable, by Lemma 1.1.11 it suffices to show that the pullback $M \times_{\mathcal{X}} N$ is a smooth manifold. This follow by definition of representable submersion. $\square$

1.2 Lie groupoids and differentiable stacks

The theory of groupoids, and in particular of Lie groupoids (see Appendix E), is closely related to the theory of differentiable stacks. These tools provide a way to construct differentiable stacks from classical constructions (e.g., the action of a Lie groupoid on a manifold). More than that, we will see that there is an equivalence between the ∞ -category of differentiable stacks equipped with a choice of atlas and the category of Lie groupoids.

This is a subtle point, since a given differentiable stack may admit different atlases. One way to eliminate this ambiguity is to consider Lie groupoids up to *Morita equivalence*. Although we will not develop this notion in the present thesis, the interested reader is referred to [16, Section 2.3.1 and Appendix D.3].

Definition 1.2.1 (Classifying stack). Let $\mathcal{G} = (\mathcal{G}_1 \rightrightarrows \mathcal{G}_0)$ be a Lie groupoid. We define its *classifying stack* $\mathbb{B}\mathcal{G} \in \mathbf{Shv}(\mathbf{Mfd})$ as the geometric realization of its nerve (see Definition E.1.4 and Remark E.1.16) in $\mathbf{Shv}(\mathbf{Mfd})$:

$$\mathbb{B}\mathcal{G} := \operatorname{colim}_{n \in \Delta^{op}} N\mathcal{G}_n$$

If some kind of universal justice exists this should be a differentiable stack. This is indeed the case.

Lemma 1.2.2 (cf. [16, Lemma 2.3.3]). *Let $\mathcal{G}$ be a Lie groupoid. Then the canonical map $\mathcal{G}_0 \rightarrow \mathbb{B}\mathcal{G}$ is a representable atlas for $\mathbb{B}\mathcal{G}$. In particular, $\mathbb{B}\mathcal{G}$ is a differentiable stack.*

Proof. Since $\mathcal{G}_0$ is a smooth manifold, it is enough to prove that the map $\mathcal{G}_0 \rightarrow \mathbb{B}\mathcal{G}$ is a representable submersion. By Proposition 1.1.11, we can verify representability after pulling back along an effective epimorphism.

Now observe that, by Corollary F.2.5, groupoid objects in an ∞ -topos are effective. In particular, the map $\mathcal{G}_0 \rightarrow \mathbb{B}\mathcal{G}$ is itself an effective epimorphism. Hence, to prove that $\mathcal{G}_0 \rightarrow \mathbb{B}\mathcal{G}$ is a representable submersion, it suffices to show that the map

$$\operatorname{pr}_1 : \mathcal{G}_0 \times_{\mathbb{B}\mathcal{G}} \mathcal{G}_0 \rightarrow \mathcal{G}_0$$

is a representable submersion. We note that this morphism is equivalent to the source map $s : \mathcal{G}_1 \rightarrow \mathcal{G}_0$ (see Proposition F.2.3), which is a representable submersion by assumption. This concludes the proof. $\square$

Example 1.2.3 (Classifying stack of Lie group). Every Lie group G gives rise to a one-object Lie groupoid $G \rightrightarrows \operatorname{pt}$ and thus we can consider the associated differentiable stack $\mathbb{B}G$ called *classifying stack of G* . Notice that in this case the isotropy group of $\mathbb{B}G$ is G (see Remark F.2.4).

We can now explain how differentiable stack added quotients to our original category of manifold.¹

Example 1.2.4 (Global quotient stack). Consider a manifold M and G a Lie group acting on M . We can construct a Lie groupoid $G \ltimes M = (G \times M \rightrightarrows M)$ called the *action groupoid* (see Example E.1.13). We use the suggestive notation $M//G$ for the associated differentiable stack $\mathbb{B}(G \ltimes M)$ and we call it the *quotient stack of M by G* . Since the action groupoid $G \ltimes M$ is functorial in M , we obtain a functor

$$-//G : \mathbf{Mfd}_G \rightarrow \mathbf{DiffStk}$$

Where $\mathbf{Mfd}_G$ denotes the category of smooth G -manifold. Note that $\operatorname{pt}//G$ for the point is precisely the classifying stack $\mathbb{B}G$ of G .

In the introduction to the thesis, we discussed the idea that quotients should be allowed to have *internal symmetries*. We now see how Example 1.2.4 precisely illustrates this principle in the case where the action of G is proper. In general, given a differentiable stack $\mathcal{X}$, each point carries additional structure. More precisely, we have the following pullback squares:

$$\begin{array}{ccc} G_x & \longrightarrow & \operatorname{pt} \\ \downarrow & \lrcorner & \downarrow x \\ \operatorname{pt} & \xrightarrow{x} & \mathcal{X} \end{array}$$

where G_x is a Lie group (see [16, Lemma 3.4.2]). Now consider the case $\mathcal{X} \simeq M//G$, where a Lie group G acts properly on a smooth manifold M . In this context, the nontriviality of G_x reflects the failure of the action to be free. Indeed, one expects that when the action of G is free, there are no internal symmetries (i.e., $G_x \simeq 0$), and in that case the quotient stack $M//G$ is equivalent to the smooth manifold M/G ordinary quotient. This intuition is formalized in the following proposition.

Proposition 1.2.5. *Let M be a manifold, and let G be a Lie group acting properly on M . If the action is free then:*

$$M//G \simeq M/G$$

Proof. This is a consequence of [16, Proposition 3.4.5] together with [16, Example 3.3.2]. $\square$

¹With the right notion of separation one can prove that every separated differentiable stack locally looks like such quotients, see [16, Theorem 3.7.2]

Definition 1.2.6. We denote by $\text{Alt}^{\text{rep}}(\text{Shv}(\mathbf{Mfd})) \subseteq \text{Fun}([1], \text{Shv}(\mathbf{Mfd}))$ the full subcategory of the arrow of $\text{Shv}(\mathbf{Mfd})$ spanned by the representable atlases $M \rightarrow \mathcal{X}$ in the sense of Definition 1.1.12.

Proposition 1.2.7. *Sending a Lie groupoid $\mathcal{G}$ to the representable atlas $\mathcal{G}_0 \rightarrow \mathbb{B}\mathcal{G}$ defines an equivalence of ∞ -categories $\text{LieGrpd} \xrightarrow{\sim} \text{Alt}^{\text{rep}}(\text{Shv}(\mathbf{Mfd}))$*

Proof. This proof is verbatim from [16, proof of Proposition 2.3.7]. Recall from Proposition F.2.3 the equivalence

$$\text{Grpd}(\text{Shv}(\mathbf{Mfd})) \simeq \text{Atl}(\text{Shv}(\mathbf{Mfd}))$$

between the ∞ -category of groupoid objects in $\text{Shv}(\mathbf{Mfd})$ and the ∞ -category of atlases (i.e. effective epimorphisms) in $\text{Shv}(\mathbf{Mfd})$. This equivalence is given by sending a groupoid object $\mathcal{G}$ to the effective epimorphism $\mathcal{G}_0 \rightarrow \mathbb{B}\mathcal{G}$, and its inverse sends an effective epimorphism to its Čech nerve. We claim that this equivalence restricts to the desired equivalence

$$\text{LieGrpd} \simeq \text{Atl}^{\text{rep}}(\text{Shv}(\mathbf{Mfd}))$$

of full subcategories. By Lemma 1.2.2, the map $\mathcal{G}_0 \rightarrow \mathbb{B}\mathcal{G}$ is indeed a representable atlas if $\mathcal{G}$ is a Lie groupoid. Conversely, if $p: M \rightarrow \mathcal{X}$ is a representable atlas, its Čech nerve $\check{C}(p)$ is given at level $[n] \in \Delta$ by the n -fold fiber product

$$\check{C}(p)_n \simeq M \times_{\mathcal{X}} M \times_{\mathcal{X}} \cdots \times_{\mathcal{X}} M,$$

which is a smooth manifold by the assumptions that M is a manifold and that the map p is a representable submersion. It follows that $\check{C}(p)$ is the nerve of a Lie groupoid $\mathcal{G} = (M \times_{\mathcal{X}} M \rightrightarrows M)$, finishing the proof. $\square$

Corollary 1.2.8 (cf. [16, Corollary 2.3.8]). *Every differentiable stack $\mathcal{X}$ is equivalent to the classifying stack $\mathbb{B}\mathcal{G}$ of some Lie Groupoid $\mathcal{G}$*

Proof. This is a consequence of Proposition 1.2.7: given an atlas $M \rightarrow \mathcal{X}$, there exists a Lie groupoid $\mathcal{G}$ such that $M \simeq \mathcal{G}_0$ and $\mathcal{X} \simeq \mathbb{B}\mathcal{G}$. $\square$

Warning 1.2.9 (cf. [16, Warning 2.3.9]). Let $\mathcal{G}$ and $\mathcal{H}$ be Lie groupoids. Then Proposition 1.2.7 makes it clear that a morphism of stacks $\mathbb{B}\mathcal{G} \rightarrow \mathbb{B}\mathcal{H}$ is not, in general, equivalent to a morphism of Lie groupoids $\mathcal{G} \rightarrow \mathcal{H}$. What is true, however, is that a morphism of stacks $\mathbb{B}\mathcal{G} \rightarrow \mathbb{B}\mathcal{H}$ *together* with a map $\mathcal{G}_0 \rightarrow \mathcal{H}_0$ compatible with the canonical atlases $\mathcal{G}_0 \rightarrow \mathbb{B}\mathcal{G}$ and $\mathcal{H}_0 \rightarrow \mathbb{B}\mathcal{H}$ is equivalent to a morphism of Lie groupoids $\mathcal{G} \rightarrow \mathcal{H}$.

1.3 Vector bundles

We now aim to extend the notion of a vector bundle over a manifold to the setting of stacks. To that end, we recall that one of the historical motivations for introducing stacks was to find geometric objects whose points classify certain constructions. It turned out that most of these *classifying objects* could not be constructed internally within a given geometric context (e.g., differential geometry or scheme theory). However, it was possible to prove that they are sheaves over the relevant geometric site, allowing mathematicians to work with them without requiring an explicit internal construction. In some cases, explicit descriptions of such objects were later found.

One of the natural questions that arose was to find a classifying object for principal $\mathcal{G}$ -bundles, where $\mathcal{G}$ is a Lie groupoid. It was shown that the appropriate classifying object is the classifying stack $\mathbb{B}\mathcal{G}$ of the groupoid $\mathcal{G}$ (see [16, Section II.2.3.3]).

Returning to our main goal of defining vector bundles, recall that there is an equivalence between isomorphism classes of vector bundles and those of principal $\text{GL}(n)$ -bundles (see [46, Section 55]). Following the discussion above, it is therefore natural to use the classifying object $\mathbb{B}\text{GL}(n)$ to formulate the following definition.

Definition 1.3.1 (Vector bundle). Let $\mathcal{X}$ be a stack on $\mathbf{Mfd}$. For $n \geq 0$, we define an *n -dimensional vector bundle* π on $\mathcal{X}$ to be a morphism of stacks $\pi: \mathcal{X} \rightarrow \mathbb{B}\text{GL}(n)$. The *total space* $\mathcal{E}_\pi$ of the bundle π is defined as the following pullback of stack:

$$\begin{array}{ccc} \mathcal{E}_\pi & \longrightarrow & \mathbb{R}^n // \text{GL}(n) \\ \downarrow & \lrcorner & \downarrow \\ \mathcal{X} & \xrightarrow{\pi} & \mathbb{B}\text{GL}(n) \end{array}$$

We will often abuse notation by referring to a map $\mathcal{E} \rightarrow \mathcal{X}$ as a vector bundle, leaving the classifying map $\pi : X \rightarrow \mathbb{B}\mathrm{GL}(n)$ implicit. We denote the groupoid of n -dimensional vector bundles over $\mathcal{X}$ by

$$\mathrm{Vect}_n^{\simeq}(\mathcal{X}) := \mathrm{Hom}_{\mathrm{Shv}(\mathbf{Mfd})}(\mathcal{X}, \mathbb{B}\mathrm{GL}(n)).$$

More generally, we define

$$\mathrm{Vect}^{\simeq}(\mathcal{X}) := \mathrm{Hom}_{\mathrm{Shv}(\mathbf{Mfd})}\left(\mathcal{X}, \prod_{n=0}^{\infty} \mathbb{B}\mathrm{GL}(n)\right).$$

This construction determines a limit-preserving functor

$$\mathrm{Vect}^{\simeq} : \mathrm{Shv}(\mathbf{Mfd})^{\mathrm{op}} \rightarrow \mathrm{Grpd}.$$

Construction 1.3.2. Define the category **VB** of vector bundles as the category whose:

- objects are couples smooth manifold (E, M) together with a morphism $p : E \rightarrow M$ which p exhibits E as a vector bundle over the smooth manifold M .
- morphisms $(F, N) \rightarrow (E, M)$ are commutative diagrams of the form:

$$\begin{array}{ccc} F & \xrightarrow{g} & E \\ q \downarrow & & \downarrow p \\ N & \xrightarrow{f} & M \end{array}$$

where g is a morphism of vector bundles covering the smooth map f .

The induced functor $F : \mathbf{VB} \rightarrow \mathbf{Mfd}$, which sends a vector bundle $p : E \rightarrow M$ to its base M , is a Cartesian fibration. To see this, observe that for any vector bundle $p : E \rightarrow M$ and any smooth map $f : N \rightarrow M$, the pullback vector bundle $f^*E \rightarrow N$ together with the canonical commutative square covering f , defines an F -Cartesian lift of f . We denote the straightening of the cartesian fibration $\mathbf{VB} \rightarrow \mathbf{Mfd}$ by $\mathrm{Vect} : \mathbf{Mfd}^{\mathrm{op}} \rightarrow \mathrm{Cat}_{\infty}$.

Remark 1.3.3. Notice that Vect satisfies descent with respect to open covers. More precisely, let M be a smooth manifold and let $\{U_i\}_{i \in I}$ be an open cover such that each U_i is equipped with a vector bundle $p_i : E_i \rightarrow U_i$. If the p_i agree on pairwise intersections and satisfy the cocycle condition, then they determine a global vector bundle $p : E \rightarrow M$, and conversely, every vector bundle over M can be locally described in this way.

Thanks to the above observation together with Theorem F.3.11, the functor Vect extends uniquely (up to equivalence) to a limit preserving functor on $\mathrm{Shv}(\mathbf{Mfd})^{\mathrm{op}}$. With a slight abuse of notation, we will continue to denote this extended functor by $\mathrm{Vect} : \mathrm{Shv}(\mathbf{Mfd})^{\mathrm{op}} \rightarrow \mathrm{Cat}_{\infty}$.

Definition 1.3.4 (Category of vector bundles). For a stack $\mathcal{X}$ on $\mathbf{Mfd}$, we refer to the resulting category $\mathrm{Vect}(\mathcal{X})$ as the *category of vector bundles over $\mathcal{X}$* .

Since the functor $(-)^{\simeq} : \mathrm{Cat} \rightarrow \mathrm{Grpd}$ which assigns to an ordinary category $\mathcal{C}$ its underlying groupoid $\mathcal{C}^{\simeq}$, is a right adjoint (and thus preserves limits), we obtain a natural equivalence of groupoids:

$$\mathrm{Vect}(\mathcal{X})^{\simeq} \simeq \mathrm{Vect}^{\simeq}(\mathcal{X}).$$

Remark 1.3.5. The usual homotopy invariance of vector bundles also holds in the context of separated differentiable stacks (see [16, Section II.2.5.1] for such a result, and [16, Definition 3.3.1] for the definition of separated stacks).

1.3.1 VB-groupoids

Given a Lie groupoid $\mathcal{G}$, we would like to have a way to construct vector bundles over $\mathbb{B}\mathcal{G}$ starting from some structures over $\mathcal{G}$. This can be achieved using VB-groupoid, which can be seen as the vector bundles analogue for Lie groupoids.

Definition 1.3.6 (VB-groupoid, [23]). Given a Lie Groupoid $\mathcal{G}$, a *VB-groupoid over $\mathcal{G}$* is a morphism $\pi : \mathcal{E} \rightarrow \mathcal{G}$ of Lie groupoid such that the maps $\pi_0 : \mathcal{E}_0 \rightarrow \mathcal{G}_0$ and $\pi_1 : \mathcal{E}_1 \rightarrow \mathcal{G}_1$ are vector bundles and the structure maps of $\mathcal{E}$ are morphisms of vector bundles.

Definition 1.3.7 (Cartesian morphism of Lie groupoid). A morphism $f : \mathcal{H} \rightarrow \mathcal{G}$ of Lie groupoid is called *cartesian* if the commutative squares

$$\begin{array}{ccc} \mathcal{H}_1 & \xrightarrow{s} & \mathcal{H}_0 \\ f_1 \downarrow & & \downarrow f_0 \\ \mathcal{G}_1 & \xrightarrow{s} & \mathcal{G}_0 \end{array} \quad \text{and} \quad \begin{array}{ccc} \mathcal{H}_1 & \xrightarrow{t} & \mathcal{H}_0 \\ f_1 \downarrow & & \downarrow f_0 \\ \mathcal{G}_1 & \xrightarrow{t} & \mathcal{G}_0 \end{array}$$

are pullback squares.²

Notation. For the next proof, we denote by $\Gamma_I^{\text{cc}}(E)$ the cocartesian sections of a cocartesian fibration $p : E \rightarrow I$.

Lemma 1.3.8 (cf. [16, Lemma 2.5.11.]). *Let $\pi : \mathcal{E} \rightarrow \mathcal{G}$ be a VB-groupoid, and assume that the morphism π is Cartesian. Then the induced map $\mathbb{B}\pi : \mathbb{B}\mathcal{E} \rightarrow \mathbb{B}\mathcal{G}$ is naturally a vector bundle of differentiable stacks.*

Proof. Consider the functor

$$H : \Delta \xrightarrow{(N\mathcal{G})^{\text{op}}} \mathbf{Mfd}^{\text{op}} \xrightarrow{\text{Vect}} \text{Cat}_{\infty}$$

and recall that

$$\lim H \simeq \Gamma_{\Delta}^{\text{cc}}(\text{Un}^{\text{cc}}(H)).$$

By passing to nerves, the groupoid map $\pi : \mathcal{E} \rightarrow \mathcal{G}$ induces a morphism $N\pi : N\mathcal{E} \rightarrow N\mathcal{G}$ of simplicial manifolds. As π is cartesian, by Lemma F.2.6 $N\pi$ is a cartesian natural transformation of functors $\Delta^{\text{op}} \rightarrow \text{Diff}$. In particular, each $N\mathcal{E}_n$ is a vector bundle over $N\mathcal{G}_n$. It follows that the simplicial map $N\pi$ defines a section of $\text{Un}^{\text{cc}}(H)$, and thus determines an object in the limit $\lim H \simeq \lim_{[n] \in \Delta} \text{Vect}(\mathcal{G}_n)$.

Since the assignment $\mathcal{X} \mapsto \text{Vect}(\mathcal{X})$ satisfies descent, we see that this limit is $\text{Vect}(\mathbb{B}\mathcal{G})$, giving a vector bundle over $\mathbb{B}\mathcal{G}$. Since the forgetful functors $\text{Vect}(\mathcal{X}) \rightarrow \text{Sub}_{/\mathcal{X}}$ are compatible with pullback, it follows that the underlying submersion of this vector bundle is the map $\mathbb{B}\mathcal{E} \rightarrow \mathbb{B}\mathcal{G}$, finishing the proof. $\square$

1.4 Relative tangent bundle

We have now developed enough theory to introduce the relative tangent bundle and finally see the problem mentioned in the introduction. First let us see how we define the tangent groupoid.

Definition 1.4.1 (Tangent groupoid). Let $\mathcal{G}$ be a Lie groupoid. we define its *tangent groupoid* $T\mathcal{G}$ as the groupoid:

$$T\mathcal{G} := (T\mathcal{G}_1 \overset{ds}{\underset{dt}{\rightrightarrows}} T\mathcal{G}_0),$$

where the structure maps are the derivatives of the structure morphism of $\mathcal{G}$. It comes with a natural map of Lie groupoids $\pi : T\mathcal{G} \rightarrow \mathcal{G}$.

The morphism $\pi : T\mathcal{G} \rightarrow \mathcal{G}$ is naturally a VB-groupoid, but in general it is not Cartesian and so does not necessarily define a vector bundle over $\mathbb{B}\mathcal{G}$. We can solve the problem by looking at the relative version of the above construction.

Definition 1.4.2 (Relative tangent groupoid). Let $f : \mathcal{H} \rightarrow \mathcal{G}$ be a smooth submersion of Lie groupoid, meaning that $f_0 : \mathcal{H}_0 \rightarrow \mathcal{G}_0$ and $f_1 : \mathcal{H}_1 \rightarrow \mathcal{G}_1$ are smooth submersions. Passing to tangent groupoid morphism $df : T\mathcal{H} \rightarrow T\mathcal{G}$ which is fiberwise linear. As it lives over f , it correspond to a morphism $df : T\mathcal{H} \rightarrow f^*T\mathcal{G}$ of VB-groupoid over $\mathcal{H}$. We define the *relative tangent groupoid* Tf as the fiberwise kernel of this map:

$$Tf := \ker(df : T\mathcal{H} \rightarrow f^*T\mathcal{G})$$

Lemma 1.4.3 (cf. [16, Lemma 3.5.3.]). *Assume the map $f : \mathcal{H} \rightarrow \mathcal{G}$ is a cartesian morphism (see Definition 1.3.7), then the VB-groupoid $Tf \rightarrow \mathcal{H}$ is also cartesian.*

²Actually we just need one of them.

We now see that, combining the above lemma with Lemma 1.3.8, the relative tangent groupoid defines a vector bundle $\mathbb{B}(Tf) \rightarrow \mathbb{B}\mathcal{H}$. In [16, Proposition 3.5.4] it is proven that this vector bundle is Morita invariant.

Construction 1.4.4 (cf. [16, Definition 3.5.5]). Let $f : \mathcal{Y} \rightarrow \mathcal{X}$ be representable submersion of differentiable stacks.

- Choose a representable atlas $p : M \twoheadrightarrow \mathcal{X}$ and let $\mathcal{G}$ denote the associated Lie groupoid given as the Čech nerve of p . We obtain an equivalence $\mathcal{X} \simeq \mathbb{B}\mathcal{G}$.
- Define an atlas $q : N \twoheadrightarrow \mathcal{Y}$ for $\mathcal{Y}$ as in Lemma 1.1.13 by pulling back p along f :

$$\begin{array}{ccc} N & \xrightarrow{f_0} & M \\ q \downarrow & \lrcorner & \downarrow p \\ \mathcal{Y} & \xrightarrow{f} & \mathcal{X} \end{array}$$

Let $\mathcal{H}$ denote the Čech nerve of q , so that $\mathcal{Y} \simeq \mathbb{B}\mathcal{H}$.

- There is a canonical smooth submersion of Lie groupoids $f_\bullet : \mathcal{H} \rightarrow \mathcal{G}$, which is cartesian by Lemma F.2.6. We let T_f be the classifying stack of the relative tangent groupoid $Tf_\bullet$:

$$T_f := \mathbb{B}T(f_\bullet).$$

- The structure map $Tf_\bullet \rightarrow \mathcal{H}$ induces a morphism of stacks $T_f \rightarrow \mathcal{Y}$, which by Lemma 1.4.3 and Lemma 1.3.8 admits a canonical structure of a vector bundle.

Definition 1.4.5 (Relative tangent bundle). Let $f : \mathcal{Y} \rightarrow \mathcal{X}$ be representable submersion of differentiable stack. The vector bundle $T_f \rightarrow \mathcal{Y}$ is called the *relative tangent bundle*.

Lemma 1.4.6 (cf. [16, Lemma 3.5.6]). *Up to Morita equivalence, the definition of the vector bundle $T_f \rightarrow \mathcal{Y}$ is independent of the choice of atlas $M \twoheadrightarrow \mathcal{X}$.*

The limitations of the construction of the relative tangent bundle T_f are highlighted in [16, Remark 3.5.7]:

The construction of the relative tangent bundle T_f we have provided is somewhat unsatisfactory, as it is not defined intrinsically in terms of the differentiable stacks but requires a (non-coherent) choice of atlas.

In this thesis, we address this problem by resolving the following issues:

- independence from the choice of an atlas,
- lack of intrinsic structure,
- absence of a relation to a universal property.

The answer to the first two issues can be easily obtained from the following description.

Construction 1.4.7. Consider the composite

$$\mathrm{Fun}([1], \mathbf{Mfd})^{\mathrm{op}} \xrightarrow{\mathrm{ev}_0^{\mathrm{op}}} \mathbf{Mfd}^{\mathrm{op}} \xrightarrow{\mathrm{Vect}(-)} \mathrm{Cat}_\infty$$

By unstraightening we obtain the following cartesian fibration:

$$\widetilde{\mathrm{Vect}} \longrightarrow \mathrm{Fun}([1], \mathbf{Mfd}).$$

We may describe $\widetilde{\mathrm{Vect}}$ as the category whose:

- objects are pairs $(M \rightarrow N, E)$, where $M \rightarrow N$ is a smooth map of manifolds and E is a vector bundle over M ,

- morphisms $(g : M \rightarrow N, E) \rightarrow (f : P \rightarrow W, F)$ consist of a commutative square

$$\begin{array}{ccc} M & \xrightarrow{k} & P \\ g \downarrow & & \downarrow f \\ N & \xrightarrow{h} & W \end{array} \quad (1.1)$$

together with a compatible morphism of vector bundles $E \rightarrow k^*F$ over M .

Definition 1.4.8. Let M be a smooth manifold. We define the category $\text{Vect}/_M$ as the category whose:

- objects are pairs $(N \rightarrow M, E)$, where $N \rightarrow M$ is a smooth submersion of smooth manifolds and E is a vector bundle over N ,
- morphisms $(g : N \rightarrow M, E) \rightarrow (f : P \rightarrow M, F)$ consist of commutative triangles

$$\begin{array}{ccc} N & \xrightarrow{h} & P \\ g \searrow & & \swarrow f \\ & M & \end{array}$$

together with a compatible morphism of vector bundles $E \rightarrow h^*F$ over N .

Remark 1.4.9. Notice that a smooth map $f : M \rightarrow N$ induces a functor $f^* : \text{Vect}/_N \rightarrow \text{Vect}/_M$, given by pullback. More precisely, for an object $(P \rightarrow N, E)$, we set

$$f^*(P \rightarrow N, E) := (P \times_N M \rightarrow M, g^*E),$$

where $g : P \times_N M \rightarrow P$ is the natural projection. On a morphism $(P \rightarrow N, E) \rightarrow (Q \rightarrow N, F)$, the functor f^* acts by applying the corresponding universal property of the pullback construction.

Notation. We denote by Sub the full subcategory of $\text{Fun}([1], \mathbf{Mfd})$ spanned by the submersions. Furthermore, we denote by $\text{Vect}/_-$ the functor

$$\text{Vect}/_- : \mathbf{Mfd}^{\text{op}} \longrightarrow \text{Cat} \hookrightarrow \text{Cat}_{\infty}$$

defined as follows:

- to a smooth manifold M it assigns the category $\text{Vect}/_M$;
- to a morphism $f : M \rightarrow N$ it assigns the functor $f^* : \text{Vect}/_N \rightarrow \text{Vect}/_M$.

Remark 1.4.10. Construction 1.4.7 applies equally well if we replace $\text{Fun}([1], \mathbf{Mfd})$ with Sub . By abuse of notation, we continue to denote by $\widetilde{\text{Vect}}$ the induced Cartesian fibration over Sub .

We are now in the following situation

$$\begin{array}{ccc} \widetilde{\text{Vect}} & \xrightarrow{\quad} & \text{Sub} \\ & \searrow & \swarrow \text{ev}_1 \\ & \mathbf{Mfd} & \end{array} \quad (1.2)$$

where the top functor is cartesian. By applying the straightening equivalence, we obtain a natural transformation

$$\alpha : \text{Vect}/_- \Rightarrow \text{Sub}/_-.$$

Both $\text{Vect}/_-$ and $\text{Sub}/_-$ satisfy descent. Hence, by Theorem F.3.11, we obtain an induced natural transformation

$$\alpha^+ : \text{Vect}/_-^+ \Rightarrow \text{Sub}/_-^+$$

between the corresponding functors from $\text{Shv}(\mathbf{Mfd})^{\text{op}}$. The cartesian unstraightening of these two functors, together with the natural transformation α^+ , yields a diagram analogous to (1.2), where we replace $\mathbf{Mfd}$ with $\text{Shv}(\mathbf{Mfd})$, $\widetilde{\text{Vect}}$ with $\widetilde{\text{Vect}}^+$, and Sub with Sub^+ .

Construction 1.4.11. Remark A.2.12 provides a way to construct a section σ_T of the top map of diagram (1.2). Explicitly, the section σ_T sends a morphism $f: N \rightarrow M$ to the object (f, Tf) , and a commutative square (1.1) to the morphism $(g, Tg) \rightarrow (f, Tf)$ where $g \rightarrow f$ is given by the square above, and $Tg \rightarrow Tf$ is the map constructed in Remark A.2.12.

By descent, we can lift the section from the manifold case to the stack case. Formally, the argument is the same as the one given above Construction 1.4.11.

$$\begin{array}{ccc}
 \widetilde{\text{Vect}}^+ & \xrightleftharpoons{\sigma_T^+} & \text{Sub}^+ \\
 & \searrow \quad \swarrow & \\
 & \text{Shv}(\mathbf{Mfd}) &
 \end{array}$$

Definition 1.4.12. Let $f: \mathcal{X} \rightarrow \mathcal{Y}$ be representable submersion of differentiable stacks. The *relative tangent bundle* T_f of f is defined to be the value of σ_T^+ at the submersion f .

In the discrete case over $\mathbf{Mfd}$, we have an identification of vector fields with the dual of differential forms. If differential forms were characterized by a universal property, this would provide the starting point for a deeper discussion. Motivated by these considerations, in Subsection 3.5.3 we present an alternative description of the relative tangent bundle via the cotangent complex. We show that one is the dual of the other, and that over $\mathbf{Mfd}$, this recovers the aforementioned perspective.

Chapter 2

Derived $\mathcal{C}^\infty$ -Geometry

This chapter is devoted to presenting the geometric framework of derived differential geometry, with a particular focus on the universal property of the ∞ -category of derived manifolds. For an introductory exposition on derived geometry, see the Introduction of this thesis.

In Section 2.1, we recall the theory of discrete $\mathcal{C}^\infty$ -rings, which leads naturally to their derived analogue, $\mathcal{C}^\infty$ -rings, in Section 2.2. As in the algebraic case, in Section 2.3 (together with the beginning of Section 2.5) we associate to each $\mathcal{C}^\infty$ -ring a topological space equipped with a sheaf of $\mathcal{C}^\infty$ -rings. In Section 2.3, we focus on finitely presented $\mathcal{C}^\infty$ -rings, as they will be needed in the next section. Section 2.4 presents the main result of the joint work of Carchedi and Steffens [12], where the universal ∞ -category obtained by adding non-transverse pullbacks to $\mathbf{Mfd}$ is constructed. Finally, in Section 2.5 we introduce derived stacks.

To avoid redundancy with Appendix D.2, we assume familiarity with the theory developed therein. The main references for this chapter are [47] for discrete $\mathcal{C}^\infty$ -rings, and [12], [9], [10], [57], [58], and [50] for the remaining material.

2.1 Discrete $\mathcal{C}^\infty$ -rings

Following [47], we provide a brief introduction to discrete $\mathcal{C}^\infty$ -rings, which will aid our understanding of certain constructions in the context of $\mathcal{C}^\infty$ -rings and serve as a useful tool in some proofs.

Warning 2.1.1. Usually in the literature discrete $\mathcal{C}^\infty$ -rings are called $\mathcal{C}^\infty$ -rings and what we call $\mathcal{C}^\infty$ -rings are called simplicial/homotopical $\mathcal{C}^\infty$ -rings. However in this thesis we want to emphasize the fact that our main object of study are simplicial/homotopical $\mathcal{C}^\infty$ -rings. So we will call them $\mathcal{C}^\infty$ -rings.

Notation. We denote by $\mathbf{C}^\infty$ the full subcategory of $\mathbf{Mfd}$ spanned by manifolds of the form $\mathbb{R}^n$ for some $n \geq 0$ and by $\mathbf{Com}_{\mathbb{R}}$ the full subcategory of $\mathbb{R}$ -schemes spanned by those schemes of the form $\mathbb{A}_{\mathbb{R}}^n$ for some $n \geq 0$.

Definition 2.1.2. A *discrete $\mathcal{C}^\infty$ -ring* is a 0-truncated $\mathbf{C}^\infty$ -algebra (see Definition D.2.4). The category of discrete $\mathcal{C}^\infty$ -rings will be denoted by $\mathbf{disC}^\infty\mathbf{Ring}$.

We may regard discrete $\mathcal{C}^\infty$ -rings as richer algebraic structures than ordinary rings, encoding an operation for each smooth map $\mathbb{R}^n \rightarrow \mathbb{R}$, with addition and multiplication arising as special cases. More precisely, let A be a discrete $\mathcal{C}^\infty$ -ring. The smooth maps $\text{sum} + : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $\text{product} \cdot : \mathbb{R}^2 \rightarrow \mathbb{R}$ on $\mathbb{R}$ encode respectively the analogue of a sum

$$A(+): A(\mathbb{R}) \times A(\mathbb{R}) \simeq A(\mathbb{R}^2) \rightarrow A(\mathbb{R})$$

and a product

$$A(\cdot): A(\mathbb{R}) \times A(\mathbb{R}) \simeq A(\mathbb{R}^2) \rightarrow A(\mathbb{R})$$

on $A(\mathbb{R})$. Then for every $f: \mathbb{R}^m \rightarrow \mathbb{R}$ we get a new operation:

$$A(f): A(\mathbb{R}) \times \cdots \times A(\mathbb{R}) \simeq A(\mathbb{R}^m) \rightarrow A(\mathbb{R}).$$

Note that this last requirement fully determines the structure of the discrete $\mathcal{C}^\infty$ -ring. Recall, in fact, that a 0-truncated $\mathbf{C}^\infty$ -algebra A is a product-preserving functor $A: \mathbf{C}^\infty \rightarrow \mathbf{Set}$. In particular, one would a priori need to specify a morphism $A(f): A^n \rightarrow A^m$ for every smooth map $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$. However, working componentwise, we see that it is sufficient to specify how the functor acts on morphisms of the form $f: \mathbb{R}^n \rightarrow \mathbb{R}$ for $n \geq 0$.

Convention. It is customary to denote a discrete $\mathcal{C}^\infty$ -ring A using its value $A(\mathbb{R})$ at $\mathbb{R}$.

Example 2.1.3. The prototypical example of a $\mathcal{C}^\infty$ -ring is given by the ring of smooth $\mathcal{C}^\infty(M)$ function over a smooth manifold M . Where the $\mathcal{C}^\infty$ structure is given as follows: to every smooth map $f = (f_1, \dots, f_m) : \mathbb{R}^n \rightarrow \mathbb{R}^m$ we associate:

$$\begin{aligned} \mathcal{C}^\infty(M)(f) : \mathcal{C}^\infty(M)^n &\rightarrow \mathcal{C}^\infty(M)^m \\ (g_1(x), \dots, g_n(x)) &\mapsto (f_1(g_1(x), \dots, g_n(x)), \dots, f_m(g_1(x), \dots, g_n(x))). \end{aligned}$$

Remark 2.1.4. To every discrete $\mathcal{C}^\infty$ -rings we may associate its underlying (classical) algebra essentially forgetting all the maps that do not come from sum or products. For a precise statement see Example D.2.23

Proposition 2.1.5 (cf. [47, Proposition 1.1]). *The discrete $\mathcal{C}^\infty$ -ring $\mathcal{C}^\infty(\mathbb{R}^n)$ of smooth functions in n variables is the free discrete $\mathcal{C}^\infty$ -ring on n generators, given by the projection functions.*

Proof. Call π_i the projection $\mathbb{R}^n \rightarrow \mathbb{R}$ on the i^{th} coordinate. n elements of an arbitrary discrete $\mathcal{C}^\infty$ -ring A determines a morphism of discrete $\mathcal{C}^\infty$ -rings $\varphi : \mathcal{C}^\infty(\mathbb{R}^n) \rightarrow A$ with $\varphi(\pi_i) = a_i$, by:

$$\varphi(f) = A(f)(a_1, \dots, a_n).$$

The uniqueness of the morphism φ follows by the commutativity of the following diagram:

$$\begin{array}{ccc} \mathcal{C}^\infty(\mathbb{R}^n)^n & \xrightarrow{\varphi^n} & A^n \\ \mathcal{C}^\infty(\mathbb{R}^n)(f) \downarrow & & \downarrow A(f) \\ \mathcal{C}^\infty(\mathbb{R}^n) & \xrightarrow{\varphi} & A \end{array}$$

□

Corollary 2.1.6. *We have the following equivalences:*

$$\text{Hom}_{\mathbf{Alg}_{\mathcal{C}^\infty}(\mathbf{Set})}(\mathcal{C}^\infty(\mathbb{R}^n), A) \simeq \text{Hom}_{\mathbf{Alg}_{\mathbf{Com}_{\mathbb{R}}}(\mathbf{Set})}(\mathbb{R}[x_1, \dots, x_n], A(\mathbb{R})) \simeq \text{Hom}_{\mathbf{Set}}(\{x_1, \dots, x_n\}, A(\mathbb{R}))$$

We now address the problem of inverting elements of discrete $\mathcal{C}^\infty$ -rings.

Definition 2.1.7. Let A be a discrete $\mathcal{C}^\infty$ -ring and let a be an element of A . We say that a morphism $A \rightarrow A\{a^{-1}\}$ exhibits $A\{a^{-1}\}$ as a localization of A at the element a if for every discrete $\mathcal{C}^\infty$ -ring C the map

$$\text{Hom}_{\mathbf{Alg}_{\mathcal{C}^\infty}(\mathbf{Set})}(A\{a^{-1}\}, C) \rightarrow \text{Hom}_{\mathbf{Alg}_{\mathcal{C}^\infty}(\mathbf{Set})}(A, C)$$

induces an isomorphism with the subset $\text{Hom}_{\mathbf{Alg}_{\mathcal{C}^\infty}(\mathbf{Set})}^a(A, C) \subseteq \text{Hom}_{\mathbf{Alg}_{\mathcal{C}^\infty}(\mathbf{Set})}(A, C)$ of those morphism that sends a to invertible elements in C .

We now aim to present an explicit construction of the localization.

Proposition 2.1.8 (cf. [47, Proposition 1.6]). *Let $a \in \mathcal{C}^\infty(\mathbb{R}^n)$, and let $U = a^{-1}\{\mathbb{R} - 0\}$. Then*

$$\mathcal{C}^\infty(\mathbb{R}^n)\{a^{-1}\} \simeq \mathcal{C}^\infty(U).$$

An important result in the theory of discrete $\mathcal{C}^\infty$ -ring is the following:

Theorem 2.1.9 (cf [47, Theorem 2.8]). *The functor $\mathcal{C}^\infty : \mathbf{Mfd} \rightarrow \mathbf{dis}\mathcal{C}^\infty\mathbf{Ring}^{op}$ sending a manifold M to the discrete $\mathcal{C}^\infty$ -ring of smooth functions on M is fully faithful, factors through finitely presented discrete $\mathcal{C}^\infty$ -rings (see Definition D.2.17), and preserves transverse pullbacks.*

Combining Theorem 2.1.9 with Proposition 2.1.8, we deduce that the localization of $\mathcal{C}^\infty(\mathbb{R}^n)$ at an element a is given by the following pushout square:

$$\begin{array}{ccc} \mathcal{C}^\infty(\mathbb{R}) & \xrightarrow{q_a} & \mathcal{C}^\infty(\mathbb{R}^n) \\ \downarrow & & \downarrow \\ \mathcal{C}^\infty(\mathbb{R} \setminus \{0\}) & \longrightarrow & \mathcal{C}^\infty(U). \end{array}$$

Corollary 2.1.10. *Let A be a discrete $\mathcal{C}^\infty$ -ring, and let $a \in A$. Then the following square is a pushout in $\mathbf{dis}\mathcal{C}^\infty\mathbf{Ring}$:*

$$\begin{array}{ccc} \mathcal{C}^\infty(\mathbb{R}) & \xrightarrow{q_a} & A \\ \downarrow & & \downarrow \\ \mathcal{C}^\infty(\mathbb{R} \setminus \{0\}) & \longrightarrow & A[1/a]. \end{array}$$

2.2 $\mathcal{C}^\infty$ -Rings

We now introduce the main object of this chapter: $\mathcal{C}^\infty$ -rings. These are the derived analogues of discrete $\mathcal{C}^\infty$ -rings; in particular, one may think of them as discrete $\mathcal{C}^\infty$ -rings in which every structural property, such as associativity or commutativity, is encoded as additional data (i.e., homotopies). For a deeper understanding of this idea, we refer to Appendix D.1, in particular Examples D.1.6 and D.1.13.

In this section, we first give the definition and present the main properties of $\mathcal{C}^\infty$ -rings. Subsection 2.2.2 briefly discusses $\mathbf{C}^\infty$ -completion. Our approach will be model-independent; however, having a concrete model is helpful for performing explicit computations. For this reason, Subsection 2.2.3 is devoted to presenting such a model. In Subsection 2.2.4, we discuss cell objects, a useful tool that will come in handy in Chapter 3. Finally, Subsection 2.2.5 deals with localization, laying the groundwork for introducing a topology on $\mathcal{C}^\infty$ -rings.

Throughout this section, we follow [12, Section 4], [10, Sections 3 and 5], and [57].

2.2.1 Definitions and first properties

Most of the material in this subsection will not be proved; instead, we refer to [12, section 4].

Definition 2.2.1 ($\mathcal{C}^\infty$ -rings). We define the ∞ -category $\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}$ of $\mathcal{C}^\infty$ -rings as the ∞ -category of $\mathbf{C}^\infty$ -algebras (see Definition D.2.4). In other words: $\mathbf{d}\mathcal{C}^\infty\mathbf{Ring} = \mathbf{Alg}_{\mathbf{C}^\infty}(\mathbf{Spc})$.

Notation. We will denote pushouts in $\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}$ with $\otimes^\infty$.

Lemma 2.2.2 ([12] Lemma 4.1). *The functor*

$$\mathcal{C}^\infty : \mathbf{Mfd} \rightarrow \mathbf{d}\mathcal{C}^\infty\mathbf{Ring}^{op}$$

is fully faithful.

Proof. This follows directly from Theorem 2.1.9, together with the fully faithful inclusion $\mathbf{dis}\mathcal{C}^\infty\mathbf{Ring} \hookrightarrow \mathbf{d}\mathcal{C}^\infty\mathbf{Ring}$. $\square$

The first significant result arises from the theory of unramified transformations (see appendix D.2.3) and will prove useful when attempting to extract information about pushouts from the algebraic setting.

Lemma 2.2.3 (cf. [12, Lemma 4.6]). *The transformation of algebraic theories $\mathbf{Com}_{\mathbb{R}} \rightarrow \mathbf{C}^\infty$ is unramified.*

Corollary 2.2.4 (cf. [12, Corollary 4.7]). *$(-)^{\text{alg}}$ preserves pushouts along effective epimorphism.*

Proof. Apply Proposition D.2.27 to the unramified transformation $\mathbf{Com}_{\mathbb{R}} \rightarrow \mathbf{C}^\infty$. $\square$

We now present a series of results aimed at understanding the behavior of the essential image of the functor $\mathcal{C}^\infty : \mathbf{Mfd} \rightarrow \mathbf{d}\mathcal{C}^\infty\mathbf{Ring}^{op}$, particularly in relation to the formation of pushouts in $\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}$.

Lemma 2.2.5 (cf. [12, Proposition 4.9]). *Let U be an open subset of $\mathbb{R}^m$. Then U is the fiber $f^{-1}(0)$ of a smooth map $f : \mathbb{R}^{m+1} \rightarrow \mathbb{R}$ which is transverse to 0.*

Proof. Set $C = \mathbb{R}^m \setminus U$. Define $f(x_1, \dots, x_m, t) = t \cdot \chi_C(x_1, \dots, x_m) - 1$ where $\chi_C : \mathbb{R}^m \rightarrow \mathbb{R}$ is a smooth function such that $C = \chi_C^{-1}(0)$. Observing that:

$$f(x_1, \dots, x_m, t) = 0 \iff \chi_C(x_1, \dots, x_m) \neq 0, \quad t = \frac{1}{\chi_C(x_1, \dots, x_m)}$$

we deduce $U \cong f^{-1}(0)$. Moreover, for $(x_1, \dots, x_m, t) \in f^{-1}(0)$ we have:

$$\frac{d}{dt} = f_* \left(\frac{1}{\chi_C(x_1, \dots, x_m)} \frac{\partial}{\partial t} \right).$$

As a consequence f is transverse to 0. □

Recall that, in our conventions, smooth manifolds are assumed to be second countable, Hausdorff, and of bounded dimension (see Notations and Conventions).

Lemma 2.2.6. *Every smooth manifold can be written as a retract of a transverse pullback of Cartesian manifold.*

Now recall that any manifold is a retract of an open $U \subseteq \mathbb{R}^k$ for some k (see Appendix A.1). In particular, putting together the above lemma and Proposition 2.2.12 we find the following.

Corollary 2.2.7. *For all M $\mathcal{C}^\infty(M)$ is a retract of a pushout of the form:*

$$\begin{array}{ccc} \mathcal{C}^\infty(\mathbb{R}) & \longrightarrow & \mathbb{R} = \mathcal{C}^\infty(\mathbb{R}^0) \\ \downarrow & & \downarrow \\ \mathcal{C}^\infty(\mathbb{R}^{n+1}) & \longrightarrow & \mathcal{C}^\infty(U). \end{array} \quad \lrcorner$$

Lemma 2.2.8 (cf. [12, Lemma 4.11]). *Let M and N be smooth manifolds, then the natural map*

$$\mathcal{C}^\infty(M) \otimes^\infty \mathcal{C}^\infty(N) \rightarrow \mathcal{C}^\infty(M \times N)$$

is an equivalence

Remark 2.2.9 (cf. [12, Remark 4.12]). Inspecting the proof of the aforementioned lemma (see the cited article), one observes that the functor $\mathcal{C}^\infty : \mathbf{Mfd} \rightarrow \mathbf{dC}^\infty\mathbf{Ring}^{op}$, factors through the full subcategory of finitely presented algebras.

Lemma 2.2.10 (cf. [12, Lemma 4.13]). *The functor $\mathcal{C}^\infty : \mathbf{Mfd} \rightarrow \mathbf{dC}^\infty\mathbf{Ring}^{op}$ sending a manifold M to the discrete $\mathcal{C}^\infty$ -ring of smooth functions on M preserves transverse pullbacks of the form:*

$$\begin{array}{ccc} N \times_U M & \longrightarrow & N \\ \downarrow & \lrcorner & \downarrow \\ M & \longrightarrow & U \end{array}$$

where U is an open submanifold of $\mathbb{R}^n$ for $n \geq 0$.

Remark 2.2.11. The proof strategy of the above propositions is largely similar across cases. Since the statements concern pushouts, the main tool employed is the spectral sequence provided in [40, 7.2.1.19], which is particularly useful in this context. Another key ingredient is the use of a Koszul algebra, as constructed in of [12, Lemma 4.5], which is instrumental in the computation of the relevant Tor groups. Finally, the theory of unramified morphisms plays a crucial role, allowing one to transfer information from the algebraic setting and apply it effectively in the context of the proofs.

The following proposition may be viewed as a derived analogue of Theorem 2.1.9.

Proposition 2.2.12 (cf. [12, Lemma 4.52]). *The functor $\mathcal{C}^\infty : \mathbf{Mfd} \rightarrow \mathbf{dC}^\infty\mathbf{Ring}^{op}$ preserves transverse pullback.*

Although this result is more general than Lemma 2.2.10, its proof relies on that lemma. In particular, to avoid any circular reasoning in what follows, we have chosen to state both results explicitly.

2.2.2 $\mathbf{C}^\infty$ -completion

We now introduce a pair of adjoint functors that will serve as a bridge between the algebraic and the differential frameworks.

Denote by $\tau : \mathbf{Com}_{\mathbb{R}} \rightarrow \mathbf{C}^\infty$ the obvious morphism of algebraic theories. By Corollary D.2.25 we get an adjunction between their associated categories of algebras:

$$\begin{array}{ccc} & \xrightarrow{(-)} & \\ \mathbf{dRing}_{\mathbb{R}} & \perp & \mathbf{dC}^\infty\mathbf{Ring} \\ & \xleftarrow{(-)^{\text{alg}}} & \end{array}$$

where $(-)^{\text{alg}} = \tau^*$ and $\widehat{(-)} = \tau_!$.

Definition 2.2.13. Let A be $\mathbf{C}^\infty$ -ring, we define the *underlying algebra* as A^{alg}

Definition 2.2.14. Let A be a commutative $\mathbb{R}$ -algebra, we define its $\mathbf{C}^\infty$ -completion as $\widehat{A}$.

Example 2.2.15. The $\mathbf{C}^\infty$ -completion of the free $\mathbb{R}$ -algebra on n generators, $\mathbb{R}[x_1, \dots, x_n]$, is the free $\mathbf{C}^\infty$ -ring on n -element, namely $\mathbf{C}^\infty(\mathbb{R}^n)$.

Example 2.2.16 (cf [10, Example 3.10]). The $\mathbf{C}^\infty$ -completion may be destructive. Consider the complex number as an $\mathbb{R}$ -algebra: $\mathbb{C} \simeq \mathbb{R}[x]/(x^2 + 1)$. Although $x^2 + 1$ is not a unit in $\mathbb{R}[x]$, it become one after $\mathbf{C}^\infty$ -completion, since $x^2 + 1$ is a non-vanishing $\mathbf{C}^\infty$ -function on $\mathbb{R}$. In particular the $\mathbf{C}^\infty$ -completion of $\mathbb{C}$ is the zero $\mathbf{C}^\infty$ -ring.

We conclude this brief subsection with a remark about the functor $(-)^{\text{alg}}$.

Remark 2.2.17. Notice that the functor $(-)^{\text{alg}}$ is conservative and preserves sifted colimits. To see this, consider the following commutative diagram:

$$\begin{array}{ccc} \mathbf{dC}^\infty\mathbf{Ring} & \xrightarrow{(-)^{\text{alg}}} & \mathbf{dRing}_{\mathbb{R}} \\ & \searrow \theta_{\mathbb{R}} & \swarrow \theta_{\mathbb{R}} \\ & \mathbf{Spc} & \end{array}$$

Here, both functors $\theta_{\mathbb{R}}$ denote evaluation at the object $\mathbb{R}$. By Remark D.2.14, both are conservative. In particular, this implies that $(-)^{\text{alg}}$ is also conservative. The claim that $(-)^{\text{alg}}$ preserves sifted colimits follows from Lemma D.2.24.

2.2.3 The model structure

As previously mentioned we are going to give also a model presentation of the category $\mathbf{dC}^\infty\mathbf{Ring}$. The exposition closely follow the [10, section 5.1.]

Definition 2.2.18 (cf. [11, Definition 4.14]). A *differential graded $\mathbf{C}^\infty$ -algebra* (or dg- $\mathbf{C}^\infty$ -algebra), is a non-negatively graded differential $\mathbb{R}$ -algebra $(A_\bullet, d)$ together with a compatible structure of discrete $\mathbf{C}^\infty$ -ring on the set A_0 . A morphism of dg- $\mathbf{C}^\infty$ -algebras $(A_\bullet, d) \rightarrow (B_\bullet, d')$ is a morphism is a morphism of dg-algebras such that $A_0 \rightarrow B_0$ is a morphism of discrete $\mathbf{C}^\infty$ -Rings. We will denote this category with $\mathbf{dgC}^\infty\mathbf{Alg}$.

Lemma 2.2.19 ([11]). *The forgetful functor $U : \mathbf{dgC}^\infty\mathbf{Alg} \rightarrow \mathbf{Ch}_{\geq 0}(\mathbb{R})$ admits a left adjoint.*

Definition 2.2.20. Denote by $\text{Sym}_{\mathbf{C}^\infty}$ the left adjoint in Lemma 2.2.19. For every $V_\bullet \in \mathbf{Ch}_{\geq 0}(\mathbb{R})$ we will refer to $\text{Sym}_{\mathbf{C}^\infty}(V_\bullet)$ as the *free differential graded algebra generated by $V_\bullet$* .

We now present an explicit description of the left adjoint. Consider $(V^\bullet, d) \in \mathbf{dgVect}$ and consider for all V^i a basis $\{e_i^\alpha\}$. We can describe the differential $d^i : V^i \rightarrow V^{i+1}$ in term of the basis:

$$d_i(e_i^\alpha) = \sum_{\beta} M_{\alpha\beta}^i e_{i+1}^\beta$$

Then we can describe the free dg- $\mathbb{R}$ -algebra $\text{Sym}((V^\bullet, d))$ as $\text{Sym}(V^\bullet)$ equipped with differential:

$$D := \sum_{\beta, \alpha} M_{\alpha\beta}^i e_{i+1}^\beta \frac{\partial}{\partial e_i^\alpha} \quad (2.1)$$

Similarly, $\text{Sym}_{\mathbf{C}^\infty}(V^\bullet)$ is the free dg- $\mathbf{C}^\infty$ -algebra on the variables e_i^α . In particular if $V^\bullet = V$ is concentrated in degree zero of dimension n then:

$$\text{Sym}_{\mathbf{C}^\infty}(V) \simeq \mathcal{C}^\infty(\mathbb{R}^n).$$

On the other hand if $V^0 = 0$ then:

$$\text{Sym}_{\mathbf{C}^\infty}(V^\bullet) \simeq \text{Sym}(V^\bullet).$$

Putting this two together we get:

$$\text{Sym}_{\mathbf{C}^\infty}(V^\bullet) \simeq \text{Sym} \left(\bigoplus_{i \neq 0} V^i[-i] \right)_{\text{Sym}(V^0)} \otimes_{\text{Sym}(V^0)} \mathcal{C}^\infty(V^0)$$

The differential is given by 2.1. For more details see [57, Remark 4.1.5.3 and Constriction 4.1.5.4].

We now outline the construction of a model structure on differential graded $\mathcal{C}^\infty$ -algebras. Recall that the category $\text{Ch}_{\geq 0}(\mathbb{R})$ admits a model structure (called the projective model structure) in which weak equivalences are quasi-isomorphisms and fibrations are surjections in all positive degrees. Moreover, by 2.2.19 we have a forgetful-free adjunction:

$$\begin{array}{ccc} & \xrightarrow{\text{Sym}_{\mathbf{C}^\infty}} & \\ \text{Ch}_{\geq 0}(\mathbb{R}) & \begin{array}{c} \perp \\ \xleftarrow{U} \end{array} & \mathbf{dgC}^\infty \mathbf{Alg}. \end{array}$$

As a consequence, we can apply transfer theorem to obtain the following:

Proposition 2.2.21 (cf. [11, Theorem 6.5]). *The exists a cofibrantly generated, almost simplicial, model category structure on the category of dg- $\mathbf{C}^\infty$ -algebras, unique with the property that $(A_\bullet, d) \rightarrow (B_\bullet, d')$ is a weak equivalence (respectively fibration) if and only if the induced morphism of the underlying chain complex is, with respect to the projective model structure on $\text{Ch}_{\geq 0}(\mathbb{R})$.*

Definition 2.2.22. Denote by $\mathbf{dgC}^\infty \mathbf{Alg}[W^{-1}]$ the ∞ -category associated to the above model category. Similarly we define $\mathbf{dgC}^\infty \mathbf{Alg}_{\geq 0}[W^{-1}]$.

Definition 2.2.23. A morphism $f: (A, d) \rightarrow (B, d)$ between dg- $\mathcal{C}^\infty$ -algebras is *quasi-free* if there exists a $\mathbb{N}$ -graded super vector space $W^\bullet$ such that f is isomorphic to the inclusion of A into $A \otimes_{\mathbb{R}} \text{Sym}_{\mathbf{C}^\infty}(W^\bullet)$, after forgetting the differentials.

Proposition 2.2.24 ([11]). *A morphism $f: (A, d) \rightarrow (B, d)$ between non-positively graded differential $\mathcal{C}^\infty$ -algebras is a cofibration if and only if it is a retract of a quasi-free extension.*

Corollary 2.2.25. *Cofibrant objects of $\mathbf{dgC}^\infty \mathbf{Alg}_{\geq 0}$ are retracts of quasi-free dg- $\mathcal{C}^\infty$ -algebras. In particular, each algebra $\mathcal{C}^\infty(\mathbb{R}^n)$, regarded as a dg- $\mathcal{C}^\infty$ -algebra, is cofibrant, and the collection of these algebras is closed under coproducts.*

Remark 2.2.26. let U be an open in $\mathbb{R}^n$. Then the above Lemma plus Lemma 2.2.5 and Proposition 2.2.12 implies that $\mathcal{C}^\infty(U)$ is given by a pushout of cofibrant discrete objects.

Remember that our original goal for this section was to provide a model description of $\mathbf{dC}^\infty \mathbf{Ring}$. In particular we now want to present an analogue of the Dold-Kan correspondence in the case of $\mathcal{C}^\infty$ -Rings.

Construction 2.2.27 (cf. [10, below Proposition 5.8]). By Corollary 2.2.25 coproducts of free algebras coincide with the derived $\mathcal{C}^\infty$ -tensor products. It follows that the canonical functor

$$\begin{aligned} \mathbf{C}^\infty &\longrightarrow (\mathbf{dgC}^\infty \mathbf{Alg}_{\geq 0}[W^{-1}])^{\text{op}} \\ \mathbb{R}^n &\longmapsto \mathcal{C}^\infty(\mathbb{R}^n) \end{aligned}$$

preserves finite products. Hence, by Theorem D.2.8, we deduce that this extends to a unique colimit-preserving functor of ∞ -categories

$$N_{\mathbf{C}^\infty}: \mathbf{dC}^\infty\mathbf{Ring} \longrightarrow \mathbf{dgC}^\infty\mathbf{Alg}_{\geq 0}[W^{-1}]$$

which sends each algebra of the form $\mathcal{C}^\infty(\mathbb{R}^n)$ to itself. By the adjoint functor (see Theorem B.3.7), it follows that there exists a right adjoint $\Gamma^{\mathcal{C}^\infty}$. Even more, it follows from the Yoneda lemma that $\Gamma^{\mathcal{C}^\infty}$ sends a $\mathbf{dgC}^\infty$ -algebra A to

$$\Gamma^{\mathcal{C}^\infty}(A): \mathbf{C}^\infty \longrightarrow \mathbf{Spc}, \quad \mathbb{R}^n \longmapsto \mathrm{Hom}_{\mathbf{dgC}^\infty\mathbf{Alg}_{\geq 0}}(\mathcal{C}^\infty(\mathbb{R}^n), A).$$

Theorem 2.2.28 ($\mathbf{C}^\infty$ -Dold-Kan correspondence [50] Corollary 2.2.10). *The above adjunction is an adjoint equivalence of ∞ -categories.*

Corollary 2.2.29 (cf. [9, Corollary 3.23]). *For any $\mathbf{dgC}^\infty$ -algebra $(A^\bullet, d)$,*

$$\mathrm{Hom}_{\mathbf{dgC}^\infty\mathbf{Alg}_{\geq 0}}(\mathcal{C}^\infty(\mathbb{R}), A) \simeq |\Gamma(A)|,$$

where $|\Gamma(A)|$ is the homotopy type of the simplicial set corresponding to the underlying chain complex A via the Dold-Kan correspondence. In particular, we have isomorphisms

$$\pi_n \mathrm{Hom}_{\mathbf{dgC}^\infty\mathbf{Alg}_{\geq 0}}(\mathcal{C}^\infty(\mathbb{R}), A) \cong H^{-n}(A).$$

2.2.4 Cell-object decomposition

We now introduce a powerful tool that will be useful in Chapter 3: cell objects. The reference for this subsection is [57, section 4.1.3].

Definition 2.2.30 (cf. [57, Notation 4.1.3.30]). Let V be a real vector space, possibly of infinite dimension. We write

$$\mathrm{Sym}_{\mathbf{C}^\infty}(V) := \mathrm{colim}_{V' \subseteq V, \dim V' < \infty} \mathcal{C}^\infty((V')^\vee)$$

for the free $\mathcal{C}^\infty$ -ring on V (see [19]). Evaluation at $0 \in V^\vee$ yields a morphism $\mathcal{C}^\infty(V^\vee) \rightarrow \mathbb{R}$ of $\mathcal{C}^\infty$ -rings. Thus, $\mathcal{C}^\infty(V^\vee)$ is augmented over the initial object in $\mathcal{C}^\infty$ -rings, and we may consider its n -fold suspension $\Sigma^n \mathcal{C}^\infty(V^\vee)$ with respect to this augmentation.

Remark 2.2.31 (cf. [58, Remark 3.1.15]). The results in this subsection are stated for $\mathcal{C}^\infty$ -rings, but it is easy to see that they hold for Lawvere theory $\mathbf{Com}_k$.

Definition 2.2.32 (cf. [57, Definition 4.1.3.31]). Let $A \rightarrow B$ be morphism of $\mathcal{C}^\infty$ -rings. We say that B is a *good A -cell object* if B is a colimit of a sequential diagram

$$A = A_{-1} \xrightarrow{\phi_{-1}} A_0 \xrightarrow{\phi_0} A_1 \longrightarrow \cdots,$$

where ϕ_{-1} is a pushout along a map of the form $A \rightarrow A \otimes^\infty \mathrm{Sym}_{\mathbf{C}^\infty}(V_{-1})$ for V_{-1} a possibly infinite dimensional vector space, and ϕ_n for $n \geq 0$ is given by a pushout of the form

$$\begin{array}{ccc} A \otimes^\infty \Sigma^n \mathrm{Sym}_{\mathbf{C}^\infty}(V_n) & \longrightarrow & A_n \\ \downarrow & \lrcorner & \downarrow \phi_n \\ A & \longrightarrow & A_{n+1} \end{array}$$

for V_n a possibly infinite dimensional vector space.

Proposition 2.2.33 (cf. [57, Proposition 4.1.3.32]). *Let $A \rightarrow B$ be a morphism of $\mathcal{C}^\infty$ -rings, then B is equivalent to a good A -cell object.*

Remark 2.2.34 (cf. [57, Remark 4.1.3.37]). Being a left adjoint, the $\mathbf{C}^\infty$ -completion $\widehat{(-)}$ preserves colimits, so we have: $\Sigma^n \mathrm{Sym}_{\mathbf{C}^\infty}(V) \simeq \widehat{\mathrm{Sym}(V[n])}$ for each real vector space $\mathbb{R}$. In particular we get the following:

$$\mathrm{Hom}_{\mathbf{dC}^\infty\mathbf{Ring}}(\Sigma^n \mathrm{Sym}_{\mathbf{C}^\infty}(V^\vee), A) \simeq \mathrm{Hom}_{\mathbf{dMod}_{\mathbb{R}}}(V[n], A^{\mathrm{alg}})$$

for all $A \in \mathbf{dC}^\infty\mathbf{Ring}$.

Lemma 2.2.35 (cf. [57, Lemma 4.1.3.38]). *Let V be a real vector space and $n > 0$. The map $V[n] \rightarrow (\Sigma^n \mathrm{Sym}_{\mathbf{C}^\infty}(V))^{\mathrm{alg}}$ corresponding to the identity $\Sigma^n \mathrm{Sym}_{\mathbf{C}^\infty}(V) \rightarrow \Sigma^n \mathrm{Sym}_{\mathbf{C}^\infty}(V)$ via the above equivalence induces an equivalence $\mathrm{Sym}(V[n]) \rightarrow \Sigma^n \mathrm{Sym}_{\mathbf{C}^\infty}(V)$ of $\mathbf{dRing}_{\mathbb{R}}$.*

2.2.5 Localization of $\mathcal{C}^\infty$ -rings

In algebraic geometry the basic opens of an affine scheme $\mathbf{Spec}(A)$ are given by localizations at some element a of the ring A . In analogy with the algebraic case we introduce the concept of a localization for $\mathcal{C}^\infty$ -rings and present a characterization for a morphism to be a localization. The analogy goes further as this localization will give us a basis (and not only that) for a topology on $\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}$. We closely follow [12, Section 4.2].

Definition 2.2.36. Let A and B be $\mathcal{C}^\infty$ -rings and let $a \in \pi_0(A)$. We say that a map $f : A \rightarrow B$ *exhibits B as a localization of A with respect to a* if:

1. $f(a) \in \pi_0(B)$ is invertible,
2. for every $C \in \mathbf{d}\mathcal{C}^\infty\mathbf{Ring}$, the map

$$\mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}(B, C) \rightarrow \mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}(A, C)$$

is equivalent to the inclusion of those connected components of $\mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}(A, C)$ spanned by those morphisms that sends a to an invertible element of $\pi_0(C)$.

Restricting to discrete $\mathcal{C}^\infty$ -rings, the definition above agrees with Definition 2.1.7. Recall from Corollary 2.1.10 that the localization of a discrete $\mathcal{C}^\infty$ -ring A at an element $a \in A$ is, up to equivalence, given by the following pushout diagram:

$$\begin{array}{ccc} \mathcal{C}^\infty(\mathbb{R}) & \xrightarrow{q_a} & A \\ \downarrow & & \downarrow \\ \mathcal{C}^\infty(\mathbb{R} \setminus \{0\}) & \longrightarrow & A[1/a]. \end{array}$$

We now present an analogous characterization in the general setting of $\mathcal{C}^\infty$ -rings.

Proposition 2.2.37 (cf. [12, Proposition 4.17]). *Let $f : A \rightarrow B$ be a map of $\mathcal{C}^\infty$ -rings and let $a \in \pi_0(A)$. The following equivalent:*

1. The map $f : A \rightarrow B$ exhibits B as a localization with respect to a .
2. B fits into a pushout diagram

$$\begin{array}{ccc} \mathcal{C}^\infty(\mathbb{R}) & \xrightarrow{q_a} & A \\ \downarrow & & \downarrow \\ \mathcal{C}^\infty(\mathbb{R} \setminus \{0\}) & \longrightarrow & B \end{array}$$

where q_a is the unique up to homotopy map associated to $a \in \pi_0(A)$.

Remark 2.2.38. An immediate consequence of the above Proposition point *ii*) is that the localization always exists.

Proof Proposition 2.2.37. First consider the case A is finitely generated. By Proposition D.2.18, there exists an effective epimorphism $p : \mathcal{C}^\infty(\mathbb{R}^n) \rightarrow A$ for some n . In particular, the map $q_a : \mathcal{C}^\infty(\mathbb{R}) \rightarrow A$ corresponding to the element $a \in \pi_0(A)$ factors through p . This implies that a is the image of some element $\hat{a} \in \mathcal{C}^\infty(\mathbb{R}^n)$ under p . Consider the diagram

$$\begin{array}{ccccc} \mathcal{C}^\infty(\mathbb{R}) & \xrightarrow{\hat{a}} & \mathcal{C}^\infty(\mathbb{R}^n) & \xrightarrow{p} & A \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{C}^\infty(\mathbb{R} \setminus \{0\}) & \longrightarrow & \mathcal{C}^\infty(\mathbb{R}^n)[\hat{a}^{-1}] & \longrightarrow & B. \end{array}$$

The left square is a pushout by Corollary 2.1.10 and Lemma 2.2.10, so the right square is a pushout if and only if the outer rectangle is a pushout. However, the right square being a pushout is equivalent to B being the

localization of A at a . In fact, let C be an $\mathcal{C}^\infty$ -ring. Then:

$$\begin{aligned}
\mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}(B, C) &= \mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}\left(A \otimes_{\mathcal{C}^\infty(\mathbb{R}^n)}^\infty \mathcal{C}^\infty(\mathbb{R}^n)[\hat{a}^{-1}], C\right) \\
&= \mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}(A, C) \times_{\mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}(\mathcal{C}^\infty(\mathbb{R}^n), C)} \mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}(\mathcal{C}^\infty(\mathbb{R}^n)[\hat{a}^{-1}], C) \\
&= \mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}(A, C) \times_{\mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}(\mathcal{C}^\infty(\mathbb{R}^n), C)} \mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}^{\hat{a}}(\mathcal{C}^\infty(\mathbb{R}^n), C) \\
&= \mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}^{\hat{a}}(A, C).
\end{aligned}$$

This conclude case A is finitely generated.

Now let A be a general $\mathcal{C}^\infty$ -ring, and choose some $a \in \pi_0(A)$. By a cofinality argument, we can write A as a directed colimit of finitely generated subrings

$$\mathrm{colim}_{i \in \mathcal{J}} A_i \simeq A$$

such that $a \in \pi_0(A_i)$ for all $i \in \mathcal{J}$. We claim that the map

$$\varphi : A \rightarrow \mathrm{colim}_{i \in \mathcal{J}} (A_i[a^{-1}])$$

is a localization. To prove this, take a $\mathcal{C}^\infty$ -Ring C and $g \in \mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}(A, C)$ and then consider the following pullback:

$$\begin{array}{ccc}
K_g & \longrightarrow & \lim_{i \in \mathcal{J}} \mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}(A_i[a^{-1}], C) \\
\downarrow & & \downarrow \\
\{g\} & \longrightarrow & \mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}(A, C).
\end{array}$$

The map φ is a localization if and only if, for every $\mathcal{C}^\infty$ -ring C and every morphism $g : A \rightarrow C$, the space K_g is contractible when $g(a)$ is invertible, and empty otherwise. Let g_i denote the maps $A_i \rightarrow C$ induced by g , and define K_{g_i} as the pullback:

$$\{g_i\} \times_{\mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}(A_i, C)} \mathrm{Hom}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Ring}}(A_i[a^{-1}], C).$$

Since limits commute with each other, we have an equivalence between the limit $\lim_{i \in \mathcal{J}} K_{g_i}$ and K_g . If $g(a)$ (and therefore $g_i(a)$) is not invertible, K_g is a limit of empty spaces and also empty. On the other hand, if $g(a)$ (and therefore $g_i(a)$) is invertible, then each K_{g_i} is contractible, and since limits of contractible spaces are again contractible, it follows that K_g is contractible. Now if $A_i \rightarrow A_i[a^{-1}]$ satisfies ii) for all $i \in \mathcal{J}$, then

$$A \rightarrow \mathrm{colim}_{i \in \mathcal{J}} (A_i[a^{-1}])$$

satisfies ii) and vice versa. Since the A_i are finitely generated, by the first case we conclude. $\square$

Remark 2.2.39. Combining Remark 2.2.9, Lemma D.2.19 and Proposition 2.2.37 shows that localization of finitely presented $\mathcal{C}^\infty$ -ring with respect to any element of $\pi_0(A)$ is again finitely presented.

2.3 Spectra of $\mathcal{C}^\infty$ -rings

Continuing the parallel with algebraic geometry, we now turn to the definition of the *affines* in our setting. In algebraic geometry, one starts with an ordinary commutative ring and associates to it its spectrum, which is a topological space equipped with a sheaf of rings. Guided by this analogy, we begin a discussion that will lead to the association, for each finitely presented $\mathcal{C}^\infty$ -ring A , of a topological space $\mathbf{Spec}^\infty(A)$ together with a sheaf of $\mathcal{C}^\infty$ -rings $\mathcal{O}_A$. In this section, we restrict to finitely presented $\mathcal{C}^\infty$ -rings in order to better study their properties and because they will be needed in Section 2.4. However, we will see in Subsection 2.5.1 that most of the construction can be carried out for general $\mathcal{C}^\infty$ -rings.

The main reference for this section is [12] section 4.3. Unless otherwise stated, theorems and proofs are from *loc. cit.* The original construction in the discrete setting is due to [21].

2.3.1 The functor $\mathbf{Spec}^\infty$

Our first goal is to associate a topology to each finitely presented $\mathcal{C}^\infty$ -ring. The notations follow those introduced in Appendix D.2. In particular we denote by $j^{\mathbf{fp}}$ the corepresented functor from $\mathbf{C}^\infty$ to $(\mathbf{d}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}})^{\mathbf{op}}$. The forgetful functor $u : \mathbf{C}^\infty \rightarrow \mathbf{Top}$ preserves finite products. As a consequence by Theorem D.2.20 the right Kan extension

$$\begin{array}{ccc} \mathbf{C}^\infty & \xrightarrow{j^{\mathbf{fp}}} & (\mathbf{d}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}})^{\mathbf{op}} \\ \downarrow u & \swarrow \text{Ran}_{j^{\mathbf{fp}}} u & \\ \mathbf{Top} & & \end{array}$$

preserves finite limits.

Definition 2.3.1. Let $A \in (\mathbf{d}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}})^{\mathbf{op}}$. We define the *underlying topological space* $\mathbf{Spec}^\infty(A)$ of A as $\text{Ran}_{j^{\mathbf{fp}}} u(A)$.

By Proposition D.2.22 we have that: $\mathbf{Spec}^\infty \simeq \mathbf{Spec}_0^\infty \circ \pi_0$, where $\mathbf{Spec}_0^\infty$ is the right Kan extension of u along the functor $j^{\mathbf{fp}} : \mathbf{C}^\infty \rightarrow (\mathbf{dis}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}})^{\mathbf{op}}$. In particular to describe $\mathbf{Spec}^\infty$ it suffices to describe $\mathbf{Spec}_0^\infty$. Now notice that the forgetful functor

$$\underline{(-)} : \mathbf{Top} \rightarrow \mathbf{Set}$$

preserves limits; hence, again by right Kan extension we obtain

$$\begin{array}{ccc} \mathbf{C}^\infty & \xrightarrow{j^{\mathbf{fp}}} & (\mathbf{d}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}})^{\mathbf{op}} \\ \downarrow \underline{u} & \swarrow \underline{\mathbf{Spec}_0^\infty} := \text{Ran}_{j^{\mathbf{fp}}} \underline{u} & \\ \mathbf{Set} & & \end{array}.$$

Furthermore the functor $\underline{u} : \mathbf{C}^\infty \rightarrow \mathbf{Set}$ preserves finite products so it defines a $\mathcal{C}^\infty$ -ring, which, unwinding definitions, turns out to be $\mathcal{C}^\infty(\mathbb{R}^0) = \mathbb{R}$. Consider a finitely presented $\mathcal{C}^\infty$ -ring A . Since $A \simeq \lim_{\mathbf{C}_{A/}^\infty} j^{\mathbf{fp}}(X)$, we have that

$$\begin{aligned} \underline{\mathbf{Spec}_0^\infty}(A) &= \lim_{\mathbf{C}_{A/}^\infty} \underline{u}(j^{\mathbf{fp}}(X)) \\ &\simeq \lim_{\mathbf{C}_{A/}^\infty} \text{Hom}_{(\mathbf{d}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}})^{\mathbf{op}}}(\mathbb{R}, j^{\mathbf{fp}}(X)) \\ &\simeq \text{Hom}_{(\mathbf{d}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}})^{\mathbf{op}}}(\mathbb{R}, A) \\ &\simeq \text{Hom}_{\mathbf{d}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}}}(A, \mathbb{R}). \end{aligned}$$

Thanks to the above calculation we find the following formula:

$$\underline{\mathbf{Spec}_0^\infty}(-) = \text{Hom}_{\mathbf{Alg}_{\mathcal{C}^\infty}(\mathbf{Spc})^{\mathbf{fp}}}(-, \mathbb{R}) \simeq \text{Hom}_{\mathbf{Alg}_{\mathcal{C}^\infty}(\mathbf{Set})^{\mathbf{fp}}}(\pi_0(-), \mathbb{R})$$

Hence, if A is a finitely presented $\mathcal{C}^\infty$ -ring, the underlying set of $\mathbf{Spec}^\infty(A)$ is

$$\text{Hom}_{\mathbf{dis}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}}}(\pi_0(A), \mathbb{R}).$$

Since $\mathbf{Spec}^\infty = \text{Ran}_{j^{\mathbf{fp}}} u$ can be computed via the pointwise Kan extension formula, it follows that $\mathbf{Spec}^\infty(A)$ carries the initial topology with respect to the family of maps

$$\text{Hom}_{\mathbf{dis}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}}}(\pi_0(A), \mathbb{R}) \rightarrow \text{Hom}_{\mathbf{dis}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}}}(\pi_0(\mathcal{C}^\infty(\mathbb{R}^n)), \mathbb{R}) = \mathbb{R}^n,$$

induced by all morphisms $\mathcal{C}^\infty(\mathbb{R}^n) \rightarrow A$, where $\mathbb{R}^n$ is equipped with the standard topology. This is the coarsest topology making each map

$$\text{Hom}_{\mathbf{dis}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}}}(\pi_0(A), \mathbb{R}) \rightarrow \text{Hom}_{\mathbf{dis}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}}}(\pi_0(\mathcal{C}^\infty(\mathbb{R})), \mathbb{R}) = \mathbb{R}$$

continuous, for every morphism $\mathcal{C}^\infty(\mathbb{R}) \rightarrow \pi_0(A)$. Now, since $\mathcal{C}^\infty(\mathbb{R})$ is the free $\mathcal{C}^\infty$ -ring on one generator (see Proposition 2.1.5), we have a canonical isomorphism

$$\mathrm{Hom}_{\mathbf{dis}\mathcal{C}^\infty\mathbf{Ring}^{\mathrm{fp}}}(\mathcal{C}^\infty(\mathbb{R}), \pi_0(A)) \simeq \pi_0(A).$$

In particular, for each element $a \in \pi_0(A)$, we obtain a map

$$\mathrm{Hom}_{\mathbf{dis}\mathcal{C}^\infty\mathbf{Ring}^{\mathrm{fp}}}(\pi_0(A), \mathbb{R}) \rightarrow \mathbb{R},$$

given by evaluating morphisms at the element a . Summarizing, we have that:

$$\mathbf{Spec}^\infty(A) = \mathrm{Hom}_{\mathbf{Alg}_{\mathcal{C}^\infty}(\mathbf{Set})^{\mathrm{fp}}}(\pi_0(-), \mathbb{R})$$

equipped with the coarsest topology making each valuation map

$$ev_a : \mathrm{Hom}_{\mathbf{Alg}_{\mathcal{C}^\infty}(\mathbf{Set})^{\mathrm{fp}}}(\pi_0(A), \mathbb{R}) \rightarrow \mathbb{R}$$

continuous, for each $a \in \pi_0(A)$. In particular this means that the subsets of $\mathbf{Spec}^\infty(A)$ of the form $ev_a^{-1}(U)$ constitute a subbasis for the topology.

Lemma 2.3.2. *Each open subset of the form $ev_a^{-1}(U)$ is also of the form $W_b := ev_b^{-1}(\mathbb{R} \setminus 0)$. Moreover, these open subset are closed under finite intersection. In particular, they form a basis.*

Proof. Let $U \subseteq \mathbb{R}$ be open. Then there exists a smooth function

$$\chi_U : \mathbb{R} \rightarrow \mathbb{R}$$

such that $U = \chi_U^{-1}(\mathbb{R} \setminus 0)$ (see [47] Lemma 1.4). But this implies that

$$\begin{aligned} ev_a^{-1}(U) &= (\chi_U \circ ev_a)^{-1}(\mathbb{R} \setminus 0) \\ &= ev_{\chi_U(a)}^{-1}(\mathbb{R} \setminus 0). \end{aligned}$$

For the second statement notice that $\mathbb{R}$ has no zero-divisors, in particular $W_a \cap W_b = W_{ab}$. □

Proposition 2.3.3. *Every open subset of $\mathbf{Spec}^\infty(A)$ is of the form W_a .*

Proof. Since $\pi_0(A)$ is finitely presented, it is in particular finitely generated, so we may find a surjective homomorphism

$$\pi : \mathcal{C}^\infty(\mathbb{R}^n) \rightarrow \pi_0(A),$$

and $\mathbf{Spec}^\infty(A) = \mathbf{Spec}^\infty(\pi_0(A))$ is naturally a closed subset of $\mathbf{Spec}^\infty(\mathcal{C}^\infty(\mathbb{R}^n)) = \mathbb{R}^n$. Let U be any open subset of $\mathbf{Spec}^\infty(A)$, and denote its complement by C . Since C is also closed in $\mathbb{R}^n$, there exists a smooth function $g : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $g^{-1}(0) = C$. It follows that $C = ev_{\pi(g)}^{-1}(0)$, and hence $U = W_{\pi(g)}$. □

2.3.2 Homotopically (locally) $\mathcal{C}^\infty$ -ringed spaces

Construction 2.3.4. Consider the functor:

$$\mathrm{Shv}(-, \mathbf{d}\mathcal{C}^\infty\mathbf{Ring})^{\mathrm{op}} : \mathbf{Top} \rightarrow \mathbf{Cat}_\infty$$

(see Construction F.3.23) and the induced coCartesian fibration

$$\mathbf{Top}_{\mathcal{C}^\infty} := \mathrm{Un}^{\mathrm{cc}}(\mathrm{Shv}(-, \mathbf{d}\mathcal{C}^\infty\mathbf{Ring})^{\mathrm{op}}) \rightarrow \mathbf{Top}.$$

Definition 2.3.5 (Homotopically $\mathcal{C}^\infty$ -ringed spaces). We define the category of *homotopically $\mathcal{C}^\infty$ -ringed topological spaces* to be $\mathbf{Top}_{\mathcal{C}^\infty}$.

Informally, we can think of $\mathbf{Top}_{\mathcal{C}^\infty}$ as the ∞ -category whose:

- objects are pairs $(X, \mathcal{O}_X)$ with X topological space and $\mathcal{O}_X$ a sheaf of $\mathcal{C}^\infty$ -rings,
- morphisms are pairs $(f, f_\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$, with f continuous map and $f_\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ morphism of sheaves of $\mathcal{C}^\infty$ -rings over Y .

As in scheme theory, we are particularly interested in those ringed spaces that exhibit “nice” local behavior. More specifically, we would like to adapt the condition that the stalk of the structure sheaf at every point of the topological space X be a local ring. For objects in $\mathbf{dC}^\infty\mathbf{Ring}$, this condition does not make sense in the usual way, due to the derived structure involved. However, it has a whole different flavor on π_0 .

Definition 2.3.6 (Homotopically locally $\mathcal{C}^\infty$ -ringed spaces). We denote by $\mathbf{Loc}_{\mathbf{C}^\infty}$ the full subcategory of $\mathbf{Top}_{\mathbf{C}^\infty}$ spanned by those objects $(X, \mathcal{O}_X)$ such that each stalk of $\pi_0(\mathcal{O}_X)$ is a local discrete $\mathcal{C}^\infty$ -ring (see [47, Chapter I section 3]) with residue field $\mathbb{R}$. Objects of $\mathbf{Loc}_{\mathbf{C}^\infty}$ are called *homotopically locally $\mathcal{C}^\infty$ -ringed spaces*.

Remark 2.3.7. In general we should also ask for the induced morphism on stalks to be morphisms of local rings. This is not needed since all stalks are $\mathbb{R}$.

Remark 2.3.8. Notice that by Remark F.3.24 $\mathbf{Top}_{\mathbf{C}^\infty} \rightarrow \mathbf{Top}$ is a bifibration.

Lemma 2.3.9. *The ∞ -category $\mathbf{Top}_{\mathbf{C}^\infty}$ has finite limits and the functor to $\mathbf{Top}$ preserves them.*

Proof. By Remark F.3.26 we have an equivalence:

$$\mathbf{Top}_{\mathbf{C}^\infty} \simeq (\mathrm{Un}^{\mathrm{cc}}(\mathrm{Shv}(-, \mathbf{dC}^\infty\mathbf{Ring})^*))^{\mathrm{op}}$$

over $\mathbf{Top}$. So it suffices to prove that $\mathrm{Un}^{\mathrm{cc}}(\mathrm{Shv}(-, \mathbf{dC}^\infty\mathbf{Ring})^*)$ has finite colimits and the functor to $\mathbf{Top}^{\mathrm{op}}$ preserves them. Consider a continuous map $f : X \rightarrow Y$ and the following commutative diagram:

$$\begin{array}{ccccc} \mathbf{Alg}_{\mathbf{C}^\infty}(\mathrm{Shv}(Y)) & \xrightarrow{\sim} & \mathrm{Shv}(\mathrm{Shv}(Y), \mathbf{Alg}_{\mathbf{C}^\infty}(\mathbf{Spc})) & \xrightarrow{f^*} & \mathrm{Shv}(\mathrm{Shv}(X), \mathbf{Alg}_{\mathbf{C}^\infty}(\mathbf{Spc})) & \xrightarrow{\sim} & \mathbf{Alg}_{\mathbf{C}^\infty}(\mathrm{Shv}(X)) \\ U_Y \downarrow & & & & & & \downarrow U_X \\ \mathrm{Shv}(Y) & \xrightarrow{\hspace{10em}} & & \xrightarrow{f^*} & & & \mathrm{Shv}(X) \end{array} .$$

Where the horizontal equivalences are from Lemma F.3.12 and the vertical maps are the forgetful functors. Since $\mathbf{C}^\infty$ is an algebraic theory, the forgetful functor

$$\mathbf{Alg}_{\mathbf{C}^\infty}(\mathrm{Shv}(X)) \rightarrow \mathrm{Shv}(X)$$

is conservative and preserves limits, and hence f^* preserves finite limits, since the functor

$$f^* : \mathrm{Shv}(Y) \rightarrow \mathrm{Shv}(X)$$

does. The result now follows from [37, Corollary 4.3.1.11]. $\square$

Remark 2.3.10. This remark is verbatim from [12, Remark 4.25]. Unwinding the definitions, we see that moreover, given a functor from a finite ∞ -category J

$$F : J \rightarrow \mathbf{Top}_{\mathbf{C}^\infty},$$

the limit may be computed as follows: first, consider the fibration

$$p : \mathbf{Top}_{\mathbf{C}^\infty} \rightarrow \mathbf{Top}$$

and denote by X the limit of $p \circ F$. Let $\rho : \Delta_X \Rightarrow pF$ be a limiting cone for X . Consider the functor

$$F_X : J \rightarrow \mathrm{Shv}(X, \mathbf{dC}^\infty\mathbf{Ring})$$

$$j \mapsto \rho_j^* F(j),$$

and let $\mathcal{O}_X := \varprojlim_{j \in J} F_X(j)$. Then $(X, \mathcal{O}_X)$ is a limit of F .

Proposition 2.3.11 (cf. [12, Proposition 4.26]). *The ∞ -category $\mathbf{Top}_{\mathbf{C}^\infty}$ is idempotent complete (see [37, Section 4.4.5]).*

Remark 2.3.12 (cf. [12, Remark 4.28]). For every $X \in \mathbf{Top}$, we have a natural inclusions

$$i_{0,X} : \mathrm{Shv}(X, \mathbf{disC}^\infty \mathbf{Ring}) \hookrightarrow \mathrm{Shv}(X, \mathbf{dC}^\infty \mathbf{Ring}),$$

such that for every morphism $X \rightarrow Y$, we obtain a commutative square:

$$\begin{array}{ccc} \mathrm{Shv}(X, \mathbf{disC}^\infty \mathbf{Ring}) & \xrightarrow{f_*} & \mathrm{Shv}(Y, \mathbf{disC}^\infty \mathbf{Ring}) \\ \downarrow i_{0,X} & & \downarrow i_{0,Y} \\ \mathrm{Shv}(X, \mathbf{dC}^\infty \mathbf{Ring}) & \xrightarrow{f_*} & \mathrm{Shv}(Y, \mathbf{dC}^\infty \mathbf{Ring}). \end{array}$$

By [53, Proposition A.1], this yields a natural transformation:

$$i_{0,-} : \mathrm{Shv}(-, \mathbf{disC}^\infty \mathbf{Ring}) \Rightarrow \mathrm{Shv}(-, \mathbf{dC}^\infty \mathbf{Ring}).$$

Therefore, by passing to opposite categories and applying unstraightening, we get an induced functor

$$i_0 : \mathbf{Top}_{\mathbf{disC}^\infty \mathbf{Ring}} \rightarrow \mathbf{Top}_{\mathbf{C}^\infty}$$

over $\mathbf{Top}$, where $\mathbf{Top}_{\mathbf{disC}^\infty \mathbf{Ring}}$ denotes the category of $\mathbf{disC}^\infty \mathbf{Ring}$ -structured spaces (see Definition F.3.25). By [37, Proposition 5.5.6.16], each $i_{0,X}$ admits a left adjoint $\pi_{0,X}$, so by [40, Proposition 7.3.2.6], we obtain a left adjoint:

$$t_0 : \mathbf{Top}_{\mathbf{C}^\infty} \rightarrow \mathbf{Top}_{\mathbf{disC}^\infty \mathbf{Ring}}.$$

Since f^* is left exact colimit preserving, it follows from [37, Proposition 5.5.6.28] that t_0 is cartesian. Passing to the opposite categories, we define $i'_0 := i_0^{\mathrm{op}}$ and $t'_0 := t_0^{\mathrm{op}}$, so that i'_0 is left adjoint to t'_0 . Concretely, we have:

$$t'_0(X, \mathcal{O}_X) = (X, \pi_0 \mathcal{O}_X).$$

Lemma 2.3.13 (cf. [12, Lemma 4.29]). *The ∞ -category $\mathrm{Loc}_{\mathbf{C}^\infty}$ has finite limits and the inclusion*

$$\mathrm{Loc}_{\mathbf{C}^\infty} \hookrightarrow \mathbf{Top}_{\mathbf{C}^\infty}$$

preserves them. Moreover, $\mathrm{Loc}_{\mathbf{C}^\infty}$ is idempotent complete.

Proposition 2.3.14 (cf. [12, Proposition 4.30]). *Let $(\mathbb{R}, \mathcal{C}_{\mathbb{R}}^\infty)$ be the smooth manifold $\mathbb{R}$ regarded as an element of $\mathrm{Loc}_{\mathbf{C}^\infty}$. Then, for all $n \geq 0$, $(\mathbb{R}, \mathcal{C}_{\mathbb{R}}^\infty)^n \simeq (\mathbb{R}^n, \mathcal{C}_{\mathbb{R}^n}^\infty)$*

Proof. For $n = 0$ there's nothing to prove. For $n = 2$ we can follow Remark 2.3.10 to get:

$$(\mathbb{R}, \mathcal{C}_{\mathbb{R}}^\infty) \times (\mathbb{R}, \mathcal{C}_{\mathbb{R}}^\infty) \simeq (\mathbb{R}^2, pr_1^* \mathcal{C}_{\mathbb{R}}^\infty \otimes^\infty pr_2^* \mathcal{C}_{\mathbb{R}}^\infty).$$

Where the sheaf $pr_1^* \mathcal{C}_{\mathbb{R}}^\infty \otimes^\infty pr_2^* \mathcal{C}_{\mathbb{R}}^\infty$ over $\mathbb{R}^2$ is the sheafification of the presheaf:

$$W \mapsto pr_1^* \mathcal{C}_{\mathbb{R}}^\infty(W) \otimes^\infty pr_2^* \mathcal{C}_{\mathbb{R}}^\infty(W).$$

We now wish to prove that the sheafification is equivalent to $\mathcal{C}_{\mathbb{R}^2}^\infty$. For this, it suffices to show that for all $U, V \subseteq \mathbb{R}$ opens we have $pr_1^* \mathcal{C}_{\mathbb{R}}^\infty(U \times V) \otimes^\infty pr_2^* \mathcal{C}_{\mathbb{R}}^\infty(U \times V) \simeq \mathcal{C}^\infty(U \times V)$. Since pr_1 and pr_2 are open maps, the above $\mathcal{C}^\infty$ -ring is equivalent to:

$$\mathcal{C}_{\mathbb{R}}^\infty(U) \otimes^\infty \mathcal{C}_{\mathbb{R}}^\infty(V).$$

Now the result follows from Lemma 2.2.8. For other n the proof is similar. $\square$

Corollary 2.3.15 (cf. [12, Corollary 4.31]). *$(\mathbb{R}, \mathcal{C}_{\mathbb{R}}^\infty)$ has the canonical structure of a $\mathcal{C}^\infty$ -ring object in $\mathrm{Loc}_{\mathbf{C}^\infty}$.*

Construction 2.3.16. The above corollary provide a $\mathcal{C}^\infty$ -ring in $\mathrm{Loc}_{\mathbf{C}^\infty}$ that we are going to denote $\underline{\mathbb{R}} : \mathbf{C}^\infty \rightarrow \mathrm{Loc}_{\mathbf{C}^\infty}$. By Theorem D.2.20 we obtain a new functor

$$\mathrm{Ran}_{j_{\mathrm{fp}}}(\underline{\mathbb{R}}) : (\mathbf{dC}^\infty \mathbf{Ring}^{\mathrm{fp}})^{\mathrm{op}} \rightarrow \mathrm{Loc}_{\mathbf{C}^\infty}$$

through right Kan extension.

Definition 2.3.17. We define $\mathbf{Spec}_{\mathcal{C}^\infty} := \mathbf{Ran}_{j_{\mathbf{fp}}}(\mathbb{R})$, and for every finitely presented $\mathcal{C}^\infty$ -ring A , we refer to $\mathbf{Spec}_{\mathcal{C}^\infty}(A)$ as the $\mathcal{C}^\infty$ -spectrum of A .

We can also give a 0-truncated version of the above construction:

Definition 2.3.18. Denote by $\mathbf{Loc}_{\mathcal{C}^\infty}^0$ the full subcategory of $\mathbf{Loc}_{\mathcal{C}^\infty}$ spanned by those object $(X, \mathcal{O}_X)$ such that $\mathcal{O}_X$ is 0-truncated.

We denote by $\mathbb{R}^0$ the factorization of the functor $\mathbb{R}$ through $\mathbf{Loc}_{\mathcal{C}^\infty}^0$. As above we may right Kan extend and obtain:

$$\mathbf{Ran}_{j_{\mathbf{fp}}}(\mathbb{R}^0) : \left(\mathbf{dis}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}} \right)^{\mathrm{op}} \rightarrow \mathbf{Loc}_{\mathcal{C}^\infty}^0.$$

We will denote the above functor by $\mathbf{Spec}_{\mathcal{C}^\infty}^0$.

Remark 2.3.19 (cf. [12, Remark 4.33]). We now want to relate the two functor $\mathbf{Spec}_{\mathcal{C}^\infty}$ and $\mathbf{Spec}_{\mathcal{C}^\infty}^0$. This can be done by means of Proposition D.2.22, in fact the functor $t'_0 : \mathbf{Loc}_{\mathcal{C}^\infty} \rightarrow \mathbf{Loc}_{\mathcal{C}^\infty}^0$ preserves finite limits and $\mathbf{Loc}_{\mathcal{C}^\infty}^0$ is a 1-category, so we have that for all finitely presented $\mathcal{C}^\infty$ -rings A in $\mathbf{Spc}$,

$$t'_0(\mathbf{Spec}_{\mathcal{C}^\infty}(A)) \simeq \mathbf{Spec}_{\mathcal{C}^\infty}^0(\pi_0(A))$$

Remark 2.3.20 (Explicit description of $\mathbf{Spec}_{\mathcal{C}^\infty}$, cf. [12, Remark 4.34]). Since $p : \mathbf{Loc}_{\mathcal{C}^\infty} \rightarrow \mathbf{Top}$ preserves finite limits (by Lemma 2.3.9 and Lemma 2.3.13), it follows that $p \circ \mathbf{Spec}_{\mathcal{C}^\infty} \simeq \mathbf{Spec}^\infty$. Hence we can write $\mathbf{Spec}_{\mathcal{C}^\infty}(A) = (\mathbf{Spec}^\infty(A), \mathcal{O}_A)$ for some sheaf $\mathcal{O}_A$.

Theorem 2.3.21 ([21] Theorem 16). *The natural functor $\mathbf{Mfd} \rightarrow \mathbf{Loc}_{\mathcal{C}^\infty}^0$ factors as*

$$\mathbf{Mfd} \xrightarrow{\mathcal{C}^\infty} \left(\mathbf{dis}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}} \right)^{\mathrm{op}} \xrightarrow{\mathbf{Spec}_{\mathcal{C}^\infty}^0} \mathbf{Loc}_{\mathcal{C}^\infty}^0$$

Theorem 2.3.22 ([21] Theorem 14). *The functor*

$$\mathbf{Spec}_{\mathcal{C}^\infty}^0 : \left(\mathbf{dis}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}} \right)^{\mathrm{op}} \rightarrow \mathbf{Loc}_{\mathcal{C}^\infty}^0$$

is fully faithful, and for any finitely presented discrete $\mathcal{C}^\infty$ -ring we have

$$A \simeq \Gamma(\mathcal{O}_A)$$

where Γ is the global section functor.

Corollary 2.3.23 (cf. [12, Corollary 4.38]). *For A a finitely presented discrete $\mathcal{C}^\infty$ -ring, and $W_a \subset \mathbf{Spec}^\infty(A)$ an arbitrary open subset, with $a \in A$, we have*

$$\mathcal{O}(W_a) \simeq A[a^{-1}]$$

Remark 2.3.24. In the proof of the above proposition (see [12]), it is observed that for a discrete $\mathcal{C}^\infty$ -ring A and an element $a \in A$, we have an equivalence

$$\mathbf{Spec}^\infty(A[a^{-1}]) \simeq W_a.$$

Now, let A be a (possibly non-discrete) finitely presented $\mathcal{C}^\infty$ -ring and $a \in \pi_0(A)$. Then we obtain

$$\mathbf{Spec}^\infty(A[a^{-1}]) \simeq \mathbf{Spec}^\infty(\pi_0(A[a^{-1}])) \simeq \mathbf{Spec}^\infty(\pi_0(A)[a^{-1}]) \simeq W_a.$$

In particular, since by Proposition 2.2.37 the localization $A[a^{-1}]$ is finitely presented, it follows that the Grothendieck topology on the essential image of $\mathbf{Spec}^\infty$ restricts to a Grothendieck topology on $\left(\mathbf{d}\mathcal{C}^\infty \mathbf{Ring}^{\mathbf{fp}} \right)^{\mathrm{op}}$.

At this point, for a detailed discussion of the following results we refer to [12, Sections 4.3.1 and 4.3.2]. Here, we present only the results that will be needed.

Proposition 2.3.25 ([12] Corollary 4.46, Corollary 4.49). *Let A be a finitely presented $\mathcal{C}^\infty$ -algebra in $\mathbf{Spc}$. Then the presheaf*

$$\tilde{\mathcal{O}}_A : W_a \mapsto A[a^{-1}]$$

is a sheaf. Moreover.

$$\mathcal{O}_A \simeq \tilde{\mathcal{O}}_A$$

Proposition 2.3.26 ([12] Theorem 4.50 and Proposition 4.51). *The functor*

$$\mathbf{Spec}_{\mathbf{C}^\infty} : (\mathbf{dC}^\infty\mathbf{Ring}^{\mathbf{fp}})^{op} \rightarrow \mathbf{Loc}_{\mathbf{C}^\infty}$$

is fully faithful. Moreover, if M is a smooth manifold, then

$$\mathbf{Spec}_{\mathbf{C}^\infty}(\mathcal{C}^\infty(M)) \simeq (M, \mathcal{C}_M^\infty)$$

Proposition 2.3.27 (cf. [12, Proposition 4.52]). *The functor $\mathcal{C}^\infty : \mathbf{Mfd} \hookrightarrow \mathbf{dC}^\infty\mathbf{Ring}^{op}$ sending a manifold M to the discrete $\mathcal{C}^\infty$ -ring of smooth functions on M is a fully faithful, and preserves transverse pullback.*

2.4 Derived manifolds

In this section, we finally turn our attention to derived manifolds. As previously noted, the ∞ -category of derived manifolds should be understood as an enlargement of the category $\mathbf{Mfd}$, where we formally add all pullbacks while preserving the transverse ones already present in $\mathbf{Mfd}$. The first goal of this section is to make this idea precise.

We begin by introducing the category of derived manifolds via a universal property, and we examine the first consequences of this property. We then proceed to study a concrete model for this category.

Unless otherwise specified, theorems and proofs are taken from Sections 2 and 5.1 of the joint work of Carchedi and Steffens [12].

2.4.1 Universal property and consequences

Construction 2.4.1. Let $\mathcal{D}$ denote the subcategory of $\mathbf{Cat}_\infty$ whose

- objects are small ∞ -categories that have finite limits and are idempotent complete,
- morphisms are left exact functors.

Denote by $F : \mathcal{D} \rightarrow \mathbf{Cat}_\infty$ the functor that to every ∞ -category $\mathcal{C}$ in $\mathcal{D}$ assigns the full subcategory $\mathbf{Fun}^\natural(\mathbf{Mfd}, \mathcal{C})$ of $\mathbf{Fun}(\mathbf{Mfd}, \mathcal{C})$ on those functors preserving transverse pullback and the terminal object. Denote by $\mathcal{E} \rightarrow \mathcal{D}$ the associated CoCartesian fibration. $\mathcal{E}$ can be described as the category whose:

- objects are finitely complete idempotent complete ∞ -categories equipped with a functor from $\mathbf{Mfd}$ which preserves transverse pullback and terminal object.
- morphisms are left exact functor which are compatible with the distinguished functor from $\mathbf{Mfd}$.

Definition 2.4.2. The ∞ -category $\mathbf{dMfd}$ of *derived manifolds* is defined to be an initial object of $\mathcal{E}$, if it exists.

Unwinding the definition we find that $\mathbf{dMfd}$ has finite limits, is idempotent complete and receive a transverse pullback and terminal object preserving functor i from the category $\mathbf{Mfd}$ of smooth manifold. Even more in this context the universal property of the initial object says that for any other idempotent complete ∞ -category $\mathcal{C}$ with finite limits, composition with i induces an equivalence of ∞ -categories

$$\mathbf{Fun}^{\mathbf{lex}}(\mathbf{dMfd}, \mathcal{C}) \xrightarrow{\sim} \mathbf{Fun}^\natural(\mathbf{Mfd}, \mathcal{C}).$$

It can be proven that this category exists by pure abstract reasoning, however in the next section we are going to present an explicit model of such a category.

We now present some immediate consequence of the universal property.

Example 2.4.3 (Topology on $\mathbf{dMfd}$). We can consider the analogue of the forgetful functor $u : \mathbf{Mfd} \rightarrow \mathbf{Top}$ for the ∞ -category $\mathbf{dMfd}$. More precisely the forgetful functor u preserves transverse pullback and terminal object and so by universal property we obtain a functor $U : \mathbf{dMfd} \rightarrow \mathbf{Top}$ such that the following diagram commutes:

$$\begin{array}{ccc} \mathbf{Mfd} & \xrightarrow{i} & \mathbf{dMfd} \\ \downarrow u & \swarrow U & \\ \mathbf{Top} & & \end{array}$$

We will call the functor U *the underlying space functor*. Through this construction, we can give $\mathbf{dMfd}$ a Grothendieck topology $J_{\mathbf{dMfd}}$ that mimics the one on $\mathbf{Top}$. Explicitly, we declare a collection of maps $\{f_\alpha : V_\alpha \rightarrow \mathcal{M}\}_\alpha$ to be a cover of a derived manifold $\mathcal{M}$ if and only if $\{U(f_\alpha) : U(V_\alpha) \rightarrow U(\mathcal{M})\}$ is an open cover of topological spaces (formally, we are using [37, Proposition 6.2.4.2] with the left exact morphism given by the composition $\mathbf{dMod} \rightarrow \mathbf{Top} \hookrightarrow \mathbf{Shv}(\mathbf{Top})$).

Example 2.4.4 (The functor $\mathcal{O}_{\mathbf{dMfd}}$). Since $i : \mathbf{Mfd} \rightarrow \mathbf{dMfd}$ preserves transverse pullback and terminal object, the composite $\mathbf{C}^\infty \rightarrow \mathbf{dMfd}$ preserves finite products. In particular for $\mathcal{M}$ any derived manifold, the functor

$$\mathrm{Hom}_{\mathbf{dMfd}}(\mathcal{M}, i(-)) : \mathbf{C}^\infty \rightarrow \mathbf{Spc}$$

preserves finite products, and hence defines an object of $\mathbf{dC}^\infty \mathbf{Ring}$. Therefore we get an induced functor:

$$\mathcal{O}_{\mathbf{dMfd}} : \mathbf{dMfd}^{op} \rightarrow \mathbf{dC}^\infty \mathbf{Ring}.$$

Notice that $\mathcal{O}_{\mathbf{dMfd}}$ sends a smooth manifold to the $\mathcal{C}^\infty$ -ring $\mathcal{C}^\infty(M)$ described in Example 2.1.3.

2.4.2 Explicit construction

We now present some of the main results from [12]. All the material in this subsection is drawn from Section 5.1 of *loc. cit.*

Lemma 2.4.5. *Consider $q : \mathbf{C}^\infty \hookrightarrow \mathbf{Mfd}$ be the fully faithful inclusion. Let $\mathcal{C}$ be an idempotent complete ∞ -category with finite limits. Then the following conditions are equivalent for a functor $F : \mathbf{C}^\infty \rightarrow \mathcal{C}$:*

1. *F preserves finite products*
2. *The right kan extension $\mathrm{Ran}_q F$ exists and preserves transverse pullback and the terminal object.*

Proof. Suppose ii) holds then by fully faithfulness of q we have that $q^* \mathrm{Ran}_q \simeq id$, in particular i) holds. To prove the converse we first consider the case $\mathcal{C}$ complete and cocomplete. Since assumption F preserves finite products we have that by Theorem D.2.20 the right Kan extension

$$\mathrm{Ran}_{j^{\mathrm{fp}}} F : \left(\mathbf{dC}^\infty \mathbf{Ring}^{\mathrm{fp}} \right)^{op} \rightarrow \mathcal{C}$$

exists and preserves finite limit. Consider the following commutative diagram:

$$\begin{array}{ccc} \mathbf{C}^\infty & & \\ \downarrow q & \searrow j^{\mathrm{fp}} & \\ \mathbf{Mfd} & \xrightarrow{\mathbf{C}^\infty} & \left(\mathbf{dC}^\infty \mathbf{Ring}^{\mathrm{fp}} \right)^{op} \end{array}$$

and recall from Lemma 2.2.2 that $\mathbf{C}^\infty$ is fully faithful. We can then use Lemma D.2.10 to see that the right Kan extension $\mathrm{Ran}_q F$ exists and is given by:

$$(\mathrm{Ran}_{j^{\mathrm{fp}}} F) \circ \mathbf{C}^\infty.$$

By Theorem 2.2.12 $\mathrm{Ran}_q F$ preserves transverse pullback and the terminal object.

Now consider $\mathcal{C}$ as in the hypothesis of the proposition. Then, the composite

$$\mathbf{C}^\infty \xrightarrow{F} \mathcal{C} \xrightarrow{y} \mathbf{PSh}(\mathcal{C})$$

preserves finite product and so by the previous case the right Kan extension $\mathrm{Ran}_q(y \circ F)$ exists and it preserves transverse pullback and terminal object.

Claim: $\mathrm{Ran}_q(y \circ F)$ takes values in representable.

By completeness of $\mathbf{PSh}(\mathcal{C})$ we can compute Kan extension through pointwise formula. More precisely, the value of the functor $\mathrm{Ran}_q(y \circ F)$ at a smooth manifold M can be computed as the limit of the following diagram:

$$\mathbf{C}_{M/}^\infty \xrightarrow{t} \mathbf{C}^\infty \xrightarrow{F} \mathcal{C} \xrightarrow{y} \mathbf{PSh}(\mathcal{C}).$$

Now, using that the Yoneda embedding is fully faithful, together with the claim, we conclude that the limit required to compute the pointwise formula for $\text{Ran}_q(F)$ exists in $\mathcal{C}$. Moreover, we can deduce that $\text{Ran}_q(F)$ preserves transverse pullbacks and the terminal object.

Proof of Claim: Recall that by Lemma 2.2.6 a smooth manifold M can be written as a retract of a transverse pullback of Cartesian manifolds. For any Cartesian manifold, $\mathbb{R}^n$, $\text{Ran}_q(y \circ F)(\mathbb{R}^n) \simeq y(F(\mathbb{R}^n))$ is in the essential image of y . Consequently, since $\text{Ran}_q(y \circ F)$ preserves transverse pullback (and retracts) we have that $\text{Ran}_q(y \circ F)(M)$ is a retract of a pullback of a representable. The representability of $\text{Ran}_q(y \circ F)(M)$ now follows by the fact that $\mathcal{C}$ has finite limits and retracts and the Yoneda embedding preserves them. $\square$

Corollary 2.4.6. *Let $\mathcal{C}$ be an idempotent complete ∞ -category with finite limits. Then*

$$q^* : \text{Fun}(\mathbf{Mfd}, \mathcal{C}) \rightarrow \text{Fun}(\mathbf{C}^\infty, \mathcal{C})$$

restricts to an equivalence of ∞ -categories

$$q^* : \text{Fun}^\natural(\mathbf{Mfd}, \mathcal{C}) \rightarrow \mathbf{Alg}_{\mathbf{C}^\infty}(\mathcal{C})$$

between the ∞ -category of functors which preserves transverse pullback and terminal object and functors which preserves finite products.

Proof. By Lemma 2.4.5, there exist a global right Kan extension functor

$$\text{Ran}_q : \mathbf{Alg}_{\mathbf{C}^\infty}(\mathcal{C}) \rightarrow \text{Fun}^\natural(\mathbf{Mfd}, \mathcal{C})$$

which is right adjoint to q^* by universal property of right Kan extension. We now prove that Ran_q exhibits an inverse of q^* . First notice that since q is fully faithful, the counit $q^* \text{Ran}_q \rightarrow \text{id}$ is an equivalence. It follows that to prove the equivalence we only need to prove that the unit is an equivalence and this can be checked on objects. Let $F : \mathbf{Mfd} \rightarrow \mathcal{C}$ be in $\text{Fun}^\natural(\mathbf{Mfd}, \mathcal{C})$. Again by 2.2.6 every manifold is a retract of a transverse pullback and both F and $\text{Ran}_q q^* F$ preserve transverse pullbacks (and retracts) we conclude. $\square$

Theorem 2.4.7. *The ∞ -category $\mathbf{dMfd}$ exists. The functor $\mathcal{O}_{\mathbf{dMfd}}$ constructed in Example 2.4.4 induces a canonical equivalence of ∞ -categories:*

$$\mathbf{dMfd} \simeq \left(\mathbf{dC}^\infty \mathbf{Ring}^{\text{fp}} \right)^{\text{op}}.$$

Furthermore, for all idempotent complete ∞ -categories $\mathcal{C}$ with finite limits, composition with

$$i \circ q : \mathbf{C}^\infty \rightarrow \mathbf{dMfd}$$

induces an equivalence of ∞ -categories

$$\text{Fun}^{\text{lex}}(\mathbf{dMfd}, \mathcal{C}) \xrightarrow{\sim} \mathbf{Alg}_{\mathbf{C}^\infty}(\mathcal{C})$$

between the ∞ -category of left exact functors from derive manifold to $\mathcal{C}$, and the ∞ -category of $\mathbf{C}^\infty$ -rings in $\mathcal{C}$.

Proof. The existence of an equivalence is essentially a consequence of the universal property since the couple $(\mathbf{dMfd}, i \circ q)$ satisfy the same universal property as $\left(\left(\mathbf{dC}^\infty \mathbf{Ring}^{\text{fp}} \right)^{\text{op}}, j^{\text{fp}} \right)$. In particular, $i \circ q$ is a universal (see D.2.29) $\mathbf{C}^\infty$ -ring. It follows that by Remark D.2.32 the equivalence can be concretely realized as:

$$\Gamma_{i \circ q} = \mathcal{O}_{\mathbf{dMfd}}^{\text{op}} : \mathbf{dMfd} \rightarrow \left(\mathbf{dC}^\infty \mathbf{Ring}^{\text{fp}} \right)^{\text{op}}$$

with inverse given by $\text{Ran}_{j^{\text{fp}}}(i \circ q)$.

For the second part, consider the composition:

$$\text{Fun}^{\text{lex}}(\mathbf{dMfd}, \mathcal{C}) \xrightarrow{i^*} \text{Fun}^\natural(\mathbf{Mfd}, \mathcal{C}) \xrightarrow{q^*} \mathbf{Alg}_{\mathbf{C}^\infty}(\mathcal{C})$$

where both functor are equivalences, the first one by definition of $\mathbf{dMfd}$ and the second one by Corollary 2.4.6. $\square$

Proposition 2.4.8. *The Grothendieck topology $J_{\mathbf{dMfd}}$ on $\mathbf{dMfd}$ of Example 2.4.3 is subcanonical.*

Proof. Since both U and $\mathbf{Spec}^\infty$ are Kan extended from $\mathbf{Mfd}$, the following diagram commutes

$$\begin{array}{ccc}
 & \mathbf{Mfd} & \\
 i \swarrow & & \searrow \mathcal{C}^\infty \\
 \mathbf{dMfd} & \xleftarrow{\text{Ran}_{j^{\text{fp}}}(i \circ q)} & (\mathbf{dC}^\infty \mathbf{Ring}^{\text{fp}})^{\text{op}} \\
 U \downarrow & & \swarrow \mathbf{Spec}^\infty \\
 \mathbf{Top} & &
 \end{array}$$

Under the equivalence $\text{Ran}_{j^{\text{fp}}}(i \circ q)$, $J_{\mathbf{dMfd}}$ corresponds to Grothendieck topology on $(\mathbf{dC}^\infty \mathbf{Ring}^{\text{fp}})^{\text{op}}$ induced from $\mathbf{Top}$ via f (see Remark 2.3.24). Call this Grothendieck topology J . It suffices to show that J is subcanonical. We can now conclude from the fact that $\mathbf{Spec}_{\mathbf{C}^\infty}$ is fully faithful and open cover topology on $\text{Loc}_{\mathbf{C}^\infty}$ is subcanonical. $\square$

Proposition 2.4.9. *The functor*

$$\mathcal{O}_{\mathbf{dMfd}} : \mathbf{dMfd}^{\text{op}} \rightarrow \mathbf{dC}^\infty \mathbf{Ring}$$

is $J_{\mathbf{dMfd}}$ -sheaf.

Proof. Under the equivalence

$$\mathbf{dMfd} \simeq (\mathbf{dC}^\infty \mathbf{Ring}^{\text{fp}})^{\text{op}}$$

the universal $\mathcal{C}^\infty$ -ring object $i \circ q$ corresponds to j^{fp} . Recall now that for a derived manifold M we have:

$$\mathcal{O}_{\mathbf{dMfd}}(M) = \text{Hom}(M, i \circ q).$$

Notice also that for any finitely presented $\mathcal{C}^\infty$ -ring A we have that:

$$\text{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}^{\text{op}}}(A, j^{\text{fp}}) \simeq A.$$

Now the result follow by Proposition 2.4.8. $\square$

Remark 2.4.10. There's no standard terminology for derived manifold. What we refer to as derived manifolds (following [12]) are called finitely presented affines in other works (e.g., [57]). In some contexts, the term "derived manifold" is instead used to denote objects that are closer in nature to derived schemes; see, for instance, [50, Definition 5.1.9].

2.5 Derived stacks

In this section, we introduce the final geometric framework, which will encompass all the settings developed thus far. As anticipated in the introduction to the thesis, this framework allows us to rigorously discuss intersections, deformations, and quotients.

We begin in Subsection 2.5.1, where we enlarge the ∞ -category $\mathbf{dMfd}$ to the ∞ -category $\mathbf{Aff}$. This enlargement is essential for formulating the theory of deformations of $\mathbf{dMfd}$. In Subsection 2.5.2, we define derived stacks and relate them to the classical notions of stacks and differentiable stacks.

The main references for this section are [10], [57], [3], [30], and [50].

2.5.1 Complete $\mathcal{C}^\infty$ -rings

The material in this subsection is drawn from [50, Section 5.1.1] and [10, Sections 6 and 9.6].

We begin by briefly describing the behavior of the functor $\mathbf{Spec}_{\mathbf{C}^\infty}$ for $\mathcal{C}^\infty$ -rings that are not necessarily finitely presented.

First, observe that we can in fact define a functor

$$\mathbf{Spec}_{\mathbf{C}^\infty} : \mathbf{dC}^\infty \mathbf{Ring} \rightarrow \text{Loc}_{\mathbf{C}^\infty}$$

essentially in the same way as in Section 2.3.2. Given a $\mathcal{C}^\infty$ -ring A , the construction of the underlying topological space is the same as that described in Subsection 2.3.1. However, Proposition 2.3.3 still applies only in the finitely presented case.

Construction F.3.27 yields a *global sections functor*

$$\mathcal{O}: \mathbf{Top}_{\mathbf{C}^\infty} \rightarrow \mathbf{dC}^\infty\mathbf{Ring}^{\mathrm{op}}$$

which sends a homotopically $\mathcal{C}^\infty$ -ringed space $(X, \mathcal{O}_X)$ to the $\mathcal{C}^\infty$ -ring $\mathcal{O}_X(X)$. We denote the induced functor on $\mathbf{Loc}_{\mathbf{C}^\infty}$ by the same symbol, $\mathcal{O}$.

Proposition 2.5.1 (cf. [50, Proposition 5.1.8]). *The functor $\mathcal{O}$ fits into an adjunction:*

$$\begin{array}{ccc} & \xrightarrow{\mathcal{O}} & \\ \mathbf{Loc}_{\mathbf{C}^\infty} & \perp & \mathbf{dC}^\infty\mathbf{Ring}^{\mathrm{op}} \\ & \xleftarrow{\mathbf{Spec}_{\mathbf{C}^\infty}} & \end{array}$$

Definition 2.5.2. We define the ∞ -category of *affine derived $\mathcal{C}^\infty$ -schemes*, or just *affines*, $\mathbf{Aff}$ as the essential image of the functor:

$$\mathbf{Spec}_{\mathbf{C}^\infty}: \mathbf{dC}^\infty\mathbf{Ring}^{\mathrm{op}} \rightarrow \mathbf{Loc}_{\mathbf{C}^\infty}.$$

Remark 2.5.3. An analogous definition can be given in the context of discrete $\mathcal{C}^\infty$ -rings. In this case, we denote by $\mathbf{Aff}_{\mathrm{cl}}$ the subcategory spanned by the functor

$$\mathbf{Spec}_{\mathbf{C}^\infty}^0: \mathbf{disC}^\infty\mathbf{Ring} \rightarrow \mathbf{Loc}_{\mathbf{C}^\infty}^0.$$

As in the case of $\mathbf{Spec}_{\mathbf{C}^\infty}$, note that the functor $\mathbf{Spec}_{\mathbf{C}^\infty}^0$ can also be defined for discrete $\mathcal{C}^\infty$ -rings that are not necessarily finitely presented.

Unlike algebraic scheme theory, we do not obtain an equivalence between affine schemes and $\mathcal{C}^\infty$ -rings. Instead, we will see that affine schemes embed fully faithfully into the category of $\mathcal{C}^\infty$ -rings.

Proposition 2.5.4 (cf. [50, Proposition 5.1.26]). *The unit*

$$\eta: \mathrm{id}_{\mathbf{Aff}} \implies \mathbf{Spec}_{\mathbf{C}^\infty} \circ \mathcal{O}$$

is an equivalence.

Definition 2.5.5 (Complete $\mathcal{C}^\infty$ -rings, cf. [50, Definition 5.1.27]). A $\mathcal{C}^\infty$ -ring A is called *complete* if the counit if the

$$\varepsilon: \mathcal{O}(\mathbf{Spec}_{\mathbf{C}^\infty}(A)) \rightarrow A$$

is an equivalence. We denote the full subcategory of complete $\mathcal{C}^\infty$ -rings by $\mathbf{dC}^\infty\mathbf{Ring}^\wedge$.

Proposition 2.5.6 (cf. [50, Example 5.1.30]). *Any finitely presented $\mathcal{C}^\infty$ -ring is complete.*

Corollary 2.5.7 (cf. [50, Corollary 5.1.28], [10, Corollary 6.10]). *The adjunction $\mathcal{O}: \mathbf{Aff} \rightleftarrows \mathbf{dC}^\infty\mathbf{Ring}^{\mathrm{op}}: \mathbf{Spec}_{\mathbf{C}^\infty}$ realizes the following equivalence.*

$$\mathbf{Aff} \simeq (\mathbf{dC}^\infty\mathbf{Ring}^\wedge)^{\mathrm{op}}.$$

Moreover, $\mathbf{dC}^\infty\mathbf{Ring}^\wedge$ is a reflective subcategory of $\mathbf{dC}^\infty\mathbf{Ring}$ with the left adjoint to inclusion given by $\mathcal{O} \circ \mathbf{Spec}_{\mathbf{C}^\infty}$.

Corollary 2.5.8 (cf. [10, corollary 6.11]). *The above adjunction restricts to an equivalence:*

$$\mathbf{Aff}^{\mathrm{fp}} \simeq (\mathbf{dC}^\infty\mathbf{Ring}^{\mathrm{fp}})^{\mathrm{op}}$$

between affines of finite type and the opposite ∞ -category of finitely presented $\mathcal{C}^\infty$ -ring. In particular, we have:

$$\mathbf{Aff}^{\mathrm{fp}} \simeq (\mathbf{dC}^\infty\mathbf{Ring}^{\mathrm{fp}})^{\mathrm{op}} \simeq \mathbf{dMfd}.$$

2.5.2 Definition

We now introduce derived stacks. Just as the ∞ -category of stacks is defined as the ∞ -category $\mathrm{Shv}(\mathbf{Mfd})$ of sheaves in spaces, the ∞ -category of derived stacks is defined as the ∞ -category of sheaves on $\mathbf{Aff}$, equipped with a suitable Grothendieck topology.

We will not spell out the details of this construction, and instead refer to the appropriate literature (e.g., [57] and [50]). However, we will discuss the main properties of the ∞ -category $\mathrm{Shv}(\mathbf{Aff})$ and its relation to the ∞ -category of differentiable stacks.

Definition 2.5.9. A collection of maps $\{f_i : A \rightarrow A_i\}$ in $\mathbf{dC}^\infty\mathbf{Ring}$ is *open covering* if the collection $\{\mathbf{Spec}_{\mathbf{C}^\infty}(f_i) : \mathbf{Spec}_{\mathbf{C}^\infty}(A_i) \rightarrow \mathbf{Spec}_{\mathbf{C}^\infty}(A)\}$ in $\mathrm{Loc}_{\mathbf{C}^\infty}$ is an open covering.

Remark 2.5.10. Since the functor $\mathbf{Spec}_{\mathbf{C}^\infty}$ preserves pullbacks, it follows that open coverings define a Grothendieck topology on $\mathbf{dC}^\infty\mathbf{Ring}^{\mathrm{op}}$. This topology restricts to the one previously constructed on the subcategory of finitely presented $\mathbf{C}^\infty$ -rings.

Definition 2.5.11. We define the ∞ -category of *derived stacks* $\mathbf{dSt}$ to be the ∞ -category of sheaves over the site $(\mathbf{Aff}, \text{opens})$ of affines equipped with the topology given by open coverings.

Remark 2.5.12 (cf. [50, Remark 5.2.1]). Since $\mathbf{Aff}$ is not small, there is a set-theoretic issue. We will tacitly assume that $\mathbf{Aff}$ is the opposite of the (small) ∞ -category of κ -small objects in the presentable category of complete $\mathbf{C}^\infty$ -rings, for some sufficiently large κ . Recall that $\mathbf{Mfd}$ is essentially small (see [1, Remark 3.1.2]).

Proposition 2.5.13 (cf. [50, Example 5.2.2]). *The topology on $\mathbf{Aff}$ is subcanonical. In particular we get a fully faithful embedding $\mathbf{Aff} \hookrightarrow \mathbf{dSt}$.*

Proof. Let M and N be affines, and let $\{f_i : U_i \rightarrow N\}_{i \in I}$ be a covering sieve of N . We then have the following commutative diagram:

$$\begin{array}{ccc} \mathrm{Hom}_{\mathbf{Aff}}(N, M) & \xrightarrow{\quad\quad\quad} & \lim_i \mathrm{Hom}_{\mathbf{Aff}}(U_i, M) \\ \wr \downarrow & & \downarrow \wr \\ \mathrm{Hom}_{\mathbf{dC}^\infty\mathbf{Ring}}(\mathcal{O}(M), \mathcal{O}(N)) & \xrightarrow{\sim} & \lim_i \mathrm{Hom}_{\mathbf{dC}^\infty\mathbf{Ring}}(\mathcal{O}(M), \mathcal{O}(U_i)). \end{array}$$

The vertical arrows are equivalences by Corollary 2.5.7, and the bottom arrow is an equivalence because $\mathcal{O}$ is a sheaf on N . $\square$

Remark 2.5.14. In [50, Proposition 5.2.3], Nuiten proves a stronger result for derived $\mathbf{C}^\infty$ -schemes.

Construction 2.5.15. $\mathcal{O} : \mathbf{Aff}^{\mathrm{op}} \rightarrow \mathbf{dC}^\infty\mathbf{Ring}$ give rise to $\mathcal{O} : \mathbf{dSt}^{\mathrm{op}} \rightarrow \mathbf{dC}^\infty\mathbf{Ring}$ by Theorem F.3.11.

Construction 2.5.16 ($\mathrm{Shv}(\mathbf{Mfd}) \hookrightarrow \mathbf{dSt}$, cf. [3, Definition 8.60], [10, Section 7.2]). Recall from Construction D.2.21 the adjunction:

$$\begin{array}{ccc} \mathbf{Aff}_{\mathrm{cl}} & \begin{array}{c} \xrightarrow{i} \\ \perp \\ \xleftarrow{\pi_0} \end{array} & \mathbf{Aff} \end{array}$$

Both functors preserve covers. In particular, by Kan extension we obtain the following sequence of adjunctions:

$$\mathrm{Shv}(\mathbf{Aff}_{\mathrm{cl}}) \begin{array}{c} \xleftarrow{i_!} \xrightarrow{i_!} \\ \xleftarrow{i^*} \simeq (\pi_0)_! \xrightarrow{i^*} \\ \xleftarrow{i_*} \simeq (\pi_0)^* \xrightarrow{i_*} \\ \xleftarrow{(\pi_0)_*} \end{array} \mathbf{dSt}$$

where $i_!$ is fully faithful thanks to Corollary F.3.22. $\mathrm{Shv}(\mathbf{Aff}_{\mathrm{cl}})$ is coreflective subcategory of $\mathbf{dSt}$. In particular $\mathrm{Shv}(\mathbf{Mfd})$ is closed under colimits.

Definition 2.5.17 (cf. [10, Definition 7.3]). A sheaf $\mathcal{X}$ is *classical* if it is in the essential image of $i_!$.

Remark 2.5.18 (cf. [26, Below Proposition 4.3.1]). Notice that a classical stack is space-valued; in particular, it does not necessarily land in n -truncated objects for any finite n . In the case where $X = \mathbf{Spec}_{\mathbf{C}^\infty}(A)$ for some discrete $\mathbf{C}^\infty$ -ring A , the restriction of X to $\mathbf{Aff}_{\mathrm{cl}}$ is set-valued. However, this need not be true for X itself. For example, if $X = \mathbf{Spec}_{\mathbf{C}^\infty}(\mathbf{C}^\infty(\mathbb{R}))$, then X sends each affine $\mathbf{Spec}_{\mathbf{C}^\infty}(A)$ to the underlying space of A .

Remark 2.5.19. The fully faithful functor $\mathbf{Mfd} \hookrightarrow \mathbf{Aff}_{\text{cl}}$ is both continuous and cocontinuous. As before, Corollary [F.3.22](#) provides a fully faithful functor $\text{Shv}(\mathbf{Mfd}) \hookrightarrow \text{Shv}(\mathcal{C}^\infty \mathbf{Ring})$ which is a left adjoint. In particular, the category $\text{Shv}(\mathbf{Mfd})$ is closed under colimits in both $\text{Shv}(\mathbf{Aff}_{\text{cl}})$ and $\mathbf{dSt}$.

To properly define geometric derived stacks, one would need to take a detour into the theory of smooth maps and n -groupoids. However, since our main focus is on differentiable stacks, we will not explicitly require this additional material in the remainder of the thesis. We therefore refer the reader to the relevant literature, such as [\[57\]](#) and [\[50, Section 5.2\]](#). From now on, we will use the term *geometric derived stack* to refer to what Nuiten calls a *derived stack* in [\[50, Section 5.2\]](#).

Chapter 3

Cotangent complex

This chapter aims to define the cotangent complex for derived stacks and to study some of its fundamental properties. Along the way, we introduce several structurally important notions, such as quasi-coherent modules and square-zero extensions.

In more detail, we begin with Section 3.1, where we introduce the $\mathcal{C}^\infty$ -analogues of the algebraic notions of derivation, square-zero extension, and Kähler differentials. In Section 3.2, we transpose this theory to the derived setting, which naturally leads to Section 3.3, where we finally introduce the cotangent complex for $\mathcal{C}^\infty$ -rings. In this section, we also prove our main result: the identification of the module of differential forms over a smooth manifold M with the cotangent complex associated to $\mathcal{C}^\infty(M)$. In Section 3.4, we introduce the notion of quasi-coherent modules over derived stacks and study their properties. Finally, Section 3.5 is devoted to the cotangent complex of derived stacks.

The main references for this chapter are [50], [57], [29], and [10].

3.1 Discrete derivation

This section is devoted to the study of the $\mathcal{C}^\infty$ -analogue of derivations in ordinary commutative algebra. The results presented here serve both to build intuition for the derived setting and to provide background that will be useful when studying the cotangent complex.

3.1.1 Discrete module

As first step we recall the definition of the category of modules over the category **CRing** of discrete commutative rings.

Definition 3.1.1. We define the *categories of discrete module* as $\mathbf{Mod} := \mathbf{Mod}(\mathbf{Ab})$ (see also Definition D.1.9).

Using the notation of Appendix D.3, we observe that $\mathbf{Mod}$ is equipped with a forgetful map $a^*: \mathbf{Mod} \rightarrow \mathbf{CRing}$ which assigns to each pair (A, E) the underlying commutative ring A . Recall also from Remark 2.1.4 that there is a forgetful map $\mathcal{C}^\infty\mathbf{Ring} \rightarrow \mathbf{CRing}$ sending each discrete $\mathcal{C}^\infty$ -ring to its underlying commutative ring. We can now give the following definition.

Definition 3.1.2. Define the category of *discrete modules over $\mathcal{C}^\infty$ -rings* $\mathcal{C}^\infty\mathbf{Mod} := \mathbf{Mod} \times_{\mathbf{CRing}} \mathcal{C}^\infty\mathbf{Ring}$.

In other words we think of $\mathcal{C}^\infty\mathbf{Mod}$ as the category whose:

- objects are pairs (A, E) where A gets equipped with the structure of a discrete $\mathcal{C}^\infty$ -ring and E is a discrete module over A ,
- morphisms $(A, E) \rightarrow (B, D)$ are given by a morphism of discrete $\mathcal{C}^\infty$ -rings $A \rightarrow B$ and an A -linear map $E \rightarrow D$.

From the above definition, we see that the theory of discrete modules over $\mathcal{C}^\infty$ -rings is, at first glance, not fundamentally different from its purely algebraic counterpart. In fact, as we see, discrete modules over a fixed $\mathcal{C}^\infty$ -ring A are nothing more than ordinary discrete modules over the underlying $\mathbb{R}$ -algebra A^{alg} . However, the key point is that we do not entirely forget the $\mathcal{C}^\infty$ -structure. When considering a morphism $(A, E) \rightarrow (B, D)$ in $\mathcal{C}^\infty\mathbf{Mod}$, the

map $A \rightarrow B$ is not merely a morphism of the underlying algebras, but a morphism of $\mathcal{C}^\infty$ -rings. This subtlety will play a central role in what follows, leading to a modified notion of *derivation* and of *Kähler differentials*.

Notation. Let A be a discrete $\mathcal{C}^\infty$ -ring. We denote the category of discrete modules over A by Mod_A .

Remark 3.1.3. Notice that $\mathcal{C}^\infty\mathbf{Mod}$ is algebraic, generated by

$$\mathcal{C}^\infty\mathbf{PolyMod} := \{(\mathcal{C}^\infty(\mathbb{R}^n), V) \in \mathcal{C}^\infty\mathbf{Ring}^{free} \times \text{Mod}_{\mathbb{R}} \mid V \text{ free of finite rank over } \mathcal{C}^\infty(\mathbb{R}^n)\}.$$

3.1.2 $\mathcal{C}^\infty$ -Derivations

We now turn to the notion of derivation. Since, for a discrete $\mathcal{C}^\infty$ -ring A , there is no distinction between Mod_A and $\text{Mod}_{A^{\text{alg}}}$, it is tempting to adopt the purely algebraic definition of derivation. However, a closer inspection quickly reveals that something is amiss. Specifically, consider the $\mathcal{C}^\infty$ -ring $\mathcal{C}^\infty(\mathbb{R})$ and its algebraic Kähler differentials $\Omega_{\mathcal{C}^\infty(\mathbb{R})/\mathbb{R}}^{\text{alg}}$, together with the universal derivation

$$d : \mathcal{C}^\infty(\mathbb{R}) \rightarrow \Omega_{\mathcal{C}^\infty(\mathbb{R})/\mathbb{R}}^{\text{alg}}.$$

Then, one observes that

$$d(\sin(x)) \neq \cos(x) \cdot dx.$$

This indicates that the algebraic definition fails to fully capture the differential structure encoded in a $\mathcal{C}^\infty$ -ring, and thus a more refined definition is required.

Definition 3.1.4. Let A be a discrete $\mathcal{C}^\infty$ -ring and let E be a discrete module over A . A $\mathcal{C}^\infty$ -derivation of A is a map of sets $D : A \rightarrow E$ such that the following hold: for every $h \in \mathcal{C}^\infty(\mathbb{R}^n)$ and every $f_1, \dots, f_n \in A$

$$D(A(h)(f_1, \dots, f_n)) = \sum_i A\left(\frac{\partial h}{\partial x_i}\right)(f_1, \dots, f_n) \cdot Df_i \quad (3.1)$$

is satisfied. We will denote the set of $\mathcal{C}^\infty$ -derivation of A in E by $\text{Der}_{\mathcal{C}^\infty}(A, E)$.

Remark 3.1.5. Note that in the definition given above, it is essential to work with $\mathcal{C}^\infty$ -manifolds. If we were to allow $\mathcal{C}^n$ -manifolds, issues would arise in the module structure. Indeed, if we assume that $f = h(g_1, \dots, g_n)$ for some $h \in \mathcal{C}^n(\mathbb{R})$, then each partial derivative $\partial h / \partial x_i$ belongs only to $\mathcal{C}^{n-1}$. Consequently, the expressions $(\partial h / \partial x_i)(g_1, \dots, g_n) \cdot d(g_i)$ are not well-defined in general, and the identity (3.1) becomes meaningless.

Remark 3.1.6. Definition 3.1.4 recovers the usual algebraic derivation, while incorporating an additional condition that reflects the smooth structure. More precisely:

1. (Linearity) Using the function $ax + by \in \mathcal{C}^\infty(\mathbb{R}^2)$, we obtain:

$$D(af + bg) = a \cdot Df + b \cdot Dg.$$

2. (Leibniz rule) Using the function $xy \in \mathcal{C}^\infty(\mathbb{R}^2)$, we obtain:

$$D(fg) = g \cdot Df + f \cdot Dg.$$

Remark 3.1.7. Notice that $\mathcal{C}^\infty$ -derivations assemble into a functor:

$$\text{Der}_{\mathcal{C}^\infty}(A, -) : \text{Mod}_A \rightarrow \mathbf{Set}.$$

Definition 3.1.8. Let A be a discrete $\mathcal{C}^\infty$ -ring. A discrete A -module E together with a derivation d will be called a $\mathcal{C}^\infty$ -Kähler differential if the natural transformation

$$\text{Hom}_{\text{Mod}_A}(E, -) \xrightarrow{- \circ d} \text{Der}_{\mathcal{C}^\infty}(A, -)$$

is an equivalence.

We now prove that the $\mathcal{C}^\infty$ -Kähler differential always exists. It then follows immediately from the definition that it is unique up to isomorphism.

Definition 3.1.9. Let A be a discrete $\mathcal{C}^\infty$ -ring. We define the A -module Ω_A as the following quotient:

$$\frac{\bigoplus_{f \in A} A \cdot df}{\sim}$$

with the equivalence relation generated by the following condition: if $f, g_1, \dots, g_n \in A$ and $h \in \mathcal{C}^\infty(\mathbb{R}^n)$ with $f = A(h)(g_1, \dots, g_n)$ then $df \sim \sum_{i=1}^n A(\partial h / \partial x_i)(g_1, g_2, \dots, g_n) \cdot d(g_i)$.

Proposition 3.1.10. Let A be a discrete $\mathcal{C}^\infty$ -ring and let E be a A -module. Then, there exists a natural isomorphism of sets:

$$\text{Hom}_{\text{Mod}_A}(\Omega_A, E) \cong \text{Der}_{\mathcal{C}^\infty}(A, E),$$

given by assigning to every A -linear morphism $\phi : \Omega_A \rightarrow E$ the $\mathcal{C}^\infty$ -derivation $d_\phi : A \rightarrow E$ defined by $d_\phi(f) = \phi(df)$. Conversely to each $\mathcal{C}^\infty$ -derivation $D : A \rightarrow E$, we associate the A -linear morphism $\varphi_D : \Omega_A \rightarrow E$ uniquely determined by $\varphi_D(df) = D(f)$.

Proof. The first step is to verify that the assignment is well defined. Let $\phi : \Omega_A \rightarrow E$ be an A -linear morphism. Our goal is to show that d_ϕ is a $\mathcal{C}^\infty$ -derivation. That is, for any $h \in \mathcal{C}^\infty(\mathbb{R}^n)$ and $g_1, \dots, g_n \in A$ such that $f = A(h)(g_1, \dots, g_n)$, we must check that the following identity holds:

$$d_\phi(f) = \sum_{i=1}^n A(\partial h / \partial x_i)(g_1, g_2, \dots, g_n) \cdot d_\phi(g_i).$$

This follows directly from the definition of the equivalence relation $\sim$. Indeed,

$$\begin{aligned} d_\phi(f) &= d_\phi(h(g_1, \dots, g_n)) = \phi(d(h(g_1, \dots, g_n))) && \text{By def of } d_\phi \\ &= \phi\left(\sum_{i=1}^n A(\partial h / \partial x_i)(g_1, g_2, \dots, g_n) \cdot d(g_i)\right) && \text{By def } \sim \\ &= \sum_{i=1}^n A(\partial h / \partial x_i)(g_1, g_2, \dots, g_n) \cdot \phi(d(g_i)) && \text{By } A\text{-linearity} \\ &= \sum_{i=1}^n A(\partial h / \partial x_i)(g_1, g_2, \dots, g_n) \cdot d_\phi(g_i) && \text{By def of } d_\phi. \end{aligned}$$

Conversely given a $\mathcal{C}^\infty$ -derivation $D : A \rightarrow E$ we first must verify that φ_D is well defined. More precisely, if $\omega \in \Omega_A$ can be written in two equivalent ways: $\sum g_i \cdot df_i$ and $\sum g'_i \cdot df'_i$, then we will show that

$$\varphi_D\left(\sum g_i \cdot df_i\right) = \varphi_D\left(\sum g'_i \cdot df'_i\right).$$

It suffices to check that the relation $\sim$ is preserved. In particular, if $f = A(h)(g_1, \dots, g_n)$ for some $h \in \mathcal{C}^\infty(\mathbb{R}^n)$ and $g_1, \dots, g_n \in A$, then we must have:

$$\varphi_D(df) = \sum_{i=1}^n A(\partial h / \partial x_i)(g_1, g_2, \dots, g_n) \cdot \varphi_D(dg_i)$$

But this now follows essentially by definition:

$$\begin{aligned} \varphi_D(df) &= D(f) = D(h(g_1, \dots, g_n)) \\ &= \sum_{i=1}^n A(\partial h / \partial x_i)(g_1, g_2, \dots, g_n) \cdot D(g_i) && D \text{ is a } \mathcal{C}^\infty\text{-derivation} \\ &= \sum_{i=1}^n A(\partial h / \partial x_i)(g_1, g_2, \dots, g_n) \cdot \varphi_D(dg_i) && \text{definition of } \varphi_D. \end{aligned}$$

Moreover, φ_D is A -linear by construction. Finally, to verify that the two constructions are inverse to each other, observe that:

$$\varphi_{d_\phi}(df) = d_\phi(f) = \phi(df), \quad \text{and} \quad d_{\varphi_D}(f) = \varphi_D(df) = D(f),$$

as required. $\square$

3.1.3 $\mathcal{C}^\infty$ -Kähler differentials for smooth manifolds

Theorem 3.1.11. *Let M be a smooth manifold, then there is an isomorphism of $\mathcal{C}^\infty(M)$ -modules:*

$$\Omega^1(M) \cong \Omega_{\mathcal{C}^\infty(M)}$$

This subsection is devoted to the proof of the above theorem.

Lemma 3.1.12 (Extension lemma for differential 1-forms). *Let M be a properly embedded submanifold of $\mathbb{R}^m$ (see Definition A.1.1) and let $\omega \in \Omega^1(M)$, then there exist $\tilde{\omega} \in \Omega^1(\mathbb{R}^m)$ such that $\tilde{\omega}|_M = \omega$.*

Proof. Denote with $i : M \hookrightarrow \mathbb{R}^m$ the natural inclusion. Our goal is to construct $\tilde{\omega} \in \Omega^1(\mathbb{R}^m)$ such that $\tilde{\omega}|_M = i^*(\tilde{\omega}) = \omega$. To construct such an extension, we first need to extend ω to an open neighborhood of M in $\mathbb{R}^m$. To achieve the result, consider a tubular neighborhood U of M , and let $\pi : U \rightarrow M$ be the retraction given by Proposition A.1.6. Then we can consider $\omega' = \pi^*(\omega) \in \Omega^1(U)$. We can now take a bump function ψ in $\mathbb{R}^m$ with support contained in U and equal 1 on M and construct:

$$\tilde{\omega} = \begin{cases} \psi \cdot \omega' & \text{on } U \\ 0 & \text{on } \mathbb{R}^m \setminus \text{Supp}(\psi). \end{cases}$$

It is now clear that $\tilde{\omega}|_M = \omega$. □

Remark 3.1.13. The above results holds true more generally for p -forms with the same proof.

The following fact is stated in [20] but no proof can be found there and in the literature I couldn't find an actual proof.

Lemma 3.1.14. *Let M be a smooth manifold, then $\Omega^1(M)$ is generated by $\{df\}_{f \in \mathcal{C}^\infty(M)}$.*

Proof. By the Whitney embedding theorem (see Theorem A.1.2 and Remark A.1.3) we have a proper smooth embedding $i : M \hookrightarrow \mathbb{R}^m$ for some m . Consider a 1-differential form $\omega \in \Omega^1(M)$ and construct with Lemma 3.1.12 its extension $\tilde{\omega} \in \Omega^1(\mathbb{R}^m)$ to $\mathbb{R}^m$. We can now use the global coframe $dx_1, \dots, dx_m \in \Omega^1(\mathbb{R}^m)$ associated to the standard coordinates of $\mathbb{R}^m$ to uniquely write $\tilde{\omega}$ as follow: $\tilde{\omega} = \sum f_j \cdot dx_j$ where f_j are smooth function over $\mathbb{R}^m$. Denoting $g_j = i^*(x_j) \in \mathcal{C}^\infty(M)$, we can conclude that the set $\{dg_j\}_{j=1}^m$ generates $\Omega^1(M)$ as a $\mathcal{C}^\infty(M)$ -module by pulling back:

$$\omega = i^*(\tilde{\omega}) = i^*(\sum f_j \cdot dx_j) = \sum i^*(f_j) \cdot di^*(x_j) = \sum i^*(f_j) \cdot dg_j.$$
□

A more general statement in the case of finitely presented discrete $\mathcal{C}^\infty$ -rings may be found in [29, Proposition 5.6].

Lemma 3.1.15. *Let M be a smooth manifold. Given*

$$\omega = \sum_{j=1}^l g_j \cdot df_j = \sum_{j=1}^m g'_j \cdot df'_j$$

then there exist $h_1, \dots, h_l, h'_1, \dots, h'_m \in \mathcal{C}^\infty(\mathbb{R}^k)$ and $t_1, \dots, t_k \in \mathcal{C}^\infty(M)$ such that $f_j = h_j(t_1, \dots, t_k)$ and $f'_j = h'_j(t_1, \dots, t_k)$ for all j and

$$\sum_{j=1}^l g_j \cdot \frac{\partial h_j}{\partial x_p}(t_1, \dots, t_k) = \sum_{j=1}^m g'_j \cdot \frac{\partial h'_j}{\partial x_p}(t_1, \dots, t_k) \quad \text{for } 1 \leq p \leq k$$

Proof. Consider $i : M \hookrightarrow \mathbb{R}^k$ a proper smooth embedding and consider the extension $\tilde{\omega}$ of ω to $\mathbb{R}^k$, constructed as in Lemma 3.1.12. Using the same notations from the lemma, we have:

$$\tilde{\omega} = \begin{cases} \sum_{j=1}^l \psi \cdot \pi^*(g_j) \cdot d\pi^*(f_j) & \text{on } U \\ 0 & \text{on } \mathbb{R}^k \setminus \text{Supp}(\psi) \end{cases} = \begin{cases} \sum_{j=1}^m \psi \cdot \pi^*(g'_j) \cdot d\pi^*(f'_j) & \text{on } U \\ 0 & \text{on } \mathbb{R}^k \setminus \text{Supp}(\psi) \end{cases}.$$

Notice that for every j we have $\pi^*(f_j) = h_j(x, \dots, x_k)$ and $\pi^*(f'_j) = h'_j(x, \dots, x_k)$ for some $h_j, h'_j \in \mathcal{C}^\infty(\mathbb{R}^k)$. So on U we get:

$$\sum_{j=1}^l \psi \cdot \pi^*(g_j) \cdot d(h_j(x_1, \dots, x_k)) = \sum_{j=1}^m \psi \cdot \pi^*(g'_j) \cdot d(h'_j(x_1, \dots, x_k)).$$

Using the chain rule, we may rewrite this as:

$$\sum_{p=1}^k \left(\sum_{j=1}^l \psi \cdot \pi^*(g_j) \cdot \frac{\partial h_j}{\partial x_p}(x_1, \dots, x_k) \right) dx_p = \sum_{p=1}^k \left(\sum_{j=1}^m \psi \cdot \pi^*(g'_j) \cdot \frac{\partial h'_j}{\partial x_p}(x_1, \dots, x_k) \right) dx_p.$$

In particular, given a global coframe $(dx_1, \dots, dx_k)$, we see that for every $p = 1, \dots, k$

$$\sum_{j=1}^l \psi \cdot \pi^*(g_j) \cdot \frac{\partial h_j}{\partial x_p}(x_1, \dots, x_k) = \sum_{j=1}^m \psi \cdot \pi^*(g'_j) \cdot \frac{\partial h'_j}{\partial x_p}(x_1, \dots, x_k)$$

Now pulling back everything via i^* we get the lemma. $\square$

We can finally prove Theorem. 3.1.11

Proof of Theorem 3.1.11. Consider the $\mathcal{C}^\infty$ -linear map $\varphi : \bigoplus_{f \in \mathcal{C}^\infty(M)} \mathcal{C}^\infty(M) df \rightarrow \Omega^1(M)$ defined on generators as $\varphi(df) = df$. By lemma 3.1.14 we have that φ is surjective. We now need to prove that the kernel is exactly the submodule defined by the relation $\sim$. Consider elements $g_1, \dots, g_l, f_1, \dots, f_l \in \mathcal{C}^\infty(M)$ such that the relation

$$\sum_{j=1}^l g_j df_j = 0$$

holds in $\Omega^1(M)$. We need to show that this relation is already satisfied in the quotient $\Omega_{\mathcal{C}^\infty(M)}^\#$. Using Lemma 3.1.15, we may find $h_1, \dots, h_l \in \mathcal{C}^\infty(\mathbb{R}^k)$ and $t_1, \dots, t_k \in \mathcal{C}^\infty(M)$ such that: $f_j = h_j(t_1, \dots, t_k)$ for all j and

$$\sum_{j=1}^l g_j \cdot d(h_j(t_1, \dots, t_k)) = 0.$$

Then, by the chain rule, we have

$$\sum_{p=1}^k \left(\sum_{j=1}^l g_j \cdot \frac{\partial h_j}{\partial x_p}(t_1, \dots, t_k) \right) dt_p = 0$$

and hence

$$\sum_{j=1}^l g_j \cdot \frac{\partial h_j}{\partial x_p}(t_1, \dots, t_k) = 0 \quad \forall p.$$

This shows the claim. $\square$

3.1.4 Square-zero extensions of discrete $\mathcal{C}^\infty$ -rings

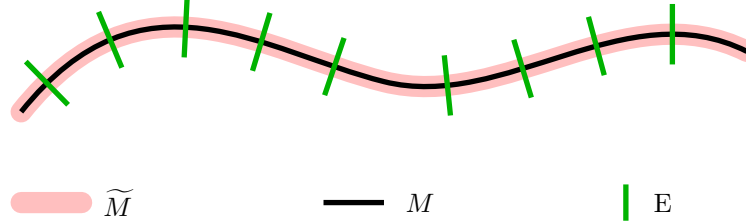
We now introduce square-zero extensions in order to present an alternative perspective on $\mathcal{C}^\infty$ -derivations. This viewpoint will help build intuition for the derived setting, where we will define derivations via the derived version of square-zero extensions.

Definition 3.1.16. Let A be a discrete $\mathcal{C}^\infty$ -ring and let N be a discrete A -module. The *square-zero extension* of A by N is defined to be the discrete $\mathcal{C}^\infty$ -ring $A \oplus N$ with the structure induced by the following assignment:

$$\begin{aligned} f \in \mathcal{C}^\infty(\mathbb{R}^k) & \mapsto (A \oplus N)^k \xrightarrow{f_*} A \oplus N \\ ((a_1, n_1), \dots, (a_k, n_k)) & \longmapsto \left(f(a_1, \dots, a_k), \sum_{i=1}^k \frac{\partial f(a_1, \dots, a_k)}{\partial x_i} \cdot n_i \right) \end{aligned}$$

Remark 3.1.17. Notice that $A \oplus N$ comes equipped with a morphism of discrete $\mathcal{C}^\infty$ -rings to A given by the projection onto the first coordinate.

Square-zero extensions can be viewed as formal thickenings. More precisely, let A be the $\mathcal{C}^\infty$ -ring of smooth functions on a smooth manifold M , and let N be the A -module of global sections of a vector bundle $E \rightarrow M$. Then we may interpret the square-zero extension $A \oplus N$ as the $\mathcal{C}^\infty$ -ring of functions on the infinitesimal thickening $\widetilde{M}$ of M in the E -direction, which can be heuristically understood as the space obtained by adjoining to M all the *vectors of infinitesimal length* in the vector bundle E .



Definition 3.1.18. Let A be a discrete $\mathcal{C}^\infty$ -ring and let N be a discrete A -module. We define a *discrete derivation* to be a morphism of $\mathcal{C}^\infty$ -rings over A from A to the square-zero extension $A \oplus N$. That is, a morphism of $\mathcal{C}^\infty$ -rings $\phi : A \rightarrow A \oplus N$ such that the projection onto the first component satisfies $pr_A(\phi(a)) = a$ for all a .

Proposition 3.1.19. *There is a 1-1 correspondence between $\mathcal{C}^\infty$ -derivation of definition 3.1.4 and the discrete derivation from 3.1.18.*

Proof. Let $\phi : A \rightarrow A \oplus N$ be a discrete derivation in the sense of Definition 3.1.18. By assumption, we may write ϕ as (id_A, δ) for some $\delta : A \rightarrow N$. We claim that δ is a $\mathcal{C}^\infty$ -derivation in the sense of Definition 3.1.4. By definition of morphism of $\mathcal{C}^\infty$ -rings for every $f \in \mathcal{C}^\infty(\mathbb{R}^n)$ there exists a commutative square of the form:

$$\begin{array}{ccc} A^n & \xrightarrow{(\text{id} \oplus \delta)^n} & (A \oplus N)^n \\ f_* \downarrow & & \downarrow f_* \\ A & \xrightarrow{\text{id} \oplus \delta} & A \oplus N \end{array}$$

from which we get:

$$\begin{array}{ccc} (a_1, \dots, a_n) & \xrightarrow{\quad \quad \quad} & ((a_1, \delta a_1), \dots, (a_n, \delta a_n)) \\ \downarrow & & \downarrow \\ f(a_1, \dots, a_n) & \xrightarrow{\quad \quad \quad} & (f(a_1, \dots, a_n), \delta f(a_1, \dots, a_n)) = (f(a_1, \dots, a_n), \sum_{i=1}^n \frac{\partial f(a_1, \dots, a_n)}{\partial x_i} \delta a_i) \end{array}$$

In particular:

$$\delta f(a_1, \dots, a_n) = \sum_{i=1}^n \frac{\partial f(a_1, \dots, a_n)}{\partial x_i} \delta a_i.$$

Reversing the logic shows that for every $\mathcal{C}^\infty$ -derivation δ , the map $\phi := (\text{id}_A, \delta)$ is a discrete derivation. □

Corollary 3.1.20. *Let M be a smooth manifold and N be a discrete $\mathcal{C}^\infty(M)$ -module, then:*

$$\text{Hom}_{\mathcal{C}^\infty \mathbf{Ring} / \mathcal{C}^\infty(M)}(\mathcal{C}^\infty(M), \mathcal{C}^\infty(M) \oplus N) \cong \text{Der}_{\mathcal{C}^\infty}(\mathcal{C}^\infty(M), N) \cong \text{Hom}_{\text{Mod}_{\mathcal{C}^\infty(M)}}(\Omega^1(M), N).$$

3.2 Modules and square-zero extensions

The aim of this section is to prepare the ground for a definition of the cotangent complex. Building on intuition from the previous sections we will talk about modules in the derived setting and use them to construct square-zero extensions. More precisely, in subsection 3.2.1 and subsection 3.2.2, we define and characterize the category of modules over $\mathcal{C}^\infty$ -rings. Section 3.2.3 is devoted to the study of the tangent category, a powerful tool generalizing the category of modules, which we will use to define square-zero extensions in Subsection 3.2.4.

3.2.1 Modules

We now need to revise our notion of a “module”. The ones introduced in the previous section are merely sets; in particular, they fail to capture any meaningful homotopical information beyond what is concentrated in degree zero. To address this issue, in the following subsection we adopt a new perspective and take as the basic definition of a module over a $\mathcal{C}^\infty$ -ring an object in the derived category.

Let R be a commutative ring. In Appendix C, we briefly introduce the derived category $\mathcal{D}(R)$ of R -modules as the localization of the category $\text{Ch}(R)$ of unbounded chain complexes at the class of quasi-isomorphisms. In Appendix D.3, we define a different category, denoted $\mathbf{dMod}$, which leads to an alternative notion of the derived category associated to R , namely $\mathbf{dMod}_{HR}$. Theorem D.3.6 then provides an equivalence

$$\mathbf{dMod}_{HR} \simeq \mathcal{D}(R).$$

In the remainder of this thesis, we will use both models interchangeably, choosing between them as appropriate in order to take advantage of the respective features of each framework.

Definition 3.2.1. Define the ∞ -category of modules over $\mathcal{C}^\infty$ -rings as the category

$$\mathbf{dC}^\infty\mathbf{Mod} := \mathbf{dMod} \times_{\mathbf{CAlg}} \mathbf{dC}^\infty\mathbf{Ring}.$$

In other words we think of $\mathbf{dC}^\infty\mathbf{Mod}$ as the category whose:

- objects are couples (A, E) where A gets equipped with the structure of a $\mathcal{C}^\infty$ -ring and E is a module over A ,
- morphism $(A, E) \rightarrow (B, D)$ are given by a morphism of $\mathcal{C}^\infty$ -rings $A \rightarrow B$ and a A -linear map $E \rightarrow D$.

Definition 3.2.2. We define the category of $\mathbf{dC}^\infty\mathbf{Mod}^{\text{cn}}$ of *connective modules over $\mathcal{C}^\infty$ -rings* as the full subcategory of $\mathbf{dC}^\infty\mathbf{Mod}$ spanned by objects (A, M) for which M is a connective A^{alg} -module.

Remark 3.2.3. Recall from Remark 3.1.3 that the category $\mathcal{C}^\infty\mathbf{Mod}$ is algebraic. In particular, by the theory developed in Appendix D.1.2, it is meaningful to define the ∞ -category of connective derived $\mathcal{C}^\infty$ -modules as

$$\mathbf{dC}^\infty\mathbf{Mod}'^{\text{cn}} := \mathcal{P}_\Sigma(\mathcal{C}^\infty\mathbf{PolyMod}),$$

where $\mathcal{P}_\Sigma$ denotes the ∞ -category of sifted-colimit-preserving presheaves on the category of $\mathcal{C}^\infty\mathbf{PolyMod}$.

Thanks to [57, Corollary 4.3.2.6], we have that the two introduced notions of connective modules coincide. Consequently, we will use the notation $\mathbf{dMod}^{\text{cn}}$ uniformly to refer to the ∞ -category of connective derived $\mathcal{C}^\infty$ -modules.

3.2.2 Characterizing the category of derived modules

It is a classical observation due to Beck [5] that, for a commutative ring A , there is an equivalence:

$$\text{Ab}(\mathbf{CRing}/A) \simeq \text{Mod}_A$$

between the category of abelian group objects in the category of commutative rings over A , and the category of A -modules. More precisely, to an object $(\pi : B \rightarrow A)$ in $\text{Ab}(\mathbf{CRing}/A)$, one assigns the kernel $\ker(\pi) \in \text{Mod}_A$, while to an A -module M , one assigns the square-zero extension $A \oplus \epsilon M$, where $\epsilon^2 = 0$.

We now wish to transport this idea to the ∞ -categorical setting. To that end, we recall that the ∞ -categorical analogue of the functor $\text{Ab}(-)$ is given by the stabilization functor $\mathbf{Stab}$, while the counterpart of the category of A -modules is the derived category $\mathbf{dMod}_A$ (see Appendices C and D.3 for details). In light of the above discussion we present the following result:

Theorem 3.2.4 (cf. [57, Corollary 4.3.1.4]). *For any $A \in \mathbf{dC}^\infty\mathbf{Ring}$, there is a canonical equivalence of ∞ -categories:*

$$\mathbf{Stab}(\mathbf{dC}^\infty\mathbf{Ring}/_A) \simeq \mathbf{dMod}_A$$

The proof will requires some preliminaries.

Augmentation ideal and symmetric algebras

In [40, Definition 7.3.4.3], the augmentation ideal functor is defined. However, in this thesis we have aimed to minimize the use of the language of ∞ -operads. In particular, although our treatment is less general, we hope that it will be more transparent to non-experts.

We begin by recalling the following proposition:

Proposition 3.2.5 (cf. [40, Proposition 3.4.1.3.]). *Let $\mathcal{C}$ be a symmetric monoidal ∞ -category. Assume that $\mathcal{C}$ admits geometric realizations and that the tensor product of $\mathcal{C}$ preserves geometric realizations in both variables. Let R be a commutative algebra in $\mathcal{C}$. Then the forgetful functors $\mathbf{CAlg}(\mathbf{dMod}_R(\mathcal{C})) \rightarrow \mathbf{CAlg}(\mathcal{C})$ lift to equivalence of ∞ -categories*

$$\mathbf{CAlg}(\mathbf{dMod}_R(\mathcal{C})) \xrightarrow{\sim} \mathbf{CAlg}(\mathcal{C})_{R/}.$$

Definition 3.2.6. We define the *forgetful functor* ρ_A as the composite

$$\mathbf{dRing}_{A//A} \rightarrow \mathbf{dRing}_{A/} \rightarrow \mathbf{CAlg}(\mathbf{dMod}_A^{\text{cn}}) \rightarrow \mathbf{dMod}_A^{\text{cn}}$$

Remark 3.2.7. The functor ρ_A factors as:

$$\begin{array}{ccc} \mathbf{dRing}_{A//A} & \xrightarrow{\rho_A} & \mathbf{dMod}_A^{\text{cn}} \\ & \searrow F & \uparrow \\ & & (\mathbf{dMod}_A^{\text{cn}})_{A//A}. \end{array}$$

In particular, the functor F defines an object in

$$\text{Fun}\left(\mathbf{dRing}_{A//A}, (\mathbf{dMod}_A^{\text{cn}})_{A//A}\right).$$

Now recall that $(\mathbf{dMod}_A^{\text{cn}})_{A//A}$ is the (non-full) subcategory of the pullback

$$\begin{array}{ccc} \bullet & \xrightarrow{\quad} & \text{Fun}([2], \mathbf{dMod}_A^{\text{cn}}) \\ \downarrow & \lrcorner & \downarrow \text{ev}_{\{0 \rightarrow 2\}} \\ * & \xrightarrow{\text{id}_A} & \text{Fun}([1], \mathbf{dMod}_A^{\text{cn}}) \end{array}$$

where morphism at the bottom are trivial:

$$\begin{array}{ccccc} & & N & & \\ & \nearrow & & \searrow & \\ A & & & & E \\ & \searrow & & \nearrow & \\ & & A & & \\ & & \parallel & & \\ & & A & & \\ & & \parallel & & \\ & & A & & \\ & & \parallel & & \\ & & A & & \end{array}$$

By applying the currying and uncurrying equivalences, we obtain the following sequence of equivalences:

$$\text{Fun}\left(\mathbf{dRing}_{A//A}, \text{Fun}([2], \mathbf{dMod}_A^{\text{cn}})\right) \simeq \text{Fun}\left(\mathbf{dRing}_{A//A} \times [2], \mathbf{dMod}_A^{\text{cn}}\right) \simeq \text{Fun}\left([2], \text{Fun}\left(\mathbf{dRing}_{A//A}, \mathbf{dMod}_A^{\text{cn}}\right)\right)$$

In particular, through these equivalences F correspond to the following diagram of functors:

$$\begin{array}{ccc} & \rho_A & \\ \underline{A} & \nearrow & \searrow \\ & \underline{A} & \end{array}$$

which exhibits $\underline{A}$ as a retract of ρ_A .

Definition 3.2.8. In the above situation, we define the *augmentation ideal functor* I_A as the fiber of the morphism $\rho_A \rightarrow \underline{A}$.

Remark 3.2.9. Notice that I_A is also the fiber of the morphism $\rho_A \rightarrow \underline{A}$ in $\text{Fun}(\mathbf{dRing}_{A//A}, \mathbf{dMod}_A)$. This follows from the associated long exact sequence, the existence of a splitting $\underline{A} \rightarrow \rho_A \rightarrow \underline{A}$, and the fact that both ρ_A and $\underline{A}$ take values in connective objects. By [40, Remark 1.4.2.9], $\text{Fun}(\mathbf{dRing}_{A//A}, \mathbf{dMod}_A)$ is stable. In particular, the middle term in the split short exact sequence

$$I_A \rightarrow \rho_A \rightarrow \underline{A}$$

is equivalent to the direct sum $I_A \oplus \underline{A}$.

Construction 3.2.10. Consider the free algebra functor

$$\begin{aligned} \text{Sym} : \mathbf{PolyMod} &\rightarrow \mathbf{Poly} \\ (R, M) &\mapsto \bigoplus_{n \geq 0} \left(M^{\otimes_R^n} \right)_{\Sigma_n}. \end{aligned}$$

We can now consider the composition $\mathbf{PolyMod} \xrightarrow{\text{Sym}} \mathbf{Poly} \rightarrow \mathbf{dRing}$ to construct the derived functor:

$$\text{LSym} : \mathbf{dMod}^{\text{cn}} \rightarrow \mathbf{dRing}.$$

The induced functor on $\mathbf{dMod}_A^{\text{cn}}$ *symmetric A algebra* and is denoted Sym_A .

Warning 3.2.11. In general, $\text{LSym}(M)$ differs substantially from

$$\text{hSym}(M) := \bigoplus_n \left(M^{\otimes_R^n} \right)_{h\Sigma_n},$$

where $(M^{\otimes_R^n})_{h\Sigma_n}$ denotes the homotopy orbits. The same holds for $\text{LSym}_R^n(M)$ and $(M^{\otimes_R^n})_{h\Sigma_n}$. However, when R is a $\mathbb{Q}$ -algebra, one can construct an equivalence. We briefly outline the argument. In general, there is a map $\text{hSym}(P) \rightarrow \text{LSym}(P)$ arising from the canonical map

$$(P^{\otimes n})_{h\Sigma_n} \rightarrow H_0((P^{\otimes n})_{h\Sigma_n}) \simeq \text{LSym}^n(P)$$

defined for finitely presented projective R -modules P and then left Kan extended (see also [3, Remark 2.72]). Moreover, one has the description

$$(P^{\otimes n})_{h\Sigma_n} \simeq P^{\otimes n} \otimes_{R[\Sigma_n]}^{\mathbb{L}} R.$$

When $R = \mathbb{Q}$, Maschke's theorem implies that $\mathbb{Q}[\Sigma_n]$ is semisimple for each n . In particular, by [62, Theorem 4.2.2], every module over $\mathbb{Q}[\Sigma_n]$ is projective, and hence

$$P^{\otimes n} \otimes_{\mathbb{Q}[\Sigma(n)]}^{\mathbb{L}} \mathbb{Q} \simeq P^{\otimes n} \otimes_{\mathbb{Q}[\Sigma(n)]} \mathbb{Q} \simeq (P^{\otimes n})_{\Sigma_n} \simeq \text{LSym}^n(P).$$

Remark 3.2.12. Notice that

$$\text{LSym}_A \simeq \prod_{n \geq 0} \text{LSym}_A^n.$$

Proposition 3.2.13 (cf. [40, Proposition 7.3.4.5]). *Let A be an animated commutative $\mathbb{R}$ -algebra then:*

$$\begin{array}{ccc} \mathbf{dMod}_A & \xrightarrow{\text{LSym}_A} & \mathbf{dRing}_{A//A} \\ & \perp & \\ & \xleftarrow{I_A} & \end{array}$$

Remark 3.2.14. By [40, Proposition 7.3.4.5. and Remark 4.3.4.6], the unit of the adjunction is given by:

$$\text{id} = \text{LSym}_A^1 \rightarrow \prod_{n \geq 1} \text{LSym}_A^n.$$

Remark 3.2.15. By construction LSym_A is reduced (see Definition B.4.1), since it sends 0 to A .

Remark 3.2.16 (cf. [40, Remark 7.4.3.13.]). Let A be a $\mathcal{C}^\infty$ -ring. If $M \in \mathbf{dMod}_A^{\text{cn}}$ is n -connective then $\text{LSym}^n(M)$ is nm -connective.

Proof of the theorem

The proof presented here may be found in [57, section 4.3.1].

Notation. We will write the suspension functor on $\mathbf{dRing}_{A//A}$ as Σ_A and the loop functor as Ω_A . The forgetful functor $(-)^{\text{alg}} : \mathbf{dC}^\infty\mathbf{Ring}_{A//A} \rightarrow \mathbf{dRing}_{A^{\text{alg}}//A^{\text{alg}}}$ has a left adjoint that we denote:

$$\widehat{(-)}_A : \mathbf{dRing}_{A^{\text{alg}}//A^{\text{alg}}} \rightarrow \mathbf{dC}^\infty\mathbf{Ring}_{A//A},$$

which is given by the composition

$$\mathbf{dRing}_{A^{\text{alg}}//A^{\text{alg}}} \xrightarrow{\widehat{(-)}} \mathbf{dC}^\infty\mathbf{Ring}_{(\widehat{A^{\text{alg}}})//(\widehat{A^{\text{alg}}})} \xrightarrow{\epsilon_!} \mathbf{dC}^\infty\mathbf{Ring}_{A//A}$$

where $\widehat{(-)}$ is the left adjoint of the underlying algebra functor $(-)^{\text{alg}} : \mathbf{dC}^\infty\mathbf{Ring} \rightarrow \mathbf{dRing}_{\mathbb{R}}$ and $\epsilon_!$ is the functor taking pushouts along the counit map $\epsilon : (\widehat{A^{\text{alg}}}) \rightarrow A$ [37, Proposition 5.2.5.1].

Remark 3.2.17. Let A be an $\mathbb{R}$ -algebra. Since $- \otimes_{\mathbb{R}} A$ preserves colimits we have the following commuting diagram:

$$\begin{array}{ccccc} \mathbf{Mod}_{\mathbb{R}} & \xrightarrow{\text{Sym}} & \mathbf{dRing}_{\mathbb{R} // \mathbb{R}} & \xrightarrow{\Sigma_{\mathbb{R}}} & \mathbf{dRing}_{\mathbb{R} // \mathbb{R}} \\ & \searrow & \downarrow - \otimes_{\mathbb{R}} A & & \downarrow - \otimes_{\mathbb{R}} A \\ & & \mathbf{dRing}_{A // A} & \xrightarrow{\Sigma_A} & \mathbf{dRing}_{A // A} \end{array}$$

In particular:

$$(A \otimes_{\mathbb{R}} \Sigma_{\mathbb{R}}^k \text{Sym}(V)) \simeq \Sigma_A^k (A \otimes_{\mathbb{R}} \text{Sym}(V))$$

Lemma 3.2.18 (cf. [57, Proposition 4.3.1.2.]). *Let A be a $\mathcal{C}^\infty$ -ring, then the functor*

$$(-)^{\text{alg}} : \mathbf{dC}^\infty\mathbf{Ring}_{A//A} \rightarrow \mathbf{dRing}_{A^{\text{alg}}//A^{\text{alg}}}$$

induces an equivalence

$$\mathbf{Stab}(\mathbf{dC}^\infty\mathbf{Ring}_{A//A}) \simeq \mathbf{Stab}(\mathbf{dRing}_{A^{\text{alg}}//A^{\text{alg}}}).$$

Proof. Thanks to Proposition B.4.7, it suffices to show that $(-)^{\text{alg}}$ exhibits $(\mathbf{dC}^\infty\mathbf{Ring}_{\mathbb{R}})_{A//A}$ as monadic over $(\mathbf{dRing}_{\mathbb{R}})_{A^{\text{alg}}//A^{\text{alg}}}$, and that the unit map induces an equivalence $\partial \text{id} \rightarrow \partial((-)^{\text{alg}} \circ \widehat{(-)}_A)$. Note that we already know that $\widehat{(-)}_A$ is reduced. By the ∞ -Categorical Barr-Beck Theorem (see [40, Theorem 4.7.3.5]), monadicity is reduced to verifying that $(-)^{\text{alg}}$ is conservative and preserves sifted colimits. The conservativity of $(-)^{\text{alg}}$ is precisely the content of Remark 2.2.17. Since the category $\mathbf{dRing}_{A//A}$ consists of objects in which A appears as both a source and a target (i.e., as a retract), we need to consider separately the behavior of colimits in the over and undercategories. Fortunately Proposition B.2.2 addresses the case of overcategories, while Remark B.2.5 handles the undercategory case.

It remains to show that the unit induces an equivalence $\partial \text{id} \rightarrow \partial((-)^{\text{alg}} \circ \widehat{(-)}_A)$. From Proposition B.4.6, we recall the description of the derivatives:

$$\partial \text{id} \simeq \text{colim}_i \Omega_{A^{\text{alg}}}^i \circ \Sigma_{A^{\text{alg}}}^i \quad \text{and} \quad \partial \left((-)^{\text{alg}} \circ \widehat{(-)}_A \right) \simeq \text{colim}_i \Omega_{A^{\text{alg}}}^i \circ (-)^{\text{alg}} \circ \widehat{(-)}_A \circ \Sigma_{A^{\text{alg}}}^i.$$

In particular, it suffices to produce the equivalence on the essential image of $\Sigma_{A^{\text{alg}}}$ so that we get the equivalences:

$$\Omega_{A^{\text{alg}}}^i \circ \Sigma_{A^{\text{alg}}}^i \simeq \Omega_{A^{\text{alg}}}^i \circ (-)^{\text{alg}} \circ \widehat{(-)}_A \circ \Sigma_{A^{\text{alg}}}^i$$

for all $i \geq 1$, which is enough to prove the proposition. Any animated A^{alg} -ring B may be written as a good A^{alg} -cell object, i.e. a sequential colimit:

$$A^{\text{alg}} = A_{-1}^{\text{alg}} \rightarrow A^{\text{alg}} \otimes_{\mathbb{R}} \text{Sym}(V_{-1}) = A_0^{\text{alg}} \rightarrow A_1^{\text{alg}} \rightarrow A_2^{\text{alg}} \rightarrow \dots$$

of pushouts of the form:

$$\begin{array}{ccc} A^{\text{alg}} \otimes_{\mathbb{R}} \Sigma_{\mathbb{R}}^k \text{Sym}(V_k) & \longrightarrow & A_{k-1}^{\text{alg}} \\ \varphi_k \downarrow & \lrcorner & \downarrow \phi_k \\ A^{\text{alg}} & \longrightarrow & A_k^{\text{alg}} \end{array}$$

for V_k a vector space and $k \geq 0$. Since $\Sigma_{A^{\text{alg}}}$ preserves sequential colimit, we have $\Sigma_{A^{\text{alg}}} B \simeq \text{colim}_i (\Sigma_{A^{\text{alg}}} A_i^{\text{alg}})$. In particular, Remark 3.2.17 implies that the essential image of $\Sigma_{A^{\text{alg}}}$ consist of good A^{alg} -cell object with $A_{-1}^{\text{alg}} = A_0^{\text{alg}} = A^{\text{alg}}$ and $A_1 = A^{\text{alg}} \otimes_{\mathbb{R}} \Sigma_{\mathbb{R}} \text{Sym}(V)$. The functor $(-)_A$ preserves sequential colimits (being a left adjoint), as does $(-)^{\text{alg}}$ (as shown above). Moreover, thanks to Corollary D.2.16 we see that $(-)_A$ sends the map φ_k to an effective epimorphism:

$$A \otimes^{\infty} \Sigma^k \text{Sym}_{\mathbb{C}^{\infty}}(V^{\vee}) \rightarrow A,$$

so pushouts along this map are also preserved by $(-)^{\text{alg}}$ by unramifiedness (see Corollary 2.2.4). It follows that we only need to check that the unit is an equivalence on objects of the form $A^{\text{alg}} \otimes_{\mathbb{R}} \Sigma_{\mathbb{R}}^k \text{Sym}(V)$ for $k \geq 1$. This follows from Lemma 2.2.35 and unramifiedness applied to the effective epimorphism $\mathbb{R} \rightarrow \Sigma^k \text{Sym}_{\mathbb{C}^{\infty}}(V^{\vee})$. $\square$

Lemma 3.2.19 (cf. [57, Proposition 4.3.1.3]). *Let A be an animated commutative $\mathbb{R}$ -algebra, then the functor:*

$$I_A : \mathbf{dRing}_{A//A} \rightarrow \mathbf{dMod}_A^{\text{cn}}$$

induces an equivalence

$$\mathbf{Stab}(\mathbf{dRing}_{A//A}) \simeq \mathbf{Stab}(\mathbf{dMod}_A^{\text{cn}}) \simeq \mathbf{dMod}_A$$

Proof. Thanks to Proposition B.4.7, it suffices to show that I_A exhibits $\mathbf{dRing}_{A//A}$ as monadic over $\mathbf{dMod}_A^{\text{cn}}$, and that the unit map induces an equivalence $\partial \text{id} \rightarrow \partial(I_A \circ \text{LSym}_A)$, since by Remark 3.2.15 we already know that the left adjoint is reduced. Thanks to [40, Theorem 4.7.3.5] we may, again, reduce the monadicity condition to verifying that I_A is conservative and preserves sifted colimits. Since the inclusion $\mathbf{dMod}_A^{\text{cn}} \hookrightarrow \mathbf{dMod}_A$ is conservative and preserves sifted colimits, it suffices to check that:

$$I_A : \mathbf{dRing}_{A//A} \rightarrow \mathbf{dMod}_A$$

is conservative and preserves sifted colimits. This follows from Lemma B.2.6. Indeed, unwinding Definition 3.2.8, we see that I_A is given by the composition of the forgetful $\mathbf{dRing}_{A//A} \rightarrow (\mathbf{dMod}_A)_{/A}$, which is conservative and preserves sifted colimit, and the pullback functor along $0 \rightarrow A$, which again is conservative and preserves sifted colimit thanks to Corollary B.4.10. To prove that the unit induces an equivalence on derivatives, we first recall its explicit form from Remark 3.2.14:

$$\text{id} = \text{LSym}_A^1 \rightarrow \coprod_{n \geq 1} \text{LSym}_A^n.$$

Lemma B.4.8 implies that the natural transformation $\coprod_{n \geq 1} \partial \text{LSym}_A^n \rightarrow \partial \coprod_{n \geq 1} \text{LSym}_A^n$ is an equivalence, so we only need to show that ∂LSym_A^n is nullhomotopic for $n \geq 2$. Using Proposition B.4.6, it suffices to verify that the functors $\Omega^i \circ \text{LSym}_A^n \circ \Sigma^i$ takes value in i -connective modules for $n \geq 2$. But, by Remark 3.2.16, these functors take values in $i(n-1)$ -connective modules. $\square$

Proof of Theorem 3.2.4. This is now immediate from Lemmas 3.2.19 and 3.2.18 plus right completeness of $\mathbf{dMod}_{A^{\text{alg}}}$. $\square$

Remark 3.2.20. For another proof see [10, Theorem 9.3], however the proof relies on [55] which has a strong model categorical flavor, *de gustibus non disputandum est*.

Remark 3.2.21. Noticing that $\mathbf{Stab}(\mathcal{C}) \simeq \mathbf{Stab}(\mathcal{C}_{C/})$ for any ∞ -category $\mathcal{C}$ with finite limits and an object $C \in \mathcal{C}$, we deduce a stronger result. Namely, for any pair of $\mathcal{C}^{\infty}$ -rings B and A , we obtain the equivalence

$$\mathbf{Stab}(\mathbf{dC}^{\infty} \mathbf{Ring}_{B//A}) \simeq \mathbf{dMod}_A$$

3.2.3 Tangent category

Our next goal is to introduce the tangent category of a presentable ∞ -category and then specialize to the case of $\mathcal{C}^\infty$ -rings. The tangent category can be viewed as a generalization of the presentable fibration $\mathbf{dMod} \rightarrow \mathbf{CRing}$. In particular, it provides a framework for defining differentiation and “infinitesimal directions” in a general presentable ∞ -category.

To better understand the general construction, suppose we wish to make the assignment

$$A \mapsto \mathbf{dMod}_A$$

functorial for A a $\mathcal{C}^\infty$ -ring, without knowing that $\mathbf{dC}^\infty\mathbf{Mod} \rightarrow \mathbf{dC}^\infty\mathbf{Ring}$ is a fibration. A common strategy for constructing functors from an ∞ -category is to use the theory of coCartesian fibrations, a technique that has already appeared in this thesis. A particularly useful example is the codomain fibration

$$\mathrm{Fun}([1], \mathcal{C}) \rightarrow \mathcal{C},$$

given by evaluation at 1. The fiber over an object $A \in \mathcal{C}$ is given by the slice category $\mathcal{C}_{/A}$. Returning to our context, we take $\mathcal{C}$ to be the category $\mathbf{dC}^\infty\mathbf{Ring}$ of derived $\mathcal{C}^\infty$ -rings. Building on the theory developed in the previous section, we observe that if we manage to perform a sort of *fiberwise stabilization* on the codomain fibration, we obtain a promising candidate for constructing the desired functor. More precisely, we would get a cocartesian fibration $p : \mathcal{E} \rightarrow \mathcal{C}$ such that the fiber over each object $A \in \mathcal{C}$ is given by $\mathbf{Stab}(\mathcal{C}_{/A}) \simeq \mathbf{dMod}_A$. Then, straightening p should recover the assignment $A \mapsto \mathbf{dMod}_A$.

In what follows, we present in detail one method to achieve this goal for a general presentable ∞ -category. We then provide some intuition for alternative approaches, before focusing on the case $\mathcal{C} = \mathbf{dC}^\infty\mathbf{Ring}$, where we present a result establishing an equivalence between the tangent category of $\mathbf{dC}^\infty\mathbf{Ring}$ and the ∞ -category $\mathbf{dC}^\infty\mathbf{Mod}$.

Most of the proofs in this section rely on substantial additional theory. Rather than reproducing these technical details in full, we refer the reader to the existing literature, primarily [40, Section 7.3], [35], and [57, 4.3.2].

Stable and pointed envelope

To make precise the above idea we first need to clarify what we mean by fiberwise stabilization.

Definition 3.2.22 (Stable envelope [40, Definition 7.3.1.1]). Let $\mathcal{C}$ be a presentable ∞ -category. A *stable envelope* (or *pointed envelope*) of $\mathcal{C}$ is a categorical fibration $u : \mathcal{C}' \rightarrow \mathcal{C}$ with the following properties:

- (i) The ∞ -category $\mathcal{C}'$ is stable (pointed) and presentable.
- (ii) The functor u admits a left adjoint.
- (iii) For every presentable stable (pointed) ∞ -category $\mathcal{E}$, composition with u induces an equivalence of ∞ -categories

$$\mathrm{Fun}^R(\mathcal{E}, \mathcal{C}') \rightarrow \mathrm{Fun}^R(\mathcal{E}, \mathcal{C}).$$

Where $\mathrm{Fun}^R(\mathcal{E}, \mathcal{C}')$ denotes the full subcategory of $\mathrm{Fun}(\mathcal{E}, \mathcal{C}')$ spanned by those functor which admits left adjoint, similarly for $\mathrm{Fun}^R(\mathcal{E}, \mathcal{C})$.

More generally, suppose that $p : \mathcal{C} \rightarrow \mathcal{D}$ is a presentable fibration. A *stable envelope* (or *pointed envelope*) of p is a functor $u : \mathcal{C}' \rightarrow \mathcal{C}$ with the following properties:

1. The composition $p \circ u$ is a presentable fibration.
2. The functor u carries $(p \circ u)$ -Cartesian morphisms of $\mathcal{C}'$ to p -Cartesian morphisms of $\mathcal{C}$.
3. For every object $D \in \mathcal{D}$, the induced map $\mathcal{C}'_D \rightarrow \mathcal{C}_D$ is a stable envelope (pointed envelope) of $\mathcal{C}'_D$.

Example 3.2.23 (cf. [40, Example 7.3.1.4]). Let $\mathcal{C}$ be presentable ∞ -category and $\mathbf{Stab}(\mathcal{C})$ its stabilization. Then the composite $\mathbf{Stab}(\mathcal{C}) \xrightarrow{\Omega_{\mathcal{C}}^\infty} \mathcal{C}_* \rightarrow \mathcal{C}$ exhibit $\mathbf{Stab}(\mathcal{C})$ as a stable envelope of $\mathcal{C}$.

Proposition 3.2.24 (cf. [40, Proposition 7.3.1.7]). Let $p : \mathcal{C} \rightarrow \mathcal{D}$ be a presentable fibration of ∞ -categories. Then there exists a functor $u : \mathcal{C}' \rightarrow \mathcal{C}$ which is a stable (pointed envelope) of the presentable fibration p .

Proof. Straightening the functor p we get a functor $\chi : \mathcal{D}^{\text{op}} \rightarrow \text{Pr}^{\text{R}}$. Now recall from Remark B.3.11 the adjunction:

$$\begin{array}{ccc} \text{Pr}_{\text{st}}^{\text{R}} & \begin{array}{c} \xrightarrow{\quad} \\ \perp \\ \xleftarrow{\quad} \end{array} & \text{Pr}^{\text{R}} \\ & \text{Stab}(-) & \end{array}$$

Where $\text{Pr}_{\text{st}}^{\text{R}}$ denotes the full subcategory of stable presentable ∞ -categories. Call η the counit $\text{Stab}(-) \circ \chi \rightarrow \chi$ of the above adjunction and $\mathcal{C}'$ denote the cartesian unstraightening $\text{Un}^{\text{ct}}(\text{Stab}(-) \circ \chi)$. By Unstraightening the whole situation we get a morphism $u : \mathcal{C}' \rightarrow \mathcal{C}$ of presentable fibration over $\mathcal{D}$. Now by construction u exhibits $\mathcal{C}'$ as a stable envelope of $\mathcal{C}$. $\square$

We now wish to present an alternative construction that one may implement. In particular, it would be desirable to exhibit an explicit construction of the fiberwise stabilization internal to a presentable fibration, without relying on its straightening.

Definition 3.2.25 (Fiberwise stabilization cf. [38, Defintion 5.2.1]). Let $p : \mathcal{C} \rightarrow \mathcal{D}$ be a functor of ∞ -category. We define the following:

$$\text{Stab}(p) \subseteq \widetilde{\text{PSP}}(p) \rightarrow \mathcal{D}$$

as follows:

- $\widetilde{\text{PSP}}(p)$ is defined as the following pullback:

$$\begin{array}{ccc} \widetilde{\text{PSP}}(p) & \longrightarrow & \text{Fun}(\mathbb{Z} \times \mathbb{Z}, \mathcal{C}) \\ \downarrow & \lrcorner & \downarrow \\ \coprod_{s \in \mathcal{D}} * & \xrightarrow{\prod_{s \in \mathcal{D}} \text{const}_s} & \text{Fun}(\mathbb{Z} \times \mathbb{Z}, \mathcal{D}) \end{array}$$

In particular, we can identify objects of $\widetilde{\text{PSP}}(p)$ with pairs (s, F) , where s is an object of $\mathcal{D}$ and $F : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathcal{C}_s$ is a functor.

- We let $\text{Stab}(p)$ denote the full ∞ -category of $\widetilde{\text{PSP}}(p)$ spanned by those objects (s, F) such that F is a spectrum object of $\mathcal{C}_s$.

Example 3.2.26 (cf. [40, Example 7.3.1.5]). Let $p : \mathcal{C} \rightarrow \mathcal{D}$ be a presentable fibration, and suppose that each fiber $\mathcal{C}_D$ is pointed. The ∞ -category $\text{Stab}(p)$ of Definition 3.2.25 is a stable envelope of $\mathcal{C}$. This follows by [38, Proposition 5.2.5]

Remark 3.2.27. Finally one may construct the stable envelope through *reduced excisive* functors (see [35, Example 1.1.6]).

The stable envelope enjoys the following universal property.

Proposition 3.2.28 (cf. [40, Proposition 7.3.1.7, Remark 7.3.18]). Let $p : \mathcal{E} \rightarrow \mathcal{C}$ be a presentable fibration, and let $r : \mathcal{E}' \rightarrow \mathcal{E}$ be a stable envelope of p . Then for each presentable fibration $\mathcal{D} \rightarrow \mathcal{C}$ whose fibers are stable, composition with r induces an equivalence

$$\text{Fun}_{\mathcal{C}}^{\text{R}}(\mathcal{D}, \mathcal{E}') \xrightarrow{\sim} \text{Fun}_{\mathcal{C}}^{\text{R}}(\mathcal{D}, \mathcal{E})$$

where $\text{Fun}_{\mathcal{C}}^{\text{R}}(\mathcal{D}, \mathcal{E}')$ denotes the full subcategory spanned by those functors $\mathcal{D} \rightarrow \mathcal{E}'$ over $\mathcal{C}$ that preserve Cartesian edges and admit a left adjoint on each fiber, and similarly for $\text{Fun}_{\mathcal{C}}^{\text{R}}(\mathcal{D}, \mathcal{E})$.

As a consequence the various presentation of the stable envelope of a presentable fibration are all equivalent.

The tangent category

We can now give the following definition.

Definition 3.2.29 (Tangent category [35, Defintion 1.1.12]). Let $\mathcal{C}$ be a presentable ∞ -category. A *tangent category* (or *tangent bundle*) to $\mathcal{C}$ is a functor $T_{\mathcal{C}} \rightarrow \text{Fun}([1], \mathcal{C})$ which exhibits $T_{\mathcal{C}}$ as the stable envelope of the codomain presentable fibration $\text{Fun}([1], \mathcal{C}) \rightarrow \mathcal{C}$.

We may think of objects of $T_{\mathcal{C}}$ as pairs (A, M) where A is an object of $\mathcal{C}$ and M is an infinite loop spectrum object of $\mathcal{C}/A$. In the case $\mathcal{C}$ is the ∞ -category of $\mathbf{dC}^\infty\mathbf{Ring}$ we can identify M with an A -module (see Theorem 3.2.4). In this case, the functor $T_{\mathcal{C}} \rightarrow \mathrm{Fun}([1], \mathcal{C})$ associate to (A, M) the projection $A \oplus M \rightarrow A$ the projection from the square-zero extension of A by M (see the next subsection). Recalling that we may think of square-zero extensions as *infinitesimal thickenings* (see Remark 3.1.17), we may interpret this morphism as a *tangent vector* in $\mathcal{C}$, relating A to the *infinitesimally* nearby object $A \oplus M$. This analogy is already present in [40, below Definition 7.3.1.9].

Proposition 3.2.30 (Tangent category of $\mathbf{dC}^\infty\mathbf{Ring}$ [57, Theorem 4.3.2.8]). *There is a canonical equivalence*

$$T_{\mathbf{dC}^\infty\mathbf{Ring}} \rightarrow \mathbf{dC}^\infty\mathbf{Mod}$$

of presentable fibrations over $\mathbf{dC}^\infty\mathbf{Ring}$.

3.2.4 Square-zero extensions

We are now ready to discuss square-zero extensions in the derived setting. As already demonstrated in the discrete context, these extensions can be used to define derivations. In fact, this is precisely the perspective we adopt here. Rather than beginning with a specific class of morphisms and subsequently showing that they can be described via square-zero extensions, as is often done in the classical setting, we take a different approach. In the derived framework, such an operation is not readily available or unambiguously defined. Therefore, we will *define* derivations directly as morphisms whose codomain is a square-zero extension.

We now present two definitions of square-zero extensions, which will agree on the ∞ -category $\mathbf{dC}^\infty\mathbf{Mod}^{\mathrm{cn}}$. The first is a direct generalization of the classical (discrete) notion, while the second, though less immediate, is better suited for the developments that follow.

Construction 3.2.31. Consider the functor $\mathcal{C}^\infty\mathbf{PolyMod} \rightarrow \mathcal{C}^\infty\mathbf{Ring}$ given by associating to each (A, E) the square zero-extension $A \oplus E$ of Definition 3.1.16. By universal property of the animation we get a sifted colimit preserving functor:

$$F_{SZ} : \mathbf{dC}^\infty\mathbf{Mod}^{\mathrm{cn}} \rightarrow \mathbf{dC}^\infty\mathbf{Ring}$$

Remark 3.2.32. Notice that we have $\pi_0(F_{SZ}(A, E)) = \pi_0(A) \oplus \pi_0(E)$. For this reason we are going to denote $F_{SZ}(A, E)$ by $A \oplus E$.

Given $E \in \mathbf{dMod}_A^{\mathrm{cn}}$, we now want to regard $A \oplus E$ as an object in $\mathbf{dC}^\infty\mathbf{Ring}_{/A}$. This is possible thanks to the functoriality of the assignment $(A, E) \mapsto A \oplus E$. More precisely, the canonical morphism

$$(A, E) \longrightarrow (A, 0)$$

induces a morphism of $\mathbf{dC}^\infty\mathbf{Ring}$:

$$A \oplus E \longrightarrow A \oplus 0 \cong A.$$

It follows that we get an induced functor:

$$F'_{SZ} : \mathbf{dC}^\infty\mathbf{Mod}^{\mathrm{cn}} \rightarrow \mathrm{Fun}([1], \mathbf{dC}^\infty\mathbf{Ring}).$$

We deliberately refrain from referring to either F_{SZ} or F'_{SZ} as the *square-zero extension functor*. This choice is not due to any inadequacy of these constructions, but rather because we reserve this terminology for another definition that will be introduced shortly. We will then prove that the two constructions coincide.

Definition 3.2.33 (Square-zero extension). Let A be a $\mathcal{C}^\infty$ -ring and consider. We define the *square-zero extension functor* from A -modules as the composite

$$\mathbf{dMod}_A \simeq \mathbf{Stab}\left(\mathbf{dC}^\infty\mathbf{Ring}_{A//A}\right) \xrightarrow{\Omega_A^\infty} \mathbf{dC}^\infty\mathbf{Ring}_{A//A}.$$

For M an A -module we define the *square-zero extension of A by M* as the value of the square-zero functor from A -modules at M .

Notation. Let A be a $\mathcal{C}^\infty$ -ring and let M be an A -module. In what follows, we shall abuse notation slightly by writing $\Omega_A^\infty(M)$ to denote the square-zero extension of A by M .

Remark 3.2.34 (cf. [57, Remark 4.3.1.5]). Consider A a $\mathcal{C}^\infty$ -ring. The chain rule gives a commutative diagram:

$$\begin{array}{ccc} \mathrm{Stab}\left(\mathrm{d}\mathcal{C}^\infty\mathbf{Ring}_{A//A}\right) & \xrightarrow{\partial(I_A \circ (-)^{\mathrm{alg}})} & \mathrm{Stab}(\mathrm{dMod}_A^{\mathrm{cn}}) \\ \Omega_A^\infty \downarrow & & \downarrow \Omega^\infty \\ \mathrm{d}\mathcal{C}^\infty\mathbf{Ring}_{A//A} & \xrightarrow{I_A \circ (-)^{\mathrm{alg}}} & \mathrm{dMod}_A^{\mathrm{cn}} \end{array} \quad .$$

By Remark 3.2.9, the bottom horizontal arrow fits into the split fiber sequence:

$$I_{A^{\mathrm{alg}}} \circ (-)^{\mathrm{alg}} \rightarrow \rho_{A^{\mathrm{alg}}} \circ (-)^{\mathrm{alg}} \rightarrow \underline{A^{\mathrm{alg}}}$$

where we recall that $\underline{A^{\mathrm{alg}}}$ is the constant functor with value A^{alg} and

$$\rho_{A^{\mathrm{alg}}} : (\mathrm{dRing}_{\mathbb{R}})_{A^{\mathrm{alg}}//A^{\mathrm{alg}}} \rightarrow (\mathrm{Mod}_{A^{\mathrm{alg}}})_{/A^{\mathrm{alg}}}$$

is the underlying A^{alg} -module functor. Using the commutative diagram above and the fact the functor $\Omega^\infty : \mathrm{Stab}(\mathrm{dMod}_{A^{\mathrm{alg}}}^{\mathrm{cn}}) \rightarrow \mathrm{dMod}_{A^{\mathrm{alg}}}^{\mathrm{cn}}$ is identified with the connective cover functor, we deduce that the functor $\rho_{A^{\mathrm{alg}}} \circ (-)^{\mathrm{alg}} \circ \Omega_A^\infty$ fits into the following split short exact sequence:

$$\tau_{\geq 0} \rightarrow \rho_{A^{\mathrm{alg}}} \circ (-)^{\mathrm{alg}} \circ \Omega_A^\infty \rightarrow \underline{A^{\mathrm{alg}}}.$$

It follows that $\Omega_A^\infty(M) \simeq A \oplus \tau_{\geq 0}M$ as an A -module. The algebra structure on this object, which justifies the terminology square-zero extension, is explained in detail in [35, Remark 1.5.17].

Remark 3.2.35. It is noted in [57, Remark 4.3.1.6.] that $\Omega_A^\infty|_{\mathrm{dMod}_A^{\mathrm{cn}}}$ preserves sifted colimits. In particular using Proposition D.1.17 it can be checked that the functor $F|_{\mathrm{dMod}_A^{\mathrm{cn}}}$ and $\Omega_A^\infty|_{\mathrm{dMod}_A^{\mathrm{cn}}}$ are equivalent. Even more is true: we can identify the functor F'_{SZ} with the restriction of the tangent bundle functor $\mathrm{dMod}^{\mathrm{cn}} \subset T\mathrm{d}\mathcal{C}^\infty\mathbf{Ring} \rightarrow \mathrm{Fun}([1], \mathrm{d}\mathcal{C}^\infty\mathbf{Ring})$. For this result, we refer to the discussion in [57] starting from Corollary 4.3.2.10; the statement itself is given in [57, Corollary 4.3.2.12].

3.3 The cotangent complex

We have now developed the natural framework for defining and working with derivations and the cotangent complex.

In Subsection 3.3.1, we begin by providing a definition of derivations that naturally generalizes the one given in Definition 3.1.18. We then exploit the full power of the tangent category to functorially define the cotangent complex. Finally, in Subsection 3.3.2, we prove our main result: the identification of the cotangent complex L_M of a differentiable manifold M with its module of differential forms $\Omega^1(M)$.¹ We conclude with Subsection 3.3.3, where we introduce the relative cotangent complex and present some of its main properties.

The main references for this section are [35], [40], and [57, Section 5.1].

3.3.1 Definition and functoriality

Definition 3.3.1 (Derivation for connective modules). For $A \in \mathrm{d}\mathcal{C}^\infty\mathbf{Ring}$ and $E \in \mathrm{d}\mathcal{C}^\infty\mathbf{Mod}_A^{\mathrm{cn}}$, we define the space of A -derivations of E as:

$$\mathrm{Der}(A, E) := \mathrm{Hom}_{\mathrm{d}\mathcal{C}^\infty\mathbf{Ring}_{/A}}(\mathrm{id}_A, (A \oplus E \rightarrow A))$$

Remark 3.3.2. Notice that we have a natural fiber sequence:

$$\mathrm{Der}(A, M) \longrightarrow \mathrm{Hom}_{\mathrm{d}\mathcal{C}^\infty\mathbf{Ring}}(A, A \oplus E) \longrightarrow \mathrm{Hom}_{\mathrm{d}\mathcal{C}^\infty\mathbf{Ring}}(A, A)$$

where the fiber is taken over the identity (see [37, Lemma 5.5.5.12]).

¹While proving this result, we came across an alternative argument given in [50, Example 2.2.18].

Remark 3.3.3. Let A be a $\mathcal{C}^\infty$ -ring and let E be a connective A -module. A choice of A -derivation $\tilde{d}$ of E induces a natural transformation between the functor corepresented by E and $\text{Der}(A, -)$. More precisely, this transformation is given by the following composition:

$$\text{Hom}_{\mathbf{d}\mathcal{C}^\infty \mathbf{Mod}_A^{\text{cn}}}(E, -) \xrightarrow{A \oplus -} \text{Hom}_{\mathbf{d}\mathcal{C}^\infty \mathbf{Ring}_{/A}}(A \oplus E, A \oplus -) \xrightarrow{- \circ \tilde{d}} \text{Hom}_{\mathbf{d}\mathcal{C}^\infty \mathbf{Ring}_{/A}}(A, A \oplus -) .$$

Definition 3.3.4 (Cotangent complex). Let $A \in \mathbf{d}\mathcal{C}^\infty \mathbf{Ring}$. A *cotangent complex* for A is a connective A -module L_A together with an A -derivation d_A of L_A such that the canonical map induced by d_A

$$\text{Hom}_{\mathbf{d}\mathcal{C}^\infty \mathbf{Mod}_A^{\text{cn}}}(L_A, -) \rightarrow \text{Der}(A, -).$$

is natural equivalence of functors.

While it is close to our definitions in the discrete case, this approach makes it difficult to discuss functoriality. For this reason, we now employ the machinery developed in the previous section to provide a *global* definition, which will restrict to the one given above.

Construction 3.3.5. Recall we that for a presentable ∞ -category $\mathcal{C}$ we defined the tangent category as the stable envelope of the presentable codomain fibration (see Definition 3.2.29)

$$\begin{array}{ccc} T_{\mathcal{C}} & \xrightarrow{G} & \text{Fun}([1], \mathcal{C}) \\ & \searrow p & \swarrow \text{ev}_1 \\ & \mathcal{C} & \end{array} .$$

At each $A \in \mathcal{C}$, the fiber of the functor G at A can be identified with $\Omega_A^\infty : \mathbf{Stab}(\mathcal{C}_{/A}) \rightarrow \mathcal{C}_{/A}$, which admits a left adjoint $(\Sigma_+^\infty)_A$. By [40, Proposition 7.3.2.6] we may conclude that G admits a left adjoint relative to $\mathcal{C}$, which will be denoted by F .

Definition 3.3.6 (cf. [40, Definition 7.3.2.14]). Let $\mathcal{C}$ be a presentable ∞ -category, then we define the *absolute cotangent complex* $\mathbb{L} : \mathcal{C} \rightarrow T_{\mathcal{C}}$ as the composition :

$$\mathcal{C} \xrightarrow{\Delta} \text{Fun}([1], \mathcal{C}) \xrightarrow{F} T_{\mathcal{C}},$$

where Δ is the diagonal embedding. The value of $\mathbb{L}$ at an object A will be denoted by $\mathbb{L}_A \in \mathbf{Stab}(\mathcal{C}_{/A})$, and we will refer to $\mathbb{L}_A$ as the *cotangent complex* of A .

Remark 3.3.7 (cf. [40, Remark 7.3.2.17]). Let $\mathcal{C}$ be a presentable ∞ -category containing an object A . Unwinding the definition we see that the object $\mathbb{L}_A \in \mathbf{Stab}(\mathcal{C}_{/A})$ corresponds to the image of $\text{id}_A \in \mathcal{C}_{/A}$ under the suspension spectrum functor

$$(\Sigma_+^\infty)_A : \mathcal{C}_{/A} \rightarrow \mathbf{Stab}(\mathcal{C}_{/A}).$$

Remark 3.3.8 (cf. [40, Remark 7.3.2.15]). Let $\mathcal{C}$ be a presentable ∞ -category. Since the diagonal embedding $\mathcal{C} \rightarrow \text{Fun}([1], \mathcal{C})$ is a left adjoint to the evaluation map $\text{Fun}([1], \mathcal{C}) \rightarrow \text{Fun}(\{0\}, \mathcal{C}) \simeq \mathcal{C}$, we deduce that the absolute cotangent complex functor $\mathbb{L} : \mathcal{C} \rightarrow T_{\mathcal{C}}$ is left adjoint to the composition

$$T_{\mathcal{C}} \xrightarrow{G} \text{Fun}([1], \mathcal{C}) \xrightarrow{\text{ev}_0} \text{Fun}(\{0\}, \mathcal{C}) \simeq \mathcal{C}.$$

Returning to our context, let $\mathcal{C}$ denote the category $\mathbf{d}\mathcal{C}^\infty \mathbf{Ring}$ of derived $\mathcal{C}^\infty$ -rings, let A be a $\mathcal{C}^\infty$ -ring, and let $E \in \mathbf{dMod}_A$ be an A -module. Recall that Ω_A^∞ encodes the square-zero extension functor. Since

$$\begin{array}{ccc} \text{Hom}_{\mathbf{dMod}_A}(\mathbb{L}_A, E) & \xrightarrow{\quad} & \text{Hom}_{\mathbf{d}\mathcal{C}^\infty \mathbf{Mod}}((A, \mathbb{L}_A), (A, E)) \\ \downarrow & \lrcorner & \downarrow \\ * & \xrightarrow{\text{id}_A} & \text{Hom}_{\mathbf{d}\mathcal{C}^\infty \mathbf{Ring}}(A, A) \end{array}$$

The preceding remarks yield an equivalence:

$$\text{Hom}_{\mathbf{d}\mathcal{C}^\infty \mathbf{Ring}}(A, A \oplus \tau_{\geq 0} E) \simeq \text{Hom}_{\mathbf{d}\mathcal{C}^\infty \mathbf{Mod}}(\mathbb{L}_A, E).$$

In particular, we obtain an equivalence:

$$\mathrm{Hom}_{\mathbf{dMod}_A}(\mathbb{L}_A, E) \simeq \mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}_A}(\mathrm{id}_A, (A \oplus \tau_{\geq 0} E \rightarrow A))$$

The above discussion essentially proves the following proposition.

Proposition 3.3.9 (cf. [10, Proposition 9.26]). *For $A \in \mathbf{dC}^\infty \mathbf{Ring}$,*

$$\mathbb{L}_A \simeq (A, L_A).$$

Corollary 3.3.10. *Given $A \in \mathbf{dC}^\infty \mathbf{Ring}$ the cotangent complex L_A exist and it is unique (up to homotopy).*

Thanks to the above discussion we have constructed the cotangent complex by abstract nonsense. Our goal is to prove that for $A = \mathcal{C}^\infty(M)$ where M is a smooth manifold we can explicitly realize the cotangent complex as $\Omega^1(M)$ the module of differential forms. The next subsection is devoted to the proof of this fact.

3.3.2 Universal property of differential forms

Theorem 3.3.11 (Universal property of differential forms). *Let M be a smooth manifold, then the cotangent complex of $\mathcal{C}^\infty(M)$ is given by the module of differential forms $\Omega^1(M)$ together with the usual differential d .*

For the entire proof, set $A := \mathcal{C}^\infty(M)$. let $E \in \mathbf{dMod}_A^{\mathrm{cn}}$ be a connective module over A . The strategy of the proof is to reduce the problem to the discrete case.

Step 1. Reduce to the case $\tau_{\leq n} E$.

Thanks to Corollary C.2.10 we have that $E \simeq \lim_n \tau_{\leq n} E$, in particular we have the following sequence of equivalences:

$$\begin{aligned} \mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}_A}(A, A \oplus E) &\cong \mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}_A}(A, A \oplus (\lim_n \tau_{\leq n} E)) \\ &\cong \lim_n \mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}_A}(A, A \oplus \tau_{\leq n} E). \end{aligned}$$

Where the second equivalence is due to the fact that $\mathrm{Der}(A, -)$ preserve limits by Corollary 3.3.10. Similarly, we have:

$$\mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{\mathrm{cn}}}(\Omega^1(M), E) \simeq \lim_n \mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{\mathrm{cn}}}(\Omega^1(M), \tau_{\leq n} E).$$

This means that we can restrict to the case E is of the form $\tau_{\leq n} E$.

Step 2. Reduce to the case $\tau_{\geq n} \tau_{\leq n} E$.

Take $E \in \mathbf{dMod}_A^{\mathrm{cn}}$ and suppose that we have proved the statement for $\pi_n(E)[n] \simeq \tau_{\geq n} \tau_{\leq n} E$ and $\tau_{\leq n-1} E$. We want to deduce the case $\tau_{\leq n} E$. By applying Remark C.1.5 to $\tau_{\leq n} E$, we obtain the following fiber sequence:

$$\tau_{\geq n} \tau_{\leq n} E \rightarrow \tau_{\leq n} E \rightarrow \tau_{\leq n-1} E.$$

Since both functors $\mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}_A}(A, A \oplus -)$ and $\mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{\mathrm{cn}}}(\Omega^1(M), -)$ preserve limits, they send fiber sequences to fiber sequences. Hence, we obtain two induced sequences that we would like to compare. More precisely, we need a natural map

$$\mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{\mathrm{cn}}}(\Omega^1(M), -) \longrightarrow \mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}_A}(A, A \oplus -).$$

To do so, we apply Remark 3.3.3 to the A -derivation $\tilde{d}: A \rightarrow A \oplus \Omega^1(M)$, defined by the identity on the first component and the differential d on the second. This yields the following commutative diagram:

$$\begin{array}{ccccc} \mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{\mathrm{cn}}}(\Omega^1(M), \tau_{\geq n} \tau_{\leq n} E) & \longrightarrow & \mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{\mathrm{cn}}}(\Omega^1(M), \tau_{\leq n} E) & \longrightarrow & \mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{\mathrm{cn}}}(\Omega^1(M), \tau_{\leq n-1} E) \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}_A}(A, A \oplus \tau_{\geq n} \tau_{\leq n} E) & \longrightarrow & \mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}_A}(A, A \oplus \tau_{\leq n} E) & \longrightarrow & \mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}_A}(A, A \oplus \tau_{\leq n-1} E) \end{array}.$$

We now introduce the following notation:

- $B' = \text{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}_{/A}}(A, A \oplus \tau_{\leq n-1} E)$ and $B = \text{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{cn}}(\Omega^1(M), \tau_{\leq n-1} E)$,
- $C' = \text{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}_{/A}}(A, A \oplus \tau_{\leq n} E)$ and $C = \text{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{cn}}(\Omega^1(M), \tau_{\leq n} E)$,
- $D' = \text{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}_{/A}}(A, A \oplus \tau_{\geq n} \tau_{\leq n} E)$ and $D = \text{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{cn}}(\Omega^1(M), \tau_{\geq n} \tau_{\leq n} E)$.

Then we consider the long exact sequence on homotopy groups associated with the two fiber sequences (note that all spaces involved are pointed spaces):

$$\begin{array}{ccccccccc}
\pi_{n+1}(B) & \longrightarrow & \pi_n(D) & \longrightarrow & \pi_n(C) & \longrightarrow & \pi_n(B) & \longrightarrow & \pi_{n-1}(D) \\
\downarrow \wr & & \downarrow \wr & & \downarrow & & \downarrow \wr & & \downarrow \wr \\
\pi_{n+1}(B') & \longrightarrow & \pi_n(D') & \longrightarrow & \pi_n(C') & \longrightarrow & \pi_n(B') & \longrightarrow & \pi_{n-1}(D')
\end{array}$$

By the Five Lemma, we conclude that the morphism $C \rightarrow C'$ is a weak equivalence. This shows that the desired equivalence holds for $\tau_{\leq n} E$.

Step 3. Reduce to the discrete case.

Without loss of generality, we may assume that E is a discrete A -module. Want to prove:

$$\text{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{cn}}(\Omega^1(M), E[n]) \simeq \text{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}_{/A}}(A, A \oplus E[n]).$$

If we can prove that the two spaces are connected, then we may deduce the equivalence at the level of loop spaces. This is desirable, as it allows us to reduce to the case concentrated in degree 0, where the statement has already been proven.

Lemma 3.3.12. $\text{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{cn}}(\Omega^1(M), E[n])$ is connected.

Proof. First notice that the product $\prod_m \text{Hom}_{\mathbf{C}^\infty \mathbf{Mod}_A}(\Omega^1(M)_m, E[n]_{m+p})$ is non zero only when $p = n$, where it is equal to $\text{Hom}_{\mathbf{C}^\infty \mathbf{Mod}_A}(\Omega^1(M), E)$. By looking at Definition C.2.4 we see that the Hom complex $\underline{\text{Hom}}_{\text{Ch}(A)}(\Omega^1(M), E[n])$ has the form:

$$\cdots \longrightarrow 0 \longrightarrow 0 \longrightarrow \text{Hom}_{\mathbf{C}^\infty \mathbf{Mod}_A}(\Omega^1(M), E) \longrightarrow 0 \longrightarrow 0 \longrightarrow \cdots$$

where the non zero term is in degree n . In particular

$$H_p(\underline{\text{Hom}}_{\text{Ch}(A)}(\Omega^1(M), E[n])) = \begin{cases} 0 & p \neq n, \\ \text{Hom}_{\mathbf{C}^\infty \mathbf{Mod}_A}(\Omega^1(M), E) & p = n. \end{cases}$$

Now by Serre-Swan theorem (see Proposition A.2.3) $\Omega^1(M)$ is projective and so thanks to the above calculation and Corollary C.2.22 we see that $\text{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{cn}}(\Omega^1(M), E[n])$ is connected. $\square$

Remark 3.3.13. In the above we proved something stronger, i.e.

$$\pi_p(\text{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{cn}}(\Omega^1(M), E[n])) = \begin{cases} 0 & p \neq n, \\ \text{Hom}_{\mathbf{C}^\infty \mathbf{Mod}_A}(\Omega^1(M), E) & p = n. \end{cases}$$

Remark 3.3.14. Before proving the statement for $\text{Der}(A, E[n])$, we recall from Subsection 2.2.3 the presentation of $\mathbf{dC}^\infty \mathbf{Ring}$ as the localization $\mathbf{dgC}^\infty \mathbf{Alg}[W^{-1}]$. In particular, we observe two basic facts:

1. Since A fibrant in $\mathbf{dgC}^\infty \mathbf{Alg}$ (recall that all object are fibrant) by Proposition B.1.5 for all $B \in \mathbf{dgC}^\infty \mathbf{Alg}$ we have:

$$\text{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}_{/A}}(A, B) \simeq \text{Hom}_{(\mathbf{dgC}^\infty \mathbf{Alg}_{/A})[W^{-1}]}(A, B)$$

2. For all $E \in \text{Ch}_{\geq 0}(A)$ we have that $A \oplus E$ is fibrant in $\mathbf{dgC}^\infty \mathbf{Alg}_{/A}$ since the map $A \oplus E \rightarrow A$ given by the projection is surjective, in particular is a fibration by Proposition 2.2.21.

Remark 3.3.15. Let A' be a discrete cofibrant object in $\mathbf{dgC}^\infty \mathbf{Alg}$ with a morphism to A . Then

$$\pi_0 \operatorname{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}/A} (A', A \oplus E[n]) \simeq *.$$

To prove this, first observe that, by Remark B.1.4, A' is cofibrant in the slice category $\mathbf{dgC}^\infty \mathbf{Alg}_{/A}$, and that, by the above remark, $A \oplus E[n]$ is fibrant in $\mathbf{dgC}^\infty \mathbf{Alg}_{/A}$. Then, Proposition B.1.6 implies that we have the following surjection:

$$\operatorname{Hom}_{\mathbf{dgC}^\infty \mathbf{Alg}_{/A}} (A', A \oplus E[n]) \simeq \pi_0 \operatorname{Hom}_{\mathbf{dgC}^\infty \mathbf{Alg}_{/A}} (A', A \oplus E[n]) \twoheadrightarrow \pi_0 \operatorname{Hom}_{(\mathbf{dgC}^\infty \mathbf{Alg}_{/A})[W^{-1}]} (A', A \oplus E[n]).$$

However we have an inclusion of sets:

$$\operatorname{Hom}_{\mathbf{dgC}^\infty \mathbf{Alg}_{/A}} (A', A \oplus E[n]) \subseteq \operatorname{Hom}_{\mathbf{Ch}(\mathbb{R})/A} (A', A \oplus E[n])$$

where the right hand side consists of one element. This proves the statement.

Lemma 3.3.16. *$\operatorname{Der}(A, E[n])$ is connected.*

Proof. Suppose that $A = \mathcal{C}^\infty(U)$ for U open in $\mathbb{R}^k$ for some k . By Remark 2.2.26 A is a pushout of cofibrant discrete objects $B_i = \mathcal{C}^\infty(\mathbb{R}^{k_i})$ in $\mathbf{dC}^\infty \mathbf{Ring}$. By applying the functor $\operatorname{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}/A}(-, A \oplus E[n])$ we get a pullback:

$$\begin{array}{ccc} C = \operatorname{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}/A} (A, A \oplus E[n]) & \longrightarrow & C_2 = \operatorname{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}/A} (B_2, A \oplus E[n]) \\ \downarrow & \lrcorner & \downarrow \\ C_1 = \operatorname{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}/A} (B_1, A \oplus E[n]) & \longrightarrow & C_0 = \operatorname{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}/A} (B_0, A \oplus E[n]) \end{array}$$

from which we can deduce a long exact sequence (see [22]):

$$\cdots \longrightarrow \pi_1 C_1 \times \pi_1 C_2 \longrightarrow \pi_1 C_0 \longrightarrow \pi_0 C \longrightarrow \pi_0 C_1 \times \pi_0 C_2$$

Now notice that:

$$\begin{aligned} \pi_1 C_j &= \pi_1 \operatorname{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}/A} (B_j, A \oplus E[n]) \\ &= \pi_0 \operatorname{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}/A} (B_j, A \oplus E[n-1]) \\ &= \begin{cases} 0 & n \geq 2 \\ \operatorname{Hom}_{\mathbf{C}^\infty \mathbf{Ring}/A} (B_j, A \oplus E) & n = 1 \end{cases}. \end{aligned}$$

Where the case $n \geq 2$ is deduce from Remark 3.3.15, while the case $n = 1$ comes from the fact that the objects we are dealing with are all discrete $\mathcal{C}^\infty$ -rings. As a consequence, we see that for $n \geq 2$ the sequence has the following form:

$$\cdots \longrightarrow 0 \longrightarrow \pi_0 \operatorname{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}/A} (A, A \oplus E[n]) \longrightarrow 0.$$

In particular $\pi_0 \operatorname{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}/A} (A, A \oplus E[n]) = 0$. The case $n = 1$ is more subtle because in general $\operatorname{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}/A} (B_j, A \oplus E)$ is non zero. However to prove that $\pi_0 \operatorname{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}/A} (A, A \oplus E[1]) = 0$ we only need to prove that the map

$$\operatorname{Hom}_{\mathbf{C}^\infty \mathbf{Ring}/A} (\mathcal{C}^\infty(\mathbb{R}^{m+1}), A \oplus E) \oplus \operatorname{Hom}_{\mathbf{C}^\infty \mathbf{Ring}/A} (\mathbb{R}, A \oplus E) \longrightarrow \operatorname{Hom}_{\mathbf{C}^\infty \mathbf{Ring}/A} (\mathcal{C}^\infty(\mathbb{R}), A \oplus E)$$

is surjective. We defer the check to the end of the subsection, since it is straightforward but requires some explicit computations.

We now treat the general case $A = \mathcal{C}^\infty(M)$ for M smooth manifold. Recall that M can be realized as a retract of an open $U \subseteq \mathbb{R}^k$ for some k , in particular A is a retract of $B = \mathcal{C}^\infty(U)$. More precisely, there are morphism $g : A \rightarrow B$ and $f : B \rightarrow A$ such that $f \circ g \simeq \operatorname{id}_A$. As a consequence $\operatorname{Der}(A, E)$ is a retract of $\operatorname{Der}(B, f_* E[n])$ which is connected; in particular $\operatorname{Der}(A, E)$ is connected. $\square$

As previously noted thanks to the above two lemmas we can now prove the equivalence

$$\mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{cn}}(\Omega^1(M), E[n]) \simeq \mathrm{Der}(A, E[n]).$$

at the level of loop space. But now observe that:

$$\Omega \mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{cn}}(\Omega^1(M), E[n]) \simeq \mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{cn}}(\Omega^1(M), \Omega E[n]) \simeq \mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{cn}}(\Omega^1(M), E[n-1]).$$

Since the functor $\mathrm{Der}(A, -)$ preserves limits we can do something similar with the right hand side, i.e. :

$$\Omega \mathrm{Der}(A, E[n]) \simeq \mathrm{Der}(A, E[n-1]).$$

Proceeding inductively we reduce to the case:

$$\mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Mod}_A^{cn}}(\Omega^1(M), E[0]) \cong \mathrm{Der}(A, E[0])$$

but this is the statement from Corollary 3.1.20. This concludes the proof modulo the check in the proof of lemma 3.3.16.

Proof surjectivity. To do this we first need to recall how we realize U as a transverse pullback.

$$\begin{array}{ccc} U & \xrightarrow{\mathrm{const}} & * \\ \downarrow i & \lrcorner & \downarrow 0 \\ \mathbb{R}^{m+1} & \xrightarrow{f} & \mathbb{R} \end{array}$$

We follow the notation used in the proof of Lemma 2.2.5. From this, we deduce that it is sufficient to prove the surjectivity of the following map:

$$\mathrm{Hom}_{\mathbf{C}^\infty \mathbf{Ring}/A}(\mathbf{C}^\infty(\mathbb{R}^{m+1}), A \oplus E) \oplus \mathrm{Hom}_{\mathbf{C}^\infty \mathbf{Ring}/A}(\mathbb{R}, A \oplus E) \xrightarrow{f^* \oplus \mathrm{ev}_0^*} \mathrm{Hom}_{\mathbf{C}^\infty \mathbf{Ring}/A}(\mathbf{C}^\infty(\mathbb{R}), A \oplus E).$$

Now consider $\gamma \in \mathrm{Hom}_{\mathbf{dC}^\infty \mathbf{Ring}/A}(\mathbf{C}^\infty(\mathbb{R}), A \oplus E)$. Observe that the structural morphism $\mathbf{C}^\infty(\mathbb{R}) \rightarrow \mathbf{C}^\infty(U)$ is given by sending a smooth function g on $\mathbb{R}$ to the constant function $\mathrm{const}_{g(0)}$ on U ; Therefore, as $\mathbf{C}^\infty(\mathbb{R})$ is the free discrete $\mathbf{C}^\infty$ -ring generated by x (see Proposition 2.1.5), we deduce that γ is the unique morphism of $\mathbf{C}^\infty$ -rings sending x to $(0, h(x))$ for some $h(x) \in E$. To conclude, it suffices to produce a morphism $\gamma' \in \mathrm{Hom}_{\mathbf{C}^\infty \mathbf{Ring}/A}(\mathbf{C}^\infty(\mathbb{R}^{m+1}), A \oplus E)$ such that $f^* \circ \gamma' = \gamma$. First we observe that given $a_1, \dots, a_k \in \mathbf{C}^\infty(\mathbb{R}^{m+1})$ and $\varepsilon \in \mathbf{C}^\infty(\mathbb{R}^k)$ any $\gamma' \in \mathrm{Hom}_{\mathbf{C}^\infty \mathbf{Ring}/A}(\mathbf{C}^\infty(\mathbb{R}^{m+1}), A \oplus E)$ must take the following form:

$$\begin{array}{ccc} (a_1, \dots, a_k) & \longmapsto & ((i^* a_1, h' a_1), \dots, (i^* a_k, h' a_k)) \\ \downarrow & & \downarrow \\ \varepsilon(a_1, \dots, a_k) & \longmapsto & \left(i^* \varepsilon(a_1, \dots, a_k) = \varepsilon(i^* a_1, \dots, i^* a_k), h'(\varepsilon(a_1, \dots, a_k)) = \sum_{i=1}^k \frac{\partial \varepsilon(i^* a_1, \dots, i^* a_k)}{\partial x_i} h' a_i \right). \end{array}$$

Using the explicit description of $f = t \cdot \chi_C(x_1, \dots, x_m) - 1$ (see proof of lemma 2.2.5) we see that we need to define γ' such that:

$$\begin{aligned} h(x) &= h'(f(x_1, \dots, x_m, t)) \\ &= \chi_C(i^* x_1, \dots, i^* x_m) h'(t) + \sum_{i=1}^m \frac{\partial f}{\partial x_i}(i^* x_1, \dots, i^* x_m, i^* t) \cdot h' x_i. \end{aligned}$$

As $\mathbf{C}^\infty(\mathbb{R}^{m+1})$ is the free discrete $\mathbf{C}^\infty$ -ring generated by $x_1, \dots, x_m, t$, we can set:

$$\gamma'(x_i) = (i^* x_i, 0) \quad \gamma'(t) = \left(i^* t, \frac{1}{\chi_C(i^* x_1, \dots, i^* x_m)} \cdot h(x) \right),$$

and then extend γ' to a morphism in $\mathrm{Hom}_{\mathbf{C}^\infty \mathbf{Ring}/A}(\mathbf{C}^\infty(\mathbb{R}^{m+1}), A \oplus E)$ such that $f^* \gamma' = \gamma$. □

Remark 3.3.17. Nuiten proved the same statement in [50, Example 2.2.18]. However, we believe that the value of the above proof lies in the way it makes explicit the deep connection between derivations and differential forms.

3.3.3 Relative cotangent complex

In this subsection, we briefly recall the definition and main properties of the relative cotangent complex for $\mathcal{C}^\infty$ -rings.

Definition 3.3.18 (cf. [40, Definition 7.3.3.1]). Let $\mathcal{C}$ be a presentable ∞ -category and let $p : T_{\mathcal{C}} \rightarrow \mathcal{C}$ be a tangent bundle to $\mathcal{C}$. A *relative cofiber sequence* in $T_{\mathcal{C}}$ is a diagram σ :

$$\begin{array}{ccc} X & \longrightarrow & Y \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & Z \end{array}$$

in $T_{\mathcal{C}}$ with the following properties:

1. The map $p \circ \sigma$ factors through the projection $[1] \times [1] \rightarrow [1]$, so that the vertical arrows above become degenerate in $\mathcal{C}$.
2. The diagram σ is a pushout square.

Remark 3.3.19 (cf. [40, Definition 7.3.3.1]). Since condition i) implies that $p \circ \sigma$ is a pushout square, condition ii) is equivalent to the requirement that σ be a p -colimit diagram; see [40, Proposition 7.3.1.12].

The main reference for the forthcoming constructions is [40, below Definition 7.3.3.1]; however, we follow the approach of [57, Construction 5.1.0.1].

Let $\mathcal{E}$ be the full subcategory of

$$\mathrm{Fun}([1] \times [1], T_{\mathcal{C}}) \times_{\mathrm{Fun}([1] \times [1], \mathcal{C})} \mathrm{Fun}([1], \mathcal{C})$$

spanned by the relative cofiber sequences. We denote by ψ the forgetful functor $\mathcal{E} \rightarrow \mathrm{Fun}([1], T_{\mathcal{C}})$ given by restriction to the upper half of the diagram. In *loc.it.* it is observed that ψ is an equivalence. Let s be a section of ψ , and let $\mathrm{ev}_\infty : \mathcal{E} \rightarrow T_{\mathcal{C}}$ denote evaluation at $(1, 1) \in [1] \times [1]$.

Definition 3.3.20. The *relative cotangent complex functor* is defined as the composition

$$\mathrm{Fun}([1], \mathcal{C}) \xrightarrow{\mathbb{L}} \mathrm{Fun}([1], T_{\mathcal{C}}) \xrightarrow{s} \mathcal{E} \xrightarrow{\mathrm{ev}_\infty} T_{\mathcal{C}}.$$

We denote the image of a morphism $f : A \rightarrow B$ under the relative cotangent complex functor by

$$L_{B/A} \in T_{\mathcal{C}} \times_{\mathcal{C}} \{B\} \simeq \mathbf{Stab}(\mathcal{C}_{B/}).$$

Remark 3.3.21 (cf. [40, Remark 7.3.3.2]). The following exposition is taken from [57, Remark 5.1.0.3]. By definition, the relative cotangent complex for a morphism $A \rightarrow B$ of $\mathcal{C}^\infty$ -rings sits in the p -colimit diagram (see [37, Definition 4.3.1.1]):

$$\begin{array}{ccc} \mathbb{L}_A & \longrightarrow & \mathbb{L}_B \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbb{L}_{B/A}. \end{array}$$

Let $K^\triangleright := [1] \times [1]$, and let $q : K^\triangleright \times [1] \rightarrow K^\triangleright$ be the natural transformation that collapses $K^\triangleright$ to its cocone. Consider the composition

$$K^\triangleright \times [1] \xrightarrow{q} K^\triangleright \xrightarrow{\mathcal{J}} T_{\mathbf{dC}^\infty \mathbf{Ring}} \xrightarrow{p} \mathbf{dC}^\infty \mathbf{Ring}.$$

Then we have a commuting diagram

$$\begin{array}{ccc} K^\triangleright & \xrightarrow{\mathcal{J}} & T_{\mathbf{dC}^\infty \mathbf{Ring}} \\ \downarrow & \nearrow & \downarrow p \\ K^\triangleright \times [1] & \longrightarrow & \mathbf{dC}^\infty \mathbf{Ring} \end{array}$$

and there is a (unique up to contractible ambiguity) dotted cocartesian lift $K^\triangleright \times [1] \rightarrow T_{\mathbf{dC}^\infty \mathbf{Ring}}$ such that $K^\triangleright \times \{0\}$ is the diagram $\mathcal{J}$ (see [37, Corollary 4.3.1.11]). This lift exhibits a cocartesian transformation between $\mathcal{J}$ and a

diagram $\mathcal{J}' : [1] \times [1] \rightarrow T_{\mathbf{dC}^\infty \mathbf{Ring}} \times_{\mathbf{dC}^\infty \mathbf{Ring}} \{B\} \simeq \mathbf{Mod}_B$. By [37, Propositon 4.3.1.9], the diagram $\mathcal{J}'$ is a cofibre sequence

$$\begin{array}{ccc} f^! \mathbb{L}_A & \longrightarrow & \mathbb{L}_B \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbb{L}_{B/A} \end{array}$$

in $\mathbf{Mod}_B$.

Proposition 3.3.22 (cf. [40, Corollary 7.3.3.6]). *Consider the following commutative triangle of $\mathcal{C}^\infty$ -Rings:*

$$\begin{array}{ccc} & B & \\ A \nearrow & & \searrow f \\ & C & \end{array}$$

Then commutative square:

$$\begin{array}{ccc} f^* \mathbb{L}_{B/A} & \longrightarrow & \mathbb{L}_{C/A} \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbb{L}_{C/B} \end{array}$$

is a cofiber sequence.

Corollary 3.3.23. *Let $f : M \rightarrow N$ be a smooth submersion then $L_{M/N} \simeq \Omega^1(M/N)$ (see Definition A.2.13).*

Proof. This follows from Proposition 3.3.22 and Theorem 3.3.11. $\square$

Proposition 3.3.24 (cf. [40, Proposition 7.3.3.7]). *Consider the following pullback square commutative of $\mathcal{C}^\infty$ -rings*

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow f \\ A' & \longrightarrow & B' \end{array}$$

Then

$$f^* \mathbb{L}_{B/A} \simeq \mathbb{L}_{B'/A'}.$$

3.4 Quasi-coherent modules

This section is devoted to the construction and understanding of the category of quasi-coherent sheaves of modules. We start with subsection 3.4.1 where collecting facts from the theory presented in [29] we build an intuition that will aid our understanding in the derived case. In subsection 3.4.2 we discuss quasi-coherent modules in the affine case, then restricting to the case of manifold we relate vector bundles over a manifold M and quasi-coherent sheaves over the associated $\mathcal{C}^\infty$ -affine. Finally, in Subsection 3.4.3, by means of a descent argument, we extend the discussion to derived stacks. At the same time, we endow quasi-coherent sheaves with a monoidal structure and describe some of their local properties.

The main references for this section are [10, Section 9], [50, section 5.1.3 and 5.2.4], [57, Section 4.3] and [29].

3.4.1 Discrete setting

In this subsection we recall some of the theory of $\mathcal{C}^\infty$ -schemes emphasizing the properties of quasi-coherent modules in this setting.

As in usual algebraic geometry where we can build schemes out of affines the same happens in the $\mathcal{C}^\infty$ -case (see [29, Definition 4.23]). Similarly given a $\mathcal{C}^\infty$ -schemes $(X, \mathcal{O}_X)$ we can define (discrete) $\mathcal{O}_X$ -modules (see [29, Definition 5.10]) and their pullbacks (see [29, Definition 5.13]). We denote by $\mathbf{disShv}_{\mathcal{O}_X}(X)$ the associated category of $\mathcal{O}_X$ -modules.

Remark 3.4.1. For a $(X, \mathcal{O}_X)$ a $\mathcal{C}^\infty$ -scheme we may define the category of $\mathcal{O}_X$ -modules through the following pullback:

$$\begin{array}{ccc} \mathrm{disShv}_{\mathcal{O}_X}(X) & \longrightarrow & \mathrm{Shv}(X, \mathcal{C}^\infty \mathbf{Mod}) \\ \downarrow & \lrcorner & \downarrow \\ * & \xrightarrow{\mathcal{O}_X} & \mathrm{Shv}(X, \mathbf{dis}\mathcal{C}^\infty \mathbf{Ring}) \end{array}$$

Theorem 3.4.2 (cf. [29, Theorem 5.19]). *Let A be a discrete $\mathcal{C}^\infty$ -ring, and consider $(X, \mathcal{O}_X) = \mathbf{Spec}_{\mathcal{C}^\infty} A$. Then we have an adjunction*

$$\begin{array}{ccc} & \xrightarrow{\mathrm{MSpec}} & \\ \mathrm{Mod}_A & \perp & \mathrm{disShv}_{\mathcal{O}_X}(X) \\ & \xleftarrow{\Gamma} & \end{array}$$

where Γ is the global section functor, while MSpec is defined in [29, Definition 5.16].

Proposition 3.4.3 (cf. [29, Proposition 5.20]). *Let A be a discrete $\mathcal{C}^\infty$ -ring, and $(X, \mathcal{O}_X) = \mathbf{Spec}_{\mathcal{C}^\infty}(A)$, we have a natural isomorphism*

$$\Psi : \mathrm{MSpec} \circ \Gamma \Rightarrow \mathrm{id}_{\mathrm{disShv}_{\mathcal{O}_X}(X)}$$

of functors $\mathrm{disShv}_{\mathcal{O}_X}(X) \rightarrow \mathrm{disShv}_{\mathcal{O}_X}(X)$.

Remark 3.4.4. Let A be a discrete $\mathcal{C}^\infty$ -ring, and $(X, \mathcal{O}_X) = \mathbf{Spec}_{\mathcal{C}^\infty}(A)$. In contrast to the algebraic case where the opposite happens, $\mathrm{disShv}_{\mathcal{O}_X}(X)$ can be realized as a full subcategory of Mod_A .

Definition 3.4.5 (cf. [29, Definition 5.21]). Let $X = (X, \mathcal{O}_X)$ be a $\mathcal{C}^\infty$ -scheme and let $\mathcal{F}$ be a discrete $\mathcal{O}_X$ -module. We say that an discrete $\mathcal{O}_X$ -module $\mathcal{F}$ is *quasi-coherent* if there exist an open covering $\{U_i\}_{i \in I}$ of X such that $(U, \mathcal{O}_X|_U) \simeq \mathbf{Spec}_{\mathcal{C}^\infty} A$ and $\mathcal{F}|_{U_i} \simeq \mathrm{MSpec} N$ for some discrete $\mathcal{C}^\infty$ -ring A and $N \in \mathrm{Mod}_A$. We denote the category of discrete quasi-coherent sheaves on X by $\mathbf{disQC}(X)$.

An immediate consequence of Proposition 3.4.3 is the following.

Corollary 3.4.6 (cf. [29, Corollary 5.22.]). *Let $X = (X, \mathcal{O}_X)$ be a $\mathcal{C}^\infty$ -scheme. Then every discrete $\mathcal{O}_X$ -module E is quasi-coherent, so that $\mathbf{disQC}(X) = \mathrm{disShv}_{\mathcal{O}_X}(X)$.*

Because of the above corollary in [29, Section 8.1] the author does not distinguish between sheaves of $\mathcal{O}_X$ -modules and quasi-coherent sheaf over Deligne-Mumford $\mathcal{C}^\infty$ -stack (see [29, Definition 7.5]). Below [29, Theorem 8.6] it is noted that for $\mathcal{X}$ a Deligne-Mumford $\mathcal{C}^\infty$ -stack that is equivalent to a quotient $M//G$ by a finite group G we can identify quasi-coherent sheaves on $\mathcal{X}$ with G -equivariant sheaves on M .

The conclusion that we draw from the above discussion and that we might expect in the derived case are:

- There's no distinction between quasi-coherent modules and sheaves of modules.
- Given A a discrete $\mathcal{C}^\infty$ -ring, quasi-coherent sheaves over $\mathbf{Spec}_{\mathcal{C}^\infty}(A)$ are identified with a full subcategory of Mod_A .

3.4.2 Sheaves of modules

Definitions and main properties

A $\mathbf{d}\mathcal{C}^\infty \mathbf{Mod}$ -valued sheaf on a topological space X amount to the following data:

- a sheaf $\mathcal{O} : \mathrm{Op}(X)^{op} \rightarrow \mathcal{C}^\infty \mathbf{Ring}$ of $\mathcal{C}^\infty$ -rings.
- an $\mathcal{O}$ -modules sheaf F , i.e. the data of an $\mathcal{O}(U)$ -module $F(U)$ for every open U , such that the restriction maps $F(U) \rightarrow F(V)$ are $\mathcal{O}(U)$ -linear.

Notice that the category $\mathbf{d}\mathcal{C}^\infty \mathbf{Mod}$ comes with a functor a^* that send each $N \in \mathbf{d}\mathcal{C}^\infty \mathbf{Mod}_A$ to the underlying algebra $a^*(N) = A$. The functor $a^* : \mathbf{d}\mathcal{C}^\infty \mathbf{Mod} \rightarrow \mathbf{d}\mathcal{C}^\infty \mathbf{Ring}$ lift to a functor

$$\pi : \mathrm{Shv}(X, \mathbf{d}\mathcal{C}^\infty \mathbf{Mod}) \rightarrow \mathrm{Shv}(X, \mathbf{d}\mathcal{C}^\infty \mathbf{Ring}).$$

Definition 3.4.7 ($\mathcal{O}$ -modules sheaf). If $(X, \mathcal{O})$ is an homotopically $\mathcal{C}^\infty$ -ringed topological space, we define the ∞ -category $\mathrm{Shv}_{\mathcal{O}}(X)$ of $\mathcal{O}$ -modules sheaves to be the fiber of the functor π over $\mathcal{O}$.

Consistently with the discrete case, we do not distinguish between quasi-coherent modules and sheaves of modules.

Definition 3.4.8 (Quasi-coherent sheaves [50, Definition 5.1.34]). Let $(M, \mathcal{O})$ with M an affine. We define $\mathbf{QC}(M)$ to be the ∞ -category of $\mathcal{O}$ -modules sheaves and refer to them as *quasi-coherent sheaves* on M .

Construction F.3.27 gives rise to a global section functor

$$\Gamma : \mathbf{Top}_{\mathbf{d}\mathcal{C}^\infty\mathbf{Mod}} \rightarrow \mathbf{d}\mathcal{C}^\infty\mathbf{Mod}; (X, \mathcal{O}, F) \mapsto (\mathcal{O}_X(X), F(X))$$

from the ∞ -category of $\mathbf{d}\mathcal{C}^\infty\mathbf{Mod}$ -structured spaces (see Definition F.3.25) and the ∞ -category of modules over $\mathcal{C}^\infty$ -rings.

Lemma 3.4.9 (cf. [50, lemma 5.1.33]). *Let A be a complete $\mathcal{C}^\infty$ -ring, then $\Gamma : \mathrm{Shv}_{\mathcal{O}_A}(\mathbf{Spec}^\infty(A)) \rightarrow \mathbf{dMod}_A$ is a fully faithful right adjoint functor.*

Remark 3.4.10. For a general construction of the left adjoint in the case of $\mathbf{C}^\infty$ -ringed topoi see [10, Section 9.5].

Definition 3.4.11. Let A be a complete $\mathcal{C}^\infty$ -ring. We refer to the essential image of the global sections functor Γ as the *complete A -modules*, and to its left adjoint $\mathrm{Mod}_A \rightarrow \mathbf{QC}(\mathbf{Spec}^\infty(A))$ as the *associated sheaf* functor.

Following Nuiten [50], on affines we will not distinguish between quasi-coherent sheaves and complete A -modules over the corresponding complete $\mathcal{C}^\infty$ -ring. In particular, we will use the notation $\mathbf{QC}(A)$ to denote $\mathbf{QC}(\mathbf{Spec}^\infty(A))$; depending on the context, it will be clear whether we are referring to quasi-coherent sheaves or to complete modules.

Remark 3.4.12 (cf. [50, below Definition 5.1.34]). Let A be a complete $\mathcal{C}^\infty$ -ring and let $M = \mathbf{Spec}_{\mathbf{C}^\infty}(A)$. Let $f : M \rightarrow N$ be a morphism of affine derived manifolds, and let $f^* : \mathcal{O}(N) \rightarrow \mathcal{O}(M)$ be the associated morphism of $\mathcal{C}^\infty$ -rings. Then there is a left adjoint $f^* : \mathbf{QC}(N) \rightarrow \mathbf{QC}(M)$, which equivalently sends

- (a) an $\mathcal{O}_N$ -module sheaf E to the inverse image sheaf $f^*E = f^{-1}E \otimes_{f^{-1}\mathcal{O}_N} \mathcal{O}_M$.
- (b) a complete $\mathcal{O}(N)$ -module E to the associated sheaf of the $\mathcal{O}(M)$ -module $E \otimes_{\mathcal{O}(N)} \mathcal{O}(M)$.

For a more formal treatment, see [10, below Remark 9.42].

Remark 3.4.13 (Monoidal structure on $\mathbf{QC}$, cf. [50, Remark 5.1.35]). Let $(M, \mathcal{O})$ be an affine. We want to prove that $\mathbf{QC}(M)$ inherits a symmetric monoidal structure from the one in $\mathbf{dMod}_{\mathcal{O}(M)}$. A useful criterion is provided by [40, Proposition 2.2.1.9], which, together with [40, Example 2.2.1.7], reduces the problem to verifying the following: for every morphism $f : E \rightarrow E'$ of $\mathcal{O}(M)$ -modules such that the induced morphism on the associated sheaves is an equivalence, and for every other $\mathcal{O}(M)$ -module F , the induced map $E \otimes_{\mathcal{O}(M)} F \rightarrow E' \otimes_{\mathcal{O}(M)} F$ also induces an equivalence on the associated sheaves. This follows immediately by inspecting the corresponding map on stalks:

$$(E \otimes_{\mathcal{O}(M)} \mathcal{O}_x) \otimes_{\mathcal{O}_x} (F \otimes_{\mathcal{O}(M)} \mathcal{O}_x) \xrightarrow{f \otimes \mathrm{id}} (E' \otimes_{\mathcal{O}(M)} \mathcal{O}_x) \otimes_{\mathcal{O}_x} (F \otimes_{\mathcal{O}(M)} \mathcal{O}_x)$$

for every $x \in M$.

Remark 3.4.14 ($\mathbf{QC}$ is presentable, cf. [10, Remark 9.40]). Let A be a complete $\mathcal{C}^\infty$ -ring. Since $\mathbf{QC}(A)$ is reflective subcategory of the presentable ∞ -category $\mathbf{dMod}_A$, $\mathbf{QC}(A)$ is also presentable.

Remark 3.4.15 (cf. [10, Remark 9.41] and cf. [50, Example 5.1.36]). Let A be a complete $\mathcal{C}^\infty$ -ring. Since $\mathbf{QC}(A)$ is reflective a subcategory it is closed under limits and retracts. Moreover, by completeness of A we have that $A \in \mathbf{QC}(A)$. It follows that $\mathbf{QC}(A)$ contains all finitely presented A -modules.

Remark 3.4.16 (cf. [57, Remark 4.3.3.15]). Let A be a complete $\mathcal{C}^\infty$ -ring. The stable ∞ -category $\mathbf{QC}(A)$ (see [57, Proposition 4.3.3.8]) inherits a t -structure from $\mathrm{Shv}_{\mathcal{O}}(\mathbf{Spec}^\infty(A))$ via the spectrum-global sections equivalence. Both the inclusion functor $\mathbf{QC}(A) \hookrightarrow \mathbf{dMod}_A$ and the localization functor $\mathbf{dMod}_A \rightarrow \mathbf{QC}(A)$ are t -exact. It follows that the t -structure on $\mathbf{QC}(A)$ is simply given by $(\mathbf{QC}(A) \cap \mathbf{dMod}_A^{\leq 0}, \mathbf{QC}(A) \cap \mathbf{dMod}_A^{\geq 0})$.

Definition 3.4.17 (Perfect complexes [50, Definition 5.2.28]). Let $(M, \mathcal{O}_M)$ be an affine. we will say that an object $E \in \mathbf{QC}(M)$ is a *perfect complex* if it is locally presentable: every point of M admits an open neighborhood U such that the restriction $E|_U$ can be obtained from the structure sheaf $\mathcal{O}_{\mathcal{C}^\infty(M)}|_U$ by finite limits and colimits. We denote with $\mathbf{Perf}(M) \subseteq \mathbf{QC}(M)$ the full subcategory of perfect complexes on M .

Definition 3.4.18 (cf. [57, definition 4.3.4.2]). Let $\mathbf{Spec}_{\mathcal{C}^\infty} A$ be an affine M a complete A -module.

1. M is *n-connective* if $\pi_k(M) = 0$ for $k < n$.
2. M is *eventually connective* if there exists some $n \ll 0$ such that M is n -connective.
3. M is *truncated* if there exists some $n \gg 0$ such that M is n -truncated.
4. M is *dualizable* if the sheaf of modules $\mathcal{F}_M$ associated to M is a dualizable $\mathcal{O}_{\mathbf{Spec} A}$ -module.
5. M is *locally of finite rank* if the sheaf of modules $\mathcal{F}_M$ is locally equivalent to a finite sum $\mathcal{O}_{\mathcal{C}^\infty(M)}^{\oplus n}$.

Remark 3.4.19 (Embedding of vector bundles). We now want to relate vector bundles over a smooth manifold M with quasi-coherent sheaves over the associated affine $(M, \mathcal{C}_M^\infty)$. This can be achieved by considering the composition

$$\ell_M : \mathbf{Vect}(M) \simeq \Gamma \mathbf{Vect}(M) \hookrightarrow \mathbf{disQC}(M) \hookrightarrow \mathbf{QC}(M),$$

where the first equivalence is given by Corollary A.2.4 and the last functor arises from the fully faithful functor $H : \mathbf{Ab} \hookrightarrow \mathbf{Spt}$. Even further, vector bundles are locally of finite rank and hence perfect, so the embedding factors as

$$\ell_M : \mathbf{Vect}(M) \longrightarrow \mathbf{Perf}(M).$$

Remark 3.4.20. A first consequence of the above embedding is that $\ell_M(T^*M) \simeq \Omega^1(M) \simeq L_M$ defines a perfect complex.

3.4.3 Descent

Construction 3.4.21. In Subsection 3.2.3, we introduced the presentable fibration $\mathbf{dC}^\infty \mathbf{Mod} \rightarrow \mathbf{dC}^\infty \mathbf{Ring}$, whose cartesian straightening yields a functor:

$$\mathbf{dMod}_- : \mathbf{dC}^\infty \mathbf{Ring}^{\mathrm{op}} \rightarrow \mathbf{Pr}^{\mathbf{R}}.$$

Given a morphism $f : A \rightarrow B$ of complete $\mathcal{C}^\infty$ -rings, we obtain an induced adjunction $f^* : \mathbf{dMod}_A \rightleftarrows \mathbf{dMod}_B : f_*$ where f_* preserves quasi-coherent sheaves. In particular, this yields a commutative diagram:

$$\begin{array}{ccc} \mathbf{QC}(A) & \overset{f_*}{\dashrightarrow} & \mathbf{QC}(B) \\ \downarrow & & \downarrow \\ \mathbf{dMod}_A & \xrightarrow{f_*} & \mathbf{dMod} \end{array}.$$

which, thanks to Remark 3.4.12, lives not just in $\mathbf{Cat}_\infty$ but in $\mathbf{Pr}^{\mathbf{R}}$. By [53, Proposition A.1], this gives rise to a functor

$$\mathbf{QC}' : \mathbf{Aff} \rightarrow \mathbf{Pr}^{\mathbf{R}}.$$

Passing to opposites, we obtain the functor

$$\mathbf{QC}(-) := \mathbf{QC}'^{\mathrm{op}}(-) : \mathbf{Aff}^{\mathrm{op}} \rightarrow \mathbf{Pr}^{\mathbf{L}}$$

Lemma 3.4.22 (cf. [50, Lemma 5.2.25]). *The functor $\mathbf{QC} : \mathbf{Aff}^{\mathrm{op}} \rightarrow \mathbf{Pr}^{\mathbf{L}}$ satisfies descent and therefore determines a limit-preserving functor $\mathbf{QC} : \mathbf{dSt}^{\mathrm{op}} \rightarrow \mathbf{Pr}^{\mathbf{L}}$.*

Proof. Let M be an affine and consider the cartesian square consisting of presheaves of (large) ∞ -categories on M :

$$\begin{array}{ccc} \mathbf{Shv}_{\mathcal{O}}(-) & \longrightarrow & \mathbf{Shv}(-, \mathbf{dC}^\infty \mathbf{Mod}) \\ \downarrow & & \downarrow \\ * & \longrightarrow & \mathbf{Shv}(-, \mathbf{dC}^\infty \mathbf{Ring}) \end{array}.$$

The pullback $\mathrm{Shv}_{\mathcal{O}}(-)$ sends each open subspace $U \subseteq M$ to the ∞ -category of $\mathcal{O}$ -modules sheaves over U and each inclusion $V \subseteq U$ is simply given by the restriction functor $\mathrm{Shv}_{\mathcal{O}}(U) \rightarrow \mathrm{Shv}_{\mathcal{O}}(V)$. By Lemma F.3.28, the presheaf $\mathrm{Shv}_{\mathcal{O}}(-)$ is a sheaf. By Lemma 3.4.9, the restriction of $\mathrm{Shv}_{\mathcal{O}}(-)$ to the full subcategory of affine opens is equivalent to the functor sending an affine open $U \subseteq M$ to $\mathbf{QC}(U)$. In particular, since M is an affine, this shows that $\mathbf{QC} : \mathbf{Aff}^{\mathrm{op}} \rightarrow \widehat{\mathbf{Cat}}_{\infty}$ satisfies descent. $\square$

Remark 3.4.23 (cf. [57, Remark 4.3.3.19]). For $\mathcal{X}$ a derived stack, we call $\mathbf{QC}(\mathcal{X})$ the ∞ -category of quasi-coherent sheaves on $\mathcal{X}$. By definition of the right Kan extension, we have

$$\mathbf{QC}(\mathcal{X}) = \lim_{\mathbf{Spec}_{C^{\infty}} A \rightarrow \mathcal{X}} \mathbf{QC}(\mathbf{Spec}^{\infty} A).$$

An object in the limit is the data of a complete module M_A for each $\mathbf{Spec}_{C^{\infty}} A \in \mathbf{Aff}$ together with, for each commutative diagram

$$\begin{array}{ccc} \mathbf{Spec}_{C^{\infty}} A & \xrightarrow{f} & \mathbf{Spec}_{C^{\infty}} B \\ & \searrow & \downarrow \\ & & X \end{array}$$

of derived stacks, an equivalence $f^* M_B \simeq M_A$, and these equivalences are themselves compatible up to coherent higher homotopies.

Lemma 3.4.22 says that the functor

$$\mathbf{QC} : \mathbf{dSt}^{\mathrm{op}} \rightarrow \mathbf{Pr}^{\mathrm{L}}$$

takes colimits of derived stacks to limits of ∞ -categories, so for many sheaves, $\mathbf{QC}$ admits a simpler description than the formula above. In particular when dealing with a differentiable stack $\mathcal{X}$ we can consider a 1-Lie groupoid $\mathcal{G}_1 \rightrightarrows \mathcal{G}_0$ presenting $\mathcal{X}$ to obtain

$$\mathbf{QC}(\mathcal{X}) \simeq \lim_{\Delta} \mathbf{QC}(N(\mathcal{G})_{\bullet}).$$

In this case, we will refer to the pullback $p_0^* \mathcal{F}$ of a quasi-coherent sheaf $\mathcal{F}$ over $\mathcal{X}$, along the canonical map $p_0 : \mathcal{G}_0 \rightarrow \mathcal{X}$, as the *underlying sheaf* of $\mathcal{F}$. Note that this definition implicitly assumes a choice of atlas.

Remark 3.4.24 (Monoidal structure on $\mathbf{QC}$ for derived stacks [50, remark 5.2.27] and [8, Definition B.5.12]). Let $\mathcal{X}$ be a derived stack and let M and N be affines. We aim to endow $\mathbf{QC}(\mathcal{X})$ with a symmetric monoidal structure using descent. First, recall that by Remark 3.4.13, the (closed) symmetric monoidal structure on $\mathbf{dMod}_{\mathcal{O}(M)}$ descends to a unique closed symmetric monoidal structure on $\mathbf{QC}(M)$, with the property that the localization functor $\mathrm{Mod}_{\mathcal{O}(M)} \rightarrow \mathbf{QC}(M)$ is symmetric monoidal. Moreover, every morphism between affines $f : M \rightarrow N$ induces a symmetric monoidal functor $f^* : \mathbf{QC}(N) \rightarrow \mathbf{QC}(M)$. By [40, Theorem 4.5.3.1], we have a coCartesian fibration $\mathbf{dMod}^{\otimes} \rightarrow \mathbf{CAlg} \times \mathbf{Fin}_*$ whose straightening realizes the assignments

$$\mathbf{dMod}_{-}^{\otimes} : \mathbf{CAlg} \rightarrow \mathbf{CAlg}(\mathbf{Pr}^{\mathrm{L}}).$$

Precomposing with the forgetful functor $\mathbf{dC}^{\infty} \mathbf{Ring} \rightarrow \mathbf{CAlg}$ yields an induced functor on $\mathbf{dC}^{\infty} \mathbf{Ring}$. As before, we may apply [53, Proposition A.1] to obtain the functor

$$\mathbf{QC}^{\otimes}(-) : \mathbf{Aff}^{\mathrm{op}} \rightarrow \mathbf{CAlg}(\mathbf{Pr}^{\mathrm{L}})$$

which, by [40, Corollary 3.2.2.5] together with Lemma 3.4.22, satisfies descent.

Definition 3.4.25 (cf. [57, Definition 4.3.4.4]). Let P be a property for complete modules. We say that the property P is *stable under base change* if the following condition holds:

- (*) If (A, M) has the property P and $A \rightarrow B$ is a map of complete C^{∞} -rings, then $M \otimes_A B$ has the property P .

We say that P is *local for the open cover topology* if the following conditions hold

1. If (A, M) has the property P and $f : A \rightarrow B$ is an open map, then $(B, B \otimes_A M)$ has the property P .
2. If $\{\mathbf{Spec}_{C^{\infty}} A_i \rightarrow \mathbf{Spec}_{C^{\infty}} A\}$ is an open covering, M an A -module, and for each i , $A_i \otimes_A M$ has the property P , then M has the property P .

Notation. Let A be a complete $\mathcal{C}^\infty$ -ring and let P be a property of quasi-coherent modules. we will denote the full subcategory of $\mathbf{QC}(A)$ spanned by those objects that have the property P by $\mathbf{QC}_P(A)$.

Remark 3.4.26. Let P be a property of quasi-coherent modules that is stable under base change. Then, [53, Proposition A.1] provides a subfunctor $\mathbf{QC}_P(-)$ of

$$\mathbf{QC} : \mathbf{Aff}^{\text{op}} \rightarrow \mathbf{Pr}^{\text{L}}.$$

Proposition 3.4.27 (cf. [57, Proposition 4.3.4.5]). *Let P be a property for complete modules that is stable under base change and local for the open cover topology. Let $\mathbf{QC}_P \subseteq \mathbf{QC}$ be the full subfunctor spanned by modules that have the property P , then $\mathbf{QC}_P$ is a subsheaf.*

Proof. Let $\{f_i : U_i \rightarrow \mathbf{Spec}_{\mathcal{C}^\infty} A\}$ be an open covering and consider the associated groupoid $\mathcal{U}_\bullet$ given by the Čech nerve of the morphism

$$\coprod U_i \rightarrow \mathbf{Spec}_{\mathcal{C}^\infty}(A).$$

By descent for quasi-coherent sheaf we have:

$$\mathbf{QC}(A) = \lim_i \mathbf{QC}(\mathcal{U}_i).$$

Since the collection of fully faithful functors is stable under limits, we can identify the limit $\lim_i \mathbf{QC}_P(\mathcal{U}_i)$ with the full subcategory of $\mathbf{QC}(A)$ spanned by those module E for which $f_i^* E$ has the property P for each i . Since P is local, this is precisely the subcategory of modules that have the property P . $\square$

Proposition 3.4.28 (cf. [50, Lemma 5.2.32]). *All the properties of quasi-coherent modules of Definition 3.4.18 and 3.4.17 are local for the open cover topology and stable under base changes.*

Remark 3.4.29 ($\mathbf{QC}^{\geq 0}$ cf. [57, Remark 4.3.4.12]). Let $\mathcal{X}$ be a derived stack and A a complete $\mathcal{C}^\infty$ -ring. By Corollary 3.4.28, we have a full subcategory $\mathbf{QC}^{\geq 0}(\mathcal{X})$ of $\mathbf{QC}(\mathcal{X})$. Notice that the inclusion $\mathbf{QC}^{\geq 0}(A) \subset \mathbf{QC}(A)$ is a morphism in $\mathbf{Pr}^{\text{L}}$, so the same is true for $\mathbf{QC}^{\geq 0}(\mathcal{X})$, in particular it preserves small colimits. This full subcategory is closed under extensions and thus determines an accessible t -structure on $\mathbf{QC}(\mathcal{X})$ [40, Proposition 1.4.4.11].

Remark 3.4.30. Most of the descents results above are true more generally for the étale topology (see [57]).

Remark 3.4.31. Let $\mathcal{X}$ be a derived stack. The object $\mathcal{O}(\mathcal{X})$ from Construction 2.5.15 is a perfect complex.

Proposition 3.4.32 (Dualizable is perfect complex, cf. [57, Proposition 4.3.4.11]). *Let A be a complete $\mathcal{C}^\infty$ -ring and let M be a complete A -module, then M is dualizable if and only if M is a perfect complex.*

Remark 3.4.33. By [40, Proposition 4.6.1.11], if a symmetric monoidal ∞ -category $\mathcal{C}$ can be expressed as a limit $\mathcal{C} \simeq \lim_i \mathcal{C}_i$ of symmetric monoidal ∞ -categories, then an object $X \in \mathcal{C}$ is dualizable if and only if its image in each $\mathcal{C}_i$ is dualizable. By Corollary 3.4.28, $\mathbf{Perf}(-)$ satisfies descent, and by Proposition 3.4.32, for each complete ring A the ∞ -category $\mathbf{Perf}(A)$ consists precisely of the dualizable objects. It follows that, for any derived stack $\mathcal{X}$, the ∞ -category $\mathbf{Perf}(\mathcal{X})$ consists of dualizable objects.

3.5 Cotangent complex for derived stacks

We introduce the cotangent complex for a general derived stack and then specialize to the case of differentiable stacks, with the aim of relating it to the relative tangent bundle of Definition 1.4.12. More precisely, in Subsection 3.5.1 we introduce infinitesimally cohesive sheaves and derivations for derived stacks. In Subsection 3.5.2, we define the cotangent complex derived stacks and study its properties. Unlike the $\mathcal{C}^\infty$ -ring case, the cotangent complex for a map of sheaves does not always exist. However, Lemma 3.5.19 provides equivalent conditions which, together with Lemma 3.5.23, allow one to deduce the existence of a cotangent complex from local data. Geometric derived stacks satisfy the equivalent conditions of Lemma 3.5.19. We then specialize to the case of a representable submersion $\mathcal{X} \rightarrow \mathcal{Y}$ of differentiable stacks. Finally, in Subsection 3.5.3 we relate the relative tangent bundle to the cotangent complex. We conclude the thesis with an example in Subsection 3.5.4, where we compute the underlying sheaf of the absolute cotangent complex of a differentiable stack $\mathcal{X}$.

The exposition in this section follows [50, Chapter 6] and [30, Section 8.6]; however, an extensive discussion may be found in [41, Chapter 17].

3.5.1 Infinitesimally cohesive maps

Definition 3.5.1 ([40, Definition 17.3.1.4]). A (pre)sheaf $X : \mathbf{Aff}^{\mathrm{op}} \rightarrow \mathbf{Spc}$ is *infinitesimally cohesive* if it preserves each pullback square of complete C^∞ -rings of the form

$$\begin{array}{ccc} A'_\eta & \longrightarrow & A \\ \downarrow & & \downarrow \\ A' & \xrightarrow{\eta} & A \oplus I[1] \end{array} \quad (6.1.2)$$

where I is a complete connective A -module and $\pi_0(A') \rightarrow \pi_0(A)$ is a surjection of discrete C^∞ -rings with nilpotent kernel.

More generally, a map $p : X \rightarrow S$ of (pre)sheaves is *infinitesimally cohesive* if for each $S' = \mathrm{Spec}(B) \rightarrow S$, the base change $X \times_S S'$ is infinitesimally cohesive.

In [41, Remark 17.3.1.8], it is explained that if a functor $\mathcal{X} : \mathbf{Aff}^{\mathrm{op}} \rightarrow \mathbf{Spc}$ is infinitesimally cohesive and admits a cotangent complex $L_{\mathcal{X}}$, then $L_{\mathcal{X}}$ “controls” the deformation theory of the functor $\mathcal{X}$. This observation is adapted from the commentary found just below [41, Remark 17.3.1.8].

Example 3.5.2 (cf. [50, Example 6.1.3]). If X is representable by an affine, then X is infinitesimally cohesive.

Example 3.5.3 (cf. [50, Example 6.1.4]). The class of infinitesimally cohesive sheaves is closed under limits in $\mathrm{Shv}(\mathbf{Aff})$. Consequently, a sheaf X is infinitesimally cohesive if and only if the map $X \rightarrow *$ is infinitesimally cohesive. This remark is verbatim from the reference.

Remark 3.5.4. Let A be complete C^∞ -ring. The square zero-extension functor from Definition 3.2.33 restricts to a right adjoint functor

$$\mathbf{QC}^{\geq 0}(A) \rightarrow \mathbf{dC}^\infty \mathbf{Ring}_{A//A}$$

from the ∞ -category of connective quasi-coherent sheaves to the ∞ -category of C^∞ -rings that contains A as retract. Let $U := \mathbf{Spec}_{C^\infty}(A)$, then applying the functor $(-)^{\mathrm{op}}$ and composing with $\mathbf{Spec}_{C^\infty}(-)$ we obtain:

$$\mathbf{QC}^{\geq 0}(U)^{\mathrm{op}} \rightarrow \mathbf{Aff}_{U//U}.$$

Explicitly, for each $E \in \mathbf{QC}^{\geq 0}(U)^{\mathrm{op}}$ the functor assigns the retract diagram

$$U \rightarrow U_E := \mathbf{Spec}_{C^\infty}(A \oplus E) \rightarrow U$$

dual to $A \rightarrow A \oplus E \rightarrow A$.

Construction 3.5.5 ([50] Construction 6.1.8). Remark 3.5.4 together with Proposition 2.5.13 yields the following functor:

$$U_{(-)} : \mathbf{QC}^{\geq 0}(U)^{\mathrm{op}} \rightarrow \mathbf{dSt}_{U//U}.$$

For every map $f : U \rightarrow V$ between affines, there is a natural commuting square of functors:

$$\begin{array}{ccc} \mathbf{QC}^{\geq 0}(V)^{\mathrm{op}} & \xrightarrow{V_{(-)}} & \mathbf{dSt}_{V//V} \\ f^* \downarrow & & \downarrow f^* \\ \mathbf{QC}^{\geq 0}(U)^{\mathrm{op}} & \xrightarrow{U_{(-)}} & \mathbf{dSt}_{U//U} \end{array} \quad (3.2)$$

where the left arrows are pullbacks functor along f . If $\mathcal{X} : \mathbf{Aff}^{\mathrm{op}} \rightarrow \mathbf{Spc}$ is presheaf, then taking limit of these functors over all $U \in \mathbf{Aff}/\mathcal{X}$ yields a functor:

$$\mathbf{QC}^{\geq 0}(\mathcal{X})^{\mathrm{op}} \rightarrow \mathbf{dSt}_{\mathcal{X}//\mathcal{X}}; (\mathcal{X} \rightarrow \mathcal{X}_E \rightarrow \mathcal{X})$$

Definition 3.5.6. We will refer to $\mathcal{X}_E$ as the *square-zero extension* of $\mathcal{X}$ by E .

Remark 3.5.7 (cf. [50, Remark 6.1.10]). For any map $f : U \rightarrow V$ in $\mathbf{dSt}$, there is a square as is (3.2). For every $E \in \mathbf{QC}^{\geq 0}(U)$, there is a base change morphism

$$U_E \longrightarrow V_{f_*E},$$

which arises from a map of retract diagrams

$$\begin{array}{ccccc} U & \longrightarrow & U_E & \longrightarrow & U \\ \downarrow & & \downarrow & & \downarrow \\ V & \longrightarrow & V_{(f_*E)} & \longrightarrow & V. \end{array}$$

When f arises from a map $A \rightarrow B$ of C^∞ -rings, this vertical map arises from the obvious map $A \oplus E \rightarrow B \oplus E$. This remark is quoted almost verbatim from the reference.

Construction 3.5.8 (cf. [30, Definition 8.6.1]). Let $f : \mathcal{X} \rightarrow \mathcal{Y}$ be morphism in $\mathbf{dSt}$. Given a C^∞ -ring A , an A -point $x \in \mathcal{X}(A)$, and a connective A -module $E \in \mathbf{dMod}_A^{\text{cn}}$, consider the following diagram:

$$\begin{array}{ccc} \mathcal{X}(A \oplus E) & \longrightarrow & \mathcal{X}(A) \\ \downarrow & & \downarrow \\ \mathcal{Y}(A \oplus E) & \longrightarrow & \mathcal{Y}(A) \end{array}$$

induced by the projection $A \oplus E \rightarrow A$. This gives rise to a map of spaces:

$$\mathcal{X}(A \oplus E) \rightarrow \mathcal{Y}(A \oplus E) \times_{\mathcal{Y}(A)} \mathcal{X}(A).$$

Definition 3.5.9. In the context of Construction 3.5.8 we define the space $\text{Der}_x(\mathcal{X}/\mathcal{Y}, E)$ of *derivation of f at x with values in E* as the fiber at the point (y', x) , where $y' \in \mathcal{Y}(A \oplus E)$ is the image of $y = f(x) \in \mathcal{Y}(A)$ along the trivial derivation $A \rightarrow A \oplus E$ $a \mapsto (a, 0)$.

Remark 3.5.10. When $\mathcal{Y} \simeq *$ and $\mathcal{X}$ is affine, i.e., $\mathcal{X} \simeq \mathbf{Spec}_{C^\infty}(A)$ for a complete C^∞ -ring A , and $x = \text{id}$, Definition 3.5.9 recovers Definition 3.3.1 restricted to $\mathbf{QC}^{\geq 0}(A)$. In this case, the space we define is precisely the space of dotted arrows filling the following diagrams:

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{\text{id}} & \mathcal{X} \\ \downarrow & \nearrow \text{dotted} & \downarrow \\ \mathcal{X}_E & \longrightarrow & * \end{array} \iff \begin{array}{ccc} A & \xleftarrow{\text{id}} & A \\ \uparrow & \nwarrow \text{dotted} & \uparrow \\ A \oplus E & \xleftarrow{\quad} & \mathbb{R}. \end{array}$$

Remark 3.5.11. Definition 3.5.9 may be carried out even when instead of a C^∞ -ring A we have a derived stack $\mathcal{Z}$. To do this substitute:

- $\mathcal{X}(A)$ and $\mathcal{Y}(A)$ with $\text{Hom}_{\mathbf{dSt}}(\mathcal{Z}, \mathcal{X})$ and $\text{Hom}_{\mathbf{dSt}}(\mathcal{Z}, \mathcal{Y})$.
- $\mathcal{X}(A \oplus E)$, $\mathcal{Y}(A \oplus E)$ with $\text{Hom}_{\mathbf{dSt}}(\mathcal{Z}_E, \mathcal{X})$ and $\text{Hom}_{\mathbf{dSt}}(\mathcal{Z}_E, \mathcal{Y})$.
- The projection $A \oplus E \rightarrow A$ and the trivial derivation $A \rightarrow A \oplus E$ with the maps $\mathcal{Z} \rightarrow \mathcal{Z}_E$ and $\mathcal{Z}_E \rightarrow \mathcal{Z}$ respectively (see Construction 3.5.5).

Unwinding definition we see that $\text{Der}_x(\mathcal{X}/\mathcal{Y}, E)$ may be identified as the space of dotted section fitting in the following commutative diagram:

$$\begin{array}{ccccc} \mathcal{Z} & \xrightarrow{x} & \mathcal{X} & & \\ \downarrow & & \downarrow f & & \\ \mathcal{Z}_E & \xrightarrow{\quad} & \mathcal{Z} & \xrightarrow{f(x)} & \mathcal{Y}. \end{array}$$

(A dotted arrow from $\mathcal{Z}_E$ to $\mathcal{X}$ completes the diagram.)

Notice also that, given a morphism $E \rightarrow F$ in $\mathbf{QC}^{\geq 0}(\mathcal{Z})$, we can restrict the section along $\mathcal{Z}_F \rightarrow \mathcal{Z}_E$, thereby obtaining a functor:

$$\text{Der}_x(\mathcal{X}/\mathcal{Y}, -) : \mathbf{QC}^{\geq 0}(\mathcal{Z}) \rightarrow \mathbf{Spc}.$$

Remark 3.5.12 (cf. [50, Remark 6.1.12]). Consider a pullback square of derived stacks

$$\begin{array}{ccc} \mathcal{X}' & \xrightarrow{f} & \mathcal{X} \\ \downarrow & & \downarrow \\ \mathcal{S}' & \longrightarrow & \mathcal{S}. \end{array}$$

For any map $x' : \mathcal{Y} \rightarrow \mathcal{X}'$, there is an equivalence of functors

$$\mathrm{Der}_{x'}(\mathcal{X}'/\mathcal{S}', -) \xrightarrow{\sim} \mathrm{Der}_{f^*(x')}(\mathcal{X}/\mathcal{S}, -).$$

Remark 3.5.13 (cf. [50, Remark 6.1.13]). Let $f : U \rightarrow V$ be a map of derived stacks and let $x : V \rightarrow \mathcal{X}$ be a point. Restriction along the (natural) map of square-zero extensions from Remark 3.5.7 yields a natural transformation

$$\mathrm{Der}_x(\mathcal{X}/\mathcal{S}, f_*(-)) \longrightarrow \mathrm{Der}_{f^*(x)}(\mathcal{X}/\mathcal{S}, -).$$

Lemma 3.5.14 (cf. [50, Lemma 6.1.15]). *Suppose that $\mathcal{Y} = \mathrm{colim} Y_i$, is a colimit of sheaves and let $f_i : \mathcal{Y}_i \rightarrow \mathcal{Y}$ be the canonical maps. For every $p : \mathcal{X} \rightarrow \mathcal{S}$, $x : \mathcal{Y} \rightarrow \mathcal{X}$ and $E \in \mathbf{QC}^{\geq 0}(\mathcal{Y})$, there is a natural equivalence of spaces*

$$\mathrm{Der}_x(\mathcal{X}/\mathcal{S}; E) \longrightarrow \lim \mathrm{Der}_{f_i^*(x)}(\mathcal{X}/\mathcal{S}; f_i^* E)$$

Proof. We have the following

$$\mathrm{colim} ((\mathcal{Y}_i)_{f_i^* E}) \xrightarrow{\sim} \mathrm{colim}(\mathcal{Y}_i \times_{\mathcal{Y}} \mathcal{Y}_E) \xrightarrow{\sim} \mathcal{Y}_E$$

where the second equivalence follows from the universality of colimits (see Theorem F.1.7). The result then follows from the definition of $\mathrm{Der}_x(\mathcal{X}/\mathcal{S}; E)$ and the fact that limits commute with limits. $\square$

3.5.2 Cotangent complex for derived stacks

Definition 3.5.15 (cf. [61, Definition 1.4.1.14]). Let $\mathcal{X} \rightarrow \mathcal{S}$ be a morphism of derived stacks, let A be a complete $\mathcal{C}^\infty$ -ring, and let $x : \mathbf{Spec}_{\mathbf{C}^\infty}(A) \rightarrow \mathcal{X}$ be a morphism. We say that an eventually connective quasi-coherent module $L_{\mathcal{X}/\mathcal{S}, x} \in \mathbf{QC}^-(A)$ on $\mathbf{Spec}_{\mathbf{C}^\infty}(A)$ is a *relative cotangent complex at x for $\mathcal{X} \rightarrow \mathcal{S}$* if $L_{\mathcal{X}/\mathcal{S}, x}$ corepresents the functor $\mathrm{Der}_x(\mathcal{X}/\mathcal{S}, -)$. That is, there is an isomorphism

$$\mathrm{Hom}_{\mathbf{QC}^-(A)}(L_{\mathcal{X}/\mathcal{S}, x}, E) \simeq \mathrm{Der}_x(\mathcal{X}/\mathcal{S}; E)$$

functorial in $E \in \mathbf{QC}^{\geq 0}(A)$. If $\mathcal{S} \simeq *$, we denote the cotangent complex at x simply by $L_{\mathcal{X}, x}$.

As defined, the cotangent complex may not be unique. Indeed, we only consider maps to connective modules, whereas in general $L_{\mathcal{X}/\mathcal{Y}, x}$ is non-connective. To address this issue, we present the following lemma.

Lemma 3.5.16 (cf. [50, Lemma 6.1.16]). *Let $\mathcal{X} \rightarrow \mathcal{S}$ be infinitesimally cohesive. For every $x : \mathcal{Y} \rightarrow \mathcal{X}$ the functor $\mathrm{Der}_x(\mathcal{X}/\mathcal{S}, -)$ naturally extend to an exact functor*

$$\mathrm{Der}_x(\mathcal{X}/\mathcal{S}, -) : \mathbf{QC}^-(\mathcal{Y}) \longrightarrow \mathbf{Spt}.$$

Explicitly, the value of $\mathrm{Der}_x(\mathcal{X}/\mathcal{S}, -)$ at an n -connective module $E \in \mathbf{QC}^{\geq -n}(\mathcal{Y})$ is given by

$$\Omega^n \mathrm{Der}(\mathcal{X}/\mathcal{S}, E[n]).$$

By definition, if it exists, the cotangent complex corepresents the functor from Lemma 3.5.16. In fact,

$$\Omega^n \mathrm{Der}(\mathcal{X}/\mathcal{S}, E[n]) \simeq \Omega^n \mathrm{Hom}_{\mathbf{QC}^-(\mathcal{Y})}(L_{\mathcal{X}/\mathcal{S}, x}, E[n]) \simeq \mathrm{Hom}_{\mathbf{QC}^-(\mathcal{Y})}(L_{\mathcal{X}/\mathcal{S}, x}, E).$$

This proves the following:

Corollary 3.5.17 (Uniqueness of the cotangent complex). *If it exists, the cotangent complex is unique.*

Remark 3.5.18 (cf. [50, Remark 6.1.17]). Let $q : \mathcal{X}' \rightarrow \mathcal{X}$ and $p : \mathcal{X} \rightarrow \mathcal{S}$ be infinitesimally cohesive maps of sheaves over affines. For any point $x' : \mathcal{Y} \rightarrow \mathcal{X}'$, there is a fiber sequence of functor $\mathbf{QC}^-(\mathcal{Y}) \rightarrow \mathbf{Spt}$

$$\begin{array}{ccc} \mathrm{Der}_{x'}(\mathcal{X}'/\mathcal{X}; -) & \longrightarrow & \mathrm{Der}_{x'}(\mathcal{X}'/\mathcal{S}; -) \\ \downarrow & \lrcorner & \downarrow \\ * & \xrightarrow{0} & \mathrm{Der}_{q(x')}(\mathcal{X}/\mathcal{S}; -). \end{array}$$

Lemma 3.5.19 (cf. [50, Lemma 6.1.18]). *For any map $p : \mathcal{X} \rightarrow \mathcal{S}$ of sheaves, the following are equivalent:*

1. *p is infinitesimally cohesive and for every map $f : V \rightarrow U$ between affines, the map*

$$\mathrm{Der}_x(\mathcal{X}/\mathcal{S}; f_*(-)) \longrightarrow \mathrm{Der}_{f^*(x)}(\mathcal{X}/\mathcal{S}; -)$$

from Remark 3.5.13 is an equivalence.

2. *For any map $\mathcal{S}' \rightarrow \mathcal{S}$ from an affine, the pullback $\mathcal{X} \times_{\mathcal{S}} \mathcal{S}' : \mathrm{Aff}^{\mathrm{op}} \rightarrow \mathcal{S}$ preserves all pullback squares of complete C^∞ -rings of the form*

$$\begin{array}{ccc} \tilde{A}_\eta & \longrightarrow & A \\ \downarrow & & \downarrow 0 \\ \tilde{A} & \longrightarrow & A \oplus I[1]. \end{array}$$

Definition 3.5.20 (cf. [61, Definition 1.4.1.15]). Let $\mathcal{X} \rightarrow \mathcal{S}$ be a morphism of derived stacks, and let $L_{\mathcal{X}/\mathcal{S}} \in \mathbf{QC}^-(\mathcal{X})$ an eventually connective quasi-coherent module on $\mathcal{X}$. We say that $L_{\mathcal{X}/\mathcal{S}}$ is a *relative cotangent complex* for $\mathcal{X} \rightarrow \mathcal{S}$ if the following conditions are satisfied:

1. for every complete C^∞ -ring A and every $x : \mathbf{Spec}_{C^\infty}(A) \rightarrow X$, the inverse image $x^*L_{\mathcal{X}/\mathcal{S}}$ is a cotangent complex at x for $\mathcal{X} \rightarrow \mathcal{S}$;
2. the morphism $\mathcal{X} \rightarrow \mathcal{S}$ satisfies the equivalent conditions of Lemma 3.5.19.

If $\mathcal{Y} \simeq *$, we denote the cotangent complex simply by $L_{\mathcal{X}}$ and refer to it as the *absolute cotangent complex* of $\mathcal{X}$.

Remark 3.5.21. In the case where $\mathcal{X}$ is affine, i.e., $\mathcal{X} = \mathbf{Spec}_{C^\infty}(A)$ for some (complete) C^∞ -ring A , the object $L_{\mathcal{X}}$ does not, in general, coincide with the cotangent complex of Definition 3.3.4. What we obtain instead is the completion of that module; in particular, if it is already complete, the two coincide.

Remark 3.5.22 (cf. [50, Remark 6.1.21]). Let $f : U \rightarrow V$ be a morphism of affines and let $x : V \rightarrow \mathcal{X}$. If $p : \mathcal{X} \rightarrow \mathcal{S}$ satisfies the equivalent conditions of Lemma 3.5.19 and p admits a cotangent space at x , then p admits a cotangent space at $f^*(x)$, given by $f^*L_{\mathcal{X}/\mathcal{S},x}$.

Lemma 3.5.23 (cf. [50, Lemma 6.1.24]). *Let $p : \mathcal{X} \rightarrow \mathcal{S}$ be a map satisfying the equivalent conditions of Lemma 3.5.19. Then the following are equivalent:*

1. *For any map $x : \mathcal{Y} \rightarrow \mathcal{X}$ from a sheaf $\mathcal{Y}$, there exists a (unique) cotangent space $L_{\mathcal{X}/\mathcal{S},x} \in \mathbf{QC}^{\geq -n}(\mathcal{Y})$ at x .*
2. *p admits a cotangent complex (which is $(-n)$ -connective).*
3. *For any map $x : U \rightarrow \mathcal{X}$ from an affine U , there exists a cover $\{U_i \rightarrow U\}$ such that p admits a relative cotangent space $L_{\mathcal{X}/\mathcal{S},x_i} \in \mathbf{QC}^{\geq -n}(U_i)$ at each $x_i : U_i \rightarrow U \rightarrow \mathcal{X}$.*

Proof. Obviously (1) $\Rightarrow$ (2) $\Rightarrow$ (3). The converse follows from the following assertion. Fix a point $x : \mathcal{Y} \rightarrow \mathcal{X}$ and suppose that $\mathcal{Y} \simeq \mathrm{colim} \mathcal{Y}_i$ is a colimit of affines, with canonical maps $f_i : \mathcal{Y}_i \rightarrow \mathcal{Y}$. If p admits a cotangent space at each point $x_i := f_i^*x$, then p admits a cotangent space at x . In other words, we need to prove the corepresentability of the functor

$$\mathrm{Der}_x(\mathcal{X}/\mathcal{S}; -) : \mathbf{QC}(\mathcal{Y}) \rightarrow \mathbf{Spc}.$$

we may express $\mathbf{QC}(\mathcal{Y})$ as a limit in Cat_∞ of the ∞ -categories $\mathbf{QC}^{\geq -n}(\mathcal{Y}_i)$. For each i , we have a fully faithful Yoneda embedding

$$\mathbf{QC}^{\geq -n}(\mathcal{Y}_i)^{\mathrm{op}} \rightarrow \mathrm{Fun}(\mathbf{QC}^{\geq -n}(\mathcal{Y}_i), \mathbf{Spc}),$$

and by naturality of the Yoneda embedding, this provides a fully faithful functor

$$\mathbf{QC}^{\geq -n}(\mathcal{Y})^{\mathrm{op}} \simeq \lim_i \mathbf{QC}^{\geq -n}(\mathcal{Y}_i)^{\mathrm{op}} \rightarrow \lim_i \mathrm{Fun}(\mathbf{QC}^{\geq -n}(\mathcal{Y}_i), \mathbf{Spc}).$$

Our goal is now to find a compatible family of copresheaves on the $\mathbf{QC}^{\geq -n}(\mathcal{Y}_i)$ and to show that they are representable.

Since $\mathrm{Pr}^L \simeq (\mathrm{Pr}^R)^{\mathrm{op}}$ (see Corollary B.3.9), $\mathbf{QC}^{\geq -n}(\mathcal{Y})$ is equivalently described as a colimit of the $\mathbf{QC}^{\geq -n}(\mathcal{Y}_i)$ in Pr^R . Thus, we obtain comparison maps $(f_i)_* : \mathbf{QC}(\mathcal{Y}_i) \rightarrow \mathbf{QC}(\mathcal{Y})$ which induce the following morphism

$$\theta : \mathrm{Fun}(\mathbf{QC}^{\geq -n}(\mathcal{Y}), \mathbf{Spc}) \rightarrow \lim_i \mathrm{Fun}(\mathbf{QC}^{\geq -n}(\mathcal{Y}_i), \mathbf{Spc})$$

sending a copresheaf $F : \mathbf{QC}^{\geq -n}(\mathcal{Y}) \rightarrow \mathbf{Spc}$ to its restriction it along each of the maps

$$(f_i)_* : \mathbf{QC}^{\geq -n}(\mathcal{Y}_i) \rightarrow \mathbf{QC}^{\geq -n}(\mathcal{Y}).$$

Now take $F = \mathrm{Der}_x(\mathcal{X}/\mathcal{S}; -)$. By hypothesis, $p : \mathcal{X} \rightarrow \mathcal{S}$ satisfies the equivalent conditions in Lemma 3.5.19. In particular,

$$\mathrm{Der}_x(\mathcal{X}/\mathcal{S}; -) \circ (f_i)_* \simeq \mathrm{Der}_{x_i}(\mathcal{X}/\mathcal{S}; -).$$

It follows that the cotangent spaces $L_{\mathcal{X}/\mathcal{S}, x_i}$ form a matching family of quasi-coherent sheaves over the diagram of $\mathcal{Y}_i$. More precisely, since each $\mathrm{Der}_{x_i}(\mathcal{X}/\mathcal{S}; -)$ is corepresentable by $L_{\mathcal{X}/\mathcal{S}, x_i}$, we see that

$$\theta(F) \in \lim_i \mathbf{QC}^{\geq -n}(\mathcal{Y}_i)^{\mathrm{op}} \simeq \mathbf{QC}^{\geq -n}(\mathcal{Y})^{\mathrm{op}},$$

i.e. it is representable by an object which we denote by $L_{\mathcal{X}/\mathcal{S}, x}$. To see that this object corepresents $\mathrm{Der}_x(\mathcal{X}/\mathcal{S}; -)$, note that there are natural equivalences

$$\begin{aligned} \mathrm{Hom}_{\mathbf{QC}^{\geq -n}(\mathcal{Y})}(L_{\mathcal{X}/\mathcal{S}, x}, E) &\simeq \lim \mathrm{Hom}_{\mathbf{QC}^{\geq -n}(\mathcal{Y}_i)}(f_i^* L_{\mathcal{X}/\mathcal{S}, x}, f_i^* E) \\ &\simeq \lim \mathrm{Hom}_{\mathbf{QC}^{\geq -n}(\mathcal{Y}_i)}(L_{\mathcal{X}/\mathcal{S}, x_i}, f_i^* E) \\ &\simeq \lim \mathrm{Der}_{x_i}(\mathcal{X}/\mathcal{S}; f_i^* E) \simeq \mathrm{Der}_x(\mathcal{X}/\mathcal{S}; E) \end{aligned}$$

where the last equivalence is Lemma 3.5.14. □

Proposition 3.5.24 ([50, Proposition 6.2.6]). *Any geometric derived stack $\mathcal{X}$ satisfies the equivalent conditions of Lemma 3.5.19. In particular, it is infinitesimally cohesive*

We present an analogue of the properties discussed in Subsection 3.3.3. These are classical results; see, for example, [34, Propositions 3.2.10 and 3.2.12] or [50, Remark 6.1.26] for reference.

Proposition 3.5.25. *Let $\mathcal{X} \rightarrow \mathcal{Y}$ be an infinitesimally cohesive map of derived stacks admitting a cotangent complex $L_{\mathcal{X}/\mathcal{Y}}$. The statements following hold:*

1. *Let $g : \mathcal{Z} \rightarrow \mathcal{X}$ be an infinitesimally cohesive map of derived stack. Then there is an exact triangle*

$$f^* L_{\mathcal{Y}/\mathcal{Z}} \rightarrow L_{\mathcal{X}/\mathcal{Z}} \rightarrow L_{\mathcal{X}/\mathcal{Y}}$$

in $\mathbf{QC}(\mathcal{X})$, in the sense that if either the second or third term exists, then so does the other, and the triangle above exists.

2. *For every cartesian square*

$$\begin{array}{ccc} \mathcal{X}' & \longrightarrow & \mathcal{Y}' \\ p \downarrow & & \downarrow \\ \mathcal{X} & \longrightarrow & \mathcal{Y} \end{array}$$

of derived stacks, the cotangent complex $L_{\mathcal{X}'/\mathcal{Y}'}$ exists and there is a canonical isomorphism

$$p^* L_{\mathcal{X}/\mathcal{Y}} \simeq L_{\mathcal{X}'/\mathcal{Y}'}.$$

Proof. The statements follow from the corresponding remarks addressing the analogous situation at the level of derivations, namely Remark 3.5.18 for the first assertion and Remark 3.5.12 for the second. □

Let $h : \mathcal{X} \rightarrow \mathcal{Y}$ be a representable submersion of differentiable stacks. Denote by $N_0 \rightarrow \mathcal{Y}$ an atlas for $\mathcal{Y}$ and by M_0 the pullback atlas for $\mathcal{X}$ (see Lemma 1.1.13). Denote by $N_\bullet$ and $M_\bullet$ the associated groupoids, by $N_\bullet^+$ and $M_\bullet^+$ the associated functors $\Delta_+ \rightarrow \mathrm{Shv}(\mathbf{Mfd})$, and by $h_\bullet : M_\bullet^+ \rightarrow N_\bullet^+$ the associated natural transformation. Notice that $h_\bullet$ is cartesian.

Corollary 3.5.26. *In the above setting the cotangent complex $L_{\mathcal{X}/\mathcal{Y}}$ exists and it is perfect.*

Proof. First, note that $h : \mathcal{X} \rightarrow \mathcal{Y}$ admits a cotangent complex $L_{\mathcal{X}/\mathcal{Y}}$. This can be obtained as in the proof of Lemma 3.5.23, using a presentation $\mathcal{X} \simeq \operatorname{colim}_{\Delta} M_{\bullet}$ and [50, Lemma 6.2.7] to guarantee the existence of $L_{\mathcal{X}/\mathcal{Y},f}$ for each $f : M_i \rightarrow \mathcal{X}$ in the diagram indexed by Δ .

By definition, $L_{\mathcal{X}/\mathcal{Y}}$ is characterized by the property that for every morphism $x : U \rightarrow \mathcal{X}$ with U affine, the pullback $x^*L_{\mathcal{X}/\mathcal{Y}}$ is the cotangent complex $L_{U/\mathcal{Y}}$. In our situation, for each i we have pullback squares determined by $h_{\bullet}$ of the form

$$\begin{array}{ccc} M_i & \xrightarrow{f} & \mathcal{X} \\ \downarrow & \lrcorner & \downarrow \\ N_i & \xrightarrow{g} & \mathcal{Y}. \end{array}$$

Proposition 3.5.25 implies that $f^*L_{\mathcal{X}/\mathcal{Y}} \simeq L_{M_i/N_i}$ which, by Remark 3.4.20, is perfect for each i . Finally, by Corollary 3.4.28, we may use the description

$$\mathbf{Perf}(\mathcal{X}) \simeq \lim_{\Delta} \mathbf{Perf}(M_i)$$

to conclude the proof. $\square$

Remark 3.5.27. As a consequence of the above corollary and Remark 3.4.33, the relative cotangent complex of a representable submersion $h : \mathcal{X} \rightarrow \mathcal{Y}$ between differentiable stacks is dualizable.

3.5.3 Relation with relative tangent bundle

We have now developed enough theory to relate the relative tangent bundle of a representable submersion of differentiable stacks, as defined in Definition 1.4.12, with the relative cotangent complex. Suppose we are in the setting described above Corollary 3.5.26. Roughly speaking, since

$$L_{\mathcal{X}/\mathcal{Y}} \text{ “} \simeq \text{” } \lim_{\Delta} f^*L_{\mathcal{X}/\mathcal{Y}}$$

and we know that

$$f^*L_{\mathcal{X}/\mathcal{Y}} \simeq \Omega^1(M_i/N_i) \quad \text{and} \quad \Omega^1(M_i/N_i)^{\vee} \simeq \Gamma(M_i, T_{M_i/N_i}),$$

it is tempting to write the following (informal) chain of equivalences:

$$(L_{\mathcal{X}/\mathcal{Y}})^{\vee} \simeq \lim_{\Delta} (f^*L_{\mathcal{X}/\mathcal{Y}})^{\vee} \simeq \lim_{\Delta} \Gamma(M_i, T_{M_i/N_i}) \simeq \Gamma(Th),$$

where $\Gamma(Th)$ denotes the “global sections” of the relative tangent bundle.

Before making this precise, we first set up the categories we will use and describe the relations between them.

Construction 3.5.28. First, recall the embedding $\ell_- : \mathbf{Vect}(-) \rightarrow \mathbf{Perf}(-)$ from Remark 3.4.19. Now, consider the composites

$$\operatorname{Fun}([1], \mathbf{Mfd})^{\operatorname{op}} \xrightarrow{\operatorname{ev}_0^{\operatorname{op}}} \mathbf{Mfd}^{\operatorname{op}} \begin{array}{c} \xrightarrow{\mathbf{Perf}(-)} \\ \ell_- \uparrow \\ \xrightarrow{\mathbf{Vect}(-)} \end{array} \operatorname{Cat}_{\infty} \quad \text{and} \quad \operatorname{Fun}([1], \mathbf{Mfd})^{\operatorname{op}} \xrightarrow{\operatorname{ev}_0^{\operatorname{op}}} \mathbf{Mfd}^{\operatorname{op}} \begin{array}{c} \xrightarrow{\mathbf{Perf}(-)^{\operatorname{op}}} \\ \ell_- \uparrow \\ \xrightarrow{\mathbf{Vect}(-)^{\operatorname{op}}} \end{array} \operatorname{Cat}_{\infty}.$$

Unstraightening these four composites yields two commutative triangles over $\operatorname{Fun}([1], \mathbf{Mfd})$:

$$\begin{array}{ccc} \widetilde{\mathbf{Perf}} & & \\ \uparrow \tilde{\ell} & \searrow & \\ \widetilde{\mathbf{Vect}} & \longrightarrow & \operatorname{Fun}([1], \mathbf{Mfd}) \end{array} \quad \text{and} \quad \begin{array}{ccc} \mathbf{Perf} & & \\ \uparrow \tilde{\ell} & \searrow & \\ \mathbf{Vect} & \longrightarrow & \operatorname{Fun}([1], \mathbf{Mfd}). \end{array}$$

Notice that $\widetilde{\mathbf{Vect}}$ is the same category described in Construction 1.4.7. Informally, $\widetilde{\mathbf{Perf}}$ admits a similar description. The categories $\mathbf{Vect}$ and $\mathbf{Perf}$ have analogous descriptions, except that the morphism of vector bundles (respectively, of perfect complexes) goes in the opposite direction. The functor $\tilde{\ell}$ sends a pair $(M \rightarrow N, E)$ to $(M \rightarrow N, \Gamma(M, E))$.

As objects in $\mathbf{Fun}(\mathbf{Mfd}^{\mathrm{op}}, \mathbf{Cat}_{\infty})$, we may relate $\mathbf{Vect}(-)$ (respectively, $\mathbf{Perf}(-)$) to their opposites by taking duals

$$\begin{array}{ccc} \mathbf{Perf}(-) & \xrightleftharpoons{\quad} & \mathbf{Perf}(-)^{\mathrm{op}} \\ \uparrow \tilde{\ell} & & \uparrow \tilde{\ell} \\ \mathbf{Vect}(-) & \xrightleftharpoons{\quad} & \mathbf{Vect}(-)^{\mathrm{op}}. \end{array}$$

By unstraightening, this diagram of functors induces an analogous diagram over $\mathbf{Fun}([1], \mathbf{Mfd})$:

$$\begin{array}{ccc} \widetilde{\mathbf{Perf}} & \xrightleftharpoons{\quad} & \mathbf{Perf} \\ \uparrow \tilde{\ell} & & \uparrow \\ \widetilde{\mathbf{Vect}} & \xrightleftharpoons{\quad} & \mathbf{Vect} \end{array}$$

Notation. We denote by $\mathbf{Sub}$ the full subcategory of $\mathbf{Fun}([1], \mathbf{Mfd})$ spanned by the submersions.

Remark 3.5.29. The above construction applies equally well if we replace $\mathbf{Fun}([1], \mathbf{Mfd})$ with $\mathbf{Sub}$.

We are now in the following situation (for readability, we display only the case of $\widetilde{\mathbf{Vect}}$ and $\widetilde{\mathbf{Perf}}$):

$$\begin{array}{ccc} \widetilde{\mathbf{Perf}} & & \\ \uparrow \tilde{\ell} & \searrow & \\ \widetilde{\mathbf{Vect}} & \xrightarrow{\quad} & \mathbf{Sub} \\ & \searrow & \swarrow \mathrm{ev}_1 \\ & \mathbf{Mfd} & \end{array}$$

where the functors in the in the upper triangle are cartesian over $\mathbf{Mfd}$.

By descent we can carry out the above diagram also for stacks. Formally, the argument is the same as the one given above Construction 1.4.11. We will denote all the resulting categories by the same symbols as in Construction 3.5.28, with the addition of a superscript $+$.

In Construction 1.4.11 we have seen how to define a section σ_T to the cartesian fibration $\widetilde{\mathbf{Vect}} \rightarrow \mathbf{Sub}$. In a similar way, we may carry out the above construction for the relative cotangent bundle T^*f , obtaining a section σ_{T^*} for $\mathbf{Vect}$. By Remark A.2.9, the section σ_{T^*} is dual (in the sense of Construction 3.5.28) to σ_T .

We now turn to making precise the construction of the relative tangent bundle as a perfect sheaf, and to relating it to the sheaf of relative differential forms.

On manifold

We denote $\tilde{\ell} \circ \sigma_T$ by $\tilde{\sigma}_T$. The value of $\tilde{\sigma}_T$ at a submersion $f: M \rightarrow N$ is the pair $(f, \Gamma(M, Tf))$. Similarly, we define $\tilde{\sigma}_{T^*}$; once again, these are dual to each other in the sense of Construction 3.5.28.

On stacks

By descent (again for a formal argument see above Construction 1.4.11), we can lift the sections from the manifold case to the stack case (for readability, we display only the case of tangent bundles).

$$\begin{array}{ccc}
\widetilde{\mathbf{Perf}}^+ & & \\
\tilde{\ell}^+ \uparrow & \swarrow \tilde{\sigma}_T^+ & \\
\widetilde{\mathbf{Vect}}^+ & \xleftrightarrow{\sigma_T^+} & \mathbf{Sub}^+ \\
& \searrow & \swarrow \\
& \mathbf{Shv}(\mathbf{Mfd}) &
\end{array}$$

Now that everything is in place, we can formally prove the relation mentioned at the beginning of this discussion.

Solution. Let $h: \mathcal{X} \rightarrow \mathcal{Y}$ be a smooth surjective submersion of differentiable stacks. Choose an atlas $g_0: N_0 \twoheadrightarrow \mathcal{Y}$ and let $f_0: M_0 \rightarrow \mathcal{X}$ be its pullback. Denote by $N_\bullet$ and $M_\bullet$ the associated groupoids, by $N_\bullet^+$ and $M_\bullet^+$ the associated functors $\Delta_+ \rightarrow \mathbf{Shv}(\mathbf{Mfd})$, and by $h_\bullet: M_\bullet^+ \rightarrow N_\bullet^+$ the associated natural transformation. Notice that $h_\bullet$ is cartesian.

$$\begin{aligned}
L_{\mathcal{X}/\mathcal{Y}}^\vee &:= \underline{\mathbf{Hom}}_{\mathbf{Perf}(\mathcal{X})}(L_{\mathcal{X}/\mathcal{Y}}, \mathcal{O}_{\mathcal{X}}) \\
&\simeq \lim_{f \in \Delta} \underline{\mathbf{Hom}}_{\mathbf{Perf}(M_i)}(f^* L_{\mathcal{X}/\mathcal{Y}}, f^* \mathcal{O}_{\mathcal{X}}) && \mathbf{Perf} \text{ satisfy descent} \\
&\simeq \lim_{f \in \Delta} \underline{\mathbf{Hom}}_{\mathbf{Perf}(M_i)}(L_{M_i/N_i}, \mathcal{O}_{M_i}) && h_\bullet \text{ is cartesian and Proposition 3.5.25} \\
&\simeq \lim_{f \in \Delta} \underline{\mathbf{Hom}}_{\mathbf{Perf}(M_i)}(\Omega^1(M_i/N_i), \mathcal{O}_{M_i}) && \text{Corollary 3.3.23} \\
&\simeq \lim_{f \in \Delta} \tilde{\sigma}_T^+(f) && \text{Above construction} \\
&\simeq \tilde{\sigma}_T^+(h) && \text{descent.}
\end{aligned}$$

3.5.4 A computation

This concluding subsection presents an example of the computation of a nontrivial absolute cotangent complex. The example is inspired by [30, Example 8.6.5].

Proposition 3.5.30. *Let $\mathcal{X}$ be a differentiable stack and let $p: X_0 \rightarrow \mathcal{X}$ be a smooth atlas with associated Lie groupoid $X_1 \rightrightarrows X_0$. Then the underlying sheaf of $\mathbb{L}_{\mathcal{X}}$ is represented by the complex*

$$\cdots \rightarrow 0 \rightarrow \Omega^1(X_0) \rightarrow \Gamma(X_0, A)^\vee \rightarrow 0 \rightarrow \cdots$$

where A denotes the Lie algebroid associated to the Lie groupoid $X_1 \rightrightarrows X_0$ (see Appendix E.2).

Proof. Consider the following commutative diagram

$$\begin{array}{ccccc}
X_0 & \xrightarrow{e} & X_1 & \xrightarrow{t} & X_0 \\
\downarrow p & \lrcorner & & & \downarrow p \\
\mathcal{X} & \xlongequal{\quad} & \mathcal{X} & & \mathcal{X}
\end{array}$$

By Proposition 3.5.25 we have

$$L_{X_0/\mathcal{X}} \simeq \mathrm{id}^* L_{X_0/\mathcal{X}} \simeq e^* t^* L_{X_0/\mathcal{X}}.$$

Now consider the pullback:

$$\begin{array}{ccc}
X_1 & \xrightarrow{s} & X_0 \\
\downarrow t & \lrcorner & \downarrow p \\
X_0 & \xrightarrow{p} & \mathcal{X}
\end{array}$$

Again by Proposition 3.5.25 we obtain:

$$t^*L_{X_0/\mathcal{X}} \simeq L_{X_1/X_0}$$

By Corollary 3.3.23 we know that $L_{X_1/X_0} \simeq \Omega^1(X_1/X_0)$. Thanks to Remark E.2.3 we have

$$e^*\Omega^1(X_1/X_0) \simeq \Gamma(X_0, A)^\vee$$

where A denotes the Lie algebroid associated to the Lie groupoid $X_1 \rightrightarrows X_0$ (see Example E.2.2). By Proposition 3.5.25 there is a fiber sequence in $\mathbf{QC}(X_0)$:

$$p^*L_{\mathcal{X}} \rightarrow \Omega^1(X_0) \rightarrow \Gamma(X_0, A)^\vee,$$

hence $p^*L_{\mathcal{X}}$ is (quasi-isomorphic to) the mapping cocone of the morphism $\Omega^1(X_0) \rightarrow \Gamma(X_0, A)^\vee$. Equivalently, the underlying complex is.

$$\cdots \rightarrow 0 \rightarrow \Omega^1(X_0) \rightarrow \Gamma(X_0, A)^\vee \rightarrow 0 \rightarrow \cdots .$$

□

Appendix A

Recollection on smooth manifolds

In this appendix, we recall some useful facts about smooth manifolds and vector bundles that will be used throughout the thesis.

A.1 Smooth maps and embeddings

Definition A.1.1. Let $f: M \rightarrow N$ be a smooth map. We say that f is:

- a *smooth submersion* if the differential df is surjective on fibers over every point $x \in M$;
- an *immersion* if the differential df is injective on fibers over every point $x \in M$;
- an *embedding* if it is an injective immersion and f restricts to a homeomorphism $M \rightarrow f(M)$;
- *proper* if the preimage $f^{-1}(K) \subseteq M$ of every compact subset $K \subseteq N$ is compact.

We say that M is *properly embedded* if f is a proper embedding.

Theorem A.1.2 (Whitney-embedding theorem, cf. [32, Theorem 6.15]). *Every smooth n -dimensional manifold admit a proper smooth embedding into $\mathbb{R}^{2n+1}$.*

Remark A.1.3. We will not spell out the details, but under our conventions, where smooth manifolds are assumed to have bounded dimension, we can obtain a global embedding even in the case of smooth manifolds of mixed dimension. Roughly, the idea is to decompose the smooth manifold into submanifolds

$$M = \coprod_{i=1}^k M_i$$

where each M_i has pure dimension n_i . We then properly embed each $M_i \hookrightarrow \mathbb{R}^{m_i}$ and let $K = \max_i m_i$. Finally, we can disjointly embed all the M_i into $\mathbb{R}^{K+1}$.

Definition A.1.4 (Tubular neighborhood, [32, above Theorem 6.24]). Let M be a smoothly embedded submanifold of $\mathbb{R}^n$, and let NM denote its normal bundle (see [32, above Theorem 6.23]). A *tubular neighborhood* of M is an open neighborhood U of M in $\mathbb{R}^n$ that is the image, under the normal bundle projection E , of an open subset $V \subset NM$ of the form

$$V = \{(x, v) \in NM : |v| < \delta(x)\}$$

for some positive function $\delta: M \rightarrow \mathbb{R}$.

Proposition A.1.5 (cf. [32, Theorem 6.24]). *Every embedded submanifold of $\mathbb{R}^n$ has a tubular neighborhood.*

Proposition A.1.6 (cf. [32, Proposition 6.25]). *Let $M \subseteq \mathbb{R}^n$ be a smoothly embedded submanifold. $r: U \rightarrow M$ which is both a retraction and a smooth submersion.*

A.2 Vector bundles

Given a vector bundle $p : E \rightarrow M$ we can associate to it the sheaf of section $\Gamma(-, E)$ determined by the following data:

- to every open subset $U \subseteq M$, we associate the $\mathcal{C}^\infty(U)$ -module of smooth sections of the restricted bundle $p|_U : E|_U \rightarrow U$;
- for every inclusion of open subset $V \subseteq U$ we associate the restriction map $\Gamma(U, E) \rightarrow \Gamma(V, E)$.

Definition A.2.1. Given a smooth manifold M , we denote by $\Gamma\text{Vect}(M)$ the full subcategory of discrete quasi-coherent modules (see Subsection 3.4.1) spanned by objects of the form $\Gamma(-, E)$ for some vector bundle $p : E \rightarrow M$.

Proposition A.2.2 (cf. [48, Theorem 12.29]). *For any pair $p : E \rightarrow M$, $q : F \rightarrow M$ of vector bundles over a manifold M , the section functor $\Gamma(-, M)$ determines a one-to-one correspondence*

$$\text{Hom}_{\text{Vect}(M)}(E, F) \rightarrow \text{Hom}_{\text{Mod}_{\mathcal{C}^\infty(M)}}(\Gamma(E, M), \Gamma(F, M)).$$

Proposition A.2.3 (Serre-Swan [48, Theorem 12.32]). *Let M be a connected manifold. A $\mathcal{C}^\infty(M)$ -module P is isomorphic to the module of sections $\Gamma(M, E)$ of a smooth vector bundle $p : E \rightarrow M$ over M if and only if P is finitely generated and projective.*

Corollary A.2.4. *The functor $\Gamma(-, \bullet) : \text{Vect}(M) \rightarrow \Gamma\text{Vect}(M)$ is an equivalence of categories.*

Warning A.2.5. Given a morphism of smooth manifolds $f : M \rightarrow N$, we use the notation f^* for two distinct functors. One is the pullback functor

$$f^* : \text{Vect}(N) \rightarrow \text{Vect}(M),$$

which assigns to each vector bundle $p : F \rightarrow N$ the pullback bundle $f^*p : f^*F \rightarrow M$. The other is the base-change functor

$$f^* : \text{Mod}_{\mathcal{C}^\infty(N)} \rightarrow \text{Mod}_{\mathcal{C}^\infty(M)},$$

which assigns to each $\mathcal{C}^\infty(N)$ -module E the $\mathcal{C}^\infty(M)$ -module $\mathcal{C}^\infty(M) \otimes_{\mathcal{C}^\infty(N)} E$.

Proposition A.2.6 (cf. [48, Theorem 12.43]). *Let $f : M \rightarrow N$ be a morphism between smooth manifolds. Then the following diagram is commutative:*

$$\begin{array}{ccc} \text{Vect}(N) & \xrightarrow{f^*} & \text{Vect}(M) \\ \Gamma \downarrow & & \downarrow \Gamma \\ \text{Mod}_{\mathcal{C}^\infty(N)} & \xrightarrow{f^*} & \text{Mod}_{\mathcal{C}^\infty(M)} \end{array}$$

Proposition A.2.7 (cf. [29, Proposition 5.11.]). *Let M be a smooth manifold, then $\Gamma(U, -)$ is exact for all open subsets $U \subseteq M$.*

Definition A.2.8 (Relative tangent bundle). Let $f : M \rightarrow N$ be a smooth submersion between smooth manifolds. We define the *relative tangent bundle* Tf of f as the kernel of the morphism of vector bundles

$$df : TM \rightarrow f^*TN.$$

If the morphism is clear from the context, we write $T_{M/N}$.

The *relative cotangent bundle* of f is defined as the fiberwise dual of $T_{M/N}$. If the morphism is clear from the context, we write $T_{M/N}^*$. We set

$$\Omega^1(M/N) := \Gamma(M, T_{M/N}^*),$$

and refer to it as the $\mathcal{C}^\infty(M)$ -module of *relative 1-forms*.

Remark A.2.9. As a consequence of the definition and monoidality of Serre-Swan (see [48, Theorem 12.39]) $\Omega^1(M/N)^\vee = \Gamma(M, T_{M/N})$.

Remark A.2.10. The fiber of $T_{M/N}$ over $x \in M$ may be identified with the tangent bundle of the preimage $f^{-1}(x)$, which is a smooth submanifold of M (see [32, chapter 5]). In the same way we may regard $\Omega^1(M/N)$ as smooth of 1-forms along the fibers of f .

Remark A.2.11. The relative cotangent bundle fits into the following exact sequence of vector bundles:

$$0 \rightarrow f^*T^*N \xrightarrow{(df)^*} T^*M \rightarrow T^*_{M/N} \rightarrow 0.$$

Remark A.2.12. Consider the following commutative square of smooth manifold:

$$\begin{array}{ccc} P & \xrightarrow{k} & N \\ g \downarrow & & \downarrow f \\ W & \xrightarrow{h} & M \end{array}$$

by functoriality of the tangent bundle we get an induced commutative square

$$\begin{array}{ccc} TP & \xrightarrow{dk} & TN \\ dg \downarrow & & \downarrow df \\ TW & \xrightarrow{dh} & TM \end{array}.$$

As a consequence we get an induced a morphism of vector bundles:

$$dk : Tg \rightarrow Tf$$

Definition A.2.13. Let $f : M \rightarrow N$ be a smooth submersion. We define the *1-forms along the fibers of p* as the following cokernel:

$$\tilde{\Omega}^1(M/N) := \text{coker}(f^*\Omega^1(N) \rightarrow \Omega^1(M))$$

Remark A.2.14. Note that the map $(df)^* : f^*T^*N \hookrightarrow T^*M$ induces a map on sections

$$(df)^* : \Gamma(M, f^*(T^*N)) \hookrightarrow \Gamma(M, T^*M) = \Omega^1(M).$$

By Proposition A.2.6, the left-hand side is equivalent to $f^*\Gamma(M, T^*N) = f^*\Omega^1(N)$, so that the map in question takes the form

$$(df)^* : f^*\Omega^1(N) \hookrightarrow \Omega^1(M).$$

By Proposition A.2.7, taking global sections is exact, so the exact sequence from Remark A.2.11 induces an exact sequence

$$0 \rightarrow f^*\Omega^1(N) \hookrightarrow \Omega^1(M) \rightarrow \Omega^1(M/N) \rightarrow 0.$$

In particular, $\tilde{\Omega}^1(M/N)$ and $\Omega^1(M/N)$ coincide.

Appendix B

Topics in ∞ -category theory

This appendix collects various topics in ∞ -category theory that were essential for the thesis but too brief to justify their own appendix.

B.1 Computing the Hom space

Computing the Hom space in the localization $\mathcal{C}[W^{-1}]$ of an ∞ -category at a collection of morphisms W is, in general, a challenging task. Various techniques have been developed to obtain explicit descriptions, and in certain cases, one can derive clear and elegant formulas to evaluate it. For instance, in Remark C.2.21, we present one such formula in the context of the derived ∞ -category. Following [14], Section 7, we briefly recall some of these methods for the reader's reference.

Definition B.1.1 (cf. [14, Definition 7.4.12]). An ∞ -category with weak equivalences and fibrations is a triple $(\mathcal{C}, W, \text{Fib})$, where $\mathcal{C}$ is an ∞ -category with a final object, $W \subset \mathcal{C}$ is a subcategory which has the 2-out-of-3 property, and $\text{Fib} \subset \mathcal{C}$ a class of fibrations, such that the following properties are verified:

1. For any Cartesian square of $\mathcal{C}$

$$\begin{array}{ccc} x' & \xrightarrow{u} & x \\ p' \downarrow & & \downarrow p \\ y' & \xrightarrow{v} & y \end{array}$$

in which p is a fibration between fibrant objects, and y' is fibrant, if p belongs to W , so does p' .

2. For any map with fibrant codomain $f : x \rightarrow y$ in $\mathcal{C}$, there exists a map $w : x \rightarrow x'$ in W and a fibration $p : x' \rightarrow y$ such that f is a composition of p and w .

Definition B.1.2 (cf. [14, Definition 7.10.2]). Let $\mathcal{C}$ be an ∞ -category with weak equivalences and fibrations. A class of cofibrant objects is a set Q of object of $\mathcal{C}$ such that:

- For any object x in $\mathcal{C}$, there exists a weak equivalence $x' \rightarrow x$ with x' in Q .
- For any trivial fibration between fibrant objects $p : x \rightarrow y$ and any map $v : a \rightarrow y$ in $\mathcal{C}$ with $a \in Q$, there exists a map $u : a \rightarrow x$ such that v is a composition of u and p .

Example B.1.3. Any Quillen model category give rise to ∞ -category with weak equivalences, fibration and a class of cofibrant object essentially by definition.

Remark B.1.4 (cf. [14, Remark 7.10.3]). For any fibrant object x , the slice category $\mathcal{C}/x$ has a natural structure of ∞ -category with weak equivalences and fibrations: the weak equivalences (fibrations) of $\mathcal{C}/x$ are the maps whose image in $\mathcal{C}$ is a weak equivalence (fibration). If Q is a class of cofibrant objects, then the objects of the form (a, u) in $\mathcal{C}/x$, where $u : a \rightarrow x$ is a map with domain in Q , form a class of cofibrant objects of $\mathcal{C}/x$.

Proposition B.1.5 (cf. [14, Corollary 7.6.13]). Let $\mathcal{C}$ be an ∞ -category with weak equivalences. We denote by $\gamma : \mathcal{C} \rightarrow L(\mathcal{C})$ the localization functor. For any fibrant object x of $\mathcal{C}$, the canonical functor $\mathcal{C}/x \rightarrow L(\mathcal{C})/\gamma(x)$ induces an equivalence of ∞ -categories

$$L(\mathcal{C}/x) \simeq L(\mathcal{C})/\gamma(x).$$

Proposition B.1.6 (cf. [14, Proposition 7.10.4]). *Let $\mathcal{C}$ be an ∞ -category with weak equivalences equipped with a class of cofibrant objects. For any cofibrant objects a and any fibrant object y , the set $\mathrm{Hom}_{\mathrm{ho}(L(\mathcal{C}))}(a, y)$ is a quotient of $\mathrm{Hom}_{\mathrm{ho}(\mathcal{C})}(a, y)$.*

B.2 Adjunctions and colimits in slice categories

For the reader's convenience, we recall some theorems regarding slice categories from [37] and [40].

Proposition B.2.1 (cf. [37, Proposition 5.2.5.1]). *Suppose we are given an adjunction of ∞ -categories*

$$\mathcal{C} \xrightleftharpoons[G]{F} \mathcal{D}$$

Assume that the ∞ -category $\mathcal{C}$ admits pullbacks and let C be an object of $\mathcal{C}$. Then

1. *The induced functor $f : \mathcal{C}_{/C} \rightarrow \mathcal{D}_{/FC}$ admits a right adjoint g .*
2. *The functor g is equivalent to the composition*

$$\mathcal{D}_{/FC} \xrightarrow{g'} \mathcal{C}_{/GFC} \xrightarrow{g''} \mathcal{C}_{/C},$$

where g' is induced by G and g'' is induced by pullback along the unit map $C \rightarrow GFC$.

Proposition B.2.2 (cf. [37, Proposition 1.2.13.8]). *Let $\mathcal{C}$ be an ∞ -category and let $q : T \rightarrow \mathcal{C}$ and $p : K \rightarrow \mathcal{C}_{/q}$ be two diagrams. Let p_0 denote the composition of p with the projection $\mathcal{C}_{/q} \rightarrow \mathcal{C}$. Suppose that p_0 has a colimit in $\mathcal{C}$. Then:*

1. *The diagram p has a colimit in $\mathcal{C}_{/q}$, and that colimit is preserved by the projection $\mathcal{C}_{/q} \rightarrow \mathcal{C}$.*
2. *An extension $\tilde{p} : K^\triangleright \rightarrow \mathcal{C}_{/q}$ is a colimit of p if and only if the composition*

$$K^\triangleright \rightarrow \mathcal{C}_{/q} \rightarrow \mathcal{C}$$

is a colimit of p_0 .

Proposition B.2.3 (cf. [57, Lemma 4.1.1.29]). *Let $\mathcal{D}$ be a weakly contractible ∞ -category and let $\mathcal{C}$ be an ∞ -category that admits $\mathcal{D}$ -indexed colimits. For each $A \in \mathcal{C}$, the ∞ -category $\mathcal{C}_{A/}$ admits $\mathcal{D}$ colimits and the projection $p : \mathcal{C}_{A/} \rightarrow \mathcal{C}$ preserves and reflects $\mathcal{D}$ -indexed colimits.*

Proposition B.2.4 (cf. [37, Proposition 5.5.8.7]). *Let K be a sifted ∞ -category. Then K is weakly contractible.*

Remark B.2.5. As a consequence Proposition B.2.3 holds for sifted category.

Lemma B.2.6 (cf. [40, Lemma 7.3.4.12]). *Suppose given an adjunction of ∞ -categories*

$$\mathcal{C} \xrightleftharpoons[G]{F} \mathcal{D}$$

where $\mathcal{C}$ is stable. Let C be an object of $\mathcal{C}$, and consider the induced adjunction

$$\mathcal{C}_{/C} \xrightleftharpoons[g]{f} \mathcal{D}_{/FC}$$

(see Proposition B.2.1). Then:

1. *If the functor G is conservative, then g is conservative.*
2. *Let K be a weakly contractible simplicial set. Assume that $\mathcal{C}$ and $\mathcal{D}$ admit K -indexed colimits, that the functor G preserves K -indexed colimits, and that $\mathcal{C}$ is stable. Then the ∞ -categories $\mathcal{D}_{/FC}$ and $\mathcal{C}_{/C}$ admit K -indexed colimits, and the functor g preserves K -indexed colimits.*

B.3 Presentable ∞ -categories

This section recalls some basic properties of presentable ∞ -categories, along with the important definition of their tensor product.

The exposition follows [27], [31] section 2.1 and 28, but the results are mainly from [40].

B.3.1 Definition and main properties

Definition B.3.1 (Accessible ∞ -category, cf. [37, Definition 5.4.2.1]). Let κ be a regular cardinal. An ∞ -category $\mathcal{C}$ is κ -*accessible* if there exists ∞ -category $\mathcal{C}^0$ and an equivalence:

$$\mathrm{Ind}_\kappa(\mathcal{C}^0) \rightarrow \mathcal{C}$$

We will say that $\mathcal{C}$ is *accessible* if it is κ -accessible for some regular cardinal κ .

Definition B.3.2 (κ -continuous functor, cf. [37, Definition 5.3.4.5]). Let $\mathcal{C}$ be an ∞ -category which admits small κ -filtered colimits. We will say that a functor $f : \mathcal{C} \rightarrow \mathcal{D}$ is κ -*continuous* if it preserves κ -filtered colimits.

Definition B.3.3 (accessible functor, cf. [37, Definition 5.4.2.5]). If $\mathcal{C}$ is an accessible ∞ -category, then a functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is *accessible* if it is κ -continuous for some regular cardinal κ .

Definition B.3.4 (Presentable ∞ -category cf. [37, Definition 5.5.0.1]). An ∞ -category $\mathcal{C}$ is *presentable* if $\mathcal{C}$ is accessible and admits small colimits.

Proposition B.3.5 (cf. [37, Corollary 5.5.2.4]). *A presentable ∞ -category $\mathcal{C}$ admits all (small) limits.*

Theorem B.3.6 (cf. [37, Theorem 5.5.1.1]). *The ∞ -category $\mathcal{C}$ is presentable if and only if there exists a small ∞ -category $\mathcal{D}$ such that $\mathcal{C}$ is an accessible localization of $\mathcal{P}(\mathcal{D})$.*

Theorem B.3.7 (Adjoint functor theorem [37] Corollary 5.5.2.9). *Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor between ∞ -categories, then:*

1. *F admits a left adjoint if and only if F preserves all small colimits.*
2. *F admits a right adjoint if and only if F preserves all small limits.*

Presentable categories and left adjoint (resp. right adjoint) assemble in into ∞ -categories:

Definition B.3.8 (cf. [37, Definition 5.5.3.1]). Let $\widehat{\mathrm{Cat}}_\infty$ denote the ∞ -category of large ∞ -categories. We define subcategories Pr^L and Pr^R as follows:

- The objects of both Pr^L and Pr^R are the presentable ∞ -categories.
- A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ between presentable ∞ -categories is a morphism in Pr^L if and only if F preserves small colimits.
- A functor $G : \mathcal{C} \rightarrow \mathcal{D}$ between presentable ∞ -categories is a morphism in Pr^R if and only if G is accessible and preserves small limits.

As suggested this two category are anti-equivalent. More precisely:

Corollary B.3.9 (cf. [37, Corollary 5.5.3.4]). *There is a canonical isomorphism $\mathrm{Pr}^L \simeq (\mathrm{Pr}^R)^{op}$ in the homotopy category of ∞ -categories.*

Proposition B.3.10 (cf. [40, Proposition 1.4.4.4., Proposition 1.4.4.5]). *Let $\mathcal{C}$ and $\mathcal{D}$ be a presentable ∞ -categories, with $\mathcal{D}$ stable. Then:*

1. *$\mathbf{Stab}(\mathcal{C})$ is a presentable ∞ -category.*
2. *Composition with $\Sigma_\infty^\infty : \mathcal{C} \rightarrow \mathbf{Stab}(\mathcal{C})$ induces an equivalence:*

$$\mathrm{Hom}_{\mathrm{Pr}^L}(\mathbf{Stab}(\mathcal{C}), \mathcal{D}) \rightarrow \mathrm{Hom}_{\mathrm{Pr}^L}(\mathcal{C}, \mathcal{D})$$

Remark B.3.11. Let $\mathrm{Pr}_{\mathrm{st}}^L$ ($\mathrm{Pr}_{\mathrm{st}}^R$) denote the full subcategory of Pr^L (Pr^R) of stable presentable ∞ -categories. The above proposition gives adjunctions:

$$\begin{array}{ccc} \mathrm{Pr}^L & \begin{array}{c} \xrightarrow{\mathbf{Stab}(-)} \\ \perp \\ \xleftarrow{\quad} \end{array} & \mathrm{Pr}_{\mathrm{st}}^L \\ \mathrm{Pr}_{\mathrm{st}}^R & \begin{array}{c} \xleftarrow{\quad} \\ \perp \\ \xrightarrow{\mathbf{Stab}(-)} \end{array} & \mathrm{Pr}^R \end{array}$$

B.3.2 Tensor product

Definition B.3.12 (Bilinear functors). For presentable $\mathcal{C}, \mathcal{D}, \mathcal{E}$ write $\text{Fun}^{\text{biL}}(\mathcal{C} \times \mathcal{D}, \mathcal{E})$ for the full subcategory of the functor category consisting of all functors which are colimit-preserving in both variables separately.

Notation. Given two presentable ∞ -categories $\mathcal{C}$ and $\mathcal{D}$, we denote by $\text{Fun}^L(\mathcal{C}, \mathcal{D})$ the full subcategory of $\text{Fun}(\mathcal{C}, \mathcal{D})$ spanned by the left adjoint functors.

Definition B.3.13 (Tensor product). The *tensor product* of two presentable ∞ -category $\mathcal{C}, \mathcal{D}$ is a presentable ∞ -category $\mathcal{C} \otimes \mathcal{D}$ together with a functor:

$$\mathcal{C} \times \mathcal{D} \rightarrow \mathcal{C} \otimes \mathcal{D}$$

preserving colimits in each variable separately, such that precomposition induces an equivalence

$$\text{Fun}^L(\mathcal{C} \otimes \mathcal{D}, \mathcal{E}) \rightarrow \text{Fun}^{\text{biL}}(\mathcal{C} \times \mathcal{D}, \mathcal{E})$$

for each presentable ∞ -category $\mathcal{E}$.

Lemma B.3.14 (cf. [40, Proposition 4.8.1.17]). *Let $\mathcal{C}$ and $\mathcal{D}$ be presentable ∞ -categories. Then the tensor product exist and can be described as follows:*

$$\mathcal{C} \otimes \mathcal{D} \simeq \text{Fun}^{\text{lim}}(\mathcal{C}^{\text{op}}, \mathcal{D})$$

Lemma B.3.15. *Let $\mathcal{X}$ be an ∞ -topos (see definition F.1.5) and let $\mathcal{C}$ be a presentable ∞ -category then*

$$\mathcal{X} \otimes \mathcal{C} \simeq \text{Shv}_{\mathcal{C}}(\mathcal{X})$$

Proof. Thanks to Lemma B.3.14 we have $\mathcal{X} \otimes \mathcal{D} \simeq \text{Fun}^{\text{lim}}(\mathcal{X}^{\text{op}}, \mathcal{D}) = \text{Shv}_{\mathcal{C}}(\mathcal{X})$. □

Remark B.3.16. Let X be at topological space and let $\mathcal{C}$ be a presentable ∞ -category. Lemma B.3.15 together with Theorem F.3.11 implies that:

$$\text{Shv}(X) \otimes \mathcal{C} \simeq \text{Shv}(X, \mathcal{C}).$$

B.4 Recollection on calculus of functors

For the reader's convenience, this appendix recalls the definitions and theorems from the calculus of functors that are used in the core of the thesis. The primary reference for this section is [40, Chapter 6].

Definition B.4.1 (cf. [40, Defintion 1.4.2.1]). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor between ∞ -categories.

1. If $\mathcal{C}$ admits pushouts, then we will say that F is *excisive* if F carries pushout squares in $\mathcal{C}$ to pullback squares in $\mathcal{D}$.
2. If $\mathcal{C}$ admits a final object $*$, we will say that F is *reduced* if $F(*)$ is a final object of $\mathcal{D}$.

Remark B.4.2. The notion of excisiveness can be viewed as an abstraction of the excision axiom in the definition of a homology theory (see [40, Introduction to Section 1.4.1] and [40, Remark 1.4.3.3]).

Definition B.4.3. Let $\mathcal{C}$ and $\mathcal{D}$ be ∞ -categories with finite limits, and let $F : \mathcal{C} \rightarrow \mathcal{D}$ and $f : \mathbf{Stab}(\mathcal{C}) \rightarrow \mathbf{Stab}(\mathcal{D})$ be functors. A natural transformation $\alpha : F \circ \Omega_{\mathcal{C}}^{\infty} \rightarrow \Omega_{\mathcal{D}}^{\infty} \circ f$ *exhibit f as a (Goodwillie) derivative of F* if for each $g : \mathbf{Stab}(\mathcal{C}) \rightarrow \mathbf{Stab}(\mathcal{D})$, composition with α induces an equivalence:

$$\text{Hom}_{\text{Fun}(\mathbf{Stab}(\mathcal{C}), \mathbf{Stab}(\mathcal{D}))}(f, g) \xrightarrow{\simeq} \text{Hom}_{\text{Fun}(\mathbf{Stab}(\mathcal{C}), \mathbf{Stab}(\mathcal{D}))}(F \circ \Omega_{\mathcal{C}}^{\infty}, \Omega_{\mathcal{D}}^{\infty} \circ g).$$

Definition B.4.4 (cf. [40, Definition 6.1.1.6]). Let $\mathcal{C}$ be an ∞ -category. We will say that $\mathcal{C}$ is *differentiable* if it satisfies the following conditions:

1. The ∞ -category $\mathcal{C}$ admits finite limits.
2. The ∞ -category $\mathcal{C}$ admits sequential colimits: that is, every diagram $N(\mathbb{Z}_{\geq 0}) \rightarrow \mathcal{C}$ admits a colimit in $\mathcal{C}$.
3. The colimit functor

$$\text{colim} : \text{Fun}(\mathbb{Z}_{\geq 0}, \mathcal{C}) \rightarrow \mathcal{C}$$

is left exact. More informally: the formation of sequential colimits in $\mathcal{C}$ commutes with finite limits.

Example B.4.5 (cf. [40, Example 6.1.1.28]). Let $\mathcal{C}$ be an ∞ -category which admits finite colimits and has a final object. Suppose we are given a functor $F : \mathcal{C} \rightarrow \mathcal{D}$, where $\mathcal{D}$ is differentiable. If F is reduced, then the 1-excisive approximation to F is given by

$$P_1 F \simeq \operatorname{colim}_m \Omega^m \mathcal{D} \circ F \circ \Sigma^m$$

Proposition B.4.6 (cf. [40, Proposition 6.2.1.9]). Let $\mathcal{C}$ be an ∞ -category that has finite colimits and has final object, let $\mathcal{D}$ be a differentiable ∞ -category, and let $F : \mathcal{C} \rightarrow \mathcal{D}$ be reduced that preserves filtered colimits. Then F admits a Goodwillie derivative $\partial F : \mathbf{Stab}(\mathcal{C}) \rightarrow \mathbf{Stab}(\mathcal{D})$ given by the formula $\operatorname{colim}_m \Omega_{\mathcal{D}}^m \circ F \circ \Sigma_{\mathcal{C}}^m$.

Proposition B.4.7 (cf. [40, Corollary 6.2.2.17]). Suppose given an adjunction $F : \mathcal{C} \rightleftarrows \mathcal{D} : G$ between differentiable ∞ -categories. Assume that F is reduced, G preserves sequential colimits, and that G exhibits $\mathcal{D}$ as monadic over $\mathcal{C}$. If the unit map $\operatorname{id}_{\mathcal{C}} \rightarrow G \circ F$ induces an equivalence $\partial \operatorname{id}_{\mathcal{C}} \rightarrow \partial(G \circ F)$, then G induces an equivalence of ∞ -categories $\mathbf{Stab}(\mathcal{D}) \rightarrow \mathbf{Stab}(\mathcal{C})$.

Lemma B.4.8 (cf. [40, Lemma 7.3.4.8]). Let K be an ∞ -category. Let $\mathcal{C}$ be a pointed ∞ -category which admits finite colimits, and let $\mathcal{D}$ be a stable ∞ -category which admits sequential colimits and K -indexed colimits. Then the derivative functor $\partial : \operatorname{Fun}_*(\mathcal{C}, \mathcal{D}) \rightarrow \operatorname{Exc}(\mathcal{C}, \mathcal{D})$ preserves K -indexed colimits.

Lemma B.4.9 (cf. [40, Lemma 7.3.4.11]). Let $\mathcal{C}$ be a stable ∞ -category, let $f : C \rightarrow D$ be a morphism in $\mathcal{C}$, and let $f^* : \mathcal{C}_{/D} \rightarrow \mathcal{C}_{/C}$ be the functor given by pullback along f . Then:

1. The functor f^* is conservative.
2. Let K be a weakly contractible ∞ -category, and assume that $\mathcal{C}$ admits K -indexed colimits. Then the functor f^* preserves K -indexed colimits.

Corollary B.4.10. Let $\mathcal{C}$ be a stable ∞ -category with sifted colimits, let $f : C \rightarrow D$ be a morphism in $\mathcal{C}$, and let $f^* : \mathcal{C}_{/D} \rightarrow \mathcal{C}_{/C}$ be the functor given by pullback along f . Then:

1. The functor f^* is conservative.
2. f^* preserves sifted colimits.

Appendix C

t -structures and the derived category

In a classical course on homological algebra, the derived category of an abelian category $\mathcal{A}$ is introduced via localization of the category of chain complexes $\text{Ch}(\mathcal{A})$ with respect to quasi-isomorphisms. More precisely, one first defines $\mathcal{K}(\mathcal{A})$ as the category of chain complexes modulo chain homotopy and then the derived category $\mathcal{D}(\mathcal{A})$ is obtained as the homotopy category of $\mathcal{K}(\mathcal{A})$, where weak-equivalences are precisely quasi-isomorphisms. However, this definition loses homotopical information that one may wish to retain. A refinement of this approach is the construction of the dg-derived category, where the derived category is enriched over the symmetric closed monoidal category $(\text{Ch}(\mathbb{Z}), \otimes, \mathbb{Z}[0])$. If needed, one can recover the classical derived category by taking the zeroth homology H_0 of the mapping complexes. Nevertheless, dg-categories are not ∞ -categories. To address this issue, one introduces the dg-nerve, which allows us to associate to each dg-category an ∞ -category, thus enabling the use of the full machinery of higher category theory.

In the thesis, we present various equivalent ways for talking about the derived category. Historically, however, the perspective described above was the original one adopted in the literature. Furthermore, in the development of the present work, the ability to reduce to the case of chain complexes turns out to be particularly useful. For this reason, we recall the main definition and properties of the ∞ -derived category.

In section C.2.1, we describe how to associate an ∞ -category to a dg-category and use it to define the derived ∞ -category in the case of an abelian category with enough projectives. Subsection C.2.2 is then devoted to studying important properties of this construction. We begin this appendix by discussing t -structures, which play a central role in describing the main features of the ∞ -derived category, and more generally in the thesis.

The main reference for this appendix is Section 1.2.1 of [40], which corresponds to Subsection C.1, and Section 1.3 of the same source, which is relevant for Subsections C.2.1 and C.2.2.

C.1 t -structures

Definition C.1.1. Let $\mathcal{D}$ be a triangulated category. A t -structure on $\mathcal{D}$ is defined to be a pair of full subcategory $\mathcal{D}_{\geq 0}, \mathcal{D}_{\leq 0}$ (stable under isomorphism) having the following properties:

1. For $X \in \mathcal{D}_{\geq 0}$ and $Y \in \mathcal{D}_{\leq 0}$, we have $\text{Hom}_{\mathcal{D}}(X, Y[-1]) = 0$.
2. There are inclusions $\mathcal{D}_{\geq 0}[1] \subseteq \mathcal{D}_{\geq 0}, \mathcal{D}_{\leq 0}[-1] \subseteq \mathcal{D}_{\leq 0}$.
3. For any $X \in \mathcal{D}$, there exists a fiber sequence $X' \rightarrow X \rightarrow X''$ where $X' \in \mathcal{D}_{\geq 0}$ and $X'' \in \mathcal{D}_{\leq 0}$.

If $\mathcal{D}$ is a triangulated category equipped with a t -structure, we will write $\mathcal{D}_{\geq n}$ for $\mathcal{D}_{\geq 0}[n]$ and $\mathcal{D}_{\leq n}$ for $\mathcal{D}_{\leq 0}[n]$. Observe that we use a homological indexing convention.

Definition C.1.2. Let $\mathcal{C}$ be a stable ∞ -category. A t -structure on $\mathcal{C}$ is a t -structure on the homotopy category $\text{h}\mathcal{C}$. If $\mathcal{C}$ is equipped with a t -structure, we let $\mathcal{C}_{\geq n}$ and $\mathcal{C}_{\leq n}$ denote the full subcategories of $\mathcal{C}$ spanned by those objects which belong to $(\text{h}\mathcal{C})_{\geq n}$ and $(\text{h}\mathcal{C})_{\leq n}$, respectively.

Proposition C.1.3 (cf. [40, Proposition 1.2.1.5]). *Let $\mathcal{C}$ be a stable ∞ -category equipped with a t -structure. For each $n \in \mathbb{Z}$ the full subcategory $\mathcal{C}_{\leq n}$ is a localization of $\mathcal{C}$.*

Notation. Let $\mathcal{C}$ be a stable ∞ -category equipped with a t -structure. We will denote $\tau_{\leq n}$ the left adjoint to the inclusion $\mathcal{C}_{\leq n} \subseteq \mathcal{C}$, and $\tau_{\geq 0}$ the right adjoint to the inclusion $\mathcal{C}_{\geq 0} \subseteq \mathcal{C}$.

In [40, Proposition 1.2.1.10] it is proven that if $\mathcal{C}$ is a stable ∞ -category equipped with a t -structure, then we have an equivalence $\tau_{\leq m} \circ \tau_{\geq n} \rightarrow \tau_{\geq n} \circ \tau_{\leq m}$ of functors in $\text{Fun}(\mathcal{C}, \mathcal{C}_{\leq m} \cap \mathcal{C}_{\geq n})$. Thanks to this we can give the following definition:

Definition C.1.4. Let $\mathcal{C}$ be a stable ∞ -category equipped with a t -structure. The *heart* $\mathcal{C}^\heartsuit$ of $\mathcal{C}$ is the full subcategory $\mathcal{C}_{\leq 0} \cap \mathcal{C}_{\geq 0} \subseteq \mathcal{C}$. For each $n \in \mathbb{Z}$, we let $\pi_0 : \mathcal{C} \rightarrow \mathcal{C}^\heartsuit$ denote the functor $\tau_{\leq 0} \circ \tau_{\geq n} \cong \tau_{\geq 0} \circ \tau_{\leq n}$, and we let $\pi_n : \mathcal{C} \rightarrow \mathcal{C}^\heartsuit$ denote the composition of π_0 with the shift functor $X \mapsto X[-n]$.

Remark C.1.5. Let $\mathcal{C}$ be a stable ∞ -category equipped with a t -structure. In [40, Remark 1.2.1.8] it is noted that for $X \in \mathcal{C}$ we have the following fiber sequence:

$$\tau_{\geq n} X \rightarrow X \rightarrow \tau_{\leq n} X.$$

Definition C.1.6. Let $\mathcal{C}$ be a stable ∞ -category equipped with a t -structure. We let $\mathcal{C}^+ = \bigcup \mathcal{C}_{\leq n} \subseteq \mathcal{C}$, $\mathcal{C}^- = \bigcup \mathcal{C}_{\geq -n}$, and $\mathcal{C}^b = \mathcal{C}^+ \cap \mathcal{C}^-$. We will say that $\mathcal{C}$ is *left bounded* if $\mathcal{C} = \mathcal{C}^+$, *right bounded* if $\mathcal{C} = \mathcal{C}^-$, and *bounded* if $\mathcal{C} = \mathcal{C}^b$. Finally we define the *left completion* $\widehat{\mathcal{C}}$ of $\mathcal{C}$ as the limit $\lim_n \mathcal{C}_{\leq n}$ where the maps are the truncations $\tau_{\leq n} : \mathcal{C}_{\leq n+1} \rightarrow \mathcal{C}_{\leq n}$.

C.2 Derived ∞ -category

This section follows [40] section 1.3.

C.2.1 The dg-nerve

Definition C.2.1. Let R be a commutative ring. For $X, Y \in \text{Ch}(R)$ write $(X \otimes Y)_\bullet \in \text{Ch}(R)$ for the chain complex whose component in degree n is given by:

$$(X \otimes Y)_n := \bigoplus_{i+j=n} X_i \otimes_R Y_j$$

and whose differential is given on elements (x, y) of homogeneous degree by the following formula:

$$\partial^{X \otimes Y}(x, y) := (\partial^X x, y) + (-1)^{\deg(x)}(x, \partial^Y y).$$

Its tensor unit 1_{Ch} is R considered as a chain complex concentrated in degree 0.

Definition C.2.2 (dg-category). A *dg-category* is category enriched over $(\text{Ch}(\mathbb{Z}), \otimes, \mathbb{Z}[0])$

Construction C.2.3 (Differential graded nerve). Let $\mathcal{C}$ be a differential graded category. We will associate to $\mathcal{C}$ a simplicial set $N_{\text{dg}}(\mathcal{C})$, which we call the *differential graded nerve* of $\mathcal{C}$. For each $n \geq 0$, we define

$$N_{\text{dg}}(\mathcal{C})_n \simeq \text{Hom}_{\text{Set}_\Delta}(\Delta^n, N_{\text{dg}}(\mathcal{C}))$$

to be the set of all ordered pairs $(\{X_i\}_{0 \leq i \leq n}, \{f_I\})$, where:

- (a) For $0 \leq i \leq n$, X_i denotes an object of the differential graded category $\mathcal{C}$.
- (b) For every subset $I = \{i_- < i_m < i_{m-1} < \dots < i_1 < i_+\} \subseteq [n]$ with $m \geq 0$, f_I is an element of the abelian group $\text{Hom}_{\mathcal{C}}(X_{i_-}, X_{i_+})_m$, satisfying the equation

$$df_I = \sum_{1 \leq j \leq m} (-1)^j (f_{I - \{i_j\}} - f_{\{i_j < \dots < i_1 < i_+\}} \circ f_{\{i_- < i_m < \dots < i_j\}}).$$

If $\alpha : [m] \rightarrow [n]$ is a nondecreasing function, then the induced map $N_{\text{dg}}(\mathcal{C})_n \rightarrow N_{\text{dg}}(\mathcal{C})_m$ is given by

$$(\{X_i\}_{0 \leq i \leq n}, \{f_I\}) \mapsto (\{X_{\alpha(j)}\}_{0 \leq j \leq m}, \{g_J\}),$$

where

$$g_J = \begin{cases} f_{\alpha(J)} & \text{if } \alpha|_J \text{ is injective,} \\ \text{id}_{X_{i_-}} & \text{if } J = \{j, j'\} \text{ with } \alpha(j) = \alpha(j') = i, \\ 0 & \text{otherwise.} \end{cases}$$

In [40, Proposition 1.3.1.17] it is proved that the differential graded nerve of a differential graded category $\mathcal{C}$ and the usual homotopy coherent nerve of the underlying simplicial category $\mathcal{C}_\Delta$ (see [40, Construction 1.3.1.13]) are equivalent.

Definition C.2.4 ($\text{Ch}(\mathcal{A})$ as dg-category, cf. [40, Definition 1.3.2.1]). Let $\mathcal{A}$ be an additive category, and let M_* and N_* be chain complexes with values in $\mathcal{A}$. For each integer p , we let $\underline{\text{Hom}}_{\text{Ch}(\mathcal{A})}(M_*, N_*)_p$ denote the product

$$\prod_{n \in \mathbb{Z}} \text{Hom}_{\mathcal{A}}(M_n, N_{n+p}).$$

We will regard the collection

$$\{\underline{\text{Hom}}_{\text{Ch}(\mathcal{A})}(M_*, N_*)_p\}_{p \in \mathbb{Z}}$$

as a chain complex of abelian groups, with differential given by the formula

$$(df)(x) = d(f(x)) - (-1)^p f(dx)$$

for $f \in \underline{\text{Hom}}_{\text{Ch}(\mathcal{A})}(M_*, N_*)_p$.

Given a triple of chain complexes $M_*, N_*, P_* \in \text{Ch}(\mathcal{A})$, composition gives a bilinear map

$$\underline{\text{Hom}}_{\text{Ch}(\mathcal{A})}(N_*, P_*)_p \times \underline{\text{Hom}}_{\text{Ch}(\mathcal{A})}(M_*, N_*)_q \rightarrow \underline{\text{Hom}}_{\text{Ch}(\mathcal{A})}(M_*, P_*)_{p+q}$$

satisfying the Leibniz rule

$$d(f \circ g) = df \circ g + (-1)^p f \circ dg.$$

Using this notion of composition, we can regard $\text{Ch}(\mathcal{A})$ as a differential graded category.

Lemma C.2.5. *Let $\mathcal{A}$ be an abelian category and let $M, N \in \text{Ch}(\mathcal{A})$. Then:*

$$\pi_n(\text{Hom}_{\text{N}_{\text{dg}}(\text{Ch}(\mathcal{A}))}(M, N)) \cong H_n(\underline{\text{Hom}}_{\text{Ch}(\mathcal{A})}(M, N))$$

Proof. Take $M, N \in \text{Ch}(\mathcal{A})$. By [40, Remark 1.3.1.12] we have that

$$\text{Hom}_{\text{N}_{\text{dg}}(\text{Ch}(\mathcal{A}))}(M, N) \simeq \text{DK}_\bullet(\tau_{\geq 0} \underline{\text{Hom}}_{\text{Ch}(\mathcal{A})}(M, N))$$

where $\text{DK}_\bullet$ is the functor defined in [40, Construction 1.2.3.5]. Then in [40, Remark 1.2.3.14] it is noted that given a simplicial abelian group we can compute its homotopy groups by computing the homology of the associated complex through the Dold-Kan Correspondence ([40, Theorem 1.2.3.7]). This proves the lemma. $\square$

Definition C.2.6 (Projective object). Let $\mathcal{A}$ be an abelian category. An object $P \in \mathcal{A}$ is said to be *projective* if, for every epimorphism $M \rightarrow N$ in $\mathcal{A}$, the induced map $\text{Hom}_{\mathcal{A}}(P, M) \rightarrow \text{Hom}_{\mathcal{A}}(P, N)$ is surjective. We say $\mathcal{A}$ has *enough projective objects* if, for every object $M \in \mathcal{A}$, there exist an epimorphism $P \rightarrow M$, where P is a projective object of $\mathcal{A}$. We denote by $\mathcal{A}_{\text{proj}}$ the full subcategory of $\mathcal{A}$ spanned by the projective object.

Definition C.2.7. (Derived category $\mathcal{D}^-(\mathcal{A})$) Let $\mathcal{A}$ be an abelian category with enough projective objects. We let $\mathcal{D}^-(\mathcal{A})$ denote the ∞ -category $\text{N}_{\text{dg}}(\text{Ch}^-(\mathcal{A}_{\text{proj}}))$. We will refer to $\mathcal{D}^-(\mathcal{A})$ as the *derived ∞ -category of $\mathcal{A}$* .

Proposition C.2.8 (cf. [40, Proposition 1.3.2.10]). *Let $\mathcal{A}$ be an additive category. Then $\text{N}_{\text{dg}}(\text{Ch}(\mathcal{A}))$ is stable.*

Definition C.2.9 ($\mathcal{D}_{\geq 0}^-(\mathcal{A})$). Let $\mathcal{A}$ be an abelian category with enough projective objects. We define $\mathcal{D}_{\geq 0}^-(\mathcal{A})$ as the full subcategory of $\mathcal{D}^-(\mathcal{A})$ spanned by those chain complexes A such that the homology objects $H_n(A) \in \mathcal{A}$ vanish for $n < 0$. Similarly we define $\mathcal{D}_{\leq 0}^-(\mathcal{A})$.

Proposition C.2.10 (cf. [40, Proposition 1.3.2.19, Proposition 1.3.3.16.]). *Let $\mathcal{A}$ be an abelian category with enough projective objects. Then $(\mathcal{D}_{\geq 0}^-(\mathcal{A}), \mathcal{D}_{\leq 0}^-(\mathcal{A}))$ defines a t -structure on $\mathcal{D}^-(\mathcal{A})$ that is right bounded and left complete; moreover, $\mathcal{D}^-(\mathcal{A})^\heartsuit$ is canonically equivalent to $\mathcal{A}$.*

Lemma C.2.11 (cf. [40, Lemma 1.3.2.20]). *Let $\mathcal{A}$ be an abelian category, and let $P_* \in \text{Ch}(\mathcal{A})$ be a complex of projective objects of $\mathcal{A}$ such that $P_n \simeq 0$ for $n \ll 0$. Let $Q_* \rightarrow Q'_*$ be a quasi-isomorphism in $\text{Ch}(\mathcal{A})$. Then the induced map*

$$\underline{\text{Hom}}_{\text{Ch}(\mathcal{A})}(P_*, Q_*) \rightarrow \underline{\text{Hom}}_{\text{Ch}(\mathcal{A})}(P_*, Q'_*)$$

is a quasi isomorphism.

Proposition C.2.12 (cf. [40, Proposition 1.3.2.22]). *Let $\mathcal{A}$ be an abelian category with enough projective objects and let $\mathcal{A}_{\text{proj}}$ denote the full subcategory of $\mathcal{A}$ spanned by the projective objects. Then $\mathcal{D}_{\geq 0}^-(\mathcal{A})$ is equivalent to the underlying ∞ -category of the simplicial category $\text{Fun}(\Delta^{op}, \mathcal{A}_{\text{proj}})$.*

Remark C.2.13. In the proof of the above proposition it is proven that: $\text{N}_{\text{dg}}(\text{Ch}_{\geq 0}(\mathcal{A}_{\text{proj}})) \cong \mathcal{D}_{\geq 0}^-(\mathcal{A})$.

C.2.2 The Universal Property of $\mathcal{D}^-(\mathcal{A})$

We now present the universals properties of the ∞ -category $\mathcal{D}_{\geq 0}^-(\mathcal{A})$ and $\mathcal{D}^-(\mathcal{A})$.

Theorem C.2.14 (Universal property of $\mathcal{D}_{\geq 0}^-(\mathcal{A})$, cf. [40, Theorem 1.3.3.8]). *Let $\mathcal{A}$ be an abelian category with enough projective objects, and $\mathcal{C}$ an arbitrary ∞ -category which admits geometric realizations of simplicial objects. Then:*

1. *The restriction functor*

$$\mathrm{Fun}'(\mathcal{D}_{\geq 0}^-(\mathcal{A}), \mathcal{C}) \rightarrow \mathrm{Fun}(\mathcal{A}_{\mathrm{proj}}, \mathcal{C})$$

induces an equivalence between the full subcategory of $\mathrm{Fun}(\mathcal{D}_{\geq 0}^-(\mathcal{A}), \mathcal{C})$ spanned by those functors which preserve geometric realizations and the ∞ -category $\mathrm{Fun}(\mathcal{A}_{\mathrm{proj}}, \mathcal{C})$.

2. *A functor $F \in \mathrm{Fun}'(\mathcal{D}_{\geq 0}^-(\mathcal{A}), \mathcal{C})$ preserves finite coproducts if and only if the restriction $F|_{\mathcal{A}_{\mathrm{proj}}}$ preserves finite coproducts.*

Definition C.2.15. Let $\mathcal{C}$ and $\mathcal{C}'$ be stable ∞ -categories equipped with t -structures. We will say that a functor $f : \mathcal{C} \rightarrow \mathcal{C}'$ is *right t -exact* if it is exact and carries $\mathcal{C}_{\geq 0}$ into $\mathcal{C}'_{\geq 0}$. We say that f is *left t -exact* if it is exact and carries $\mathcal{C}_{\leq 0}$ into $\mathcal{C}'_{\leq 0}$.

Proposition C.2.16 (Universal property of $\mathcal{D}^-(\mathcal{A})$, cf. [40, Proposition 1.3.3.12]). *Let $\mathcal{A}$ be an abelian category with enough projective objects, and let $\mathcal{C}$ be a stable ∞ -category equipped with a left complete t -structure. Then the restriction functor*

$$\mathrm{Fun}(\mathcal{D}^-(\mathcal{A}), \mathcal{C}) \rightarrow \mathrm{Fun}(\mathcal{A}_{\mathrm{proj}}, \mathcal{C})$$

induces an equivalence from the full subcategory of $\mathrm{Fun}(\mathcal{D}^-(\mathcal{A}), \mathcal{C})$ spanned by the right t -exact functors to the full subcategory of $\mathrm{Fun}(\mathcal{A}_{\mathrm{proj}}, \mathcal{C})$ spanned by functors which preserve finite coproducts.

Definition C.2.17. Let $\mathcal{A}$ be a small abelian category. We define to be $\mathcal{A}^\wedge$ the category of product preserving functors from $\mathcal{A}_{\mathrm{proj}}$ to $\mathbf{Set}$.

Proposition C.2.18 (cf. [40, Proposition 1.3.3.14]). *Let $\mathcal{A}$ be an abelian category with enough projective objects. Then there exists an equivalence of ∞ -categories*

$$\mathcal{D}_{\geq 0}^-(\mathcal{A}^\wedge) \cong \mathcal{P}_\Sigma(\mathcal{A}_{\mathrm{proj}})$$

whose composition with the inclusion $\mathcal{A}_{\mathrm{proj}} \simeq \mathcal{D}^-(\mathcal{A}^\wedge)^\heartsuit \subseteq \mathcal{D}_{\geq 0}^-(\mathcal{A}^\wedge)$ is equivalent to the Yoneda embedding $\mathcal{A}_{\mathrm{proj}} \rightarrow \mathcal{P}_\Sigma(\mathcal{A}_{\mathrm{proj}})$.

C.2.3 Inverting quasi-isomorphism

Now we present two results that relates the above notions with the usual idea that the derived category should be the localization of the category $\mathrm{Ch}(\mathcal{A})$ at the quasi-isomorphism (that is, those morphisms which induce isomorphism in homology).

Proposition C.2.19 (cf. [40, Proposition 1.3.4.6]). *Let $\mathcal{A}$ be an abelian category with enough projective objects and let W be the collection of all morphism in $\mathrm{N}_{\mathrm{dg}}(\mathrm{Ch}^-(\mathcal{A}))$ which are quasi-isomorphism of chain-complexes. Then we have a canonical equivalence of ∞ -categories:*

$$\mathrm{N}_{\mathrm{dg}}(\mathrm{Ch}^-(\mathcal{A}))[W^{-1}] \simeq \mathcal{D}^-(\mathcal{A})$$

Theorem C.2.20 (cf. [40, Theorem 1.3.4.4]). *Let $\mathcal{A}$ be an abelian category with enough projective objects, let $\mathbf{A} = \mathrm{Ch}^-(\mathcal{A})$ (regarded as an ordinary category), and let W be the collection of morphisms in $\mathbf{A}$ which are quasi-isomorphisms of chain complexes. Then there is a canonical equivalence of ∞ -categories $\mathbf{A}[W^{-1}] \simeq \mathcal{D}^-(\mathcal{A})$.*

Remark C.2.21. Thanks to the theorem above, it follows that to compute the $\mathrm{Hom}_{\mathrm{Ch}^-(\mathcal{A})[W^{-1}]}(M, N)$ for $M, N \in \mathrm{Ch}^-(\mathcal{A})$ we should first make a replacement of both M and N by quasi-isomorphic projective object $Q(M)$ and $Q(N)$. However notice that:

$$\underline{\mathrm{Hom}}_{\mathrm{Ch}^-(\mathcal{A}_{\mathrm{proj}})}(QM, QN) \cong \underline{\mathrm{Hom}}_{\mathrm{Ch}^-(\mathcal{A})}(QM, QN) \stackrel{(1)}{\cong} \underline{\mathrm{Hom}}_{\mathrm{Ch}^-(\mathcal{A})}(QM, N)$$

Where (1) is exactly Lemma C.2.11. This means that:

$$\mathrm{Hom}_{\mathrm{Ch}^-(\mathcal{A})[W^{-1}]}(M, N) \cong \mathrm{Hom}_{\mathrm{N}_{\mathrm{dg}}(\mathrm{Ch}^-(\mathcal{A}_{\mathrm{proj}}))}(QM, QN) \cong \mathrm{Hom}_{\mathrm{N}_{\mathrm{dg}}(\mathrm{Ch}^-(\mathcal{A}))}(QM, N).$$

In particular we only need to replace M .

Corollary C.2.22. *Let $\mathcal{A}$ be an abelian category with enough projective objects. Let $M, N \in \mathrm{Ch}^-(\mathcal{A})$ with M projective, then:*

$$\pi_n(\mathrm{Hom}_{\mathcal{D}^-(\mathcal{A})}(M, N)) \cong H_n(\underline{\mathrm{Hom}}_{\mathrm{Ch}^-(\mathcal{A})}(M, N))$$

Proof. This is essentially Remark C.2.21 together with Lemma C.2.5. □

In Section 1.3.5 of [40], the unbounded case is treated. In particular, for a Grothendieck abelian category $\mathcal{A}$, the *derived ∞ -category* $\mathcal{D}(\mathcal{A})$ is defined. It is shown that this ∞ -category is stable and can be described as the localization of the category of chain complexes $\mathrm{Ch}(\mathcal{A})$ at the class of quasi-isomorphisms. We refer the reader to that section for full details and simply cite here a result that connects the above discussion to this new construction.

Proposition C.2.23 (cf. [40, Proposition 1.3.5.24]). *let $\mathcal{A}$ be a Grothendieck abelian category with enough projectives objects, and let $L : \mathrm{N}_{\mathrm{dg}}(\mathrm{Ch}(\mathcal{A})) \rightarrow \mathcal{D}(\mathcal{A})$ be a left adjoint to the inclusion. Then the composite functor*

$$F : \mathcal{D}^-(\mathcal{A}) \hookrightarrow \mathrm{N}_{\mathrm{dg}}(\mathrm{Ch}(\mathcal{A})) \xrightarrow{L} \mathcal{D}(\mathcal{A})$$

is a fully faithful embedding, whose essential image is the subcategory $\cup_{n \geq 0} \mathcal{D}(\mathcal{A})_{-n} \subseteq \mathcal{D}(\mathcal{A})$.

Appendix D

Algebraic structures

The goal of this chapter is to introduce the main tools used throughout the thesis for constructing algebraic structures in the ∞ -categorical context. Intuitively, we can think of these algebraic structures as analogues of classical ones, but where all constructions are considered up to homotopy. Whenever we apply operations such as commutativity or associativity, we must specify homotopies, which themselves depend on the number of inputs involved, and all coherence conditions must also hold up to homotopy in a compatible way. Clearly, carrying out such constructions explicitly by hand would be infeasible; the tools introduced here will allow us to systematically circumvent this difficulty.

In Section C.1, we discuss *animation*, a process that helps in constructing *homotopical* algebraic structures from classical ones. In Section C.2, we focus on the animation of an algebraic theory, a tool that will play a central role in Chapter 2 for constructing $\mathcal{C}^\infty$ -rings. Finally, in Section C.3, we review the notion of module spectra and relate it to the derived category introduced in Appendix C.

The main references for this appendix are [30],[3], [12],[10] and [17].

D.1 Animation

The exposition follows [30, section 7] and [3, Section 2].

D.1.1 Algebraic categories

In this subsection, we introduce algebraic categories. The goal is to understand in what sense they encode algebraic structure and why they are so important.

Definition D.1.1. Let $\mathcal{C}$ be a category. We say that $\mathcal{C}$ is *algebraic* if there exists an essentially small full subcategory $\mathcal{F}_{\mathcal{C}} \subseteq \mathcal{C}$ closed under finite coproducts, and an equivalence

$$\mathrm{Fun}_{\Pi}(\mathcal{F}_{\mathcal{C}}^{op}, \mathbf{Set}) \rightarrow \mathcal{C}$$

which restricts along the Yoneda embedding to the inclusion $\mathcal{F}_{\mathcal{C}} \hookrightarrow \mathcal{C}$.

Remark D.1.2. Following [3] we'd say that $\mathcal{C}$ is the *1-sifted completion* of $\mathcal{F}_{\mathcal{C}}$. If $\mathcal{F}_{\mathcal{C}}$ are compact projective object then we'd say $\mathcal{C}$ is *1-projectively generated*.

Definition D.1.3. Let $\mathcal{C}$ be a category:

1. An object $X \in \mathcal{C}$ is *compact* if and only if $\mathrm{Hom}_{\mathcal{C}}(X, -) : \mathcal{C} \rightarrow \mathbf{Set}$ preserves filtered colimits.
2. An object $X \in \mathcal{C}$ is *projective* if and only if $\mathrm{Hom}_{\mathcal{C}}(X, -) : \mathcal{C} \rightarrow \mathbf{Set}$ preserves reflexive coequalizers.

Note that a category $\mathcal{C}$ is algebraic if and only if the condition definition D.1.1 holds for $\mathcal{F}_{\mathcal{C}}$ the full subcategory of compact projective objects. We now present two key examples in detail. The exposition closely follows the respective references and is almost verbatim in parts.

Example D.1.4 (cf. [30, Remark 7.1.3]). The category of sets is algebraic:

$$\mathbf{Set} \cong \mathrm{Fun}_{\Pi}(\mathbf{Fin}^{op}, \mathbf{Set}).$$

Here $\mathbf{Fin}$ is the full subcategory of $\mathbf{Set}$ spanned by finite sets. Notice that through Yoneda functor, every set X defines an objects of $\mathbf{Fun}_{\Pi}(\mathbf{Fin}^{op}, \mathbf{Set})$. Conversely, to each product preserving functor $F : \mathbf{Fin}^{op} \rightarrow \mathbf{Set}$ we can assign $F(\{1\})$. It can be checked that the two functors are one the inverse of the other. Moreover, compact projective object in $\mathbf{Set}$ are exactly finite sets. Let us now unpack the data contained in an object F of $\mathbf{Fun}_{\Pi}(\mathbf{Fin}^{op}, \mathbf{Set})$:

- Sets $F_0 = F(\emptyset)$, $F_1 = F(\{1\})$, $F_2 = F(\{1, 2\})$, $\dots$
- For every map $\{1, 2, 3, \dots, n\} \rightarrow \{1, 2, 3, \dots, m\}$, an induced map $F_n \rightarrow F_m$
- canonical isomorphism $F_0 \simeq \{*\}$ and $F_n \simeq F_n \simeq (F_1)^{\times n}$ for all n .

Warning D.1.5. In the above example, we wrote finite sets in the following form $\{1, 2, \dots, n\}$. However this was done for simplicity of the exposition. It is to be understood that no ordering is chosen, and different finite sets I, J with the same cardinality may be related by an isomorphism and so encoding an isomorphism between $F(I)$ and $F(J)$ for $F \in \mathbf{Fun}_{\Pi}(\mathbf{Fin}^{op}, \mathbf{Set})$.

Example D.1.6 (cf. [30, Remark 7.1.4]). The category of abelian groups is algebraic:

$$\mathbf{Ab} \cong \mathbf{Fun}_{\Pi}(\mathcal{F}_{\mathbf{Ab}}^{op}, \mathbf{Set})$$

where $\mathcal{F}_{\mathbf{Ab}}$ is the category of finitely generated free abelian groups, which is also the category of compact projective object of $\mathbf{Ab}$. Let us again unpack the data contained in an object of $\mathbf{Fun}_{\Pi}(\mathcal{F}_{\mathbf{Ab}}^{op}, \mathbf{Set})$. First, notice that $\mathcal{F}_{\mathbf{Ab}}$ is equivalent to the category whose object are natural numbers and morphism are integers matrices:

$$\mathbf{Hom}(n, m) = \mathbf{Hom}_{\mathbf{Ab}}(\mathbb{Z}^{\oplus n}, \mathbb{Z}^{\oplus m}) = \mathbf{Mat}_{n \times m}(\mathbb{Z}),$$

and composition is given by matrix multiplication. Now we see that the data of an object F in $\mathbf{Fun}_{\Pi}(\mathcal{F}_{\mathbf{Ab}}^{op}, \mathbf{Set})$ amounts to the data of sets $F_0, F_1, F_2, \dots$ with isomorphisms $F_n \simeq (F_1)^{\times n}$ for all n , and for every matrix $\varphi \in \mathbf{Mat}_{m \times n}(\mathbb{Z})$, a map $F_{\varphi} : F_m \rightarrow F_n$ subject to identities imposed by functoriality.

We can interpret F as specifying a set $M := F_1$ equipped with operations encoded by matrices. For example, the matrix

$$\phi = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$$

encodes the operation $F_{\phi} : M^{\times n} \rightarrow M$, which can be viewed as a linear combination: a_i :

$$(x_1, \dots, x_n) \mapsto \sum_i a_i x_i.$$

In particular:

- There is a (commutative) *addition map* $M \times M \rightarrow M$, $(x_1, x_2) \mapsto x_1 + x_2$, encoded by the matrix

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

- There is a zero element $0 \in M$, encoded as the map $0 : \text{pt} \rightarrow M$ coming from the empty (0×1) -matrix.
- There is an *additive inverse map* $M \rightarrow M$, $x_1 \mapsto -x_1$, encoded by the (1×1) -matrix (-1) .

In the end, this construction shows that the structure of an abelian group can be encoded by specifying all possible linear combinations of elements, via operations determined by integer matrices. However, in the 1-categorical setting, this data is redundant: the group structure is already uniquely determined by the usual group axioms.

Other examples are the following:

Example D.1.7. The category of R -modules for R a commutative ring is algebraic :

$$\mathbf{Mod}_R \cong \mathbf{Fun}_{\Pi}(\mathcal{F}_R^{op}, \mathbf{Set})$$

Where $\mathcal{F}_R$ is the category of finitely generated free R -modules. In this case the category of compact projective object is given by the full subcategory of finitely generated projective R -modules.

Example D.1.8. The category **CRing** of commutative algebra is algebraic with

$$\mathbf{CRing} \cong \mathrm{Fun}_{\Pi}(\mathbf{Poly}^{op}, \mathbf{Set})$$

Where **Poly** is the full subcategory spanned by finitely generated polynomial algebras.

Definition D.1.9. We define the category **Mod** as follow:

- object are couples (R, M) where R is a ring and M an R -module,
- morphism $(R, M) \rightarrow (R', M')$ are given by $R \rightarrow R'$ morphism of rings and $M \rightarrow M'$ morphism of R -modules.

We denote with **PolyMod** the full subcategory spanned by those object of the form (A, M) with A finitely generated polynomial algebra and $M \cong A^{\oplus n}$ for some n .

Example D.1.10. The category **Mod** is algebraic with

$$\mathbf{Mod} \cong \mathrm{Fun}_{\Pi}(\mathbf{PolyMod}^{op}, \mathbf{Set}).$$

We now present the universal property of algebraic categories.

Proposition D.1.11. *Let $\mathcal{C}$ be an algebraic category. For every category $\mathcal{D}$ admitting filtered colimits and reflexive coequalizer, the canonical functor*

$$\mathrm{Fun}'(\mathcal{C}, \mathcal{D}) \rightarrow \mathrm{Fun}(\mathcal{F}_{\mathcal{C}}, \mathcal{D})$$

from the category of functors $\mathcal{C} \rightarrow \mathcal{D}$ that preserves filtered colimits and reflexive coequalizer, is an equivalence.

D.1.2 Animation

We now present the analogue of algebraic categories in the ∞ -categorical setting. Building on the intuition developed in the preceding subsection, the goal of this subsection is to understand why it is reasonable to consider the animation of algebraic categories, and why this construction yields an algebraic structure in the ∞ -categorical setting.

Notation. We will denote $\mathrm{Fun}(\mathcal{C}, \mathbf{Spc})$ by $\mathcal{P}(\mathcal{C})$ and $\mathrm{Fun}_{\Pi}(\mathcal{C}, \mathbf{Spc})$ by $\mathcal{P}_{\Sigma}(\mathcal{C})$.

Definition D.1.12. Let $\mathcal{C}$ be an algebraic category and let $\mathcal{C}^{\omega \text{ proj}}$ denote the full subcategory of compact projective object. We define the *animation* of $\mathcal{C}$ as:

$$\mathbf{a}\mathcal{C} := \mathcal{P}_{\Sigma}(\mathcal{C}^{\omega \text{ proj}}).$$

Example D.1.13 (cf. [30, Remark 7.6.2]). We now inspect the case of $\mathbf{Mod}_R$ for R a commutative ring, in order to understand how to think about an object in the animated setting. In doing so, please keep in mind Example D.1.6. First, notice that instead of asking $\mathbf{Mod}_R^{\omega \text{ proj}}$ to be the full subcategory finitely generated projective R -module, we may restrict to the full subcategory $\mathcal{F}_{\mathbf{Mod}_R}$ of finitely generated free R -modules. Unpacking definitions, an object in $M \in \mathbf{aMod}_R$ amounts to the following data:

1. For every $n \geq 0$, a space M_n with a canonical equivalence $M_n \rightarrow (M_1)^{\times n}$. We refer to M_1 as the *underlying space*.
2. For every R -linear map $R^{\oplus n} \rightarrow R^{\oplus m}$ (or $\phi \in \mathrm{Mat}_{m \times n}(R)$) a map of space $M_{\phi} : M_m \rightarrow M_n$. In particular for $\phi \in \mathrm{Mat}_{n \times 1}(R)$ we get an operation $(M_1)^{\times n} \rightarrow M_1$. Note that this is slightly imprecise: ϕ actually determines a morphism $M_n \rightarrow M_1$, so what we really have is a composite

$$(M_1)^{\times n} \xleftarrow{\sim} M_n \xrightarrow{M_{\phi}} M_1.$$

In the following, we will ignore this subtlety. The operation encoded by the matrix

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

will be denoted by add .

3. For every pair of R -linear map $\phi : R^{\oplus n} \rightarrow R^{\oplus m}$ and $\psi : R^{\oplus m} \rightarrow R^{\oplus k}$, an homotopy $h_{\phi, \psi} : M_{\phi} \circ M_{\psi} \simeq M_{\psi \circ \phi}$ of maps $M_k \rightarrow M_n$. In particular, the associativity square

$$\begin{array}{ccc}
M_1 \times M_1 \times M_1 & \xrightarrow{\text{add} \times \text{id}} & M_1 \times M_1 \\
\text{id} \times \text{add} \downarrow & & \downarrow \text{add} \\
M_1 \times M_1 & \xrightarrow{\text{add}} & M_1
\end{array}$$

does not commute in the 1-categorical sense, but commutes up to homotopy.

4. For every triple of R -linear maps ϕ, ψ, ω , a tetrahedron-shaped diagram expressing a “higher” homotopy between the homotopies $h_{\phi, \psi}$, $h_{\psi, \omega}$, $h_{\phi, \omega \circ \psi}$, and $h_{\psi \circ \phi, \omega}$.
5. ...

Finally we have a map of R in M_1 , i.e $R \rightarrow \text{End}(M_1)$ given by:

$$R \simeq \text{Mat}_{1 \times 1}(R) \simeq \text{Hom}_{\mathcal{F}\text{Mod}_R}(R, R) \xrightarrow{M} \text{Hom}_{\mathbf{sSpc}}(M_1, M_1) = \text{End}(M_1).$$

The upshot is that an object in $\mathbf{aMod}_R$ should be thought of as an R -module in which all operations are defined only up to homotopy. This is the crucial difference with Example D.1.6. While in that case the extra conditions were redundant, in the animated setting they are necessary, since there is no such thing as strict equality.

Remark D.1.14. Let R be a commutative ring. We are now in a position to construct two versions of derived (connective) algebraic structures associated to R : namely, the connective part of the derived category $\mathcal{D}_{\geq 0}^-(R)$ and the animation $\mathbf{aMod}_R$. Thanks to Proposition C.2.18, we obtain an equivalence:

$$\mathcal{D}_{\geq 0}^-(R) \simeq \mathbf{aMod}_R.$$

Example D.1.15. By Example D.1.8, we can animate the category of commutative rings **CRing**. The category of *animated rings*, denoted by **dRing**, is defined as

$$\mathbf{dRing} := \mathbf{aCRing}.$$

We now aim to present the universal property of the animation; this is the analogue of Proposition D.1.11 in the ∞ -categorical setting.

Definition D.1.16. Let $\mathcal{C}$ be an ∞ -category. We say that $\mathcal{C}$ is *sifted* if $\mathcal{C}$ -indexed colimits in **Spc** commute with finite product.

Proposition D.1.17 (cf. [37, Proposition 5.5.8.10, Proposition 5.5.8.15.]). *Let $\mathcal{C}$ be a small ∞ -category admitting finite coproducts.*

1. *The ∞ -category $\mathcal{P}_{\Sigma}(\mathcal{C})$ is an accessible localization of $\mathcal{P}(\mathcal{C})$.*
2. *The Yoneda embedding $y : \mathcal{C} \rightarrow \mathcal{P}(\mathcal{C})$ factors through $\mathcal{P}_{\Sigma}(\mathcal{C})$. Moreover, y carries finite coproducts in $\mathcal{C}$ to finite coproducts in $\mathcal{P}_{\Sigma}(\mathcal{C})$.*
3. *For every category $\mathcal{D}$ admitting filtered colimits and geometric realization, the canonical functor*

$$\text{Fun}_{\text{sifted}}(\mathcal{P}_{\Sigma}(\mathcal{C}), \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$$

from the category of functors $\mathcal{P}_{\Sigma}(\mathcal{C}) \rightarrow \mathcal{D}$ that preserves sifted colimits, is an equivalence.

4. *For every category $\mathcal{D}$ admitting all colimits, the canonical functor*

$$\text{Fun}_{\text{colim}}(\mathcal{P}_{\Sigma}(\mathcal{C}), \mathcal{D}) \rightarrow \text{Fun}^{\sqcup}(\mathcal{C}, \mathcal{D})$$

from the category of functors $\mathcal{P}_{\Sigma}(\mathcal{C}) \rightarrow \mathcal{D}$ that preserves all small colimits to the category of functors $\mathcal{C} \rightarrow \mathcal{D}$ which preserve finite coproducts, is an equivalence.

Remark D.1.18. Putting together point *i*) of the above proposition and Theorem B.3.6 we deduce that $\mathcal{P}_\Sigma(\mathcal{C})$ is a presentable ∞ -category.

Lemma D.1.19 ([3], Lemma 2.58). *If $\mathcal{C}$ is 1-projectively generated by $\mathcal{C}^{\omega^1 \text{proj}}$, then $\mathbf{a}\mathcal{C}$ is projectively generated by $\mathbf{a}\mathcal{C}^{\omega \text{proj}}$ and $\mathbf{a}\mathcal{C}^{\omega \text{proj}} \simeq \mathcal{C}^{\omega^1 \text{proj}}$.*

Remark D.1.20 (cf. [3, Remark 2.59.]). Following notation of Proposition D.1.17, we see that if $\mathcal{D}$ has all sifted colimits, then composition with Yoneda induces an equivalence

$$\text{Fun}_{\text{sifted}}(\mathbf{a}\mathcal{C}, \mathcal{D}) \simeq \text{Fun}(\mathcal{C}^{\omega \text{proj}}, \mathcal{D}),$$

if $\mathcal{D}$ has all colimits, then we get an equivalence

$$\text{Fun}_{\text{colim}}(\mathbf{a}\mathcal{C}, \mathcal{D}) \simeq \text{Fun}^\sqcup(\mathcal{C}^{\omega \text{proj}}, \mathcal{D}).$$

From this observation let us use the words of [3]:

This is the real point of animation: it is to allow us to build functors. It is often possible to construct a functor out of $\mathcal{C}^{\omega \text{proj}}$, and left Kan extension allows us to extend such a functor to every animated object of $\mathcal{C}$.

D.2 Lawvere theory

We will now explore a method for constructing algebraic categories from what is known as a Lawvere theory. This approach will serve both as our primary example and as our main tool for building derived manifolds. The following is adapted from [12] section 2., for details and proofs see *loc. cit.*

D.2.1 $\mathbb{T}$ -algebras

Definition D.2.1. An *algebraic theory* (or *Lawvere theory*) is an ∞ -category $\mathbb{T}$ with finite product, together with a chosen object $r \in \mathbb{T}$ called the *generator* such that all objects in $\mathbb{T}$ are equivalent to r^n for some $n \geq 0$. A *morphism of algebraic theories* $(\mathbb{T}, r) \rightarrow (\mathbb{T}', r')$ is a product preserving functor $f : \mathbb{T} \rightarrow \mathbb{T}'$ carrying r to r' .

Notation. We adopt here the same notation used in the main body of the thesis.

- Let k be a commutative ring. We denote by $\mathbf{Com}_k$ the full subcategory of k -schemes spanned by those schemes of the form $\mathbb{A}_k^n$ for some $n \geq 0$.
- We denote by $\mathbf{C}^\infty$ the full subcategory of $\mathbf{Mfd}$ spanned by manifolds of the form $\mathbb{R}^n$ for some $n \geq 0$.

Example D.2.2. $\mathbf{C}^\infty$ is an algebraic theory, with $\mathbb{R}$ as its generator.

Example D.2.3. $\mathbf{Com}_k$ is an algebraic theory with generator $\mathbb{A}_k^1$. The category $\mathbf{Com}_\mathbb{R}$ is equivalently the subcategory of $\mathbf{Mfd}$ of the form $\mathbb{R}^n$ with polynomial maps.

Definition D.2.4. Let $\mathcal{C}$ be an ∞ -category with finite products. We define the ∞ -category of $\mathbb{T}$ -algebras in $\mathcal{C}$ as the full subcategory $\mathbf{Alg}_\mathbb{T}(\mathcal{C})$ of $\text{Fun}(\mathbb{T}, \mathcal{C})$ of functor that preserves finite products. In the special case $\mathcal{C} = \mathbf{Spc}, \mathbf{Set}$ we will refer to $\mathbf{Alg}_\mathbb{T}(\mathbf{Spc})$ as the ∞ -category of $\mathbb{T}$ -algebras and to $\mathbf{Alg}_\mathbb{T}(\mathbf{Set})$ as the ∞ -category of 0-truncated $\mathbb{T}$ -algebras.

Remark D.2.5. Notice that evaluation at $\mathbb{A}_k^1$ gives an equivalence between 0-truncated $\mathbf{Com}_k$ algebras and the usual category $\mathbf{CRing}_k$ of commutative k -algebras. Thus, we recover classical theory.

Proposition D.2.6. *If $\mathcal{C}$ be a presentable ∞ -category, then $\mathbf{Alg}_\mathbb{T}(\mathcal{C})$ presentable.*

Since any co-representable functor preserve finite products the Yoneda embedding factors through

$$j : \mathbb{T} \rightarrow \mathbf{Alg}_\mathbb{T}(\mathbf{Spc})^{op}.$$

We now present an adaption of Propositions from section D.1.2 to the context of a Lawvere theory.

Theorem D.2.7 ([37], Proposition 5.5.8.10). *Let $\mathbb{T}$ be an algebraic theory. Then the following hold:*

1. $\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})$ is a localization of $\mathbf{Fun}(\mathbb{T}, \mathbf{Spc})$
2. $\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})$ is compactly generated. In particular

$$\mathrm{Ind}(\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{fp}) \simeq \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})$$

3. $j : \mathbb{T} \rightarrow \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{op}$ preserves finite product.

Theorem D.2.8 ([37], Proposition 5.5.8.15). *Let $\mathcal{C}$ be an ∞ -category with sifted limits. Then, composition with j gives an equivalence of ∞ -categories:*

$$\mathbf{Fun}^{sift}(\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{op}, \mathcal{C}) \xrightarrow{\sim} \mathbf{Fun}(\mathbb{T}, \mathcal{C})$$

between the ∞ -category of functor $\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{op} \rightarrow \mathcal{C}$ which preserves sifted limits, to the ∞ -category $\mathbf{Fun}(\mathbb{T}, \mathcal{C})$. If $\mathcal{C}$ is complete it also induces an equivalence of ∞ -categories:

$$\mathbf{Fun}^R(\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{op}, \mathcal{C}) \xrightarrow{\sim} \mathbf{Fun}_{\Pi}(\mathbb{T}, \mathcal{C})$$

between the ∞ -category of functors $\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{op} \rightarrow \mathcal{C}$ which preserves small limits, to the ∞ -category of $\mathbb{T}$ -algebras in $\mathcal{C}$. The inverse of both equivalences is given by right Kan extension.

Proof. Let $(\mathbb{T}')^{op}$ denote the essential image of $\mathbb{T}$ under the functor $j^{fp} : \mathbb{T} \rightarrow \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{op}$. Observe that, since $\mathbb{T}$ is closed under products, it follows that $\mathbb{T}'$ is closed under coproducts. By definition, we have:

$$\mathcal{P}_{\Sigma}(\mathbb{T}') = \mathbf{Fun}_{\Pi}((\mathbb{T}')^{op}, \mathbf{Spc}) \simeq \mathbf{Fun}_{\Pi}(\mathbb{T}, \mathbf{Spc}) = \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc}).$$

In particular, we can apply Proposition D.1.17:

$$\mathbf{Fun}_{sifted}(\mathcal{P}_{\Sigma}(\mathbb{T}'), \mathcal{C}^{op}) \rightarrow \mathbf{Fun}(\mathbb{T}', \mathcal{C}^{op}).$$

Applying the opposite functor $(-)^{op}$ yields the statement above. $\square$

Corollary D.2.9 (cf. [10, Corollary 2.16]). *Let $\mathbb{T}$ be Lawvere theory. A functor $F : \mathbb{T} \rightarrow \mathbf{Spc}$ is a $\mathbb{T}$ -algebra if and only if F is a sifted colimit of corepresentable functors :*

$$F \simeq \mathrm{colim}_{\alpha} j(t_{\alpha})$$

Lemma D.2.10 ([12] Lemma 3.13). *Let $\mathcal{A} \xrightarrow{j} \mathbb{B} \xleftarrow{\omega} \mathcal{C}$ be functors of ∞ -categories with ω fully faithful and $\mathbb{B}$ small, and let $F : \mathcal{A} \rightarrow \mathcal{D}$ be an arbitrary functor with $\mathcal{D}$ cocomplete. Then if the right kan extension $\mathrm{Ran}_{(\omega \circ j)} F$ exists, so does $\mathrm{Ran}_j(F)$ and moreover: $\mathrm{Ran}_j(F) \simeq \mathrm{Ran}_{(\omega \circ j)} F \circ \omega$.*

Remark D.2.11 (cf. [37, Remark 5.5.8.26]). The essential image $\tau_{\leq n}(\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc}))$ of the n^{th} truncation functor

$$\tau_{\leq n} : \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc}) \rightarrow \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})$$

is precisely the full subcategory of functors $\mathbb{T} \rightarrow \mathbf{Spc}$ whose values are n -truncated objects. As a consequence, if $\mathbb{T}$ is a 1-category, then $\tau_{\leq 0}(\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc}))$ may be identified with $\mathbf{Alg}_{\mathbb{T}}(\mathbf{Set})$; moreover we get a fully faithful inclusion $\mathbf{Alg}_{\mathbb{T}}(\mathbf{Set}) \hookrightarrow \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})$.

We now see how to associate to a $\mathbb{T}$ -algebra A a collection of homotopy sets. Let r be the generator of $\mathbb{T}$, then the evaluation at r gives us a functor $\theta_r : \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc}) \rightarrow \mathbf{Spc}$. By abuse of notation, we will use the space $\theta_r(A)$ to denote the $\mathbb{T}$ -algebra A .

Definition D.2.12. Let $\mathbb{T}$ be an algebraic theory and let A be a $\mathbb{T}$ -algebra. We define the n^{th} -homotopy set $\pi_n(A)$ of A as the n^{th} -homotopy set $\pi_n(\theta_r(A))$ of $\theta_r(A)$.

Remark D.2.13 (cf. [12, Remark 3.17]). For a $\mathbb{T}$ -algebra A the homotopy set $\pi_n(A)$ is an abelian group for $n \geq 1$. By Remark D.2.11 the set $\pi_0(A)$ can be identified with $\tau_{\leq 0}A(r)$ and if $\mathbb{T}$ is a 1-category, the set $\pi_0(A)$ carries the structure of an ordinary $\mathbb{T}$ -algebra.

Remark D.2.14. From the generating properties of the object $r \in \mathbb{T}$, we deduce immediately that the functor:

$$\theta_r : \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc}) \rightarrow \mathbf{Spc}$$

is conservative.

Lemma D.2.15 (cf. [37, Corollary 7.2.1.15]). *Let $f : X \rightarrow Y$ be a morphism of spaces. Then f is an effective epimorphism if and only if the induced map $\pi_0 X \rightarrow \pi_0 Y$ is surjective.*

Corollary D.2.16. *Let $\mathbb{T}$ be a Lawvere theory and let $f : A \rightarrow B$ be a morphism of $\mathbb{T}$ -algebras. Then f is an effective epimorphism if and only if the induced map $\pi_0 A \rightarrow \pi_0 B$ is surjective.*

Proof. This is a combination of Remark D.2.14, Lemma D.2.15 and the fact that the evaluation preserves limits and geometric realizations. $\square$

Definition D.2.17. Let $\mathbb{T}$ be an algebraic theory. We say that:

- A $\mathbb{T}$ -algebra $X \in \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})$ is *finitely generated* if the functor $\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc}) \rightarrow \mathbf{Spc}$ corepresented by X preserves small filtered colimit consisting only of monomorphisms. Denote the full subcategory of finitely generated object by $\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{\text{fg}}$.
- A $\mathbb{T}$ -algebra $X \in \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})$ is *finitely presented* if the functor $\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc}) \rightarrow \mathbf{Spc}$ corepresented by X preserves small filtered colimit consisting, that is if X is compact. Denote the full subcategory of finitely presented object by $\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{\text{fp}}$.
- A $\mathbb{T}$ -algebra is *free* if it is in the essential image of j .

By Yoneda lemma and Theorem D.2.8 it follows that free algebras are finitely presented. In particular the functor j factors as:

$$j^{\text{fp}} : \mathbb{T} \hookrightarrow (\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{\text{fp}})^{\text{op}}$$

We conclude this section by presenting some important properties of finitely generated and finitely presented algebras that will be relevant in the thesis.

Proposition D.2.18 (cf. [12, Proposition 3.20]). *Let A be a $\mathbb{T}$ -algebra. The following are equivalent:*

1. *A is finitely generated.*
2. *There exists an object $t \in \mathbb{T}$ and an effective epimorphism $q : j(t) \rightarrow A$.*

Lemma D.2.19 (Characterization of finitely presented algebras, cf. [12, Lemma 3.21]). *The subcategory $\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{\text{fp}}$ is the smallest subcategory of $\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})$ containing the finitely generated free algebras and closed under finite colimits and retracts. Moreover, any finitely presented algebra is a retract of a finite colimit of finitely generated free algebras.*

Theorem D.2.20 (cf. [12, Theorem 3.22]). *Let $\mathcal{C}$ be any idempotent complete ∞ -category with finite limits. Then composition with j^{fp} induces an equivalence of ∞ -category*

$$\mathbf{Fun}^{\text{lex}} \left((\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{\text{fp}})^{\text{op}}, \mathcal{C} \right) \xrightarrow{\sim} \mathbf{Alg}_{\mathbb{T}}(\mathcal{C})$$

whose inverse is given by right Kan extension.

We now present a pair of adjoint functors relating $\mathbb{T}$ -algebras to their truncations.

Construction D.2.21 (cf. [12, below Remark 3.26]). Fix an integer $n \geq 0$. Recall that the inclusion $\tau_{\leq n} \mathbf{Spc} \hookrightarrow \mathbf{Spc}$ admits a left adjoint, which we denote by $\tau_{\leq n}$. By [37, Lemma 6.5.1.2], the functor $\tau_{\leq n}$ preserves finite products, thereby inducing a left adjoint to the inclusion:

$$\mathbf{Alg}_{\mathbb{T}}(\mathbf{Set}) \hookrightarrow \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc}).$$

By abuse of notation, we will denote this functor again by $\tau_{\leq n}$. When $n = 0$, this is the functor:

$$\pi_0 : \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc}) \rightarrow \mathbf{Alg}_{\mathbb{T}}(\mathbf{Set}).$$

Proposition D.2.22 (cf. [12, Proposition 3.29]). *Let $\mathbb{T}$ be an $(n+1)$ -categorical algebraic theory and $\mathcal{C}$ an $(n+1)$ -category with finite limits. Then the induced functor*

$$(\tau_{\leq n})^* : \mathbf{Fun}^{\text{lex}} \left((\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{\text{fp}})^{\text{op}}, \mathcal{C} \right) \rightarrow \mathbf{Fun}^{\text{lex}} \left((\mathbf{Alg}_{\mathbb{T}}(\tau_{\leq n} \mathbf{Spc})^{\text{fp}})^{\text{op}}, \mathcal{C} \right)$$

is an equivalence of $(n+1)$ -categories

D.2.2 Induced adjunctions

Notice that if f is a morphism of algebraic theories, and $A : \mathbb{T}' \rightarrow \mathcal{C}$ is a $\mathbb{T}'$ -algebra in $\mathcal{C}$, then the composition

$$\mathbb{T} \xrightarrow{f} \mathbb{T}' \xrightarrow{A} \mathcal{C}$$

is a $\mathbb{T}$ -algebra in $\mathcal{C}$, which we denote by f^*A . This induces a functor:

$$f^* : \text{Alg}_{\mathbb{T}'}(\mathcal{C}) \rightarrow \text{Alg}_{\mathbb{T}}(\mathcal{C}).$$

Example D.2.23. We have an obvious functor $\mathbf{Com}_{\mathbb{R}} \rightarrow \mathbf{C}^{\infty}$ that is a morphism of algebraic theory. As a consequence we get an induced functor: $\mathbf{Alg}_{\mathbf{C}^{\infty}}(\mathbf{Set}) \rightarrow \mathbf{Alg}_{\mathbf{Com}_{\mathbb{R}}}(\mathbf{Set})$ called *underlying $\mathbb{R}$ -algebra*.

Lemma D.2.24 (cf. [10, Lemma 2.23]). *Let $\mathcal{C}$ be an ∞ -category in which sifted colimits commute with finite products, and let $f : \mathbb{T} \rightarrow \mathbb{T}'$ be a morphism of algebraic theories. Then*

$$f^* : \text{Alg}_{\mathbb{T}'}(\mathcal{C}) \rightarrow \text{Alg}_{\mathbb{T}}(\mathcal{C})$$

preserves limits and sifted colimits.

Corollary D.2.25 (cf. [10, Corollary 2.24]). *If $\mathcal{C}$ is a presentable ∞ -category in which sifted colimits commute with finite limits, then f^* has a left adjoint*

$$f_! : \text{Alg}_{\mathbb{T}}(\mathcal{C}) \rightarrow \text{Alg}_{\mathbb{T}'}(\mathcal{C}).$$

D.2.3 Unramified transformation

We now turn to briefly introducing one of the most significant tools in the theory, one that is heavily exploited in [12, Section 4]. It is admittedly a bit unfortunate that we will not present this part in full detail, as it plays a central role in much of the work developed there. Nevertheless, we encourage the interested reader to consult the original article to appreciate its full depth and relevance. Our reason for including these ideas here is to enable their use in some key proofs that we present in the thesis. The relevant background material can be found in [12, Section 3.2].

Unramified transformations are of particular interest because they characterize when a morphism of algebraic theories $f : \mathbb{T} \rightarrow \mathbb{T}'$ induces a pullback functor f^* that preserves certain classes of pushouts.

Definition D.2.26. Let $\mathbb{T}$ and $\mathbb{T}'$ be algebraic theories. A morphism of algebraic theories $f : \mathbb{T} \rightarrow \mathbb{T}'$ is *unramified* if for each morphism $g : X \rightarrow Y$ in $\mathbb{T}'$ and each $Z \in \mathbb{T}'$, the diagram

$$\begin{array}{ccc} f^*j(X \times Y) & \xrightarrow{f^*j(\text{pr}_{X \times Y})} & f^*j(X \times Y \times Z) \\ \downarrow f^*j((\text{id}_X, g)) & & \downarrow f^*j((\text{id}_X, g \times \text{id}_Z)) \\ f^*j(X) & \xrightarrow{f^*j(\text{pr}_X)} & f^*j(X \times Z) \end{array}$$

is a pushout in $\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})$.

The following is essentially the reason why we are interested in this type of transformations.

Proposition D.2.27 (cf. [12, Theorem 3.32]). *Let $f : \mathbb{T} \rightarrow \mathbb{T}'$ be an unramified transformation of algebraic theories. Then $f^* : \mathbf{Alg}_{\mathbb{T}'}(\mathbf{Spc}) \rightarrow \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})$ preserves pushouts along effective epimorphism.*

D.2.4 Universal $\mathbb{T}$ -algebra

This subsection introduces universal $\mathbb{T}$ -algebras, a tool that will be needed in Chapter 2. Our presentation follows [12, Section 3.3].

Construction D.2.28. Let $(\mathcal{C}, S)$ be a pair with $\mathcal{C}$ an ∞ -category, and S a $\mathbb{T}$ -algebra in $\mathcal{C}$. Call

$$\tilde{\Gamma}_S : \mathcal{C}^{op} \rightarrow \text{Fun}(\mathbb{T}, \mathbf{Spc})$$

the currying of the following composite:

$$\mathcal{C}^{op} \times \mathbb{T} \xrightarrow{id_{\mathcal{C}^{op}} \times S} \mathcal{C}^{op} \times \mathcal{C} \xrightarrow{\text{Hom}_{\mathcal{C}}(-, -)} \mathbf{Spc}.$$

Given an object C of $\mathcal{C}$, the functor

$$\text{Hom}_{\mathcal{C}}(C, S(-)) : \mathbb{T} \rightarrow \mathbf{Spc}$$

preserves finite products, and hence is a $\mathbb{T}$ -algebra. This means that the functor $\tilde{\Gamma}_S$ factors through $\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})$. We now call Γ_S the opposite of this factorization, in particular:

$$\Gamma_S : \mathcal{C} \rightarrow \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{op}.$$

If in addition $\mathcal{C}$ is idempotent complete and has finite limits, by Theorem D.2.20, we get left exact functor

$$\text{Ran}_{j^{\text{fp}}} S : (\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{\text{fp}})^{op} \rightarrow \mathcal{C}$$

Through right Kan extension. Moreover, there is a canonical equivalence

$$S \simeq (j^{\text{fp}})^* \circ \text{Ran}_{j^{\text{fp}}} S.$$

The functor $\text{Ran}_{j^{\text{fp}}} S$ canonically induces a natural transformation

$$\eta : \omega \Rightarrow \Gamma_S \circ \text{Ran}_{j^{\text{fp}}} S,$$

where

$$(\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{\text{fp}})^{op} \xrightarrow{\omega} (\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc}))^{op}$$

is the canonical inclusion. More precisely by the Yoneda lemma, for all finitely presented $\mathbb{T}$ -algebras A ,

$$A \simeq \text{Hom}_{\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})}(j(-), A) \simeq \text{Hom}_{(\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{\text{fp}})^{op}}(A, j^{\text{fp}}(-)),$$

and

$$\Gamma_S \circ \text{Ran}_{j^{\text{fp}}} S(A) \simeq \text{Hom}_{\mathcal{C}}(\text{Ran}_{j^{\text{fp}}} S(A), \text{Ran}_{j^{\text{fp}}} S(j^{\text{fp}}(-))) ,$$

and under these equivalences, η is just the map on mapping spaces induced by $\text{Ran}_{j^{\text{fp}}} S$.

Definition D.2.29. We say that a $\mathbb{T}$ -algebra S in $\mathcal{C}$ is:

- *versal* if $\Gamma_S \circ \text{Ran}_{j^{\text{fp}}} S$ is left exact and the morphism $\eta_S(r) : \omega(r) \rightarrow \Gamma_S S(r)$ is an equivalence, if r is the generator of $\mathbb{T}$.
- *generating* if every object in $\mathcal{C}$ is a retract of a finite limit of $S(r)$.
- *universal* if and only if it is both versal and generating.

Proposition D.2.30 (cf. [12, Proposition 3.43]). *The following are equivalent for a $\mathbb{T}$ -algebra S in $\mathcal{C}$:*

1. *S is versal,*
2. *$\eta_S : \omega \Rightarrow \Gamma_S \circ \text{Ran}_{j^{\text{fp}}} S$ is an equivalence.*

Theorem D.2.31 (cf. [12, Theorem 3.44]). *Let S be a $\mathbb{T}$ -algebra in $\mathcal{C}$. Then*

$$\text{Ran}_{j^{\text{fp}}} S : (\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{\text{fp}})^{op} \rightarrow \mathcal{C}$$

is fully faithful if and only if S is versal. Moreover, $\text{Ran}_{j^{\text{fp}}} S$ is an equivalence if and only if S is universal.

Remark D.2.32. Suppose that S is universal. Then it follows, in particular, that Γ_S restricts to a functor

$$\Gamma_S : \mathcal{C} \rightarrow (\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{\text{fp}})^{op}$$

inverse to $\text{Ran}_{j^{\text{fp}}} S$.

D.3 Modules

The discussion in this section closely follows the lecture notes of Bastiaan Cnossen for the Stable Homotopy Theory course [17]. A general reference for the results is [40].

Definition D.3.1 (Commutative ring spectrum). A *commutative ring spectrum* R is a commutative algebra in the symmetric monoidal ∞ -category $(\mathbf{Spt}, \otimes, \mathbb{S})$. We denote by

$$\mathbf{CAlg} := \mathbf{CAlg}(\mathbf{Spt})$$

the ∞ -category of *commutative ring spectra*. We denote by

$$\mathbf{dMod} := \mathbf{Mod}(\mathbf{Spt}).$$

Given a commutative ring spectrum R , we define the ∞ -category of R -modules as

$$\mathbf{dMod}_R := \mathbf{Mod}_R(\mathbf{Spt}).$$

Theorem D.3.2 (Recognition principle). *The functor $\Omega^\infty : \mathbf{Spt} \rightarrow \mathbf{Spc}$ uniquely lifts to a functor $\Omega^\infty : \mathbf{Spt} \rightarrow \mathbf{CGrp}(\mathbf{Spc})$. This functor admits a fully faithful left adjoint*

$$\mathbb{B}^\infty : \mathbf{CGrp}(\mathbf{Spc}) \hookrightarrow \mathbf{Spt}$$

whose image is precisely the subcategory $\mathbf{Spt}_{\geq 0}$ of *connective spectra*.

Remark D.3.3. Let $\mathcal{C}$ be an algebraic category and let $\mathcal{F}_{\mathcal{C}}$ be the subcategory of generators. Thanks to the above result and the fact that we have

$$\mathbf{Fun}_{\Pi}(\mathcal{F}_{\mathcal{C}}^{\mathrm{op}}, \mathbf{Spc}) \simeq \mathbf{Fun}_{\Pi}(\mathcal{F}_{\mathcal{C}}^{\mathrm{op}}, \mathbf{CGrp}(\mathbf{Spc})) \simeq \mathbf{Fun}_{\Pi}(\mathcal{F}_{\mathcal{C}}^{\mathrm{op}}, \mathbf{Spt}_{\geq 0}),$$

thus we recover the definition of animation given in [30].

Definition D.3.4. Let R be a commutative ring seen as an element of $\mathbf{CGrp}(\mathbf{Spc})$ through $\mathbf{Ab} = \mathbf{CGrp}(\mathbf{Set}) \hookrightarrow \mathbf{CGrp}(\mathbf{Spc})$. Then we define the *Eilenberg-MacLane spectrum* of R to be $HR := \mathbb{B}^\infty R$.

Proposition D.3.5. *There is a fully faithful inclusion $H : \mathbf{CRing} \hookrightarrow \mathbf{CAlg}$ sending each commutative ring R to its Eilenberg-MacLane spectrum HR .*

Theorem D.3.6 ([40], Theorem 7.1.2.13). *Let R be a commutative ring. Then there is a canonical equivalence of symmetric monoidal-categories $\mathbf{dMod}_{HR} \cong \mathcal{D}(R)$.*

Remark D.3.7 (Putting everything together). We have encountered various constructions and notions of derived categories and algebraic structures. In this remark, we explain how they are related in the case of a commutative ring R .

By Remark D.1.14, there is an equivalence between the animation of $\mathbf{Mod}_R$ and the connective derived category:

$$\mathbf{aMod}_R \simeq \mathcal{D}_{\geq 0}^-(R).$$

Let $\mathcal{F}_R \subseteq \mathbf{Mod}_R$ denote the full subcategory spanned by finitely generated free R -modules. Then [40, Remark 1.4.2.9] provides an equivalence

$$\mathbf{Stab}(\mathbf{Fun}(\mathcal{F}_R^{\mathrm{op}}, \mathbf{Spc})) \simeq \mathbf{Fun}(\mathcal{F}_R^{\mathrm{op}}, \mathbf{Spt}),$$

which restricts to an equivalence between the full subcategories of product-preserving functors. This yields the following chain of equivalences:

$$\mathcal{D}(R) \simeq \mathbf{Stab}(\mathcal{D}_{\geq 0}^-(R)) \simeq \mathbf{Stab}(\mathbf{aMod}_R) \simeq \mathbf{Fun}_{\Pi}(\mathcal{F}_R^{\mathrm{op}}, \mathbf{Spt}).$$

Finally, by Theorem D.3.6, we have

$$\mathbf{dMod}_{HR} \simeq \mathcal{D}(R).$$

Each of these notions has its own advantages. Throughout this thesis, we will freely pass from one perspective to the other, depending on the context.

Appendix E

Groupoids, Lie Groupoids and Lie algebroids

Groupoids and Lie groupoids play a central role throughout this thesis, particularly in the context of stacks and derived stacks. For this reason, we include here a brief recollection of the relevant definitions and structures. We also provide a short discussion of Lie algebroids. For a comprehensive treatment of the subject, see [42] and [43].

Our exposition is inspired by Appendix E.2 and Appendix D of [16], as well as Section 4.1 of [30].

E.1 Groupoid objects

Definition E.1.1. A *groupoid object* in an ∞ -category $\mathcal{C}$ is a functor $\mathcal{G}_\bullet : \Delta^{\text{op}} \rightarrow \mathcal{C}$ such that for all $[n] \in \Delta$ and all partitions $[n] = S \cup S'$ into subsets S and S' such that $S \cap S'$ consist of one element $m \in [n]$, the square

$$\begin{array}{ccc} G_n = \mathcal{G}_\bullet([n]) & \longrightarrow & \mathcal{G}_\bullet(S') \\ \downarrow & & \downarrow \\ \mathcal{G}_\bullet(S) & \longrightarrow & \mathcal{G}_\bullet(m) \end{array}$$

is cartesian. Here $S, S', m \in \Delta$ with the ordering induced by $[n]$ and the maps in the diagram are given by restrictions along the inclusions. We will denote by $\mathbf{Grpd}(\mathcal{C}) \subseteq \text{Fun}(\Delta^{\text{op}}, \mathcal{C})$ the full subcategory spanned by groupoid object.

Remark E.1.2. Given a groupoid object $\mathcal{G}_\bullet$ in a ∞ -category $\mathcal{C}$, we may think of $\mathcal{G}_0$ as the *objects* of $\mathcal{G}_\bullet$ and $\mathcal{G}_1$ as the object of *morphisms* of $\mathcal{G}_\bullet$. More precisely, the maps $\mathcal{G}(d_1), \mathcal{G}(d_0) : \mathcal{G}_1 \rightarrow \mathcal{G}_0$ may be regarded as the *source* and *target*, respectively. Then, using the defining condition of a groupoid object, we can construct a *composition* as follows. Consider the partition $[2] = \{0, 1\} \cup \{1, 2\}$, then we find an equivalence $(\mathcal{G}(d_0), \mathcal{G}(d_1)) : \mathcal{G}_2 \xrightarrow{\sim} \mathcal{G}_1 \times_{\mathcal{G}_0} \mathcal{G}_1$, which allow us to construct a composition via the map:

$$\mathcal{G}_1 \times_{\mathcal{G}_0} \mathcal{G}_1 \simeq \mathcal{G}_2 \xrightarrow{\mathcal{G}(d_1)} \mathcal{G}_1.$$

We may also construct *inverses* by taking the partitions: $[2] = \{0, 1\} \cup \{0, 2\}$ and $\{0, 2\} \cup \{1, 2\}$. Finally the condition for all other n may be thought as witnessing higher coherence data among compositions and inverses.

Definition E.1.3 (cf. [30, Definition 4.2.1]). A *group object* in an ∞ -category $\mathcal{C}$ is a groupoid object $\mathcal{G}_\bullet$ such that $\mathcal{G}_0 \in \mathcal{C}$ is terminal. Given a group object $\mathcal{G}_\bullet$, we will often abuse notation by saying that $G := \mathcal{G}_1 \in \mathcal{C}$ is a group object.

We now aim to establish a connection between the classical notion of a 1-groupoid and the higher-categorical notion of a groupoid object.

Definition E.1.4 (Nerve of a groupoid). Let $\mathcal{G} = (\mathcal{G}_1 \rightrightarrows \mathcal{G}_0)$ be a 1-groupoid; The *nerve* $N\mathcal{G}$ of $\mathcal{G}$ is defined as the simplicial set $N\mathcal{G} : \Delta^{\text{op}} \rightarrow \mathbf{Set}$ whose n -simplices are given by the sets

$$\mathcal{G}_n := \mathcal{G}_1 \times_{s, \mathcal{G}_0, t} \cdots \times_{s, \mathcal{G}_0, t} \mathcal{G}_1$$

which arises as the n -fold fibred product of $\mathcal{G}_1$ over $\mathcal{G}_0$ in **Set**. The face maps of $N\mathcal{G}$ are obtained from source, target and multiplication maps of $\mathcal{G}$, while the degeneracy maps of $N\mathcal{G}$ are obtained from the unit map of $\mathcal{G}$.

The above construction defines a groupoid object in **Set**. In fact, the nerve functor $N(-)$ induces an equivalence between the 1-category of 1-groupoids and the ∞ -category of groupoid objects in **Set**. Moreover, this equivalence restricts to an equivalence between group objects on both sides.

Another way to produce groupoid objects is through the Čech nerve construction.

Construction E.1.5 (Čech nerve). Denote by Δ_+ the augmented simplex category and by $\Delta_+^{\leq 0}$ the full subcategory of Δ_+ spanned by $[-1]$ and $[0]$. Let $\mathcal{C}$ be an ∞ -category and let $f : X \rightarrow Y$ be a morphism in $\mathcal{C}$ viewed as a functor $f : \Delta_+^{\leq 0} \rightarrow \mathcal{C}$. Suppose that the right Kan extension $\text{Ran}_\iota(f)$ of f along the inclusion $\iota : \Delta_+^{\leq 0} \hookrightarrow \Delta_+$ exists.

Definition E.1.6. In the situation of the above construction we define the *Čech nerve* $\check{C}(f)$ of f as the restriction of $\text{Ran}_\iota(f)$ to Δ^{op} .

Informally one can describe the n -simplex of $\check{C}(f)$ as given by:

$$\check{C}(f)_n = X \times_Y X \times_Y \cdots \times_Y X$$

the n -fold product of X over Y

Lemma E.1.7 ([37] Proposition 6.1.2.11). *Let $f : \mathcal{X} \rightarrow \mathcal{Y}$ be a morphism in an ∞ -category $\mathcal{C}$. If the Čech nerve $\check{C}(f)$ is defined then it is a groupoid object in $\mathcal{C}$.*

Definition E.1.8 (Geometric realization). Given a simplicial diagram $\mathcal{G}_\bullet : \Delta^{\text{op}} \rightarrow \mathcal{C}$, we will refer to its colimit as its *geometric realization*, and denote it by:

$$|\mathcal{G}_\bullet| := \text{colim}_{n \in \Delta^{\text{op}}} \mathcal{G}_n$$

when it exists.

Definition E.1.9 (Effective epimorphism). A morphism $f : X \rightarrow Y$ in $\mathcal{C}$ is an *effective epimorphism* if the iterated fibred products $X \times_Y X \times_Y \cdots \times_Y X$ all exist in $\mathcal{C}$, and the augmented simplicial diagram:

$$\cdots \rightrightarrows X \times_Y X \times_Y X \rightrightarrows X \times_Y X \rightrightarrows X \longrightarrow Y$$

is a colimit diagram.

Remark E.1.10. We can think of effective epimorphisms as a generalization of the notion of covering. Indeed, given an open covering $\{U_i\}_{i \in I}$ of a topological space X , the canonical morphism:

$$\coprod U_i \rightarrow X$$

is effective, since we have the equivalence

$$X \simeq \text{coeq} \left(\coprod_{i,j} (U_i \times_X U_j) \rightrightarrows \coprod U_i \right)$$

Now consider a groupoid object $\mathcal{G}$ in an ∞ -category $\mathcal{C}$. We have a map $\mathcal{G} \rightarrow |\mathcal{G}|$ encoded by a colimit diagram $\Delta_+^{\text{op}} \rightarrow \mathcal{C}$ which in turns gives a map $\mathcal{G} \rightarrow \check{C}(\mathcal{G}_0 \rightarrow |\mathcal{G}|)$.

Definition E.1.11 (effective groupoid). A groupoid object $\mathcal{G}$ in an ∞ -category $\mathcal{C}$ is *effective* if the map

$$\mathcal{G} \rightarrow \check{C}(\mathcal{G}_0 \rightarrow |\mathcal{G}|)$$

constructed above is an equivalence.

E.1.1 Lie groupoids

Definition E.1.12 (Lie groupoid). A *Lie groupoid* $\mathcal{G}$ is a 1-groupoid $(\mathcal{G}_1 \rightrightarrows \mathcal{G}_0)$ in which $\mathcal{G}_1$ and $\mathcal{G}_0$ are smooth manifold, the structure maps $s, t : \mathcal{G}_1 \rightarrow \mathcal{G}_0$, $m : \mathcal{G}_1 \times_{s, \mathcal{G}_0, t} \mathcal{G}_1 \rightarrow \mathcal{G}_1$ and $e : \mathcal{G}_0 \rightarrow \mathcal{G}_1$ are all smooth, and the source and target maps s, t are smooth submersions. We will also say that $\mathcal{G}$ is a groupoid over $\mathcal{G}_0$.

A *morphism of Lie groupoids* $f : \mathcal{G} \rightarrow \mathcal{H}$ is a smooth functor, i.e. a pair of smooth maps $f_0 : \mathcal{G}_0 \rightarrow \mathcal{H}_0$ and $f_1 : \mathcal{G}_1 \rightarrow \mathcal{H}_1$ with the structure maps. We denote by **LieGrpd** the category of Lie groupoids and morphism of Lie groupoids.

We now see how some of the usual construction in differential geometry may be encoded in terms of Lie groupoid.

Example E.1.13 (Action groupoid). Let G be a Lie group acting on a manifold M . We may define the Lie groupoid $G \ltimes M$ as follows:

- $(G \ltimes M)_0 = M$ and $(G \ltimes M)_1 = G \times M$;
- The structure maps are given as follows: source $s(g, x) = x$, target $t(g, x) = g \cdot x$, unit $e(x) = (e, x)$ (where e is the unit of G), inversion $i(g, x) = (g^{-1}, gx)$ and multiplication $m((g', gx), (g, x)) = (g'g, x)$.

We call $G \ltimes M$ the *action groupoid* of M . In the special case $M = \text{pt}$ we call $G \ltimes \text{pt}$ the *classifying groupoid* of G .

The above construction defines a functor $G \ltimes - : \mathbf{Mfd}_G \rightarrow \mathbf{LieGrpd}$, where $\mathbf{Mfd}_G$ denotes the category of smooth G -manifold.

Example E.1.14. (Čech groupoid) Let $p : M \rightarrow N$ be a smooth submersion between smooth manifold. Then its *Čech groupoid* $\check{C}'(p)$ is define as following Lie groupoid:

- $\check{C}'(p)_0 = M$ and $\check{C}'(p)_1 = M \times_N M$;
- The structure maps are given as follows: source $s(x, x') = x$, target $t(x, x') = x'$, unit $e(x) = (x, x)$, inversion $i(x, x') = (x', x)$ and multiplication $m((x, x'), (x', x'')) = (x, x'')$.

When $N = \text{pt}$ is a point, the resulting Lie groupoid $M \times M \rightrightarrows M$ is known as the *pair groupoid* of M . Notice that the nerve $N\check{C}''(p)$ of the Lie groupoid $\check{C}''(p)$ is equivalent to the Čech nerve $\check{C}(p)$ of the morphism p , justifying the name.

Example E.1.15 (Pullback groupoid). Let $\mathcal{G}$ be a Lie groupoid over M , let N be a smooth and let $f : N \rightarrow M$ be a smooth submersion. We define the *pullback groupoid* $f^*\mathcal{G}$ by letting $(f^*\mathcal{G})_0 = N$ and defining:

$$(f^*\mathcal{G})_1 = N \times_M \mathcal{G}_1 \times_M N = \{(x, g, y) \in N \times \mathcal{G}_1 \times N \mid s(g) = f(x), t(g) = f(y)\},$$

with structure maps coming from $\mathcal{G}$. As a special case, we get for every open subspace $U \subseteq M$ a groupoid $\mathcal{G}|_U$ called the *restriction of $\mathcal{G}$ to U* :

$$\mathcal{G}|_U = \left(\{g \in \mathcal{G}_1 \mid s(g), t(g) \in U\} \xrightleftharpoons[t]{s} U \right)$$

Remark E.1.16. In the case of Lie groupoids, the nerve construction factors through the category $\text{Fun}(\Delta^{\text{op}}, \mathbf{Mfd})$. For this to hold, it is crucial that the source and target maps s, t are smooth submersions.

E.2 The Lie algebroid associated to a Lie groupoid

Heuristically, if Lie groupoids may be thought as the *many object version* of Lie groups, then Lie algebroids can be thought as the *many object version* of Lie algebras.

For a definition of a Lie algebra and Lie bracket with refer to [32, below Theorem 8.33]. Given M a smooth manifold we endow $\Gamma(M, TM)$ with the usual Lie bracket $[-, -]$ (see [32, above Lemma 8.25]).

Definition E.2.1. A *Lie algebroid* over a manifold M is a vector bundle $A \rightarrow M$, together with a morphism of vector bundles $\rho : A \rightarrow TM$ called *anchor map*, and a Lie bracket $[-, -]_A$ on the sheaf of sections $\Gamma(M, A)$, such that:

1. ρ commutes with the brackets: for all $X, Y \in \Gamma(M, A)$,

$$\rho[X, Y]_A = [\rho(X), \rho(Y)];$$

2. it satisfies the Leibniz rule: for all $X, Y \in \Gamma(M, A)$, $f \in C^\infty(M)$,

$$[X, f \cdot Y]_A = \rho(X)(f) \cdot Y + f \cdot [X, Y]_A.$$

Example E.2.2 (cf. [43, Proposition 3.2.3 and Remark 3.2.4] and [42, Definition 3.5.8]). Every Lie groupoid $\mathcal{G}_1 \rightrightarrows \mathcal{G}_0$ has an associated Lie algebroid. Explicitly, the underlying vector bundle is given by restricting the vector bundle $\ker ds$ to the unit space $\mathcal{G}_0$. More precisely,

$$A := e^* \ker(ds : T\mathcal{G}_1 \rightarrow s^*T\mathcal{G}_0).$$

The anchor map is induced by the target map t . More precisely:

$$\begin{array}{ccccccc} A & \longrightarrow & Ts & \hookrightarrow & TG & \xrightarrow{dt} & TM \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \mathcal{G}_0 & \xrightarrow{e} & \mathcal{G}_1 & \xlongequal{\quad} & \mathcal{G}_1 & \xrightarrow{t} & \mathcal{G}_0 \end{array}$$

To define the Lie bracket, we sketch the construction and refer to [42] and [43] for details. Given sections of $s, w : \mathcal{G}_0 \rightarrow A$, we may extend them in a unique way as a right-invariant vector fields $\hat{s}$ and $\hat{w}$ over $\mathcal{G}_1$ (see [45, Definition 3.2.2] and [42, above Lemma 3.5.5]). We then define the Lie bracket by

$$[s, w]_A := [\hat{s}, \hat{w}] \circ e.$$

Remark E.2.3. In the above setting, by the definition of A , its dual $A^\vee$ is precisely $e^*T_{\mathcal{G}_1/\mathcal{G}_0}^*$, the pullback along $e : M \rightarrow \mathcal{G}_1$ of the relative cotangent bundle of the submersion $s : \mathcal{G}_1 \rightarrow \mathcal{G}_0$ (see Definition A.2.8). By Proposition A.2.6, the global sections functor commutes with base change, and hence we have:

$$e^*\Omega^1(\mathcal{G}_1/\mathcal{G}_0) = e^*\Gamma(\mathcal{G}_1, T_{\mathcal{G}_1/\mathcal{G}_0}^*) \simeq \Gamma(\mathcal{G}_1, e^*T_{\mathcal{G}_1/\mathcal{G}_0}^*) = \Gamma(\mathcal{G}_1, A^\vee).$$

Appendix F

∞ -topoi

Throughout this thesis, we will encounter many examples of ∞ -topoi, which will essentially serve as the ambient setting for the geometric constructions we develop. Accordingly, the following appendix provides a brief recollection of the theory of ∞ -topoi. For a thorough and comprehensive treatment of the subject, we refer the reader to [37, Chapter 6].

To give an intuition for what an ∞ -topos is, we quote two passages from the introduction of *loc. cit.* The first draws a parallel with ordinary topos theory:

Roughly speaking, an ∞ -topos is an ∞ -category which looks like the ∞ -category of spaces, just as an ordinary topos is a category which looks like the category of sets.

The second establishes a connection with topological spaces and homotopy theory:

There are several different ways of thinking about what an ∞ -topos $\mathcal{X}$ is. On the one hand, we can view $\mathcal{X}$ as a generalized topological space; on the other hand, we can think of $\mathcal{X}$ as an alternative universe in which we can do homotopy theory.

This appendix is organized as follows. In Section F.1, we recall the definition of an ∞ -topos and present its most important characterization. In Section F.2, we examine some properties of groupoids in an ∞ -topos and introduce groupoid actions. Finally, in Section F.3, we discuss sheaf topoi.

The main references for the material in this appendix are [37, Chapter 6] and [54]; however, our exposition primarily follows [16, Appendix E].

F.1 Definition and characterization of ∞ -topoi

We now introduce some basic definitions needed for the theory of ∞ -topoi.

Definition F.1.1. (Cartesian natural transformation) Let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be two functors between ∞ -categories. A natural transformation $\alpha : F \rightarrow G$ is called a *cartesian natural transformation* if, for every morphism $f : A \rightarrow B$ in $\mathcal{C}$, the natural square

$$\begin{array}{ccc} F(A) & \xrightarrow{F(f)} & F(f) \\ \alpha(A) \downarrow & & \downarrow \alpha(B) \\ G(A) & \xrightarrow{G(f)} & G(B) \end{array}$$

is cartesian.

Definition F.1.2 (cf. [37, Definition 6.1.1.2]). Let $\mathcal{C}$ be a presentable ∞ -category. We will say that *colimits in $\mathcal{C}$ are universal* if, for all morphism $f : X \rightarrow Y$ in $\mathcal{C}$ the associated pullback functor:

$$f^* : \mathcal{C}_{/Y} \rightarrow \mathcal{C}_{/X}$$

preserves (small) colimits.

Proposition F.1.3 ([37] proposition 6.1.1.4). *Let $\mathcal{X}$ be an ∞ -category which admits finite limits. The following conditions are equivalent:*

- *The ∞ -category $\mathcal{X}$ is presentable, and colimits in $\mathcal{X}$ are universal.*
- *The Cartesian fibration $ev_1 : \text{Fun}([1], \mathcal{X}) \rightarrow \mathcal{X}$ is classified by a functor $\mathcal{X}_{/-} : \mathcal{X}^{op} \rightarrow \text{Pr}^L$.*

Definition F.1.4 (Disjoint coproducts). Let $\mathcal{C}$ be an ∞ -category finite coproducts and initial object $\emptyset$. We will say that *coproducts in $\mathcal{C}$ are disjoint* if every pushout of the form:

$$\begin{array}{ccc} \emptyset & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X & \longrightarrow & X \sqcup Y \end{array}$$

is also a pullback.

The intuition behind disjoint coproducts is analogous to the situation in the category of sets: given two objects X and Y , their coproduct $X \sqcup Y$ is such that the intersection of X and Y inside $X \sqcup Y$ is empty.

Definition F.1.5 (cf. [37, Definition 6.1.0.4.]). An ∞ -category $\mathcal{B}$ is called an ∞ -topos if it is equivalent to an accessible left exact Bousfield localization of a presheaf category $\text{PSh}(\mathcal{C})$, for some small ∞ -category $\mathcal{C}$.

Definition F.1.6 (cf. [37, Definition 6.3.1.1.]). A functor $\psi_* : \mathcal{B} \rightarrow \mathcal{B}'$ between ∞ -topos is a *geometric morphism* if it admits a left exact left adjoint $\psi^* : \mathcal{B}' \rightarrow \mathcal{B}$.

Theorem F.1.7 (cf. [37, theorem 6.1.0.6]). *An ∞ -category $\mathcal{B}$ is an ∞ -topos if and only if the following holds:*

1. *The ∞ -category $\mathcal{B}$ is presentable*
2. *Colimits in $\mathcal{B}$ are universal ;*
3. *Coproducts in $\mathcal{B}$ are disjoint;*
4. *Every groupoid object of $\mathcal{B}$ is effective.*

Theorem F.1.8 (Descent for ∞ -topoi [37, Theorem 6.1.3.9]). *Let $\mathcal{X}$ be a presentable ∞ -category. The following conditions are equivalent:*

1. *The Cartesian fibration $ev_1 : \text{Fun}([1], \mathcal{X}) \rightarrow \mathcal{X}$ is classified by a limit-preserving functor $\mathcal{X}^{op} \rightarrow \text{Pr}^L$.*
2. *Let K be a small simplicial set and $\bar{\alpha} : \bar{p} \rightarrow \bar{q}$ a natural transformation of diagrams $\bar{p}, \bar{q} : K \rightarrow \mathcal{X}$. Suppose that $\bar{q}$ is a colimit diagram, and that $\alpha = \bar{\alpha}|_K$ is a Cartesian transformation. Then $\bar{p}$ is a colimit diagram if and only if $\bar{\alpha}$ is a cartesian natural transformation*

Remark F.1.9. By [37, Proposition 6.1.3.10] the above statement holds in any ∞ -topos.

F.2 Groupoids in ∞ -topoi

We now delve deeper into the study of groupoids (see Appendix E) in the context of ∞ -topos. In particular we will see how groupoids correspond to effective epimorphisms. We then give a brief description of groupoid actions.

F.2.1 Groupoids and effective epimorphism in ∞ -topoi

Definition F.2.1 (Atlas). Let $\mathcal{B}$ be an ∞ -topos and let $f : X \rightarrow Y$ be a morphism in $\mathcal{B}$. If f is an effective epimorphism we will say that f exhibit X as an *atlas* for Y . We write $\text{Atl}(\mathcal{B}) \subseteq \text{Fun}(\Delta^1, \mathcal{B})$ for the full subcategory spanned by the effective epimorphisms.

Definition F.2.2. Let $\mathcal{G}$ be a groupoid in an ∞ -topos $\mathcal{B}$, we will call the geometric realization $|\mathcal{G}|$ of the groupoid *classifying stack* of $\mathcal{G}$ and denote it by $\mathbb{B}\mathcal{G}$.

The change of notation is justified by the fact that in the analogy in which ∞ -topos are sort of generalized topological spaces we would like to think of classifying stack as the analogue of classifying space in usual algebraic topology.

Proposition F.2.3 ([37, Discussion below Corollary 6.2.3.5]). *Let $\mathcal{B}$ be an ∞ -topos. Then the Čech nerve construction restricts to an equivalence*

$$\check{C} : \text{Alt}(\mathcal{B}) \xrightarrow{\sim} \text{Grpd}$$

with inverse given by sending a groupoid $\mathcal{G}$ to $\mathcal{G}_0 \rightarrow \mathbb{B}\mathcal{G}$

The above proposition may be seen as a generalized version of May's recognition principle. In particular, in the special case $\mathcal{G}_0 \simeq *$ and $\mathcal{B} = \mathbf{Spc}$, we recover the classical statement.

Remark F.2.4. An immediate consequence of the above proposition is that

$$\mathcal{G}_n \simeq \mathcal{G}_0^{\times_{\mathbb{B}\mathcal{G}}^n}.$$

In particular, if $\mathcal{G}$ is a group object, i.e. $\mathcal{G}_0 \simeq *$, then we have that $\Omega\mathbb{B}\mathcal{G} \simeq \mathcal{G}_1$.

Corollary F.2.5. *The structure map $\mathcal{G}_0 \rightarrow \mathbb{B}\mathcal{G}$ of the colimit is an effective epimorphism.*

Lemma F.2.6 (cf. [16, Lemma E.16]). *Let $\alpha : \mathcal{G} \rightarrow \mathcal{H}$ morphism of groupoids in an ∞ -topos $\mathcal{B}$, and let*

$$\begin{array}{ccc} \mathcal{G}_0 & \xrightarrow{\alpha} & \mathcal{H}_0 \\ \downarrow & & \downarrow \\ |\mathcal{G}| & \xrightarrow{|\alpha|} & |\mathcal{H}| \end{array}$$

be the map in $\text{Atl}(\mathcal{B})$ induced by the equivalence in F.2.3. Then α is a Cartesian natural transformation of simplicial object in $\mathcal{B}$ if and only if the above square is a pullback.

Proposition F.2.7 ([37] Proposition 6.2.3.15). *Let $\mathcal{X}$ be an ∞ -topos. Effective epimorphisms are stable under base change.*

F.2.2 Groupoids actions

We now introduce groupoid actions. Just as groupoids generalize groups by allowing for morphisms between objects, groupoid actions provide a corresponding generalization of group actions.

One of the main issues with classical quotients is that they tend to forget about the structure of the action, retaining only the equivalence classes, without recording how the objects are related. This results in a mismatch between reasoning at the level of the quotient and reasoning equivariantly. A key advantage of the theory we are about to introduce is that it resolves this issue. In particular, points in the quotient are allowed to carry internal symmetries inherited from the action. As expected, in the special case of a free actions, the homotopical quotient coincides with the ordinary quotient of the 0-truncation, as there are no internal symmetries. This is the content of Proposition F.2.15. For a more geometric perspective on this phenomenon, see the introduction to the thesis and Section 1.2. The following discussion closely follows [16, Section E.3] and [30, Section 4.2].

Action by groupoid objects

Definition F.2.8. Let $\mathcal{B}$ be an ∞ -topos, $\mathcal{G}$ a groupoid in $\mathcal{B}$ and consider an object $X \in \mathcal{B}$. An *action* of $\mathcal{G}$ on X is given by the following data:

- A groupoid object $\mathcal{G} \ltimes X$ in $\mathcal{B}$;
- an equivalence $(\mathcal{G} \ltimes X)_0 \simeq X$;
- a cartesian natural transformation $c : \mathcal{G} \ltimes X \rightarrow \mathcal{G}$ of simplicial objects in $\mathcal{B}$.

We let $\text{Act}_{\mathcal{G}}(\mathcal{B}) \subseteq \mathbf{Grpd}(\mathcal{B})_{/\mathcal{G}}$ denote the full subcategory of $\mathcal{G}$ -actions in $\mathcal{B}$.

Remark F.2.9 (cf. [30, Remark 4.2.4]). Let us unravel the above definition in the case of a group object $G_\bullet$. (See Definition E.1.3) acting on an object X . The third condition in definition F.2.8 implies that we have equivalences:

$$(G \ltimes X)_n \simeq G_n \times_{G_0} X \simeq G^{\times n} \times X.$$

Thus the simplicial diagram $G \ltimes X$ takes the form:

$$\cdots \rightrightarrows G \times G \times X \rightrightarrows G \times X \rightrightarrows X$$

Moreover, the face map $d_1 : G \times X \rightarrow X$ is the projection onto the first coordinate, since the square is cartesian by definition:

$$\begin{array}{ccc} (G \ltimes X)_1 & \xrightarrow{d_1} & X \\ \downarrow & & \downarrow \\ G_1 & \xrightarrow{d_1} & G_0 \end{array}.$$

Similarly, the degeneracy map $s_0 : X \rightarrow G \times X$ is given by (e', id) where $e' : X \rightarrow \text{pt} \rightarrow G$ where pt is the terminal object and $e = s_0 : \text{pt} \rightarrow G_0$ the *neutral element*.

Notation. Given a groupoid action $\mathcal{G} \ltimes X$ of a groupoid $\mathcal{G}$ on an object X , the face map $d_0 : (\mathcal{G} \ltimes X)_1 \rightarrow (\mathcal{G} \ltimes X)_0$ determines a map

$$a : \mathcal{G}_1 \times_{\mathcal{G}_0} X \rightarrow X$$

which we will refer to as the *action map*.

Informally, we may think of a group action $G \ltimes X$ as an ordinary action $G \times X \rightarrow X$ that satisfies higher coherences up to homotopy. Following this idea, we now examine how to arrive at the notion of a quotient starting from the ordinary definition. Given an ordinary group acting on a set X , we usually define the quotient as:

$$\text{colim} \left(G \times X \begin{smallmatrix} \xrightarrow{a} \\ \rightrightarrows \\ \xrightarrow{s} \end{smallmatrix} X \right) \simeq \text{coeq} \left(G \times X \begin{smallmatrix} \xrightarrow{a} \\ \rightrightarrows \\ \xrightarrow{s} \end{smallmatrix} X \right) \simeq X/G,$$

where a is the action map and s is the source projection. We observe that the colimit above is taken over the simplicial diagram of Remark F.2.9 truncated at the first two degrees. In the spirit of retaining the full structure of the action, it is natural to expect that in the ∞ -categorical setting we should not truncate the diagram, but instead take the colimit of the entire simplicial object. Heuristically, this motivates the following definition.

Definition F.2.10. Let $\mathcal{G} \ltimes X$ be an action of a groupoid object on X in an ∞ -topos $\mathcal{B}$ then we define the *quotient* $X//\mathcal{G}$ to be the geometric realization $|\mathcal{G} \ltimes X|$ of $\mathcal{G} \ltimes X$.

The composite:

$$\text{Act}_{\mathcal{G}}(\mathcal{B}) \subseteq \mathbf{Grpd}(\mathcal{B})_{/\mathcal{G}} \subseteq \text{Fun}([1], \mathbf{Grpd}(\mathcal{B})) \subseteq \text{Fun}([1], \text{Fun}(\Delta^{op}, \mathcal{B})) \xrightarrow{\text{Fun}([1], \text{colim}(-))} \text{Fun}([1], \mathcal{B})$$

defines a functor $-//\mathcal{G} : \text{Act}_{\mathcal{G}}(\mathcal{B}) \rightarrow \mathcal{B}_{/\mathbb{B}\mathcal{G}}$, sending the map $\mathcal{G} \ltimes X \rightarrow \mathcal{G}$ to the induced map $X//\mathcal{G} \rightarrow \mathbb{B}\mathcal{G}$.

Proposition F.2.11 (Classification of groupoid actions cf. [16, Proposition E.20]). *The quotient functor $-//\mathcal{G}$ is an equivalence of ∞ -categories:*

$$-//\mathcal{G} : \text{Act}_{/\mathcal{G}}(\mathcal{B}) \xrightarrow{\sim} \mathcal{B}_{/\mathbb{B}\mathcal{G}}$$

Free and transitive actions

Definition F.2.12 (cf. [54, Definition 3.2.89]). Let $\mathcal{G} \ltimes X$ be an action of a groupoid object G on an object X in the fixed ∞ -topos $\mathcal{B}$. We define the *shear map* of $\mathcal{G} \ltimes X$ as the map:

$$\text{shear}_1 = (a, \text{pr}_X) : \mathcal{G}_1 \times_{\mathcal{G}_0} X \rightarrow X \times X$$

where $a : \mathcal{G}_1 \times_{\mathcal{G}_0} X \rightarrow X$ is the action map and pr_X is the projection onto X .

Remark F.2.13. In [54, Definition 3.2.89] *higher* version of the shear map are defined.

Definition F.2.14 (cf. [54, Definition 3.2.91]). Let $\mathcal{B}$ be an ∞ -topos and let $\mathcal{G} \ltimes X$ be an action of a groupoid object $\mathcal{G}$ on an object $X \in \mathcal{B}$. we say that the action is:

- *free* if shear_1 is a monomorphism (i.e. $\mathcal{G}_1 \times_{\mathcal{G}_0} X \xrightarrow{\text{shear}_1} X \times X$)
- *transitive* if shear_1 is a effective epimorphism (i.e. $\mathcal{G}_1 \times_{\mathcal{G}_0} X \rightarrow X \times X$)
- *regular* if shear_1 is an equivalence (i.e. $\mathcal{G}_1 \times_{\mathcal{G}_0} X \xrightarrow[\sim]{\text{shear}_1} X \times X$).

We now state a result that justifies the intuitive idea of quotients discussed in the introduction, establishing a connection between ordinary quotients and those arising in the ∞ -categorical setting when the action is free.

Proposition F.2.15 (cf. [54, Proposition 3.2.94]). *Let $\mathcal{B}$ be an ∞ -topos, and let $\mathcal{G} \in \mathbf{Grp}(\mathcal{B})$. If $\mathcal{G} \ltimes X \in \text{Act}_{\mathcal{G}}(\mathcal{B})$ is a free action and the morphism to the terminal object $X \rightarrow \text{pt}$ is an effective epimorphism, then the quotient $X//\mathcal{G}$ coincides with the ordinary quotient of the 0-truncations:*

$$X//\mathcal{G} \simeq i_0(\tau_{\leq 0}(X)/\tau_{\leq 0}(\mathcal{G})),$$

where $\tau_{\leq 0}$ denotes the 0-truncation functor (see [37, Proposition 5.5.6.18]).

F.3 Sheaf topoi

In this section, we present our main example of an ∞ -topos, namely those arising as categories of sheaves. In Subsection F.3.1, we introduce sites and the corresponding categories of sheaves, and conclude with some useful properties. In Subsection F.3.2, we describe how to construct morphisms between sheaf categories starting from morphisms between the underlying sites. Finally, in Subsection F.3.3, for a presentable ∞ -category $\mathcal{D}$, we construct the functor $\text{Shv}(-, \mathcal{D})$ and use it to define $\mathcal{D}$ -structured spaces. The main references for this section are [16, Section E.4] and [50, Section 5.1.1].

F.3.1 Definitions and first properties

In ordinary category theory given a 1-categorical Grothendieck site $(\mathcal{C}, \tau)$ we usually want to consider presheaves over $\mathcal{C}$ that satisfy a certain gluing condition. More specifically, we say that a 1-presheaf $\mathcal{F} \in \text{Fun}(\mathcal{C}^{\text{op}}, \mathbf{Set})$ is a 1-sheaf, if for all objects X of $\mathcal{C}$ and for every covering $\mathcal{U} = \{g_i : U_i \rightarrow X\}$ in τ the comparison map in the following diagram :

$$\mathcal{F}(X) \longrightarrow \lim \left(\prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \text{Hom}_{\text{Fun}(\mathcal{C}^{\text{op}}, \mathbf{Set})} (y(U_i) \times_{y(X)} y(U_j), \mathcal{F}) \right)$$

is an equivalence. By means of Yoneda's Lemma the condition is usually rewritten as asking that the map:

$$\text{Hom}_{\text{Fun}(\mathcal{C}^{\text{op}}, \mathbf{Set})} (y(x), \mathcal{F}) \longrightarrow \text{Hom}_{\text{Fun}(\mathcal{C}^{\text{op}}, \mathbf{Set})} (h_{\mathcal{U}}, \mathcal{F})$$

is an equivalence, where:

$$h_{\mathcal{U}} = \text{coeq} (\prod_{i,j} y(U_i) \times_{y(X)} y(U_j) \rightrightarrows \prod_i y(U_i)).$$

Unwinding definitions $h_{\mathcal{U}}$ can be described as the set of maps $f : Y \rightarrow X$ that factors through some $g_i : U_i \rightarrow X$. In particular we can identify $h_{\mathcal{U}}$ as the full subcategory of $\mathcal{C}/X$ spanned by those object that has a morphism to one of the $g_i : U_i \rightarrow X$. Although more complicated, this point of view has the advantage of being easier to be generalized in the ∞ -categorical setting.

Definition F.3.1 (cf. [37, Definitio 6.2.2.1]). Let $\mathcal{C}$ be an ∞ -category. A *sieve* on $\mathcal{C}$ is a full subcategory $\mathcal{C}^{(0)} \subseteq \mathcal{C}$ having the property that if $f : X \rightarrow Y$ is a morphism in $\mathcal{C}$ and Y belongs to $\mathcal{C}^{(0)}$, then also $X \in \mathcal{C}^{(0)}$.

If $X \in \mathcal{C}$ is an object, a *sieve on* X is a sieve on the ∞ -category $\mathcal{C}/X$.

Given $f : X \rightarrow Y$ and a sieve $\mathcal{C}_{/Y}^{(0)} (= (\mathcal{C}/Y)^{(0)})$ on Y , we let $f^* \mathcal{C}_{/Y}^{(0)} \subseteq \mathcal{C}/X$ denote the subcategory spanned by those morphism $(g : Z \rightarrow X) \in \mathcal{C}/X$ such that $f \circ g \in \mathcal{C}_{/Y}^{(0)}$.

Definition F.3.2 (cf. [37, Definitio 6.2.2.1]). A *Grothendieck topology* on an ∞ -category $\mathcal{C}$ consists of a specification, for each object X in $\mathcal{C}$, of a collection of sieves on X which we will refer to as *covering sieves*. The collection of covering sieves is required to satisfy the following properties:

1. If X is an object of $\mathcal{C}$, then the sieve $\mathcal{C}_{/X} \subseteq \mathcal{C}_{/X}$ is covering sieve.
2. If $f : X \rightarrow Y$ is a morphism in $\mathcal{C}$ and $\mathcal{C}_{/Y}^{(0)}$ is covering sieve on Y , then $f^*\mathcal{C}_{/Y}^{(0)}$ is a covering sieve on X .
3. Let X be an object of $\mathcal{C}$, $\mathcal{C}_{/X}^{(0)}$ a covering sieve on X , and $\mathcal{C}_{/X}^{(1)}$ arbitrary sieve on X . Suppose that, for each $f : Y \rightarrow X$ in $\mathcal{C}_{/X}^{(0)}$, the pullback sieve $f^*\mathcal{C}_{/X}^{(1)}$ is a covering sieve of Y . Then $\mathcal{C}_{/X}^{(1)}$ is a covering sieve of X .

We call a *site* an ∞ -category equipped with a Grothendieck topology.

Remark F.3.3. It is proven in [37, Lemma 6.2.2.5] that the datum of a sieve $\mathcal{C}^0$ on $\mathcal{C}$ is equivalent to the datum of a subterminal object $R \in \text{PSh}(\mathcal{C})$. Informally the correspondence is given as follow:

$$\begin{aligned} \mathcal{C}^{(0)} &\mapsto \left(X \mapsto R(X) = \begin{cases} \text{pt} & \text{if } X \in \mathcal{C}^{(0)}, \\ \emptyset & \text{otherwise.} \end{cases} \right) \\ \{X \in \mathcal{C} \mid R(X) = \text{pt}\} &\leftarrow R. \end{aligned}$$

Since $\text{PSh}(\mathcal{C}_{/X}) \simeq \text{PSh}(\mathcal{C})_{/y(X)}$ we have that the datum of a sieve over X is the same as the datum of a monomorphism $R \hookrightarrow y(x)$.

We are now ready to define sheaves.

Definition F.3.4. Let $\mathcal{C}$ be a site with Grothendieck topology τ . Let S be the collection of all monomorphism $\{R \hookrightarrow y(X)\}$ which correspond to covering sieve on $\mathcal{C}$. An object $\mathcal{F} \in \text{PSh}(\mathcal{C})$ is called a τ -sheaf or simply a *sheaf* if it is S -local, i.e. for all $R \hookrightarrow y(X)$ covering sieve, the induced map of spaces:

$$\mathcal{F}(X) = \text{Hom}_{\text{PSh}(\mathcal{C})}(y(X), \mathcal{F}) \rightarrow \text{Hom}_{\text{PSh}(\mathcal{C})}(R, \mathcal{F})$$

is an equivalence. We will denote $\text{Shv}_\tau(\mathcal{C})$ the full subcategory of $\text{PSh}(\mathcal{C})$ spanned by ssheaves with respect to τ ; if the topology τ is clear from the context we will write $\text{Shv}(\mathcal{C})$ for $\text{Shv}_\tau(\mathcal{C})$.

Proposition F.3.5 (cf. [37, Proposition 6.2.2.7]). *The inclusion $\text{Shv}_\tau(\mathcal{C}) \subseteq \text{PSh}(\mathcal{C})$ admits a left exact accessible left adjoint $L_\tau : \text{PSh}(\mathcal{C}) \rightarrow \text{Shv}_\tau(\mathcal{C})$. In particular, $\text{Shv}_\tau(\mathcal{C})$ is an ∞ -topos.*

The functor L_τ is called *sheafification functor*.

Definition F.3.6 (Čech descent, [16, Definition E.35]). Let $\mathcal{C}$ be an ∞ -category and let $\mathcal{U} = \{U_i \rightarrow X\}_{i \in I}$ be a collections of morphism. denote with $\check{C}(\mathcal{U})$ the Čech nerve of the morphism

$$\coprod_{i \in I} y(U_i) \rightarrow y(X)$$

in $\text{PSh}(\mathcal{C})$. We say that $\mathcal{F} \in \text{PSh}(\mathcal{C})$ satisfy *Čech descent condition with respect to $\mathcal{U}$* if

$$\mathcal{F}(X) = \text{Hom}_{\text{PSh}(\mathcal{C})}(y(X), \mathcal{F}) \rightarrow \lim_{[n] \in \Delta^{op}} \text{Hom}_{\text{PSh}(\mathcal{C})}(\check{C}(\mathcal{U})_n, \mathcal{F})$$

is an equivalence.

Recalling the discussion at the beginning of the section, we now return to the classical idea of gluing local data, with the important caveat that we are no longer working in **Set**. In particular, we must now require that higher coherence conditions are satisfied. More explicitly, the diagram expressing the sheaf condition no longer terminates at pairwise intersections, but continues indefinitely, assigning data for all possible iterated intersections. Pictorially, assuming that $\mathcal{C}$ has pullbacks, the Čech descent condition can be represented as follows:

$$\mathcal{F}(X) \longrightarrow \prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \times_X U_j) \rightrightarrows \cdots$$

Proposition F.3.7 (cf. [37, Lemma 6.2.3.18]). *Let $\mathcal{U} = \{U_i \rightarrow X\}_{i \in I}$ be a collection of morphism in an ∞ -category $\mathcal{C}$ and let $R \hookrightarrow X$ be the covering sieve generated by $\mathcal{U}$. Then a presheaf $\mathcal{F}$ on $\mathcal{C}$ satisfies Čech descent with respect to $\mathcal{U}$ if and only if is local with respect to $R \hookrightarrow X$.*

The above discussion can be generalized in the case we take presheaf with value in any presentable ∞ -category. We are not going to rewrite the above statements in this appendix and instead we refer to [27] Appendix A.4.

We now continue by characterizing effective epimorphism in the case of sheaves on a site.

Definition F.3.8 (cf. [16, Definition E.37]). Let $\mathcal{C}$ be an ∞ -category equipped with a Grothendieck topology τ . Let $f : X \rightarrow Y$ be a morphism in $\mathrm{Shv}_\tau(\mathcal{C})$ and assume that $Y = y(C)$ lies in the image of the sheafified Yoneda functor $y : \mathcal{C} \rightarrow \mathrm{Shv}_\tau(\mathcal{C})$. We say that f *admits local sections* if there exists a covering family $\{U_i \rightarrow C\}_{i \in I}$ of $\mathcal{C}$ such that the base change $f' : X \times_Y y(U_i) \rightarrow y(U_i)$ of f along the map $y(U_i) \rightarrow Y$ admits a section for every $i \in I$. If $f : X \rightarrow Y$ is an arbitrary morphism in $\mathrm{Shv}_\tau(\mathcal{C})$, we say that f *admits local sections* if the base change along any map $y(C) \rightarrow Y$ from a representable admits a local sections.

Lemma F.3.9 (cf. [16, Lemma E.38]). *A morphism $f : \mathcal{X} \rightarrow \mathcal{Y}$ in $\mathrm{Shv}_\tau(\mathcal{C})$ is an effective epimorphism if and only if it admits local sections.*

We now state a key result allowing the extension of functors satisfying descent from a site to its topos of sheaves. We begin with a definition.

Definition F.3.10 (cf. [41, Definition 1.3.1.4]). Let $\mathcal{B}$ be an ∞ -topos and let $\mathcal{V}$ an ∞ -category. A $\mathcal{V}$ -valued sheaf on $\mathcal{B}$ is a functor $\mathcal{B}^{\mathrm{op}} \rightarrow \mathcal{V}$ which preserves small limits. We define the ∞ -category $\mathrm{Shv}(\mathcal{B}, \mathcal{V})$ of sheaves with value in $\mathcal{V}$ as the full subcategory ∞ -category of $\mathrm{Fun}(\mathcal{B}^{\mathrm{op}}, \mathcal{V})$ spanned by the $\mathcal{V}$ -valued sheaf.

Theorem F.3.11 (cf. [41, Proposition 1.3.1.7]). *Let $\mathcal{C}$ be a site with associated topos $\mathcal{B} = \mathrm{Shv}(\mathcal{C})$ and let $\mathcal{V}$ be a category that has small limits. Then precomposition with the functor $\mathcal{C} \rightarrow \mathcal{B}$ induces an equivalence of categories:*

$$\mathrm{Shv}(\mathcal{B}, \mathcal{V}) \rightarrow \mathrm{Shv}(\mathcal{C}, \mathcal{V})$$

We conclude the section by relating $\mathbb{T}$ -algebras in a sheaf category to the category of sheaves valued in $\mathbb{T}$ -algebras, where $\mathbb{T}$ is a Lawvere theory.

Lemma F.3.12. *Let $\mathcal{B}$ be an ∞ -topos and $\mathbb{T}$ be an algebraic theory. Then we have an equivalence:*

$$\mathbf{Alg}_{\mathbb{T}}(\mathcal{B}) \simeq \mathrm{Shv}(\mathcal{B}, \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc}))$$

Proof. By Theorem D.2.8 we have:

$$\mathbf{Alg}_{\mathbb{T}}(\mathcal{B}) \simeq \mathrm{Fun}^R(\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{\mathrm{op}}, \mathcal{B}).$$

By [36, Remark 1.1.2] we conclude that

$$\mathrm{Fun}^R(\mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})^{\mathrm{op}}, \mathcal{B}) \simeq \mathrm{Fun}^R(\mathcal{B}^{\mathrm{op}}, \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})).$$

□

Remark F.3.13. Putting Lemma F.3.12 and Theorem F.3.11 we find that for $(\mathcal{C}, \tau)$ a site:

$$\mathbf{Alg}_{\mathbb{T}}(\mathrm{Shv}_\tau(\mathcal{C})) \simeq \mathrm{Shv}_\tau(\mathcal{C}, \mathbf{Alg}_{\mathbb{T}}(\mathbf{Spc})).$$

F.3.2 Morphisms of sites

Definition F.3.14 (cf. [16, Definition E.39]). Let $(\mathcal{C}, \tau)$ and $(\mathcal{D}, \tau')$ be ∞ -categories equipped with Grothendieck topologies. A functor $u : \mathcal{C} \rightarrow \mathcal{D}$ is called *continuous* if the functor $\mathrm{PSh}(\mathcal{D}) \rightarrow \mathrm{PSh}(\mathcal{C})$ given by precomposition with u restricts to a functor

$$u^* : \mathrm{Shv}_{\tau'}(\mathcal{D}) \rightarrow \mathrm{Shv}_\tau(\mathcal{C}),$$

i.e. sends τ' -sheaves on $\mathcal{D}$ to τ -sheaves on $\mathcal{C}$.

Remark F.3.15. In the situation of the above definition we have that u_* admits a left adjoint

$$u_! : \mathrm{Shv}_\tau(\mathcal{C}) \rightarrow \mathrm{Shv}_{\tau'}(\mathcal{D})$$

given by the composite

$$\mathrm{Shv}_\tau(\mathcal{C}) \hookrightarrow \mathrm{PSh}(\mathcal{C}) \xrightarrow{\mathrm{Lan}_y u} \mathrm{PSh}(\mathcal{D}) \xrightarrow{L_{\tau'}} \mathrm{Shv}_{\tau'}(\mathcal{D}),$$

where $\mathrm{Lan}_y u$ denotes left Kan extension of u along the Yoneda embedding $y : \mathcal{C} \rightarrow \mathrm{PSh}(\mathcal{C})$.

Remark F.3.16 (cf [16, Remark E.40]). Assume that $\mathcal{C}$ and $\mathcal{D}$ admit pullbacks and that u preserves pullbacks. Then $u : \mathcal{C} \rightarrow \mathcal{D}$ is continuous if and only if u sends τ -covering sieves to τ' -covering sieves.

Definition F.3.17 (cf. [16, Defintion E.41]). Let $(\mathcal{C}, \tau)$ and $(\mathcal{D}, \tau')$ be sites. A functor $u : \mathcal{C} \rightarrow \mathcal{D}$ is called a *morphism of sites* $\mathcal{D} \rightarrow \mathcal{C}$ if it is a continuous functor and the pullback functor

$$u^! : \text{Shv}_\tau(\mathcal{C}) \rightarrow \text{Shv}_{\tau'}(\mathcal{D})$$

preserves finite limits.

Notice that if u is morphism of sites then u_* is a geometric morphism.

Definition F.3.18 (Cocontinuous functor between sites, [16, Definition E.43]). Let $(\mathcal{C}, \tau)$ and $(\mathcal{D}, \tau')$ be ∞ -categories equipped with Grothendieck topologies. A functor $\nu : \mathcal{C} \rightarrow \mathcal{D}$ is called *cocontinuous* if for every object $X \in \mathcal{C}$ and every covering sieve $\mathcal{D}_{/\nu(X)}^{(0)} \subseteq \mathcal{D}_{/\nu(X)}$ on $\nu(X) \in \mathcal{D}$, the pullback sieve

$$\nu^* \mathcal{D}_{/X}^{(0)} := \left\{ (\psi : Y \rightarrow X) \in \mathcal{C}_{/X} \mid (\nu(\psi) : \nu(Y) \rightarrow \nu(X)) \in \mathcal{D}_{/\nu(X)}^{(0)} \right\} \subseteq \mathcal{C}_{/X}$$

is a covering sieve on X .

Lemma F.3.19 ([16, Lemma E.44]). Let $\nu : (\mathcal{C}, \tau) \rightarrow (\mathcal{D}, \tau')$ be a functor between sites. Then ν is cocontinuous if and only if the right Kan extension functor

$$\text{PSh}(\mathcal{C}) \rightarrow \text{PSh}(\mathcal{D})$$

restricts to a functor

$$\nu_* : \text{Shv}_\tau(\mathcal{C}) \rightarrow \text{Shv}_{\tau'}(\mathcal{D}),$$

i.e., it sends τ -sheaves on $\mathcal{C}$ to τ' -sheaves on $\mathcal{D}$. In this case, ν_* admits a left exact left adjoint $\nu_\#$, so that ν_* is a geometric morphism of ∞ -topoi.

The functor $\nu_\#$ is defined as the composition of the following three left exact functors:

$$\text{Shv}_{\tau'}(\mathcal{D}) \hookrightarrow \text{PSh}(\mathcal{D}) \xrightarrow{- \circ \nu} \text{PSh}(\mathcal{C}) \xrightarrow{L_\tau} \text{Shv}_\tau(\mathcal{C}).$$

Remark F.3.20. In the case where ν is both continuous and cocontinuous, the lemma above yields a functor $\nu_\#$, which, by continuity, is equivalent to ν^* . In particular, we obtain the following triple adjunction:

$$\begin{array}{ccc} & \xrightarrow{\nu_!} & \\ \text{Shv}_\tau(\mathcal{C}) & \xleftarrow{\nu^*} & \text{Shv}_{\tau'}(\mathcal{D}) \\ & \xrightarrow{\nu_*} & \end{array}$$

Proposition F.3.21 ([16, Corollary E.47]). Let $\nu : (\mathcal{C}, \tau) \hookrightarrow (\mathcal{D}, \tau')$ be a functor between sites. Assume that ν is fully faithful and cocontinuous. Then the geometric morphism

$$\nu_* : \text{Shv}_\tau(\mathcal{C}) \rightarrow \text{Shv}_{\tau'}(\mathcal{D})$$

is fully faithful.

As a consequence of the above proposition and Remark F.3.20, we obtain the following corollary.

Corollary F.3.22. Let $v : (\mathcal{C}, \tau) \rightarrow (\mathcal{D}, \tau')$ be a fully faithful functor both continuous and cocontinuous, then $v_!$ is fully faithful.

F.3.3 The functor $\text{Shv}(-; \mathcal{D})$

Construction F.3.23 (The functor $\text{Shv}(-)$). Let **Top** be the site of topological spaces equipped with the open cover topology, and let

$$\mathbf{Top}_{/-} : \mathbf{Top}^{\text{op}} \rightarrow \mathbf{Site}$$

denote the straightening of the Cartesian fibration $\text{ev}_1 : \text{Fun}([1], \mathbf{Top}) \rightarrow \mathbf{Top}$ given by evaluation at 1. For every topological space $X \in \mathbf{Top}$, denote by $\text{Op}(X)$ the site of open subsets of X . Given a morphism $f : X \rightarrow Y$ in **Top**, we obtain an induced morphism of sites

$$\begin{array}{ccc}
\mathrm{Op}(Y) & \xrightarrow{f^{-1}} & \mathrm{Op}(X) \\
\downarrow & & \downarrow \\
\mathbf{Top}_Y & \longrightarrow & \mathbf{Top}_X
\end{array}$$

essentially by definition. Thanks to [53, proposition A.1] we get an induced functor:

$$\mathrm{Op}(-) : \mathbf{Top}^{\mathrm{op}} \rightarrow \mathbf{Site}$$

assigning to each space X the site $\mathrm{Op}(X)$. Next, consider the composite functor:

$$P : \mathbf{Site}^{\mathrm{op}} \rightarrow \mathrm{Cat}_{\infty}^{\mathrm{op}} \xrightarrow{\mathrm{Fun}(-, \mathbf{Spc})} \mathrm{Cat}_{\infty},$$

which assigns to a site (X, τ) its category of presheaves over X with values in spaces. We may now argue as before. More specifically, given a morphism of sites $f : (Y, \tau') \rightarrow (X, \tau)$ the induced morphism

$$\begin{array}{ccc}
\mathrm{Shv}_{\tau}(X) & \xrightarrow{\quad} & \mathrm{Shv}_{\tau'}(Y) \\
\downarrow & & \downarrow \\
P(X) & \longrightarrow & P(Y)
\end{array}$$

restricts to a morphism on sheaf categories by definition of morphism of sites. We thus obtain an induced functor:

$$\mathrm{Shv}(-) : \mathbf{Site}^{\mathrm{op}} \rightarrow \mathrm{Pr}^{\mathrm{R}}.$$

which assigns to each site (X, τ) the category of sheaves $\mathrm{Shv}_{\tau}(X)$, and to a morphism of sites $f : (Y, \tau') \rightarrow (X, \tau)$ the pushforward $f_* : \mathrm{Shv}_{\tau'}(Y) \rightarrow \mathrm{Shv}_{\tau}(X)$.

Recalling from Corollary B.3.9 the equivalence $\mathrm{Pr}^{\mathrm{R}} \simeq (\mathrm{Pr}^{\mathrm{L}})^{\mathrm{op}}$, we obtain a functor

$$\mathrm{Shv}(-)^* : \mathbf{Site} \rightarrow \mathrm{Pr}^{\mathrm{L}}$$

which assigns to each site (X, τ) the category of sheaves $\mathrm{Shv}_{\tau}(X)$, and to a morphism of sites $f : (Y, \tau') \rightarrow (X, \tau)$ the pullback $f^* : \mathrm{Shv}_{\tau'}(Y) \rightarrow \mathrm{Shv}_{\tau}(X)$.

Finally, given any presentable ∞ -category $\mathcal{D}$, we can form new functors via the tensor product:

$$\mathrm{Shv}(-, \mathcal{D}) : \mathbf{Site}^{\mathrm{op}} \rightarrow \mathrm{Pr}^{\mathrm{R}} \quad \text{and} \quad \mathrm{Shv}(-, \mathcal{D})^* : \mathbf{Site} \rightarrow \mathrm{Pr}^{\mathrm{L}}.$$

By postcomposing with the $(-)^{\mathrm{op}}$ functor, we obtain the associated functors:

$$\mathrm{Shv}(-, \mathcal{D})^{\mathrm{op}} : \mathbf{Site}^{\mathrm{op}} \rightarrow \mathrm{Cat}_{\infty} \quad \text{and} \quad (\mathrm{Shv}(-, \mathcal{D})^*)^{\mathrm{op}} : \mathbf{Site} \rightarrow \mathrm{Cat}_{\infty}.$$

Abusing notation we will denote the induced functors on $\mathbf{Top}$ in the same way. In the thesis we may refer to $\mathrm{Shv}(-, \mathcal{D})^{\mathrm{op}}$ as the *functor induced by the pushforward* and to $(\mathrm{Shv}(-, \mathcal{D})^*)^{\mathrm{op}}$ as the *functor induced by pullback*.

Remark F.3.24. Consider the cocartesian unstraightening

$$\mathbf{Top}_{\mathcal{D}} := \mathrm{Un}^{\mathrm{cc}}(\mathrm{Shv}(-, \mathcal{D})^{\mathrm{op}})$$

of the functor $\mathrm{Shv}(-, \mathcal{D})^{\mathrm{op}} : \mathbf{Top} \rightarrow (\mathrm{Pr}^{\mathrm{R}})^{\mathrm{op}}$, where $\mathcal{D}$ is a presentable ∞ -category. Since pullback functors are left adjoint to pushforward functors, it follows from [37, Corollary 5.2.2.5] that this cocartesian fibration is also a cartesian fibration. Informally, $\mathbf{Top}_{\mathcal{D}}$ may be described as the ∞ -category whose:

- objects are couple $(X, \mathcal{O}_X)$ where X is a topological space and $\mathcal{O}_X$ is a $\mathcal{D}$ -valued sheaf over X
- morphisms $(f, f_{\#}) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ consist of a continuous map $f : X \rightarrow Y$ and a morphism of sheaves $f_{\#} : \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X$.

Definition F.3.25. We will refer to $\mathbf{Top}_{\mathcal{D}}$ as the ∞ -category of $\mathcal{D}$ -structured spaces.

Remark F.3.26. Let $\mathcal{D}$ be a presentable ∞ -category, then there's an equivalence:

$$\mathbf{Top}_{\mathcal{D}} \simeq (\mathbf{Un}^{\mathrm{cc}}(\mathrm{Shv}(-, \mathcal{D})^*))^{\mathrm{op}}.$$

Construction F.3.27 (Global sections, cf. [50, Construction 5.1.5]). Let $\mathcal{D}$ be a presentable ∞ -category. Consider the pullback square:

$$\begin{array}{ccc} \mathrm{Shv}(*, \mathcal{D})^{\mathrm{op}} & \longrightarrow & \mathbf{Top}_{\mathcal{D}} \\ \downarrow & \lrcorner & \downarrow \\ * & \hookrightarrow & \mathbf{Top} \end{array}$$

Thanks to [4, Lemma 2.20], the top functor in the diagram admits a left adjoint, which we denote by $\mathcal{O}$. Explicitly, $\mathcal{O}$ assigns to a $\mathcal{D}$ -structured space $(X, \mathcal{O}_X)$ its global sections $\mathcal{O}_X(X)$.

Lemma F.3.28 (cf. [50, Lemma 5.2.4]). *For any presentable ∞ -category $\mathcal{D}$, the functor:*

$$\mathrm{Shv}(-, \mathcal{D}) : \mathbf{Top}^{\mathrm{op}} \rightarrow \mathbf{Pr}^{\mathrm{L}}$$

satisfies descent, where we are regarding the category of topological spaces as equipped with the usual open cover topology.

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